Forward and Inverse Problems in Piezoelectricity

Den Naturwissenschaftlichen Fakultäten
der Friedrich-Alexander Universität Erlangen-Nürnberg
zur
Erlangung des Doktorgrades

vorgelegt von
Dipl. Math. Tom Lahmer

aus Bonn-Bad Godesberg



Lehrstuhl für Sensorik, Universität Erlangen-Nürnberg.

Als Dissertation genehmigt
von den Naturwissenschaftlichen Fakultäten der Universität
Erlangen-Nürnberg

Tag der mündlichen Prüfung:	15. Mai 2008

Vorsitzender der Promotionskommission:	Prof. Dr. E. Bänsch
Erstberichterstatter:	Prof. Dr. B. Kaltenbacher
Zweitberichterstatter:	Prof. Dr. W. Borchers

Meinen Eltern

Abstract

For understanding and predicting the behavior of piezoelectric devices, efficient numerical simulation tools, in particular the finite element method, have been developed and are used widely. Time consuming and expensive experiments, necessary for developing new piezoelectric products, sensors and actuators, are avoided by numerically solving the mathematical formulation of the underlying physical model. This model consists of a coupling between two physical quantities, namely electric field and mechanical strain and is given by a set of partial differential equations with appropriate boundary and initial conditions. Well-posedness results concerning the solutions of the underlying PDEs are presented in this work for time-dependent, harmonic, and static computations where appropriate damping and loss mechanisms are taken into account. The accuracy of the simulation, however, relies sensitively on the material parameters governing the interaction of the physical quantities. Without the knowledge of the exact parameters no quantitative predictions can be made by numerical computations. So far, these parameters have been estimated by measurements proposed by the IEEE Standard or the European Norm CENELEC [1, 24] from well-defined test samples. The special shapes recommended there allow for simplifications, namely reduction to onedimensional problems, in the model. Explicit formulas allowing for parameter extraction from resonance characteristics and other measureable quantities are developed (cf. for loss-less models [1, 24] and [32, 60, 91, 138, 142] considering losses.). However, these results do not provide sufficiently precise information on the material coefficients. This can be seen by comparing three dimensional simulation results with parameters gained by the classical methods with measurements of the electrical impedance or mechanical displacement.

In order to overcome these insufficient exactness, efforts are invested in the inverse problem, namely the simulation-based parameter identification for piezoelectric materials. The research and methods developed in this thesis consider both the linear and the nonlinear case. For the latter functional dependencies of the materials properties from the field quantities are assumed, a model which in particular is suitable for large-signal driven resonators at high frequencies, e.g. in sonar applications. The inherent instability of the inverse problem is treated with special care by applying appropriate regularizing methods. An iterative multilevel algorithm based on modified Landweber methods is developed here. The algorithm extends ideas of [130] and shows to be very effective in combination with the detection of parameter curves as they occur in nonlinear applications. Convergence results in the case of noisy data, and regularizing properties are given for this algorithm. Sensitivity analyses and methods of optimal experiment design show the reliability of the identified parameters and give rules how to improve identification results effectively without remarkably increasing the computational effort.

Zusammenfassung

Seit geraumer Zeit werden numerische Simulationswerkzeuge eingesetzt, insbesondere die Finite Elemente Methode, um piezoelektrische Bauteile und deren Funktionsweisen genau zu verstehen und um deren Verhalten vorherzusagen. Zeitaufwändige und teure Experimente, die für die Entwicklung neuartiger piezoelektrischer Produkte, Sensoren und Aktoren, nötig sind, können durch die numerische Simulation des mathematischen Modells ersetzt werden. Jenes besteht aus der Koppelung zweier physikalischer Größen, des elektrischen Felds einerseits und der mechanischen Dehnung andererseits. Das mathematische Modell wird durch ein System von partiellen Differentialgleichungen mit entsprechenden Rand- und Anfangswerten beschrieben. Für jeweils den zeitabhängingen, harmonischen und statischen Fall werden in dieser Arbeit Existenz- und Eindeutigkeitsaussagen zu den Lösungen der zugrundeliegenden partiellen Differentialgleichungen formuliert. Passende Dämpfungs- und Verlustmechanismen werden dabei berücksichtigt. Die Genauigkeit der Simulation hängt jedoch sehr von den Materialparametern ab, welche das Zusammenspiel der physikalischen Größen steuern. Ohne eine genaue Kenntnis dieser Parameter können keine quantitativen Aussagen gemacht werden. Bislang wurden die Materialparameter nach Vorgaben des IEEE Standards und der europäischen Norm CENELEC [1, 24] aus Messungen an wohldefinierten Probekörpern bestimmt. Die vorgeschlagenen speziellen Geometrien erlauben vereinfachte eindimensionale Modelle. Explizite Formeln wurden entwickelt, welche es ermöglichen, die Parameter aus Resonanzerscheinungen und anderen messbaren Größen zu bestimmen, siehe [1, 24] und [32, 60, 91, 138, 142] für den verlustbehafteten Fall. Jedoch bieten deren Ergebnisse keine hinreichend exakten Informationen über die Materialparameter. Dies wird schnell verdeutlicht, wenn man dreidimensionale Simulationen mit den aus klassischen Methoden ermittelten Parametern durchführt und diese Ergebnisse mit gemessenen elektrischen Impedanzen und mechanischen Verschiebungen vergleicht.
Um diese unzureichende Genauigkeit zu verbessern, wird das dazugehörige inverse Problem, nämlich die simulationsbasierte Identifizierung der Materialparameter für das piezoelektrische Problem behandelt. Die Untersuchungen und Methoden, welche in dieser Arbeit entwickelt werden, berücksichtigen sowohl den linearen wie auch den nichtlinearen Fall. Im zweiten Fall wird von funktionalen Abhängigkeiten der Materialeigenschaften von den Feldgrößen ausgegangen, einem Modell, welches speziell für großsignalbetriebene Wandler in Hochfrequenzbereichen, z.B. sonaren Anwendungen, passend ist. Die inhärene Instabilität des inversen Problems wird mit spezieller Vorsicht behandelt, indem geeignete Regularisierungsmethoden angewandt werden. Ein iterativer Multilevelalgorithmus, welcher auf modifizierten Landweber Methoden basiert und die Ideen von [130] erweitert, zeigt sich als effektiv in Bezug auf die Charakterisierung von Parameterkurven, wie sie bei nichtlinearen Anwendungen auftreten. Für diesen Algorithmus werden Aussagen über die Konvergenz im Fall von verrauschten Daten und regularisierende Eigenschaften präsentiert. Sensitivitätsanalysen und Methoden der optimalen Versuchsplanung zeigen die Vertrauenswürdigkeit der ermittelten Parameter und geben Hinweise, wie die Identifizierbarkeit effektiv verbessert werden kann ohne den rechnerischen Aufwand merkbar zu erhöhen.

Acknowledgment

The thesis at hand is a resume of my work within the DFG Research group “Inverse Probleme in der Piezoelektrizität und ihren Anwendungen” under grant Ka 1778/1 at the Department of Sensor Technology, University of Erlangen-Nuremberg. It is an honor for me to write this acknowledgment now at the time when my thesis is about to finish.

There are several people whose efforts, suggestions, and impetuses are of indispensable value for my work.

In particular, I like to express my special gratitude to my first supervisor, Prof. Dr. Barbara Kaltenbacher for her patient guidance, encouragement, and excellent advice throughout this study. Especially with Barbara’s professional and methodical responsibility she could always point out ways how to tackle each open problem.

I also like to thank my secondary supervisor Prof. Dr. Wolfgang Borchers for his careful review of my work and valuable discussions during the finalization of my thesis.

I am also indebted to Prof. Dr. Reinhard Lerch for the opportunity to work at his chair, the Department of Sensor Technology. The wide-ranging subjects initiated by him made my stay at the institute an interesting scientific experience in sensor and actuator applications.

There are a lot of, no, all of my colleagues at the Department of Sensor Technology to whom I am very much indebted. Not only a friendly working atmosphere, but also the willingness to discuss arising scientific questions, in particular concerning my application piezoelectricity, resulted in a very fruitful scientific environment.

Last but not least, I want to thank my family who supported me mentally during all the years of education. Without their encouraging assistance I would certainly not have reached this point of my life. Thank you very much!

Finally, I would like to express my deepest gratitude for the constant support, understanding and love that I received from my prospective wife Daniela.

Contents

Preliminaries **i**
Abstract / Zusammenfassung i
Acknowledgment iii
Table of Contents iv
List of Figures vi
List of Tables vii
List of Symbols viii
Structure of this Thesis x
Direct and Inverse Problems in Mathematics xi
Achievements of this Thesis xii

1 The Forward Problem – Mathematical Aspects of the Piezoelectric Effect **1**
1.1 Introduction to Piezoelectricity 1
1.1.1 The Piezoelectric Effect and Fields of Applications 1
1.1.2 Historical Development of Piezoelectric Devices 2
1.1.3 Physical Background of the Piezoelectric Effect 4
1.1.4 Constitutive Equations 8
1.1.5 Energy Dissipation and Modeling of Losses 13
1.1.6 Piezoelectric Partial Differential Equations 15
1.1.7 Rotational Symmetric Case 18
1.1.8 Weak Form of Piezoelectric PDEs 19
1.2 Well-Posedness Results 20
1.2.1 Transient Case With Rayleigh-Damping 21
1.2.2 Harmonic and Static Case 33
1.3 Finite Element Modeling 36
1.3.1 Time Stepping Scheme for Transient Computations 39
1.3.2 Triangulation 40
1.4 Summary Chapter 1 43

2 The Inverse Problem – Iterative Regularization of Nonlinear Ill-Posed Problems **44**
2.1 Preliminaries on Regularization Theory 44
2.2 Regularizing Iterative Methods for Nonlinear Ill-Posed Problems . 47
2.2.1 Inexact Newton Methods 47
2.2.2 Nonlinear Landweber Iteration and Modifications 48
2.3 Modified Landweber Methods in an Iterative Multilevel Algorithm 49
2.3.1 The Multilevel Algorithm 53
2.3.2 Convergence Results 54
2.3.3 Regularization Property 61
2.3.4 Applicability to a Harmonic Identification Example 63
2.4 Summary Chapter 2 67

3 The Inverse Problem – Parameter Identification in Linear Piezoelectricity 69
3.1 Motivation . . . 69
3.2 Identification Methods - State of the Art . . . 70
3.3 PDE-Based Parameter Identification . . . 75
3.3.1 Ill-Posedness . . . 76
3.3.2 Computational Aspects . . . 77
3.3.3 Sensitivity Analysis . . . 87
3.3.4 Identification Results for a Newly Developed Ceramic . . . 89
3.3.5 Parameter Identification for Piezoelectric Composites . . . 92
3.4 Optimal Experiment Design . . . 95
3.4.1 Motivation and Approaches . . . 95
3.4.2 Implementation Issues . . . 100
3.4.3 Numerical Results . . . 102
3.5 Summary Chapter 3 . . . 104

4 Nonlinearities in Piezoelectricity – The Forward and Inverse Problem 105
4.1 Nonlinear Dependencies in Piezoelectricity . . . 105
4.2 The Nonlinear Forward Problem . . . 113
4.3 The Inverse Problem . . . 114
4.3.1 The Adjoint Operator . . . 115
4.3.2 The Degree of Ill-Posedness . . . 123
4.3.3 Numerical Results . . . 126
4.4 Summary Chapter 4 . . . 133

5 Summary and Outlook 134
5.1 Summary . . . 134
5.2 Outlook . . . 135
5.3 Zusammenfassung . . . 137
5.4 Ausblick . . . 138

References 140

List of Figures

1 Forward and inverse problems in mathematics xi
2 Piezoelectric effect 1
3 Discretization 3
4 Ferroelectric domains 4
5 Perovskite crystal structure 5
6 Heckmann diagram 9
7 Notation of axes 11
8 Computational domain, radial disc 16
9 Computational domain, thin bar 16
10 Spectrum of the piezoelectric system matrix 39
11 Simulation model of a piezoelectric stack actuator 42
12 Numerical simulation result of a piezoelectric stack actuator . . . 42
13 Convergence of modified Landweber iterations 49
14 Common resonators 72
15 Impedance analyser 73
16 Test fixture 74
17 Identification setup 76
18 Convergence of different regularization methods 78
19 Reconstruction with inexact Newton methods 80
20 Reconstruction with steepest descent 81
21 Reconstruction with the minimal error method 82
22 Reconstruction with the Landweber method 83
23 Simultaneous reconstruction of all parameters 84
24 Reconstruction of three imaginary parts, noise-free 85
25 Reconstruction of three imaginary parts, noisy 86
26 Reconstruction result with damping, magnitude 86
27 Reconstruction result with damping, phase 87
28 Identification of radial mode 90
29 Identification of longitudinal mode 91
30 Verification of identification results 92
31 Geometry of Langevin transducer 93
32 Fitting results (impedance) for Langevin transducer 93
33 Fitting results (impedance) for Langevin transducer 94
34 Fitting results (mechanical displacement) for Langevin transducer 94
35 Optimal experiment design - confidence intervals 99
36 Optimal experiment design - Number of measurements 100
37 Optimal experiment design - weight function 102
38 Optimal experiment design - confidence intervals 103
39 Nonlinear case: Amplitude saturation 106
40 Nonlinear case: Classification of nonlinearities 107
41 Nonlinear case: Effects of higher harmonics 108
42 Nonlinear case: Field dependend piezoelectric coupling 109

43 Nonlinear case: Hysteresis 111
44 Nonlinear case: Ferroelasticity 111
45 Nonlinear case: Parameter functions e_{33} and ε_{33}^{S} 126
46 Nonlinear case: Charge response 127
47 Nonlinear case: Convergence, reconstruction of permittivity . . . 128
48 Nonlinear case: Reconstructed parameter curve I 128
49 Nonlinear case: Reconstructed parameter curve II 129
50 Nonlinear case: Reconstructed parameter curve III 129
51 Nonlinear case: Simultaneous reconstruction, with data noise . . 130
52 Nonlinear case: Simultaneous reconstruction, noise free 131
53 Nonlinear case: Charge response after reconstruction, thickness resonator 132
54 Nonlinear case: Charge response after reconstruction, longitudinal resonator 132

List of Tables

1 Properties of selected materials 5
2 Physical couplings in smart materials 7
3 Conversion: Tensor to matrix notation 12
4 Overview: Well-posedness violations and their remedies 45
5 Confidence intervals of identified parameters, Pz36 88
6 Geometries of Pz36 samples 89
7 Material parameters, Pz36 according to IEEE standard 89
8 Fitting results, radial mode 90
9 Fitting results, longitudinal mode 90
10 Consistent data set for Pz36 91
11 Functional dependencies of piezoelectric material parameters . . . 109

List of Symbols

Physical quantities

Symbol	SI	Description
A	m^2	surface area
α	s^{-1}	mass damping coefficient for Rayleigh damping
β	s	stiffness damping coefficient for Rayleigh damping
C	F	capacity
$\vec{D}$	C/m^2	vector of dielectric displacement
d	m	diameter
$\vec{E}$	V/m	vector of electric field intensity
f	Hz	frequency
f_a	Hz	antiresonance frequency (zero reactance)
f_r	Hz	resonance frequency (zero susceptance)
f_p	Hz	parallel resonance frequency (maximum resistance)
f_s	Hz	series resonance frequency (maximum conductance)
f_m	Hz	frequency where impedance is maximal
f_n	Hz	frequency where impedance is minimal
G	J	Gibb's energy
H	J/m^3	electric enthalpy density
$\vec{H}_{\mathrm{mag}}$	A/m	magnetic displacement
l	m	length
ϕ	V	electric potential
ϕ_h	V	finite element solution of ϕ
ϕ_m	V	nodal vector of electric potential
ϕ^e	V	electric excitation
q^e	$\mathrm{C} = \mathrm{As}/\mathrm{m}^2$	electric charge density at surface Γ_e
$\vec{S}$	$(\delta l/l)$ dimensionless	vector of mechanical strains
$\vec{\sigma}$	N/m^2	vector of mechanical stress
$\delta\tau$	K	temperature change
t	s	time
Θ	K	absolute temperature
$\mathbf{u}$	m	vector of mechanical displacement
$\mathbf{u}_h$	m	element solution of $\mathbf{u}$
$\mathbf{u}_m$	m	nodal vector of mechanical displacement
$\dot{\mathbf{u}}$	m/s	velocity
$\ddot{\mathbf{u}}$	m/s^2	acceleration
w	m	width
ω	s^{-1}	angular frequency
Y	$S = \Omega^{-1}$	electrical admittance
Z	Ω	electrical impedance

Material parameters

Symbol	SI	Description
$\alpha^{S,E}$	$\mathrm{N}/(\mathrm{m}^2\mathrm{K})$	thermal stress (under const. strain and elec. field)
c	$\mathrm{J}/(\mathrm{m}^3\mathrm{K})$	volumetric specific heat capacity
$\mathbf{c}^E$	N/m^2	modulus of elasticity (under constant electrical field)
$\mathbf{d}$	C/N or m/V	piezoelectric strain tensor
$\mathbf{e}$	C/m^2	piezoelectric stress tensor
$\boldsymbol{\varepsilon}^S$	F/m	permittivity under constant strain
ε_0		electric constant or vacuum permittivity
$\mathbf{g}$	m^2/C	piezoelectric voltage tensor
$\mathbf{h}$	NC^{-1}	piezoelectric stress tensor
k_t		thickness coupling factor
$\mathbf{p}^{S,E}$	$\mathrm{As}/(\mathrm{m}^{\mathrm{K}})$	pyroelectric coefficient (under const. strain and elec. field)
ρ	kg/m^3	density of material
$\mathbf{s}$	m^2/N	compliance, inverse of stiffness

Mathematical terms

$\mathcal{B}$	two or three dimensional displacement to strain differential operator
∇	gradient
$\mathbb{C}$	field of complex numbers
$\nabla\cdot$	divergence
δ	data error level
F	parameter-to-solution map
Γ_e	surface covered by loaded electrode
Γ_g	surface with grounded electrode
Γ_r	parts of boundary without any electrode
$\Im()$	imaginary part of a complex number
$\mathbb{N}$	field of natural numbers
$\vec{n}$	normal unit vector
Ω	computational domain
Ω_e	finite element
p	sought for quantity
$\Re()$	real part of a complex number
$\mathbb{R}$	field of real numbers
S_h	finite element space
y^δ	measurements with data noise δ

Function Spaces

All spaces are defined on the domain Ω:

$$
\begin{aligned}
\mathcal{C}(\Omega) &:= \{u : \Omega \to \mathbb{C} \mid u \text{ is continuous}\} \\
\mathcal{C}^k(\Omega) &:= \{u : \Omega \to \mathbb{C} \mid u \text{ is } k\text{-times continuously differentiable}\} \\
\mathcal{C}^\infty(\Omega) &:= \{u : \Omega \to \mathbb{C} \mid u \text{ is infinitely differentiable}\} \\
L^2(\Omega) &:= \{u : \Omega \to \mathbb{C} \mid u \text{ is Lebesgue measurable}\} \\
L^\infty(\Omega) &:= \{u : \Omega \to \mathbb{C} \mid \text{ess sup } |\mathrm{u}| < \infty\} \\
H_{0,\Gamma}(\Omega) &:= \overline{\{w \in \mathcal{C}^\infty \ : \ w_{\Gamma_e} = w_{\Gamma_g} = 0\}}^{H^1} \\
H^1(\Omega) &:= \{u : \Omega \to \mathbb{C} \mid ||u||_{L^2} + ||\nabla u||_{L^2} < \infty\} \\
H_0^1(\Omega) &:= \{u : \Omega \to \mathbb{C} \mid ||u||_{L^2} + ||\nabla u||_{L^2} < \infty \ \wedge \ u = 0 \text{ on } \partial\Omega\} \\
H_{\mathcal{B}}^1(\Omega) &:= \overline{(C^\infty)^3}^{H_{\mathcal{B}}^1} \text{ with } ||\mathbf{v}||_{H_{\mathcal{B}}^1} := ||\mathbf{v}||_{L^2} + ||\mathcal{B}\mathbf{v}||_{L^2} \\
H^{-1}(\Omega) &:= \{f| \ \sup\{\langle f, u\rangle, u \in H_0^1(\Omega), ||u||_{H_0^1(\Omega)} \leq 1\} < \infty\} \\
H_{\mathcal{B}}^{-1}(\Omega) &:= \{f \mid \sup\{\langle f, u\rangle, \ u \in H_{\mathcal{B}}^1(\Omega), \ ||u||_{H_{\mathcal{B}}^1(\Omega)} \leq 1\} < \infty\} \\
L^\infty(0,T;X) &:= \{u : \ [0,T] \to X \mid \text{ess } \sup_{0\leq \mathrm{t} \leq \mathrm{T}} ||\mathrm{u}||_{\mathrm{X}} < \infty\} \\
L^2(0,T;X) &:= \{u : \ [0,T] \to X \mid \left(\int_0^T ||u||_X^2\right)^{\frac{1}{2}} < \infty\} \\
H^1(0,T;X) &:= \{u : \ [0,T] \to X \mid \left(\int_0^T ||u||_X^2 + ||\dot{u}||_X^2\right)^{\frac{1}{2}} < \infty\}
\end{aligned}
$$

Here $\mathcal{B}$ is the three dimensional mechanical strain to displacement differential operator as defined in (1.32) and X is a Banach space. Whenever appropriate we will consider the real valued version of these function spaces and make use of the same notation.

Structure of this Thesis

After an introduction into the theory of piezoelectricity the mathematical forward problem is derived which comes together with well-posedness theorems including the consideration of losses and energy dissipation by appropriate damping terms. The well-posedness results are indispensable in order to treat the inverse problem, here the parameter identification, correctly. A description of the finite element method applied to piezoelectric partial differential equations (PDEs) follows. Preliminaries of the theory of ill-posed problems together with a short review of existing iterative, regularizing schemes are presented in Chapter 2 giving approved numerical methods appropriate for solving parameter identification problems or more general nonlinear ill-posed problems.

A theoretical investigation of a newly developed iterative multilevel algorithm including modified Landweber iterations is performed in detail. Convergence of

this algorithm and regularizing properties are shown. Finally, the assumptions made on the nonlinearity are verified for a harmonic parameter identification example.

Chapter 3 is devoted to the parameter identification problem for linear piezoelectricity providing reconstruction results for both synthetically created data and for data from measurements performed at the Institute of Sensor Technology in Erlangen. The application of methods from optimal experiment design improves identification results by an optimal steering of frequencies at which measurements are evaluated.

In Chapter 4 both the forward and inverse problems for the case of piezoelectric nonlinearities are treated. The nonlinear problem is modelled by functional dependencies of the parameters from the physical fields. The derivation of this model is motivated by a thermodynamical approach using higher order terms which describe nonlinear effects for moderate fields. Further nonlinear physical dependencies in piezoelectricity are shortly described. In the final Chapter 4 the iterative multilevel algorithm discussed in Chapter 2 is applied to the identification of parameter curves in nonlinear piezoelectrics. Numerical results show the effectiveness of this multilevel algorithm.

Direct and Inverse Problems in Mathematics

During this thesis we repeatedly speak of the direct (forward) and inverse problem in piezoelectricity. This must not be confused with the direct and indirect (converse) piezoelectric effect which distinguishes between the direction of energy conversion in a piezoelectric material. The direct or forward problem in mathematics consists of a model, e.g. a partial differential or integral equation plus a set of parameters from which a solution is computed.

$$Forward\ Problem: \quad \underbrace{\text{Mathematical Model} + \text{parameters}}_{given} = \underbrace{\text{solution}}_{sought}$$

$$Inverse\ Problem: \quad \underbrace{\text{Mathematical Model}}_{given} + \underbrace{\text{parameters}}_{sought} = \underbrace{\text{solution}}_{given\ with\ noise}$$

Figure 1: Forward and inverse problems in mathematics

Whereas in the inverse problem one ideally knows the mathematical model together with a solution, usually measurements, and computes the parameters from them (or even parts of the model). While the so-called forward problems are in general stable problems, i.e. a continuous dependency of the solution on the input data can be ensured, this is not the case for inverse problems, where in particular error components in the data may be amplified arbitrarily strongly.

Achievements of this Thesis

+ Well-posedness results of the forward problem in case of viscous damping for the transient and the harmonic case.
+ Convergence results for modified Landweber iterations embedded into an iterative multilevel algorithm.
+ Regularization properties of this iterative multilevel algorithms.
+ Verification of the essential convergence condition for parameter estimations in a harmonic partial differential equation.
+ Simulation based characterization of elastic, dielectric, and piezoelectric moduli for small signal applications; verification by means of synthetic data.
+ Application to real world experimental data; test cases ranging from typical probes to compound transducers.
+ Improvement of measurement frequency selection by means of optimal experiment design.
+ Application of the iterative multilevel algorithm to the case of parameter curve identification for nonlinear piezoelectric applications.

1 The Forward Problem – Mathematical Aspects of the Piezoelectric Effect

We begin with the mathematical forward problem which occurs in piezoelectric applications, i.e. the derivation, solvability, and solutions of sets of partial differential equations modeling linear piezoelectricity.

1.1 Introduction to Piezoelectricity

The following sections give a brief overview over the piezoelectric effect, its fields of applications, historical development and the physical effects beyond it. Out of thermodynamical considerations appropriate constitutive equations will be derived, followed by the derivation of partial differential equations where a special focus lies in the modeling of damping terms.

1.1.1 The Piezoelectric Effect and Fields of Applications

Piezoelectric transducers convert electrical signals into mechanical ones and vice versa.

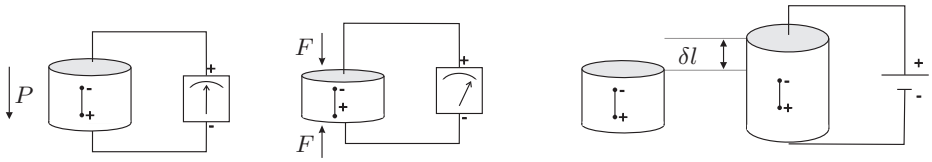


Figure 2: Piezoelectric effect. Left pictures: Direct piezoelectric effect, i.e. the generation of an electrical signal by a mechanical force. Right pictures: Converse piezoelectric effect, i.e. shape deformation due to an applied voltage or charge.

So, the term piezoelectricity (Greek: $\pi\iota\varepsilon\zeta\varepsilon\iota\nu$ = *piezein* which means to squeeze or press) is endowed with two effects: The direct effect on one hand, i.e. the conversion of a mechanical force into an electric signal which is typical for sensor applications (e.g. force and acceleration sensors). On the other hand there is the indirect or inverse piezoelectric effect, i.e. the mechanical excitation by application of an electric field (actuator applications, e.g. ultrasound generation, stack actuators) [21, 22]. The piezoelectric effect can only be observed in materials that have a non-centrosymmetric crystal structure see e.g. [134]. It occurs in

- Inorganic material
  - naturally occurring monocrystals (quartz, Seignette salt, tourmaline)
  - synthetic ceramics (lead zirconate titanate, barium titanate, lithium titanate, lithium tantalate, sodium tungstate)
- Organic material (polyvinilydene di-fluoride), biological tissues (hair and bones).

Regarding the application one could sort the electromechanical coupling phenomena as it is done in [134]

- Sensor applications - using the direct effect, e.g. pressure sensors
- Actuator applications - using the indirect effect, e.g. stack actuators, high precision positioning devices
- Sonar applications - using both effects, e.g. hydrophones, ultrasonic applications in pulse echo mode
- Energy conversion - conversion of mechanical energy into electrical energy, e.g. high voltage generator, piezotransformers, energy harvesters on vibrating systems.

For further fields of applications, see [135].

The remainder of this chapter is structured as follows: After a brief overview of the historic development starting with the discovery of the piezoelectric effect and ending with some recently developed industrial applications, the physical background of the piezoelectric phenomenon is in the focus. Then, the constitutive equations for the linear case are derived from thermodynamic potentials. We direct in particular our attention on the incorporation of damping mechanisms in order to treat losses and energy dissipation in the ceramics. With Newton's, Gauss', and Faraday's law we set up systems of differential equations for static, transient, and harmonic computations. Well-posedness results are established for the different analysis types considering damping terms. The finite element method adapted to the piezoelectric differential equations is shortly introduced in order to provide a basis for the numerical treatment of the forward problem.

1.1.2 Historical Development of Piezoelectric Devices

The first scientific paper dealing with piezoelectric phenomena and crystallographic structure was published by the brothers Pierre and Jacques Curie in 1880 [30]. They discovered a surface charge on specially prepared crystals like tourmaline, quartz, cane sugar, and Seignette salt induced by an applied mechanical stresses. They had found the direct piezoelectric effect, namely the measurable electricity from applied stress. The inverse effect however, strains due to applied voltages or charges, remained for the time unknown to them. It was Lippman in 1881 who deduced this property by studying fundamental thermodynamic principles [99]. Among others Voigt seemed to be the scientist whose contributions to the piezoelectric formulation are the most important ones [148]. The following development of scientific work with piezoelectric materials may be divided into two major parts. The first one lasted from the discovery of the piezoelectric

effect until the 1940s. In this period manufacturers exclusively worked with "natural" piezoelectric materials. One of the first serious applications was introduced by the French scientist Langevin who developed an ultrasonic submarine detector [99]. This transducer consists of thin quartz crystals glued between two steel plates which by their mass reduced the resonance frequency to some 50 kHz. It transmitted high frequency underwater signals. By measuring the time of the returning echo one detected underwater objects or the water depth.
The second important period in the development of piezoelectric transducers is the time starting from the beginning of World War II where in Japan and in the Soviet Union groups of scientists developed capacitor materials with much higher dielectric constants. This group of materials belongs to the class of ferroelectrics. Among them were the barium titanate family and later the lead zirconate titanate family which both are still commonly applied materials [117].



Figure 3: Pierre Curie (upper right) with his brother Jacques and their parents, from [112].

In the sequel, industrial groups in the U.S. among others developed the following applications: Ceramic phone cartridges, hydrophones, powerful sonar applications, small microphones, ceramic audio tone transducer, relays, and many others. Strong patents followed hoping for high benefits. Due the fact that a lot of research on this subject was conducted during wartime and in post war times, the development of new materials was still extremely difficult. Most of the groups stayed on their own and avoided knowledge transfer, [117] since reproduction was only a question of simple technical steps but development a time and money consuming procedure.

In the 1960s Japanese scientists using results of the rather competitive cooperative Barium Titanate Application Research Committee (founded in 1951) invented numbers of commercial applications, like signal filters, piezo ceramic igniter, audio buzzers, air ultrasonic transducers (e.g. used in medical application, both in diagnostics and therapy), or surface acoustic wave filters. In the recent years solid state motion, strong and fast movers for precise steering were fields of intensive research.

Nowadays active fields of research are concerned with the development of piezoelectric compounds, e.g. stack actuators which are devices consisting of several hundreds of thin piezoelectric layers. Due to their structure stack actuators allow for rather large deflections which can be used to steer injection valves in common rail diesel engines [135]. The actuality of this research is shown by the fact that the joint project by Bosch and Siemens VDO "Piezo-Injektoren: Neue Technik für saubere und sparsame Diesel- und Benzinmotoren" was awarded the “Deutschen Zukunftspreis 2005” in Berlin by the German Federal President Horst Köhler [16]. Examples for the latest application trends are actively noise absorb-

ing windows, where emerging soundwaves are compensated by artificial vibrations induced by embedded piezoelectric patches [39], and piezo-mechatronic systems embedded in skis damping vibrations during downhill ski runs on icy tracks [143]. For further reading the author refers to [21,22,66] and the webpages [65,117,118].

1.1.3 Physical Background of the Piezoelectric Effect

As already mentioned piezoelectricity is the capability of certain materials to develop electric charges on their mounted electrodes when beeing mechanically excited. On the other hand, an applied electric field results in a proportional strain. Within the 32 different crystal classes there are 20 with a non centrosymmetric structure which show a pyroelectric effect, i.e. they are able to generate an electrical potential by a change of temperature. In this subset there are ten classes showing a spontaneous polarization which makes them piezoelectric. If an external field can change the direction of the polarization the material belongs to the group of ferroelectrics [96, 134]. Below the Curie temperature, ferroelectric mate-

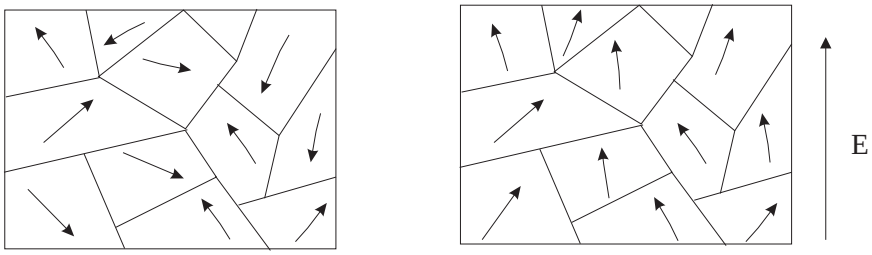


Figure 4: Domains of an unpoled piezoceramic material (left) and after poling (right). The figure is based on a source in [134].

rials consist of small regions, so called domains, which have uniform polarization. Within an unpoled material the net polarization is zero due to the stochastically distributed polarization of all domains. In this state no piezoelectric effect is present. Applying a strong electric field, the polarizations of these domains tend to line up with the field direction. Additional heating to below the Curie temperature of the material can speed up this process. An almost uniform polarization can be achieved. While removing the external electric field a few domains switch back to some random state of polarization. The majority does not, a remanent polarization remains, see Figure 4. The ceramic is said to be poled and can now be used as a piezoelectric device. An applied voltage with the same polarity as the poling voltage causes expansion along the poling axis and -by volume conservation- a contraction perpendicular to it. The minimum electric field in opposite direction that switches the polarization is called the coercive field. When a polarized piezoelectric device is heated to temperature ranges which exceed its Curie temperature, the polarizations gets lost. Above the Curie temperature the structure of the unit cells are cubic, below either tetragonal or rhomboetric. The polycrystaline materi-

als exhibit a perovskite structure, i.e. a chemical structural formula ABO_3 in which A is a big cation (Ba^{+2} or Pb^{+2}), B is a smaller cation (Ti^{+4} or Zr^{+4}) and O is oxygen. The ions are arranged as depicted in Figure 5.

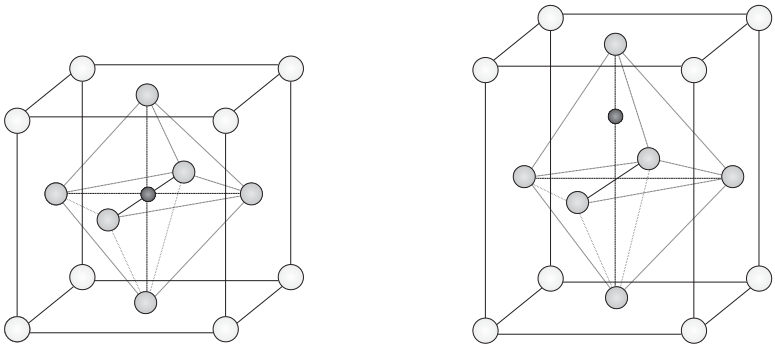


Figure 5: The perovskite structure. The left cubic structure arises above the Curie temperature, the right tetragonal structure forms a dipole by spontaneous polarization. The eight outer ions are either barium or lead. In the interior there are six oxygen and either one titanate or zirconate ion. Figure after a source in [141].

Different properties of piezoelectric materials and their fields of applications are intimately connected with each other. Naturally occurring crystals and polymers show a rather weak piezoelectric effect. They are only useable for sensor applications. Synthetically created ceramics, typically endowed with a high coupling coefficient d, are well suited materials for actuators due to their comparatively high induced strokes. In Table 1 properties of a set of piezoelectric materials are given. High permittivities $\varepsilon_{33}^{\sigma}$ result in high capacitances which require more complex

Material	$\varepsilon_{33}^{\sigma,r}$	tan δ (%)	$d_{33}(10^{-12}\,\mathrm{C/N})$	Curie T_c °C
Alkaline Niobate $(K_{0.5}Na_{0.5})NbO_3$	400	0.04	70	
Bismuth Titanate Niobate Bi_3TiNbO_9			5	940
Lithium Niobate $LiNbO_3$	84		6	1210
Tourmaline	8.2		1.83	
α-Quartz	41.03	2	d_{11} = 2.31	573
Lead Zirconate Titanate				
Pz21	3800	1.8	640	215
Pz26	1300	0.3	290	330
Pz27	1800	1.7	425	350
Pz36	610	0.3	230	350

Table 1: Physical properties of selected materials [7, 37, 134] which mainly determine the fields of application of piezoelectric sensors and actuators.

driving electronics. The quantity $\tan\ \delta$ denotes the ratio between power loss and reactive power generated. Since it is a measure of losses and electrical dissipation it is advisable to have materials with low $\tan\ \delta$. A high d_{33} guarantees high strokes in three direction with electric field in three direction. The Curie temperature gives an upper limit for the operating temperature of the materials.

Output / Input field	Strain	Charge	Magnetic field	Temperature	Light
Stress	Elasticity	Piezoelectricity	Inverse magnetostriction	Piezocaloric	Photoelasticity
Electrical field	Inverse Piezoelectricity	Permittivity	Electromagnetic effect	Electrocaloric effect	Electrooptic effect
Induction	Magnetostriction	Magnetoelectric effect	Permeability	Magnetocaloric effect	Magnetooptic effect
Heat	Thermal expansion	Pyroelectricity	Pyromagnetic effect	Specific heat cpapacity	Irradiation
Light	Photostriction	Photovoltaic effect		Solar heating	Refractive index

Table 2: Coupling effects between different physical fields, wherein the grey fileds are related to this thesis. Off-diagonal coupling is often referred to as the main property of smart materials which find their application in sensor and actuator technologies [44, 96, 116, 141].

1.1.4 Constitutive Equations

The material law which describes the piezoelectric effect in the linear (reversible) case valid in the situation of small mechanical deformations and electric fields is given for instance by [1, 97, 98, 141] and including thermal effects by [45, 136]

$$\begin{aligned} \sigma_{ij} &= \underbrace{c_{ijkl}^{E,\Theta} S_{kl}}_{elastic} - \underbrace{(e_{ijk}^{\Theta})^T E_k}_{converse\ piezo.} - \underbrace{\alpha_{ij}^{E} \Delta\tau}_{thermal\ stresses} \\ D_i &= \underbrace{e_{ijk}^{\Theta} S_{jk}}_{direct\ piezo.} + \underbrace{\varepsilon_{ij}^{S,\Theta} E_j}_{dielectric} + \underbrace{p_i^S \Delta\tau}_{pyroelectric} \\ \eta &= \underbrace{(\alpha_{ij}^{E})^T S_{ij}}_{piezocaloric} + \underbrace{(p_i^S)^T E_i}_{electrocaloric} + \underbrace{\frac{c^{S,E}}{\tau}\Delta\tau}_{heat\ capacity} . \end{aligned} \tag{1.1}$$

Here, the mechanical stress tensor σ_{ij}, the dielectric displacement D_i and the entropy η are related to the mechanical strain S_{kl} the electric field E_j and the temperature change $\Delta\tau$, respectively. The representation chosen uses Einstein's summation convention where repeated indices are automatically summed. The first equation in (1.1) can be seen as an extension of Hooke's law whereas the second line extends the electrostatic material law by a coupling to the other physical fields. The material tensors are

- $c_{ijkl}^{E,\Theta}$ - the modulus of elasticity $\mathrm{N/m^2}$
- e_{ijk}^{Θ} - the piezoelectric coupling $\mathrm{N/Vm}$
- $\varepsilon_{ij}^{S,\Theta}$ - the permittivity $\mathrm{C/(Vm)}$.
- α_{ij}^{E} - thermoelastic stress constants $\mathrm{N/(m^2K)}$
- p_i^S - pyroelectric constants $\mathrm{C/(m^2K)}$
- $c^{S,E}$ - volumetric specific heat capacity $\mathrm{J/(m^3K)}$.

The entries of the tensors $\mathbf{c}^{E,\Theta}, \mathbf{e}$ and $\boldsymbol{\varepsilon}^{S,\Theta}$ fulfill the following symmetry and positivity conditions

$$c_{ijkl}^{E,\Theta} = c_{jikl}^{E,\Theta} = c_{klij}^{E,\Theta}, \quad e_{ijk} = e_{jik}, \quad \varepsilon_{ij}^{S,\Theta} = \varepsilon_{ji}^{S,\Theta} \tag{1.2}$$

for all i, j, k, l $(1 \le i, j, k, l \le 3)$. There exist nonnegative constants c_0, ε_0 such that

$$c_{ijkl}^{E,\Theta} X_{ij} X_{kl} \ge c_0 X_{ij}^2, \quad \varepsilon_{ij}^{S,\Theta} y_i y_j \ge \varepsilon_0 y_i^2. \tag{1.3}$$

Superscripts at the parameters denote the physical field quantity which is held constant while one determines the parameter. The phenomenological theory of the piezoelectric effect is derived from thermodynamical considerations according to Lord Kelvin. In the most general setting piezoelectric interactions are related to

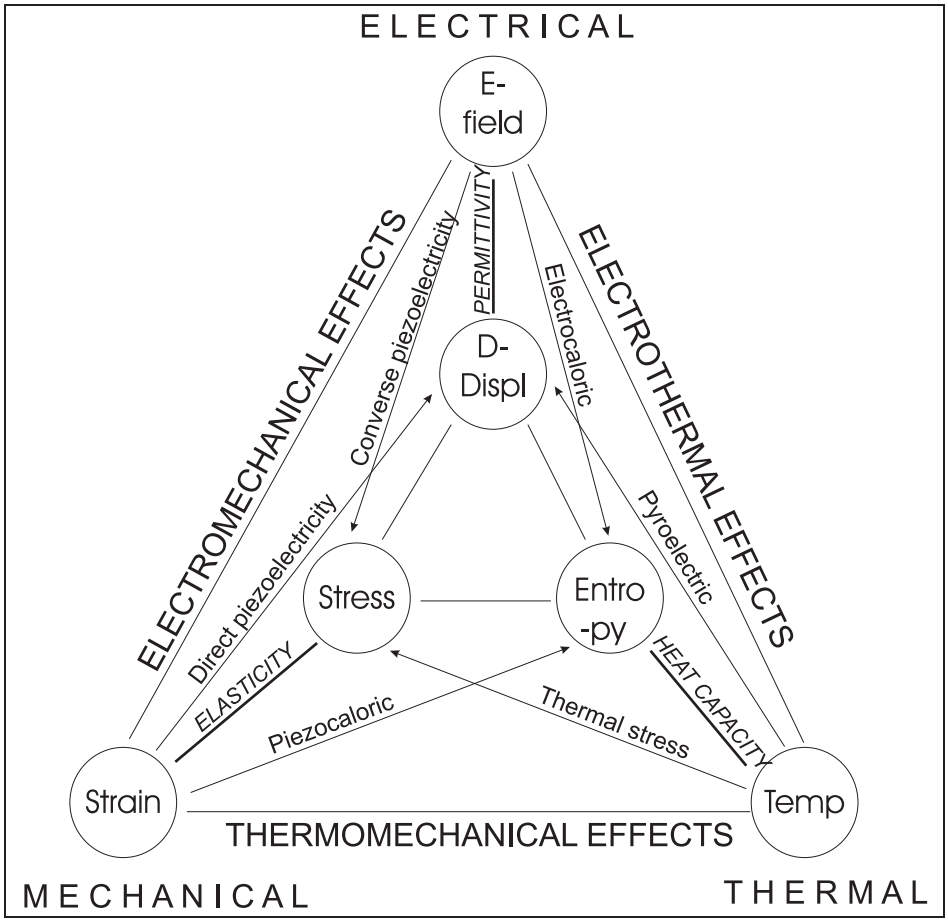


Figure 6: The Heckmann Diagram (1925) relating mechanical, electrical, and thermal variables. Intensive variables are located in the outer nodes, corresponding conjugate extensive variables form the inner triangle, see e.g. [141].

the phenomena of pyroelectric and piezocaloric effects which are due to thermal-electric-mechanical-couplings as it is depicted in Figure 6.

Since in the application under consideration these thermal coupling effects (not to be mistaken with temperature dependent material parameters) do not take effect, the representation will be restricted to pure adiabatic processes and thermal dependencies of material parameters are taken into account without explictly denoting them. In particular in the chapters concerned with the identification of material parameters the piezoelectric properties shall be evaluated at defined surroundings which include ambient temperature. Measurements have to be conducted carefully in order to avoid self warming or external heating treatment.

The different choices of independent variables, here strain and electric field, yield varying formulations for appropriate thermodynamical potentials whose min-

ima characterize equilibrium states for the systems (see, e.g. [141] for comprehensive tables relating the different choices of independent variables to the thermodynamical potentials). For the representation chosen here, the electric enthalpy density H, a measure of the energy of a thermodynamical system, is appropriate [1, 98]

$$H \;=\; U - E_i D_i, \quad i = 1, 2, 3. \tag{1.4}$$

With U the internal energy is denoted. According to the First Law of Thermodynamics stating that an increase in the internal energy of a thermodynamic system is equal to the work done by the system on the surroundings plus the amount of heat energy added to the system one has

$$dU = dW + dQ.$$

The work dW in piezoelectricity is given by [98],

$$dW = \sigma_{ij} dS_{ij} + E_i dD_i, \tag{1.5}$$

i.e. it is the sum of work per unit volume done by a stress σ_{ij} and the electric work caused by an electric field E_i. Then

$$dU = \sigma_{ij} dS_{ij} + E_i dD_i + dQ.$$

For an infinitesimal change of H it holds

$$dH = \sigma_{ij} dS_{ij} - D_i dE_i + dQ. \tag{1.6}$$

From here on solely adiabatic processes, i.e. $dQ = 0$ are considered. For works treating additionally thermal effects we refer to [9, 20, 45]. From (1.6) the two dependend variables mechanical stress σ_{ij} and dielectric displacement D_i are given by first order derivatives of H

$$\begin{aligned} \sigma_{ij} &= \left.\frac{\partial H}{\partial S_{ij}}\right|_E \quad \text{and} \\ D_i &= -\left.\frac{\partial H}{\partial E_i}\right|_S \end{aligned} \tag{1.7}$$

at constant electric field and strain, respectively. Considering the functional dependencies $\sigma_{ij}(S_{kl}, E_k)$ and $D_i(S_{kl}, E_k)$, developing them into Taylor series and truncating the result after the first term one obtains

$$\begin{aligned} \sigma_{ij} &= \sigma_{ij}(S^0_{kl}, E^0_k) + \left.\frac{\partial \sigma_{ij}}{\partial S_{kl}}\right|_E (S_{kl} - S^0_{kl}) + \left.\frac{\partial \sigma_{ij}}{\partial E_k}\right|_S (E_k - E^0_k) \\ D_i &= D_i(S^0_{kl}, E^0_k) + \left.\frac{\partial D_i}{\partial S_{kl}}\right|_E (S_{kl} - S^0_{kl}) + \left.\frac{\partial D_i}{\partial E_k}\right|_S (E_k - E^0_k) \end{aligned} \tag{1.8}$$

for the linear case. The derivation of the nonlinear case, i.e. the consideration of higher order terms, is postponed to chapter 4. Now, with (1.7) and without loss of generality setting $S^0_{kl} = 0$ and $E^0_k = 0$ as well as $\sigma_{ij}(S^0_{kl}, E^0_k) = 0$ and $D_i(S^0_{kl}, E^0_k) = 0$ one gets

$$\begin{aligned} \sigma_{ij} &= \left.\frac{\partial^2 H}{\partial S_{ij}\partial S_{kl}}\right|_E S_{kl} + \frac{\partial^2 H}{\partial E_k \partial S_{ij}} E_k, \\ D_i &= -\frac{\partial^2 H}{\partial E_i \partial S_{kl}} S_{kl} - \left.\frac{\partial^2 H}{\partial E_i \partial E_k}\right|_S E_k. \end{aligned} \tag{1.9}$$

Out of this it is apparent that the mechanical modulus, the piezoelectric coupling and permittivity tensors are given by second order derivatives of H, i.e.

$$\begin{aligned} c^E_{ijkl} &:= \left.\frac{\partial^2 H}{\partial S_{ij}\partial S_{kl}}\right|_E, \\ e_{kij} &:= -\frac{\partial^2 H}{\partial E_k \partial S_{ij}}, \\ \varepsilon^S_{ik} &:= -\left.\frac{\partial^2 H}{\partial E_i \partial E_k}\right|_S. \end{aligned} \tag{1.10}$$

Finally, the electric enthalpy density has the following form

$$H = \frac{1}{2} c^E_{ijkl} S_{ij} S_{kl} - e_{kij} E_k S_{ij} - \frac{1}{2}\varepsilon_{ij} E_i E_j. \tag{1.11}$$

The stored energy function U (see. e.g. [1]) can be determined by (1.4), (1.1) and (1.11) resulting in

$$U = \frac{1}{2} c^E_{ijkl} S_{ij} S_{kl} + \frac{1}{2}\varepsilon^E_{ij} E_i E_j. \tag{1.12}$$

The assumed positive definiteness of U imposes the positivity conditions (1.3) on the tensors $\mathbf{c}^E$ and $\boldsymbol{\varepsilon}^S$.

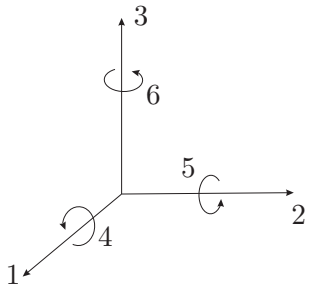


Figure 7: Notation of axes

Due to the existing symmetry the strain and stress tensor have only six different entries. Therefore a six component notation for stress and strain is suitable. Furthermore, according to the crystalline structure of piezoelectric materials a certain

ij or kl	p or q
11	1
22	2
33	3
23 or 32	4
13 or 31	5
12 or 21	6

Table 3: Conversion: Tensor to matrix notation

symmetry and sparsity in the material tensors is present. Therefore the elasticity modulus can be reduced by two dimensions. The following matrix notation (according to Voigt [80, 148]) is obtained by replacing indices ij or kl by p or q according to Table 3. The left one of the two subscripts always denotes the direction of the resulting field where the right one says in which direction the field operates. E.g. c_{13}^E symbolizes that at constant electrical field the mechanical stress has direction 1 and the mechanical strain has direction 3. Numbers $4, 5, 6$ denote shearings around the axes $1, 2$ and 3, see Figure 7. Thus the adiabatic version of (1.1) in matrix notation reads equivalently as

$$\begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \\ D_1 \\ D_2 \\ D_3 \end{pmatrix} = \begin{pmatrix} c_{11}^E & c_{12}^E & c_{13}^E & . & . & . & . & . & -e_{13} \\ c_{12}^E & c_{11}^E & c_{13}^E & . & . & . & . & . & -e_{13} \\ c_{13}^E & c_{13}^E & c_{33}^E & . & . & . & . & . & -e_{33} \\ . & . & . & c_{44}^E & . & . & . & -e_{15} & . \\ . & . & . & . & c_{44}^E & . & -e_{15} & . & . \\ . & . & . & . & . & c_{66}^E & . & . & . \\ . & . & . & . & e_{15} & . & \varepsilon_{11}^S & . & . \\ . & . & . & e_{15} & . & . & . & \varepsilon_{11}^S & . \\ e_{13} & e_{13} & e_{33} & . & . & . & . & . & \varepsilon_{33}^S \end{pmatrix} \begin{pmatrix} S_1 \\ S_2 \\ S_3 \\ S_4 \\ S_5 \\ S_6 \\ E_1 \\ E_2 \\ E_3 \end{pmatrix}, \tag{1.13}$$

where $c_{66}^E = \frac{1}{2}(c_{11}^E + c_{12}^E)$, or more compactly written

$$\vec{\sigma} = \mathbf{c}^E \vec{S} - \mathbf{e}^T \vec{E} \tag{1.14a}$$

$$\vec{D} = \mathbf{e}\vec{S} + \boldsymbol{\varepsilon}^S \vec{E}. \tag{1.14b}$$

There is a set of equivalent equations which describe the interaction of stress and strain as well as dielectric displacement and electric field

$$\begin{aligned} \vec{\sigma} &= \mathbf{c}^D \vec{S} - \mathbf{h}^t \vec{D} \\ \vec{E} &= -\mathbf{h}\vec{S} + \beta^S \vec{D}, \end{aligned} \tag{1.15}$$

$$\begin{aligned} \vec{S} &= \mathbf{s}^E \vec{\sigma} + \mathbf{d}^{\mathbf{T}} \vec{E} \\ \vec{D} &= \mathbf{d}\vec{\sigma} + \varepsilon^{\mathbf{T}} \vec{E}, \end{aligned} \tag{1.16}$$

$$\begin{aligned} \vec{S} &= \mathbf{s}^D \vec{\sigma} + \mathbf{g}^T \vec{D} \\ \vec{E} &= -\mathbf{g}^T \vec{\sigma} + \beta^T \vec{D}. \end{aligned} \tag{1.17}$$

Depending on the coupling coefficient occurring in the piezoelectric equations (1.14)-(1.17) we speak of the $\mathbf{e}, \mathbf{h}, \mathbf{d}$ or $\mathbf{g}$-form. The appropriate thermodynamical potentials for these representations are electric enthalpy density, Helmholtz free energy, Gibb's free energy, and the elastic Gibb's energy [141]. In this thesis and in general for most finite element implementations mainly the $\mathbf{e}$-form (1.14) is used since it can directly be plugged into Navier's equation of motion and Gauß's law from electromagnetics. For one dimensional parameter identification processes as explained in the standards (see Chapter 3.2) one rather works with the $\mathbf{d}$-form due to the direct accessibility of most of the parameters by analytic formulae. The other representations are only used in special cases where due to certain boundary conditions some of the variables vanish. All occurring material tensors in (1.14), (1.15), (1.16), and (1.17) can be transformed into each other by algebraic means, see e.g. [1].

Let us conclude this section with remarks on dependencies on other physical field quantities: In piezoelectric theory one usually neglects electromagnetic effects since phase velocities of piezoelectric waves are approximately five orders of magnitude less than the velocities of electromagnetic waves [1]. Piezoelectrics are polarizable but not magnetizable dielectrics. The only perceptible interaction with piezoelectricity comprises thermal effects as already mentioned and illustrated in Figure 6. It is mentioned once more, that this work is restricted to adiabatic processes, thus the lower right edge of the triangle in Figure 6 is not considered. However, heat generation is a severe reason for power dissipation. This dissipation can partially be modelled by complex-valued material tensors used in this work which will be introduced in the next section.

1.1.5 Energy Dissipation and Modeling of Losses

In general, energy conversion in piezoelectricity is never perfect, since losses occur at different parts of the system. In structural engineering viscous, velocity depending damping is a rather difficult task to visualize. Just for a few structures a finite set of damping elements exist where real viscous damping properties can be measured and be modelled correctly [29]. Numerical models of vibrating structures should be capable of accounting for different sources of energy dissipation, e.g. through nonlinear restoring forces, energy radiation and damping within the structure [51]. In this work, we will concentrate on the latter using the well-known Rayleigh model from pure elasticity problems and apply it to the piezoelectric problem [51, 62, 97, 103]. This mass and stiffness proportional viscous damping is widespreadly used in numerics avoiding the need to build damping matrices on physical properties [29].

In harmonically driven applications however in order to account for this dissipation of power we assume the material parameters to be complex (see [15, 32, 58]). With this more general approach our mathematical model describes mechanical relaxation, imperfect piezoelectric energy conversion, and dielectric dissipation. Loss

mechanisms, primarily mechanical attenuation and dielectric loss greatly impact a device's efficiency, maximum drive-level (thermal considerations) and resonant characteristics, i.e. bandwidth [119]. In particular, for a transducer which is typically driven at its electrical resonance, the dielectric loss often results in a significant increase in the internal losses accompanied by an increase of internal heat generation. In the sequel, complex material tensors are introduced as follows:

$$\mathbf{c}^E := \Re(\mathbf{c}^E) + j\Im(\mathbf{c}^E), \tag{1.18}$$

$$\mathbf{e} := \Re(\mathbf{e}) + j\Im(\mathbf{e}), \tag{1.19}$$

$$\boldsymbol{\varepsilon}^S := \Re(\boldsymbol{\varepsilon}^S) + j\Im(\boldsymbol{\varepsilon}^S), \tag{1.20}$$

where $\Re$ and $\Im$ denote the real and imaginary part of a complex number, respectively. The instantaneous influx of energy according to [15] is given by

$$\dot{W} = \frac{\partial}{\partial t}\int_\Omega \left(\frac{1}{2}\rho\dot{\mathbf{u}}^T\dot{\overline{\mathbf{u}}} + U\right) d\Omega \tag{1.21}$$

within the piezoelectric domain Ω. Together with the first law of thermodynamics and assuming that the volume Ω is arbitrary one obtains for the energy flux density

$$V = \rho\dot{\mathbf{u}}^T\ddot{\overline{\mathbf{u}}} + \vec{\sigma}^T\dot{\vec{S}} + \vec{E}^T\dot{\vec{D}}, \tag{1.22}$$

where ρ denotes the mass density ($\mathrm{kg/m^3}$), $\mathbf{u}$ is the mechanical displacement $\mathbf{u} = \mathbf{u}(\mathbf{x}, t)$, $\mathbf{x} = (x, y, z)$, and $\dot{\mathbf{u}}$ the velocity. The latter, the irreversible part of energy flux after performing a Fourier transformation and averaging over one period is

$$V_{\text{av}} = \frac{1}{2}\Re\left(-j\rho\omega^3\hat{\mathbf{u}}^T\overline{\hat{\mathbf{u}}} - j\omega\vec{\sigma}^T\vec{S} - j\omega\vec{E}^T\vec{D}\right), \tag{1.23}$$

where ω denotes the angular frequency. Since (1.23) describes the losses, V_{av} must be negative. This is however true, when the density of losses

$$V_l = \frac{1}{2}\omega\Im\left(\vec{\sigma}^T\vec{S} + \vec{E}^T\vec{D}\right) \leq 0. \tag{1.24}$$

The latter is actually the same condition Holland [58] shows in his work arguing that the complete power dissipation density in piezoelectric materials for harmonic applications can be expressed by the negative real part of the divergence of a generalized Poynting vector. Inserting now the constitutive equations from (1.14a) and (1.14b) in (1.24) one may express V_l in terms of the vectors $\vec{E}$ and $\vec{S}$ only and obtains

$$V_l = -(\frac{1}{2}\omega)\Im(\vec{S}^T\mathbf{c}^E\vec{S} + \vec{E}^T\boldsymbol{\varepsilon}^S\vec{E}). \tag{1.25}$$

Together with

$$F := (\vec{S}, \vec{E}), \text{ and } M := -\Im\begin{pmatrix} \mathbf{c}^E & 0 \\ 0 & \boldsymbol{\varepsilon}^S \end{pmatrix}$$

this gives

$$V_l = (\frac{1}{2}\omega)F^T\Im(M)F. \tag{1.26}$$

Now, as argued in [15, 58] V_l is nonnegative and since the components of F can be arbitrary $-\Im(M)$ must be nonnegative definite, which is equivalent to

$$\begin{aligned} &-\Im(\mathbf{c}^E) \text{ nonnegative definite} \\ &-\Im(\boldsymbol{\varepsilon}^S) \text{ nonnegative definite .} \end{aligned} \tag{1.27}$$

Since the piezoelectric coupling tensor $\mathbf{e}$ does not occur in (1.25) no restrictions on the piezoelectric coupling tensors can be given by thermodynamical considerations.

The connection between the Rayleigh model and the usage of complex-valued material parameters will be shown at the end of Section 1.1.6. Additionally, at this point we will formulate a transient model derived from the harmonical one using complex-valued material parameters and obtain a system which has now viscous dampers for all parts of the system, i.e. also for the piezoelectric coupling and the pure electric part.

1.1.6 Piezoelectric Partial Differential Equations (PDEs) and Boundary Conditions

In this chapter the constitutive laws discussed so far will be extended to full three dimensional sets of partial differential equations. Newton's law of motion for the mechanical behavior is

$$\mathcal{B}^T\vec{\sigma} = \rho\frac{\partial^2\mathbf{u}}{\partial t^2}. \tag{1.28}$$

Due to the fact that piezoelectric materials are insulators, Gauss' law gives

$$\nabla\cdot\vec{D} = 0. \tag{1.29}$$

Since temporal changes of the magnetic field are negligible, by Faraday's law, the electric field is the negative gradient of the electric potential [44]

$$\vec{E} = -\nabla\phi \tag{1.30}$$

with $\nabla := (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})$ and by linearized elasticity the strain is the spatial variation of the mechanical displacement $\mathbf{u}$

$$\vec{S} = \mathcal{B}\mathbf{u} \tag{1.31}$$

with the three dimensional differential operator relating mechanical strains to mechanical displacements

$$\mathcal{B} := \begin{pmatrix} \frac{\partial}{\partial x} & \cdot & \cdot & \cdot & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \cdot & \frac{\partial}{\partial y} & \cdot & \frac{\partial}{\partial z} & \cdot & \frac{\partial}{\partial x} \\ \cdot & \cdot & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & \cdot \end{pmatrix}^T . \tag{1.32}$$

With this one arrives at a system of four partial differential equations in the domain Ω where we additionally consider a Rayleigh damping model with positive factors α and β

$$\begin{aligned} \rho\frac{\partial^2\mathbf{u}}{\partial t^2} + \alpha\rho\frac{\partial\mathbf{u}}{\partial t} - \mathcal{B}^T\Big(\mathbf{c}^E\mathcal{B}\mathbf{u} + \beta\mathbf{c}^E\frac{\partial\mathcal{B}\mathbf{u}}{\partial t} + \mathbf{e}^T\nabla\phi\Big) &= 0 \quad \text{in } \Omega \\ \nabla\cdot\Big(\mathbf{e}\mathcal{B}\mathbf{u} - \boldsymbol{\varepsilon}^S\nabla\phi\Big) &= 0 \quad \text{in } \Omega\,. \end{aligned} \tag{1.33}$$

Here, additional terms containing first order time derivatives model energy dissipation within the structure. [51, 124].

The experimental setting requires to model vanishing normal stresses at the boundary and two electrodes (see Figure 8 and 9) being applied at opposite positions Γ_g and Γ_e of Ω. One of them is loaded with a prescribed potential ϕ^e, the other one is grounded. Further, on the parts of the boundary which are not covered

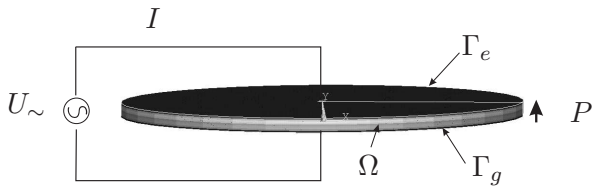


Figure 8: Piezoelectric disc. Notation of boundaries. This shape is for example used to simulate radial or thickness modes. The direction of polarization is marked with P.

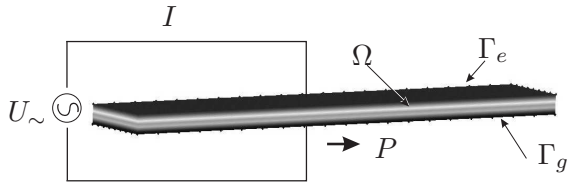


Figure 9: Piezoelectric bar. This geometry is considered while studying shear modes.

by any electrode there shall be no free charge. Together, this gives the following boundary conditions

$$\mathbf{N}^T\vec{\sigma} = 0 \quad \text{on } \partial\Omega \tag{1.34a}$$
$$\phi = 0 \quad \text{on } \Gamma_g \tag{1.34b}$$
$$\phi = \phi^e \quad \text{on } \Gamma_e \tag{1.34c}$$
$$\vec{D}\cdot\vec{n} = 0 \quad \text{on } \partial\Omega\setminus(\Gamma_e\cup\Gamma_g) \tag{1.34d}$$

where

$$\mathbf{N} = \begin{pmatrix} n_x & . & . & . & n_z & n_y \\ . & n_y & . & n_z & . & n_x \\ . & . & n_z & n_y & n_x & . \end{pmatrix}^T$$

and $\vec{n} = (n_x, n_y, n_z)$ is the outer unit normal vector. Additionally, appropriate initial conditions for an evolution analysis are given by

$$\begin{aligned} \mathbf{u}(\cdot, 0) &= \mathbf{u}_0 \\ \dot{\mathbf{u}}(\cdot, 0) &= \mathbf{u}_1. \end{aligned} \tag{1.35}$$

Since in case of constant coefficients all measurements are performed in frequency domain, the time harmonic version (excitation with periodic forcing function and response at circular frequency $\omega = 2\pi f$) of (1.1.6) will be provided here. Upon application of the Fourier transform $\hat{\mathbf{u}} \stackrel{\mathcal{F}}{=} \mathbf{u}$, $\hat{\phi} \stackrel{\mathcal{F}}{=} \phi$ the unknowns change to harmonic complex variables

$$\mathbf{u}(x,t) = e^{-i\omega t}\hat{\mathbf{u}}(x), \quad \phi(x,t) = e^{-i\omega t}\hat{\phi}(x).$$

Thus, (1.33) reads now as

$$\begin{aligned} -\rho\omega^2\hat{\mathbf{u}} + j\omega\alpha(\omega)\rho\hat{\mathbf{u}} - \mathcal{B}^T\left(\mathbf{c}^E\mathcal{B}\hat{\mathbf{u}} + j\omega\beta(\omega)\mathbf{c}^E\mathcal{B}\hat{\mathbf{u}} + \mathbf{e}^T\nabla\hat{\phi}\right) &= 0 \quad \text{in } \Omega \\ \nabla\cdot\left(\mathbf{e}\mathcal{B}\hat{\mathbf{u}} - \boldsymbol{\varepsilon}^S\nabla\hat{\phi}\right) &= 0 \quad \text{in } \Omega, \end{aligned} \tag{1.36}$$

with the same boundary conditions as in (1.34) for the Fourier transformed displacement $\hat{\mathbf{u}} = \hat{\mathbf{u}}(\mathbf{x}, \omega)$ and the electric potential $\hat{\phi} = \hat{\phi}(\mathbf{x}, \omega)$. The Rayleigh coefficients α and β in (1.36) however are functions of the frequency, i.e.

$$\alpha(\omega) = \alpha_0\omega, \quad \text{and} \quad \beta(\omega) = \frac{\beta_0}{\omega}.$$

By multiplying both lines in (1.36) with $\frac{1}{1-j\alpha_0}$ and defining the frequency dependend complex coefficients

$$\tilde{\mathbf{c}}^E := \frac{1+j\beta_0}{1-j\alpha_0}\mathbf{c}^E, \quad \tilde{\mathbf{e}} := \frac{1}{1-j\alpha_0}\mathbf{e}, \quad \tilde{\boldsymbol{\varepsilon}}^S := \frac{1}{1-j\alpha_0}\boldsymbol{\varepsilon}^S \tag{1.37}$$

the system in (1.36) transforms to the form

$$\begin{aligned} -\rho\omega^2\hat{\mathbf{u}} - \mathcal{B}^T\left(\tilde{\mathbf{c}}^E\mathcal{B}\hat{\mathbf{u}} + \tilde{\mathbf{e}}^T\nabla\hat{\phi}\right) &= 0 \quad \text{in } \Omega \\ \nabla\cdot\left(\tilde{\mathbf{e}}\mathcal{B}\hat{\mathbf{u}} - \tilde{\boldsymbol{\varepsilon}}^S\nabla\hat{\phi}\right) &= 0 \quad \text{in } \Omega. \end{aligned} \tag{1.38}$$

As discussed in Section 1.1.5 the energy dissipation is now completely described by using complex-valued material parameters. The choice of the coefficients in

(1.37) reveals that the Rayleigh model can be interpreted as a special case of complex material parameters.

Retransforming (1.38) to time domain gives a damped system of equations where also for the pure electrostatic part and the piezoelectric coupling energy dissipation is taken into account

$$\begin{aligned} \rho\frac{\partial^2\mathbf{u}}{\partial t^2} - \mathcal{B}^T\Big(\Re(\mathbf{c}^E)\mathcal{B}\mathbf{u} - \frac{1}{\omega}\Im(\mathbf{c}^E)\frac{\partial\mathbf{u}}{\partial t} + \Re(\mathbf{e})^T\nabla\phi - \frac{1}{\omega}\Im(\mathbf{e})^T\nabla\frac{\partial\phi}{\partial t}\Big) &= 0 \\ \nabla\cdot\Big(\Re(\mathbf{e})\mathcal{B}\mathbf{u} - \frac{1}{\omega}\Im(\mathbf{e})\mathcal{B}\frac{\partial\mathbf{u}}{\partial t} - \Re(\boldsymbol{\varepsilon}^S)\nabla\phi + \frac{1}{\omega}\Im(\boldsymbol{\varepsilon}^S)\nabla\frac{\partial\phi}{\partial t}\Big) &= 0 \\ &\text{in}\quad \Omega. \end{aligned} \tag{1.39}$$

The system (1.39) seems to be the more flexible model as compared to (1.36) since it takes the energy dissipation at different parts of the system into account.

In case of zero frequency (1.38) becomes the static piezoelectric system of equations

$$\begin{aligned} \mathcal{B}^T\Big(\mathbf{c}^E\mathcal{B}\mathbf{u} + \mathbf{e}^T\nabla\phi\Big) &= 0 \quad \text{in } \Omega \\ \nabla\cdot\Big(\mathbf{e}\mathcal{B}\mathbf{u} - \boldsymbol{\varepsilon}^S\nabla\phi\Big) &= 0 \quad \text{in } \Omega \end{aligned} \tag{1.40}$$

with $\mathbf{c}^E, \mathbf{e}, \boldsymbol{\varepsilon}^S$ either in $\mathbb{R}$ or $\mathbb{C}$.

1.1.7 Rotational Symmetric Case

Since many piezoelectric devices possess a circular or cylindrical shape, rotational symmetric implementations are advisable in order to reduce the computational effort drastically by reducing the space dimension from three to two. From the cylindrical coordinates (r, z, γ) with r denoting the radial, z the axial and γ the circumferential component, the latter will be set to zero, accordingly the derivative with respect to γ, i.e. $\gamma = 0$ and $\partial/\partial\gamma = 0$. Using the results of exercise A.4 in Chapter 4 of [113] the strain mechanical displacement relation changes to

$$\begin{aligned} &S_{rr} = \frac{\partial u_r}{\partial r}\,, \quad S_{\gamma\gamma} = \frac{u_r}{r}\,, \quad S_{zz} = \frac{\partial u_z}{\partial z}\,, \\ &S_{r\gamma} = S_{z\gamma} = 0\,, \quad S_{rz} = \frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}\,. \end{aligned} \tag{1.41}$$

The material law becomes

$$\begin{pmatrix} \sigma_{rr} \\ \sigma_{zz} \\ \sigma_{rz} \\ \sigma_{\gamma\gamma} \\ D_r \\ D_z \end{pmatrix} = \begin{pmatrix} c_{11}^E & c_{13}^E & . & c_{12}^E & . & -e_{31} \\ c_{13}^E & c_{33}^E & . & c_{13}^E & . & -e_{33} \\ . & . & c_{44}^E & . & -e_{15} & . \\ c_{12}^E & c_{13}^E & . & c_{11}^E & . & -e_{31} \\ . & . & e_{15} & . & \varepsilon_{11}^S & . \\ e_{13} & e_{33} & . & e_{31} & . & \varepsilon_{33}^S \end{pmatrix} \begin{pmatrix} S_{rr} \\ S_{zz} \\ S_{rz} \\ S_{\gamma\gamma} \\ E_r \\ E_z \end{pmatrix} \tag{1.42}$$

with the differential operators $\mathcal{B}$ and ∇

$$\mathcal{B} := \begin{pmatrix} \frac{\partial}{\partial r} & \cdot & \cdot\frac{\partial}{\partial z} & \frac{1}{r} \\ \cdot & \frac{\partial}{\partial z} & \frac{\partial}{\partial r} & \cdot \end{pmatrix}^T, \quad \nabla := \begin{pmatrix} \frac{\partial}{\partial r} \\ \frac{\partial}{\partial z} \end{pmatrix}. \tag{1.43}$$

Even though the space dimension is reduced, the material matrix in (1.42) shows that there are still all material parameters involved in the description of the piezoelectric effect. Thus, for the parameter identification process identifiability from axisymmetric implementations seems as adequate as from three dimensional ones. Note, however, that according to (1.43) only a first order differential operator acts on the parameter $\mathbf{c}_{12}^E$ which reduces its sensitivity in particular for thin piezoelectric discs being excited in their thickness mode, see Table 5 in Section 3.3.3.

1.1.8 Weak Form of Piezoelectric PDEs

The weak form of the piezoelectric equations in (1.33) and (1.40) will be obtained by testing each of the first lines with vector valued $\mathcal{C}^\infty$ functions $\mathbf{v} = (v_x, v_y, v_z) : \Omega \to \mathbb{R}$ and the second lines with scalar valued $\mathcal{C}^\infty$ functions w which vanish at the boundaries with electrical Dirichlet boundary conditions, i.e, Γ_e and Γ_g. Integration by parts (of the pure elastic part) yields

$$\int_\Omega (\mathcal{B}^T\vec{\sigma})^T\overline{\mathbf{v}}\, d\Omega = -\int_\Omega \vec{\sigma}^T\overline{\mathcal{B}\mathbf{v}}\, d\Omega + \int_{\partial\Omega} (\mathbf{N}^T\vec{\sigma})^T\overline{\mathbf{v}}\, d\Gamma\,, \tag{1.44}$$

where the last term vanishes due the boundary conditions assumed. In order to incorporate the inhomogeneous Dirichlet boundary condition for the electric potential the function ϕ_0 is introduced

$$\phi_0 := \phi - \phi^e\chi \in H^1_{0,\Gamma}(\Omega). \tag{1.45}$$

The function χ is supposed to have the following properties $\chi|_{\Gamma_g} \equiv 0, \quad \chi|_{\Gamma_e} \equiv 1$ and $\chi \in H^1(\Omega)$; such a χ exists if Ω is a Lipschitz domain and $\Gamma_e \cap \Gamma_g = \emptyset$. The mechanical displacement $\mathbf{u}$ is assumed to be in $H^1_{\mathcal{B}}(\Omega)$. Now, the weak form of the piezoelectric PDEs in time domain (1.33) read as

$$\begin{aligned} &\forall \mathbf{v} \in H^1_{\mathcal{B}}(\Omega) \text{ and } w \in H^1_{0,\Gamma}(\Omega)\ : \\ &\int_\Omega \Big(\rho\ddot{\mathbf{u}}^T\mathbf{v} \ + \alpha\rho\dot{\mathbf{u}}^T\mathbf{v} + \ (\mathbf{c}^E\mathcal{B}\mathbf{u})^T\mathcal{B}\mathbf{v} + \beta\,(\mathbf{c}^E\mathcal{B}\dot{\mathbf{u}})^T\mathcal{B}\mathbf{v} \\ &\qquad + (\mathbf{e}^T\nabla\phi_0)^T\mathcal{B}\mathbf{v} \ + \ (\mathbf{e}\mathcal{B}\mathbf{u})^T\nabla w - \ (\boldsymbol{\varepsilon}^S\nabla\phi_0)^T\nabla w \Big)\, d\Omega \\ &\quad = \phi^e \int_\Omega \Big(-(\mathbf{e}^T\nabla\chi)^T\mathcal{B}\mathbf{v} \ + \ (\boldsymbol{\varepsilon}^S\nabla\chi)^T\nabla w \Big)\, d\Omega. \end{aligned} \tag{1.46}$$

With spaces $H^1_B(\Omega)$ and $H^1_{0,\Gamma}(\Omega)$ of real valued functions the system in (1.39) may be brought into the weak form in a similar manner.

In frequency domain we test (1.38) with complex valued functions $\mathbf{v}$ and w and obtain

$$\begin{aligned}
&\forall \mathbf{v} \in H^1_{\mathcal{B}}(\Omega) \text{ and } w \in H^1_{0,\Gamma}(\Omega) \ : \\
&\int_\Omega \Big(-\rho\omega^2 \hat{\mathbf{u}}^T \overline{\mathbf{v}} \ + \ (\mathbf{c}^E \mathcal{B}\hat{\mathbf{u}})^T \overline{\mathcal{B}\mathbf{v}} \ + \ (\mathbf{e}^T \nabla \hat{\phi}_0)^T \overline{\mathcal{B}\mathbf{v}} \\
&\qquad\qquad + (\mathbf{e}\mathcal{B}\hat{\mathbf{u}})^T \overline{\nabla w} \ - \ (\boldsymbol{\varepsilon}^S \nabla \hat{\phi}_0)^T \overline{\nabla w} \Big) \, d\Omega \\
&= \hat{\phi}^e \int_\Omega \Big(-(\mathbf{e}^T \nabla \chi)^T \overline{\mathcal{B}\mathbf{v}} \ + \ (\boldsymbol{\varepsilon}^S \nabla \chi)^T \overline{\nabla w} \Big) \, d\Omega \, . \qquad (1.47)
\end{aligned}$$

1.2 Well-Posedness Results

Before turning to the inverse problem one should be aware whether the forward problem in its different forms is well-posed. Well-posedness results already provide constraints for the sought-for parameters and appropriate damping. Additionally, existence results show restrictions for the input data of the inverse problem, i.e. the range of the parameter-to-solution map. Even though a linear transient formulation is not used for the identification, the well-posedness results for a linearization in case of parameter curve identification for nonlinear hyperbolic problems may serve as a basis for analyzing well-posedness in the nonlinear transient case considered in Section 4.2. The latter is expected to require additional techniques from the analysis of conservation laws and is therefore not treated in this thesis.
At this point a collection of literature on well-posedness in linear piezoelectricity is presented:

- Nowacki [111] gives a uniqueness theorem for the solution of the transient differential equations describing thermopiezoelectric effects.
- Ieşan [63] proves uniqueness results and minimum principles with a derived reciprocity relation for quasi-static and dynamic piezoelectricity.
- Melnik and authors [100] give well-posedness results for non-stationary radially symmetric and axially symmetric problems.
- Miara [102] presents existence results for the strong and weak solution of the nonstationary piezoelectric problem and additionally examines controllability of the system.
- Turbé and Maugin [147] investigate in the nonstationary case and provide homogenization results considering both classical asymptotic results and Bloch-Floquet theory.
- Ciarletta and Scalia [25] derive basic equations of the linear theory of thermopiezoelectricity for materials with voids and give reciprocity and uniqueness results for isothermal processes.

- Cimatti [26] gives existence, uniqueness and periodic solutions for the stationary case using Lax-Milgram's lemma. For the non-stationary case including a mass proportional damping term he proves existence and uniqueness using the Hille-Yosida theorem.
- Akamatsu and Nakamura [3] investingate in the undamped hyperbolic case with mixed boundary conditions. They extend the works of Melnik to cases of general boundary conditions using a Galerkin approximation in the proofs.
- Geis, Sändig and Mushuris [43] show existence and uniqueness for a quasistatic problem modeling multi-layered actuators also including thin metallic electrodes.
- Mercier and Nicaise [101] give existence, uniqueness and regularity results for time-harmonic piezoelectric systems considering the full Maxwell's equations.
- Kaltenbacher, Lahmer and Mohr [77] study well-posedness results for the time-harmonic case with real valued coefficients and with complex ones where sufficiently strong damping guarantees existence and uniqueness of the solution for the complete frequency range.
- Geis [42] publishes existence and uniqueness theorems for the solution of a multistructure-multifield transmission problem modeling co-fired piezoelectric multilayer actuators where additionally thin metal inclusions, the electrodes, are taken into account together with loads caused by thermal stresses and pyroelectric effects.

1.2.1 Transient Case With Rayleigh-Damping

In the sequel we will generalize results from [3] concerning well-posedness of (1.33) to the case where the Rayleigh model is considered. Completing the results of [3] an energy estimate for the acceleration is derived. The proof of uniqueness requires an approach different to the one in [3] due to the viscous damping terms. The boundary conditions remain as introduced in Section 1.1.6, i.e. we consider the case of a mechanically unclamped piezoelectric body with voltage excitation in the sequel. Different boundary conditions can be treated analogously along the

lines of [3]

$$\begin{aligned}
\rho\ddot{\mathbf{u}} + \alpha\rho\dot{\mathbf{u}} - \mathcal{B}^T\Big(\mathbf{c}^E\mathcal{B}\mathbf{u} + \beta\mathbf{c}^E\mathcal{B}\dot{\mathbf{u}} + \mathbf{e}^T\nabla\phi\Big) &= 0 \text{ in } \Omega \\
-\nabla\cdot\Big(\mathbf{e}\mathcal{B}\mathbf{u} - \boldsymbol{\varepsilon}^S\nabla\phi\Big) &= 0 \text{ in } \Omega \\
\phi(t) &= 0 \text{ on } \Gamma_g \\
\phi(t) &= \phi^e(t) \text{ on } \Gamma_e \\
\vec{n}\cdot\Big(\mathbf{e}\mathcal{B}\mathbf{u} - \boldsymbol{\varepsilon}^S\nabla\phi\Big) &= 0 \text{ on } \Gamma_r \\
\mathbf{N}^T\Big(\mathbf{c}^E\mathcal{B}\mathbf{u} + \beta\mathbf{c}^E\mathcal{B}\dot{\mathbf{u}} + \mathbf{e}^T\nabla\phi\Big) &= 0 \text{ on } \partial\Omega \\
\mathbf{u}(t=0) &= \mathbf{u}_0 \\
\dot{\mathbf{u}}(t=0) &= \mathbf{u}_1,
\end{aligned} \tag{1.48}$$

where $\Gamma_r := \partial\Omega \setminus (\Gamma_e \cup \Gamma_g)$.

Theorem 1.1. *Let $\Omega \subset \mathbb{R}^3$ be a bounded domain with C^2 boundary $\partial\Omega$ and let the material tensors $\mathbf{c}^E, \mathbf{e}$ and $\boldsymbol{\varepsilon}^S \in \mathbb{R}$ fulfill (1.2) and (1.3). The Rayleigh coefficients α and β are assumed to be nonnegative. Let $T > 0$. Then for any $\mathbf{u}_0 \in H^1_{\mathcal{B}}(\Omega)$, $\mathbf{u}_1 \in L^2(\Omega)$ and $\phi^e \in H^1(0,T;H^{\frac{1}{2}}(\Gamma_e))$ there exists a unique solution $(\mathbf{u},\phi) \in L^\infty(0,T;H^1_{\mathcal{B}}(\Omega)) \times L^\infty(0,T;H^1_{0,\Gamma}(\Omega))$ with $\dot{\mathbf{u}} \in L^\infty(0,T;L^2(\Omega))$ and $\ddot{\mathbf{u}} \in L^2(0,T;H^{-1}_{\mathcal{B}}(\Omega))$ to (1.48) satisfying the initial conditions*

$$\mathbf{u}(0) = \mathbf{u}_0, \quad \dot{\mathbf{u}}(0) = \mathbf{u}_1 \quad \text{on } \Omega. \tag{1.49}$$

Further, we have the estimate

$$\begin{aligned}
&||\mathbf{u}||_{L^\infty(0,T,H^1_{\mathcal{B}}(\Omega))} + ||\dot{\mathbf{u}}||_{L^\infty(0,T,L^2(\Omega))} + ||\ddot{\mathbf{u}}||_{L^2(0,T,H^{-1}_{\mathcal{B}}(\Omega))} + ||\phi||_{L^\infty(0,T,H^1_{0,\Gamma}(\Omega))} \\
&\qquad \le C\left(||\mathbf{u}_0||_{H^1_{\mathcal{B}}(\Omega)} + ||\mathbf{u}_1||_{L^2(\Omega)} + ||\phi^e||_{H^1(0,T;H^{1/2}(\Gamma_e))}\right).
\end{aligned} \tag{1.50}$$

Proof. The proof consists of four parts

- A Galerkin approximation
- Energy estimates for the finite dimensional problem
- Limit to infinite dimensional case, existence of a solution
- Uniqueness of the solution

As in (1.45) we set $\phi_0 = \phi - \phi^e\chi$ and consider the weak form of (1.48)

$$\begin{aligned}\int_\Omega \rho\ddot{\mathbf{u}}^T\mathbf{v}\ d\Omega + \alpha\int_\Omega \rho\dot{\mathbf{u}}^T\mathbf{v}\ d\Omega + \int_\Omega (\mathbf{c}^E\mathcal{B}\mathbf{u})^T\mathcal{B}\mathbf{v}\ d\Omega + \beta\int_\Omega (\mathbf{c}^E\mathcal{B}\dot{\mathbf{u}})^T\mathcal{B}\mathbf{v}\ d\Omega \\ + \int_\Omega (\mathbf{e}^T\nabla\phi_0)^T\mathcal{B}\mathbf{v}\ d\Omega + \int_\Omega (\mathbf{e}\mathcal{B}\mathbf{u})^T\nabla w\ d\Omega \\ - \int_\Omega (\boldsymbol{\varepsilon}^S\nabla\phi_0)^T\nabla w\ d\Omega = \langle f,\mathbf{v}\rangle + \langle g,w\rangle,\end{aligned} \tag{1.51}$$

which is supposed to hold for all $(\mathbf{v},w) \in [H^1_{\mathcal{B}}(\Omega)]^3 \times H^1_{0,\Gamma}(\Omega)$ with $\langle f,\mathbf{v}\rangle = -\phi^e\int_\Omega(\mathbf{e}^T\nabla\chi)^T\mathcal{B}\mathbf{v}\ d\Omega$ and $\langle g,w\rangle = \phi^e\int_\Omega(\boldsymbol{\varepsilon}^S\nabla\chi)^T\nabla w\ d\Omega$. The right hand side terms of (1.51) only depend on ϕ^e with

$$\begin{aligned}||f||_{H^1(0,T;H^{-1}_{\mathcal{B}}(\Omega))} &\leq c_1(\Omega)||\phi^e||_{H^1(0,T)} \\ ||g||_{H^1(0,T;H^{-1}_{\Gamma_e}(\Omega))} &\leq c_2(\Omega)||\phi^e||_{H^1(0,T)}.\end{aligned} \tag{1.52}$$

Now, Galerkin's method is applied to (1.46) using appropriate test functions $\mathbf{v}_j \in H^1_{\mathcal{B}}(\Omega,\mathbb{R}^3)$ and $w_j \in H^1_{0,\Gamma}(\Omega)$, $j \in \mathbb{N}$. So $\mathbf{u}(t) \approx \mathbf{u}_m(t) = \sum_{j=1}^m u_m^j(t)\mathbf{v}_j$ and $\phi(t) \approx \phi_m(t) = \sum_{j=1}^m \phi_m^j(t)w_j$. All finite subsets of the test functions are assumed to be linearly independent and their linear spans to be dense in $H^1_{\mathcal{B}}(\Omega)$ and $H^1_{0,\Gamma}(\Omega)$. We define $V_m := \mathrm{span}\{\mathbf{v}_1,...\mathbf{v}_m\}$ and $W_m := \mathrm{span}\{w_1,...w_m\}$ with $\overline{\cup_{m=1}^\infty V_m} = H^1_{\mathcal{B}}(\Omega)$ and $\overline{\cup_{m=1}^\infty W_m} = H^1_{0,\Gamma}(\Omega)$. Then, for each $m \in \mathbb{N}$ there exists a unique solution
$(\mathbf{u}_m(t),\phi_m(t)) \in C^2([0,T],V_m) \times C^1([0,T],W_m)$ to (1.51) for all $(\mathbf{v},w) \in V_m \times W_m$ satisfying the initial conditions $\mathbf{u}_m(0) = \mathbf{u}_{0m}$, $\dot{\mathbf{u}}_m(0) = \mathbf{u}_{1m}$ where $\mathbf{u}_{0m}$, $\mathbf{u}_{1m}$ converge to $\mathbf{u}_0$ and $\mathbf{u}_1$ in H^1 and L^2, respectively, as $m \to \infty$.

The matrices

$$M_{uu} := (\langle\rho\mathbf{v}_j,\mathbf{v}_k\rangle)_{kj} \quad \text{and} \quad K_{\phi\phi} = (\langle\boldsymbol{\varepsilon}^S\nabla w_j,\nabla w_k\rangle)_{kj} \tag{1.53}$$

are invertible due to the linear independence of the test functions and positivity of ρ and $\boldsymbol{\varepsilon}^S$, respectively. Further, let us define

$$\begin{aligned}K_{uu} &:= (\langle\mathbf{c}^E\mathcal{B}\mathbf{v}_j,\mathcal{B}\mathbf{v}_k\rangle)_{kj}, \\ K_{u\phi} &:= (\langle\mathbf{e}^T\nabla w_j,\mathcal{B}\mathbf{v}_k\rangle)_{kj}, \\ K_{\phi u} &:= (\langle\mathbf{e}\mathcal{B}\mathbf{v}_j,\nabla w_k\rangle)_{kj}, \\ \mathbf{f}_m &:= (\langle f,\mathbf{v}_k\rangle)_k, \\ \mathbf{g}_m &:= (\langle g,w_k\rangle)_k.\end{aligned} \tag{1.54}$$

The solutions $\mathbf{u}_m(t) = (u_m^j(t))_j$ and $\phi_m(t) = (\phi_m^j(t))_j$ are formally computed by

$$\phi_m = (K_{\phi\phi})^{-1}\underbrace{(\mathbf{g}_m - K_{\phi u}\mathbf{u}_m)}_{:=F_m} \tag{1.55}$$

and

$$\ddot{\mathbf{u}}_m + \alpha\dot{\mathbf{u}}_m + \beta M_{uu}^{-1}K_{uu}\dot{\mathbf{u}}_m + M_{uu}^{-1}\left(K_{uu} + K_{u\phi}K_{\phi\phi}^{-1}K_{\phi u}\right)\mathbf{u}_m = M_{uu}^{-1}\mathbf{f}_m + M_{uu}^{-1}K_{\phi\phi}^{-1}\mathbf{g}_m. \quad (1.56)$$

In a next step one shows that the finite dimensional solution $(\mathbf{u}_m(t), \phi_m(t))$ is bounded in $L^\infty(0,T,H^1_{\mathcal{B}}(\Omega)) \times L^\infty(0,T,H^1_{0,\Gamma}(\Omega))$ and $\dot{\mathbf{u}}$ is bounded in $L^\infty(0,T,L^2(\Omega))$. Testing (1.51) with $(\dot{\mathbf{u}}_m, 0)$ gives

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\left(\langle\rho\dot{\mathbf{u}}_m(t),\dot{\mathbf{u}}_m(t)\rangle + \langle\mathbf{c}^E\mathcal{B}\mathbf{u}_m(t),\mathcal{B}\mathbf{u}_m(t)\rangle\right)\\ +\alpha\langle\rho\dot{\mathbf{u}}_m(t),\dot{\mathbf{u}}_m(t)\rangle + \beta\langle\mathbf{c}^E\mathcal{B}\dot{\mathbf{u}}_m(t),\mathcal{B}\dot{\mathbf{u}}_m(t)\rangle + \langle\mathbf{e}^T\nabla\phi_m(t),\mathcal{B}\dot{\mathbf{u}}_m(t)\rangle\\ = \langle f,\dot{\mathbf{u}}_m(t)\rangle.\end{aligned} \quad (1.57)$$

Differentiating (1.51) with respect to time and testing the result with $(0,\phi_m)$ yields

$$\langle\mathbf{e}^T\mathcal{B}\dot{\mathbf{u}}_m(t),\nabla\phi_m(t)\rangle - \frac{1}{2}\frac{d}{dt}\left(\langle\boldsymbol{\varepsilon}^S\nabla\phi_m(t),\nabla\phi_m(t)\rangle\right) = \langle\dot{g},\phi_m(t)\rangle. \quad (1.58)$$

Subtracting the two latter equations from each other gives

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\left(\langle\rho\dot{\mathbf{u}}_m(t),\dot{\mathbf{u}}_m(t)\rangle + \langle\mathbf{c}^E\mathcal{B}\mathbf{u}_m(t),\mathcal{B}\mathbf{u}_m(t)\rangle + \langle\boldsymbol{\varepsilon}^S\nabla\phi_m(t),\nabla\phi_m(t)\rangle\right)\\ +\alpha\langle\rho\dot{\mathbf{u}}_m(t),\dot{\mathbf{u}}_m(t)\rangle + \beta\langle\mathbf{c}^E\mathcal{B}\dot{\mathbf{u}}_m(t),\mathcal{B}\dot{\mathbf{u}}_m(t)\rangle = \langle f,\dot{\mathbf{u}}_m\rangle - \langle\dot{g},\phi_m\rangle.\end{aligned} \quad (1.59)$$

Integration with respect to t yields

$$\begin{aligned}&\langle\rho\dot{\mathbf{u}}_m(t),\dot{\mathbf{u}}_m(t)\rangle + \langle\mathbf{c}^E\mathcal{B}\mathbf{u}_m(t),\mathcal{B}\mathbf{u}_m(t)\rangle + \langle\boldsymbol{\varepsilon}^S\nabla\phi_m(t),\nabla\phi_m(t)\rangle\\ &+ \int_0^t \alpha\langle\rho\dot{\mathbf{u}}_m(s),\dot{\mathbf{u}}_m(s)\rangle ds + \beta\int_0^t\langle\mathbf{c}^E\mathcal{B}\dot{\mathbf{u}}_m(s),\mathcal{B}\dot{\mathbf{u}}_m(s)\rangle ds\\ &= \langle\rho\dot{\mathbf{u}}_m(0),\dot{\mathbf{u}}_m(0)\rangle + \langle\mathbf{c}^E\mathcal{B}\mathbf{u}_m(0),\mathcal{B}\mathbf{u}_m(0)\rangle + \langle\boldsymbol{\varepsilon}^S\nabla\phi_m(0),\nabla\phi_m(0)\rangle\\ &+ 2\int_0^t\langle f(s),\dot{\mathbf{u}}_m(s)\rangle ds - 2\int_0^t\langle\dot{g}(s),\phi_m(s)\rangle ds.\end{aligned} \quad (1.60)$$

Now, one estimates the following

$$\begin{aligned}\int_\Omega \mathcal{B}\mathbf{u}_m\mathbf{c}^E\mathcal{B}\mathbf{u}_m\, d\Omega &\geq \lambda_{1,mech}\int_\Omega \mathcal{B}\mathbf{u}_m\mathcal{B}\mathbf{u}_m\, d\Omega\\ &= \lambda_{1,mech}||\mathcal{B}\mathbf{u}_m||^2_{L^2}\\ &= c_{mech}\left(||\mathbf{u}_m||^2_{H^1_{\mathcal{B}}} - ||\mathbf{u}_m||^2_{L^2}\right)\end{aligned} \quad (1.61)$$

where $\lambda_{1,mech}$ is the smallest eigenvalue of $\mathbf{c}^E$ and and with Poincaré's inequality

$$\begin{aligned}\int_\Omega \nabla\phi_m \boldsymbol{\varepsilon}^S \nabla\phi_m \, d\Omega &\geq \lambda_{1,elec} \int_\Omega \nabla\phi_m \nabla\phi_m \, d\Omega \\ &= \lambda_{1,elec} ||\nabla\phi_m||^2_{L^2} \\ &\geq c_{elec} ||\phi_m||^2_{H^1} \end{aligned} \tag{1.62}$$

where $\lambda_{1,elec}$ is the smallest eigenvalue of $\boldsymbol{\varepsilon}^S$. The latter two estimations together with the positivity α and β allow to give a lower bound for the left-hand side of (1.60)

$$C_1 \left(||\dot{\mathbf{u}}_m(t)||^2_{L^2(\Omega)} + ||\mathbf{u}_m(t)||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi_m(t)||^2_{H^1_{0,\Gamma}(\Omega)} \right) - c_{mech} ||\mathbf{u}_m(t)||^2_{L^2}(\Omega). \tag{1.63}$$

With Schwartz's and Young's inequality, the right-hand side of (1.60) is bounded from above by

$$\begin{aligned} & C_2 \left(||\dot{\mathbf{u}}_m(0)||^2_{L^2(\Omega)} + ||\mathbf{u}_m(0)||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi_m(0)||^2_{H^1_{0,\Gamma}(\Omega)} \right) \\ & + \int_0^t \left(||\dot{\mathbf{u}}_m(s)||^2_{L^2(\Omega)} + ||\mathbf{u}_m(s)||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi_m(s)||^2_{H^1_{0,\Gamma}} \right) ds \\ & + ||f||^2_{L^2(0,T;L^2(\Omega))} + ||g||^2_{H^1(0,T;H^{-1}_{0,\Gamma})}. \end{aligned} \tag{1.64}$$

We are now able to apply Gronwall's inequality since for the following choices

$$\eta(t) := C_1 \left(||\dot{\mathbf{u}}_m(t)||^2_{L^2(\Omega)} + ||\mathbf{u}_m(t)||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi_m(t)||^2_{H^1_{0,\Gamma}(\Omega)} \right)$$

and

$$\begin{aligned} \xi(t) := ||\dot{\mathbf{u}}_m(0)||^2_{L^2(\Omega)} + ||\mathbf{u}_m(0)||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi_m(0)||^2_{H^1_{0,\Gamma}(\Omega)} \\ + ||f||^2_{L^2(0,T;L^2(\Omega))} + ||g||^2_{H^1(0,T;H^{-1}_{0,\Gamma}(\Omega))} \end{aligned} \tag{1.65}$$

the differential inequality $\eta'(t) \leq k(t)\eta(t) + \xi(t)$ holds. Then,

$$\begin{aligned} & ||\dot{\mathbf{u}}_m(t)||^2_{L^2(\Omega)} + ||\mathbf{u}_m(t)||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi_m(t)||^2_{H^1_{0,\Gamma}(\Omega)} \\ & \leq C_3 e^{C_3 t} \big(||\dot{\mathbf{u}}_m(0)||^2_{L^2(\Omega)} + ||\mathbf{u}_m(0)||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi_m(0)||^2_{H^1_{0,\Gamma}(\Omega)} \\ & + ||f||^2_{L^2(0,T;L^2(\Omega))} + ||g||^2_{H^1(0,T;H^{-1}_{0,\Gamma}(\Omega))} \big). \end{aligned} \tag{1.66}$$

Now, since by the finite dimensional version of the second equation in (1.48) for $t = 0$, coercivity of $\langle \boldsymbol{\varepsilon}^S \nabla\phi_m(0), \nabla\phi_m(0) \rangle$, the fact that $\Gamma_e \cup \Gamma_g \neq \emptyset$, and using Young's inequality, we have for $t = 0$

$$||\phi_m(0)||^2_{H^1_{0,\Gamma}(\Omega)} \leq C_5 (||\mathbf{u}_0||^2_{H^1_{\mathcal{B}}}(\Omega) + ||\phi^e(0)||^2_{H^{1/2}(\Gamma_e)}) \tag{1.67}$$

and considering the initial values for $\mathbf{u}$ and $\dot{\mathbf{u}}$ we get from (1.66)

$$||\dot{\mathbf{u}}_m(t)||^2_{L^\infty(0,T;L^2(\Omega))} + ||\mathbf{u}_m(t)||^2_{L^\infty(0,T;H^1_{\mathcal{B}}(\Omega))} + ||\phi_m(t)||^2_{L^\infty(0,T;H^1_{0,\Gamma}(\Omega))}$$
$$\leq C_6 \left(||\mathbf{u}_{1m}||^2_{L^2(\Omega)} + ||\mathbf{u}_0||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\phi^e||_{L^\infty(0,T;H^{1/2}(\Gamma_e))} \right). \tag{1.68}$$

In case $\beta > 0$ we get from (1.60)

$$||\mathcal{B}\dot{\mathbf{u}}_m||_{L^2(0,T;L^2(\Omega)} \leq C_7. \tag{1.69}$$

Additionally an energy estimate for the acceleration $\ddot{\mathbf{u}}$ is given in $L^2(0,T,H^{-1}_{\mathcal{B}})$. In order to show this, fix any $\mathbf{v} \in H^1_{\mathcal{B}}(\Omega)$ with $||\mathbf{v}||_{H^1_{\mathcal{B}}} \leq 1$ and $\mathbf{v} := \mathbf{v}^1 + \mathbf{v}^2$, where $\mathbf{v}^1 \in \text{span}\{\mathbf{v}_j\}_{j=1}^m$ and $(\mathbf{v}^2, \mathbf{v}_j) = 0$ for all $1 \leq j \leq m$. With $\mathbf{u}_m = \sum_{j=0}^m u_m^j(t)\mathbf{v}_j$ we have

$$\langle \ddot{\mathbf{u}}_m, \mathbf{v}^1 \rangle =$$
$$\langle f, \mathbf{v}^1 \rangle - \alpha\langle \rho\dot{\mathbf{u}}_m, \mathbf{v}^1 \rangle - \langle \mathbf{c}^E\mathcal{B}\mathbf{u}_m, \mathcal{B}\mathbf{v}^1 \rangle - \beta\langle \mathbf{c}^E\mathcal{B}\dot{\mathbf{u}}_m, \mathcal{B}\mathbf{v}^1 \rangle - \langle \mathbf{e}^T\nabla\phi_m, \mathcal{B}\mathbf{v}^1 \rangle. \tag{1.70}$$

From this we get

$$|\langle \ddot{\mathbf{u}}_m, \mathbf{v} \rangle| \leq C\big(||f||_{L^2} + ||\mathbf{u}_m||_{H^1_{\mathcal{B}}} + \langle \mathbf{e}^T\nabla(-\Delta_\epsilon)^{-1}\text{div}_0\mathbf{e}\mathcal{B}\mathbf{u}_m, \mathcal{B}\mathbf{v}^1 \rangle \big)$$
$$-\frac{d}{dt}\left(\alpha\langle \rho\mathbf{u}_m, \mathbf{v}^1 \rangle + \beta\langle \mathbf{c}^E\mathcal{B}\mathbf{u}_m, \mathcal{B}\mathbf{v}^1 \rangle \right), \tag{1.71}$$

where we have used the substitution of the electric potential

$$\phi_m(\mathbf{u}_m) = (-\Delta_\epsilon)^{-1}\text{div}_0(-\mathbf{e}\mathcal{B}\mathbf{u}_m). \tag{1.72}$$

Here, the differential operator Δ_ϵ comes from the pure electrostatic problem

$$\begin{array}{rcccl} -\Delta_\epsilon & : & H^1_{0,\Gamma}(\Omega) & \to & H^{-1}_{0,\Gamma}(\Omega)) \\ & & v & \mapsto & \left\{ \begin{array}{lccl} -\Delta_\varepsilon v & : H^1_{0,\Gamma}(\Omega) & \to & \mathbb{R} \\ & w & \mapsto & \int_\Omega (\boldsymbol{\varepsilon}^S\nabla v)^T\nabla w \, d\Omega \end{array} \right. \end{array}$$

and with div_0 the weak divergence operator is denoted

$$\begin{array}{rcccl} \text{div}_0 & : & (L^2(\Omega))^3 & \to & (H^{-1}_{0,\Gamma}(\Omega)) \\ & & \vec{f} & \mapsto & \left\{ \begin{array}{lccl} \text{div}_0\vec{f} & : H^1_{0,\Gamma}(\Omega) & \to & \mathbb{R} \\ & w & \mapsto & -\int_\Omega \vec{f}^T\nabla w \, d\Omega. \end{array} \right. \end{array}$$

Now, by

$$\begin{array}{rcl} ||\mathbf{e}^T\nabla(-\Delta_\epsilon)^{-1}\text{div}_0\mathbf{e}\mathcal{B}\mathbf{u}_m||_{L^2(\Omega)} & & \\ \leq & C_e||\text{div}_0\mathbf{e}\mathcal{B}\mathbf{u}_m||_{(H^{-1}_{0,\Gamma}(\Omega))} & \\ = & C_e \sup\limits_{w\neq 0\in H^1_{0,\Gamma}} \left| -\frac{\int_\Omega \mathbf{e}\mathcal{B}\mathbf{u}_m\nabla w}{||w||_{H^1_{0,\Gamma}}(\Omega)} \right| & \\ \leq & C_e||\mathbf{e}\mathcal{B}\mathbf{u}_m||_{L^2(\Omega)} & \\ \leq & \tilde{C}_e||\mathcal{B}\mathbf{u}_m||_{L^2(\Omega)} & \end{array} \tag{1.73}$$

we finally obtain

$$\begin{aligned}\int_0^T ||\ddot{\mathbf{u}}_m||^2_{H_{\mathcal{B}}^{-1}(\Omega)} dt &\leq C_8 \int_0^T \big(||f||_{L^2(\Omega)} + ||\mathbf{u}_m||_{H^1_{\mathcal{B}}(\Omega)} + ||\mathcal{B}\mathbf{u}_m||_{L^2}\big)dt \\ &+ \Big(\alpha||\mathbf{u}_m||_{L^\infty(0,T;H_{\mathcal{B}}^{-1}(\Omega))} + \beta||\mathcal{B}\mathbf{u}_m||_{L^\infty(0,T;L^2(\Omega))}\Big) \\ &\leq C_9\big(||\mathbf{u}_0||^2_{H^1_{\mathcal{B}}(\Omega)} + ||\mathbf{u}_1||^2_{L^2(\Omega)} + ||\phi^e||_{H^1(0,T;H^{\frac{1}{2}}(\Gamma_e))}\big),\end{aligned} \tag{1.74}$$

where we have used in case that $\beta > 0$ that (1.69) yields

$$||\mathcal{B}\dot{\mathbf{u}}_m||_{L^\infty(0,T;L^2(\Omega))} \leq \sqrt{T}||\mathcal{B}\dot{\mathbf{u}}||_{L^2(0,T;L^2(\Omega))} \leq C_8$$

and in case $\alpha > 0$ the term $||\mathbf{u}_m||_{L^\infty(0,T;H_{\mathcal{B}}^{-1}(\Omega))}$ is finite by terms in (1.63).

Thus, we have shown the boundedness of the sequences

$$\begin{aligned}\{\mathbf{u}_m, \phi_m\} &\in L^\infty(0,T;H^1_{\mathcal{B}}(\Omega)) \times L^\infty(0,T;H^1_{0,\Gamma}(\Omega)), \\ \{\dot{\mathbf{u}}_m\} &\in L^\infty(0,T;L^2(\Omega)), \\ \{\ddot{\mathbf{u}}_m\} &\in L^2(0,T;H_{\mathcal{B}}^{-1}(\Omega)).\end{aligned} \tag{1.75}$$

Existence of weak solution:
In oder to show that a weak solution to (1.48) exists, we recall that according to the energy estimates the sequence $\{\mathbf{u}_m\}_{m=0}^\infty$ is bounded in $L^\infty(0,T;H^1_B(\Omega))$, $\{\dot{\mathbf{u}}_m\}_{m=0}^\infty$ in $L^\infty(0,T;L^2(\Omega))$, $\{\ddot{\mathbf{u}}_m\}_{m=0}^\infty$ in $L^2(0,T;H_B^{-1}(\Omega))$ and $\{\phi_m\}_{m=0}^\infty$ is bounded in $L^\infty(0,T;H^1_{0,\Gamma}(\Omega))$. As a consequence there exist subsequences $\{\mathbf{u}_{m_l}\}_{l=0}^\infty \subset \{\mathbf{u}_m\}_{m=0}^\infty$ and $\{\phi_{m_l}\}_{l=0}^\infty \subset \{\phi_m\}_{m=0}^\infty$ further $\mathbf{u} \in L^\infty(0,T;H^1_B(\Omega))$, $\nu \in L^\infty(0,T;L^2(\Omega))$, $\mathbf{a} \in L^2(0,T;H_B^{-1}(\Omega))$ and $\phi \in L^\infty(0,T;H^1_{0,\Gamma}(\Omega))$ such that

$$\left\{\begin{array}{lcll}\mathbf{u}_{m_l} & \rightharpoonup & \mathbf{u} & \text{weakly star in } L^\infty(0,T;H^1_B(\Omega)) \\ \dot{\mathbf{u}}_{m_l} & \rightharpoonup & \nu & \text{weakly star in } L^\infty(0,T;L^2(\Omega)) \\ \ddot{\mathbf{u}}_{m_l} & \rightharpoonup & \mathbf{a} & \text{weakly in } L^2(0,T;H_B^{-1}(\Omega)) \\ \phi_{m_l} & \rightharpoonup & \phi_0 & \text{weakly star in } L^\infty(0,T;H^1_0(\Omega)).\end{array}\right. \tag{1.76}$$

Since d/dt is a closed operator $\nu = \dot{\mathbf{u}}$ and $\mathbf{a} = \ddot{\mathbf{u}}$. Now, we fix N and choose $\mathbf{v} \in C^1(0,T;H^1_{\mathcal{B}}(\Omega))$ and $w \in (0,T;H^1_0(\Omega))$ which are of the form

$$\mathbf{v}(t) := \sum_{j=0}^N u_m^j(t)\mathbf{v}_j, \quad w(t) := \sum_{j=0}^N \phi_m^j(t) w_j. \tag{1.77}$$

Select $m \geq N$ and substitute in (1.51) $\mathbf{u}, \dot{\mathbf{u}}, \ddot{\mathbf{u}}, \phi_0$ by $\mathbf{u}_m, \dot{\mathbf{u}}_m, \ddot{\mathbf{u}}_m$ and ϕ_m. Multiply the result with (u_m^j, ϕ_m^j) and sum from $j = 1, ..., N$. Further, integration with respect to t gives

$$\begin{aligned}
&\int_0^T \Big(\int_\Omega \rho \ddot{\mathbf{u}}_m^T \mathbf{v} \; d\Omega + \alpha \int_\Omega \rho \dot{\mathbf{u}}_m^T \mathbf{v} \; d\Omega + \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u}_m)^T \mathcal{B}\mathbf{v} \; d\Omega \\
&+\beta \int_\Omega (\mathbf{c}^E \mathcal{B}\dot{\mathbf{u}}_m)^T \mathcal{B}\mathbf{v} \; d\Omega + \int_\Omega (\mathbf{e}^T \nabla \phi_m)^T \mathcal{B}\mathbf{v} \; d\Omega \; dt \\
&+ \int_\Omega (\mathbf{e}\mathcal{B}\mathbf{u}_m)^T \nabla w \; d\Omega - \int_\Omega (\boldsymbol{\varepsilon}^S \nabla \phi_m)^T \nabla w \; d\Omega \Big) \; dt \\
&= \int_0^T \langle g, w \rangle \; dt + \int_0^T \langle f, \mathbf{v} \rangle \; dt. \qquad (1.78)
\end{aligned}$$

Setting now $m = m_l$ and using (1.76) we find for the limit that

$$\begin{aligned}
&\int_0^T \Big(\int_\Omega \rho \ddot{\mathbf{u}}^T \mathbf{v} \; d\Omega + \alpha \int_\Omega \rho \dot{\mathbf{u}}^T \mathbf{v} \; d\Omega + \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u})^T \mathcal{B}\mathbf{v} \; d\Omega \\
&+ \beta \int_\Omega (\mathbf{c}^E \mathcal{B}\dot{\mathbf{u}})^T \mathcal{B}\mathbf{v} \; d\Omega + \int_\Omega (\mathbf{e}^T \nabla \phi_0)^T \mathcal{B}\mathbf{v} \; d\Omega \; dt \\
&+ \int_\Omega (\mathbf{e}\mathcal{B}\mathbf{u})^T \nabla w \; d\Omega - \int_\Omega (\boldsymbol{\varepsilon}^S \nabla \phi_0)^T \nabla w \; d\Omega \Big) \; dt \\
&= \int_0^T \langle g, w \rangle \; dt + \int_0^T \langle f, \mathbf{v} \rangle \; dt. \qquad (1.79)
\end{aligned}$$

The latter equality then holds for all functions $\mathbf{v} \in L^2(0,T; H_B^1(\Omega))$, since functions of the form (1.77) are dense in these spaces. From (1.79) it follows that

$$\begin{aligned}
&\int_\Omega \rho \ddot{\mathbf{u}}^T \mathbf{v} \; d\Omega + \alpha \int_\Omega \rho \dot{\mathbf{u}}^T \mathbf{v} \; d\Omega + \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u})^T \mathcal{B}\mathbf{v} \; d\Omega \\
&+ \beta \int_\Omega (\mathbf{c}^E \mathcal{B}\dot{\mathbf{u}})^T \mathcal{B}\mathbf{v} \; d\Omega + \int_\Omega (\mathbf{e}^T \nabla \phi_0)^T \mathcal{B}\mathbf{v} \; d\Omega \; dt \\
&+ \int_\Omega (\mathbf{e}\mathcal{B}\mathbf{u})^T \nabla w \; d\Omega - \int_\Omega (\boldsymbol{\varepsilon}^S \nabla \phi_0)^T \nabla w \; d\Omega \\
&= \langle g, w \rangle + \langle f, \mathbf{v} \rangle \qquad (1.80)
\end{aligned}$$

for all $\mathbf{v} \in H_B^1$ and $w \in H_{0,\Gamma}^1$ and a.e $0 \le t \le T$.

We now show that the initial conditions are fulfilled: For this, choose any function $(\mathbf{v}, 0)$ with $\mathbf{v} \in C^2(0,T; H_{\mathcal{B}}^1(\Omega))$ and $\mathbf{v}(T) = \dot{\mathbf{v}}(T) = 0$ and integrate the

mass part twice and the damping terms of (1.79) once by parts with respect to time

$$\begin{aligned}
&\int_0^T \Big(\int_\Omega \rho \mathbf{u}^T \ddot{\mathbf{v}} \; d\Omega - \alpha \int_\Omega \rho \mathbf{u}^T \dot{\mathbf{v}} d\Omega \\
&+ \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u})^T \mathcal{B}\mathbf{v} \; d\Omega - \beta \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u})^T \mathcal{B}\dot{\mathbf{v}} \; d\Omega + \int_\Omega (\mathbf{e}^T \nabla \phi_0)^T \mathcal{B}\mathbf{v} \; d\Omega \Big) \; dt \\
&= \int_0^T \langle f, \mathbf{v} \rangle \; dt - \rho \langle \mathbf{u}(0), \dot{\mathbf{v}}(0) \rangle + \rho \langle \dot{\mathbf{u}}(0), \mathbf{v}(0) \rangle \\
&\quad + \alpha \rho \langle \mathbf{u}(0), \mathbf{v}(0) \rangle + \beta \langle \mathbf{c}^E \mathcal{B}\mathbf{u}(0), \mathcal{B}\mathbf{v}(0) \rangle .
\end{aligned} \tag{1.81}$$

Similarly, we obtain from (1.78)

$$\begin{aligned}
&\int_0^T \Big(\int_\Omega \rho \mathbf{u}_m^T \ddot{\mathbf{v}} \; d\Omega - \alpha \int_\Omega \rho \mathbf{u}_m^T \dot{\mathbf{v}} d\Omega \\
&\int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u}_m)^T \mathcal{B}\mathbf{v} \; d\Omega - \beta \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u}_m)^T \mathcal{B}\dot{\mathbf{v}} \; d\Omega + \int_\Omega (\mathbf{e} \nabla \phi_m)^T \mathcal{B}\mathbf{v} \; d\Omega \Big) \; dt \\
&= \int_0^T \langle f, \mathbf{v} \rangle \; dt - \rho \langle \mathbf{u}_m(0), \dot{\mathbf{v}}(0) \rangle + \rho \langle \dot{\mathbf{u}}_m(0), \mathbf{v}(0) \rangle \\
&\quad + \alpha \rho \langle \mathbf{u}_m(0), \mathbf{v}(0) \rangle + \beta \langle \mathbf{c}^E \mathcal{B}\mathbf{u}_m(0), \mathcal{B}\mathbf{v}(0) \rangle .
\end{aligned} \tag{1.82}$$

We set now $m = m_l$, recall $\mathbf{u}_{0m} \to \mathbf{u}_0 \in H^1$, $\mathbf{u}_{1m} \to \mathbf{u}_1 \in L^2$ to (1.76) and obtain

$$\begin{aligned}
&\int_0^T \Big(\int_\Omega \rho \mathbf{u}^T \ddot{\mathbf{v}} \; d\Omega - \alpha \int_\Omega \rho \mathbf{u}^T \dot{\mathbf{v}} d\Omega \\
&+ \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u})^T \mathcal{B}\mathbf{v} \; d\Omega - \beta \int_\Omega (\mathbf{c}^E \mathcal{B}\mathbf{u})^T \mathcal{B}\dot{\mathbf{v}} \; d\Omega + \int_\Omega (\mathbf{e}^T \nabla \phi_0)^T \mathcal{B}\mathbf{v} \; d\Omega \Big) \; dt \\
&= \int_0^T \langle f, \mathbf{v} \rangle \; dt - \rho \langle \mathbf{u}_0, \dot{\mathbf{v}}(0) \rangle + \rho \langle \mathbf{u}_1, \mathbf{v}(0) \rangle \\
&\quad + \alpha \rho \langle \mathbf{u}_0, \mathbf{v}(0) \rangle + \beta \langle \mathbf{c}^E \mathcal{B}\mathbf{u}_0, \mathcal{B}\mathbf{v}(0) \rangle .
\end{aligned} \tag{1.83}$$

Comparing the identities (1.81) and (1.83) we conclude that $\mathbf{u}(0) = \mathbf{u}_0$ and $\dot{\mathbf{u}}(0) + \alpha \mathbf{u}(0) = \mathbf{u}_1 + \alpha \mathbf{u}_0$, hence $\dot{\mathbf{u}}(0) = \mathbf{u}_1$.

In order to proof uniqueness we show that for the homogeneous problem, i.e. $\mathbf{u}_0 = \mathbf{u}_1 \equiv 0$, $f \equiv 0$, $g \equiv 0$, and $\phi^e \equiv 0$ the only possible solution is $\mathbf{u} \equiv 0$ and $\phi \equiv 0$. Let us recall (1.60) in the limit $l \to \infty$ along the subsequence m_l

$$\begin{aligned}
&\langle \rho \dot{\mathbf{u}}(t), \dot{\mathbf{u}}(t) \rangle + \langle \mathbf{c}^E \mathcal{B}\mathbf{u}(t), \mathcal{B}\mathbf{u}(t) \rangle + \langle \boldsymbol{\varepsilon}^S \nabla \phi(t), \nabla \phi(t) \rangle \\
&- \langle \rho \dot{\mathbf{u}}(0), \dot{\mathbf{u}}(0) \rangle + \langle \mathbf{c}^E \mathcal{B}\mathbf{u}(0), \mathcal{B}\mathbf{u}(0) \rangle + \langle \boldsymbol{\varepsilon}^S \nabla \phi(0), \nabla \phi(0) \rangle \\
&= - \left(\int_0^t \alpha \langle \rho \dot{\mathbf{u}}(s), \dot{\mathbf{u}}(s) \rangle ds + \beta \int_0^t \langle \mathbf{c}^E \mathcal{B}\dot{\mathbf{u}}(s), \mathcal{B}\dot{\mathbf{u}}(s) \rangle ds \right) .
\end{aligned} \tag{1.84}$$

Observe, that all mechanical terms at time $t = 0$ vanish by the special choice of the initial conditions and $\phi(0) = 0$ by the second equation in (1.48) and zero Dirichlet-boundary conditions for the electric potential. The term within the parenthesis on the right-hand side is positive due to the choice of the material constants and damping coefficients. By this, the only solution to the homogeneous problem is the trivial one $\mathbf{u} \equiv 0$ and $\phi \equiv 0$. □

The following theorem states well-posedness for the system in (1.39), where we have introduced viscous damping for all parts of the system. The elemination of the coupling terms by appropriate choices of test functions and a Gauss-step, see (1.59), is now not applicable anymore due to the first order derivatives in the coupling parts. Also differentiating the second line in (1.39) would lead to second order derivatives of the electric potential. Therefore, we choose a different approach where (1.39) will be rewritten in a matrix form. Before we do this we recall the PDEs, boundary and initial conditions

$$\begin{aligned}
\mathcal{B}^T\Big(\Re(\mathbf{c}^E)\mathcal{B}\mathbf{u} - \frac{1}{\omega}\Im(\mathbf{c}^E)\frac{\partial \mathbf{u}}{\partial t} + \Re(\mathbf{e})^T\nabla\phi - \frac{1}{\omega}\Im(\mathbf{e})^T\nabla\frac{\partial \phi}{\partial t}\Big) &= \rho\frac{\partial^2 \mathbf{u}}{\partial t^2} \text{ in } \Omega \\
\nabla\cdot\Big(\Re(\mathbf{e})\mathcal{B}\mathbf{u} - \frac{1}{\omega}\Im(\mathbf{e})\mathcal{B}\frac{\partial \mathbf{u}}{\partial t} - \Re(\boldsymbol{\varepsilon}^S)\nabla\phi + \frac{1}{\omega}\Im(\boldsymbol{\varepsilon}^S)\nabla\frac{\partial \phi}{\partial t}\Big) &= 0 \text{ in } \Omega \\
\phi(t) &= 0 \text{ on } \Gamma_g \\
\phi(t) &= \phi^e(t) \text{ on } \Gamma_e \\
N^T\Big(\Re(\mathbf{c}^E)\mathcal{B}\mathbf{u} - \frac{1}{\omega}\Im(\mathbf{c}^E)\frac{\partial \mathbf{u}}{\partial t} + \Re(\mathbf{e})^T\nabla\phi - \frac{1}{\omega}\Im(\mathbf{e})^T\nabla\frac{\partial \phi}{\partial t}\Big) &= 0 \text{ on } \partial\Omega \\
\vec{n}\cdot\Big(\Re(\mathbf{e})\mathcal{B}\mathbf{u} - \frac{1}{\omega}\Im(\mathbf{e})\mathcal{B}\frac{\partial \mathbf{u}}{\partial t} - \Re(\boldsymbol{\varepsilon}^S)\nabla\phi + \frac{1}{\omega}\Im(\boldsymbol{\varepsilon}^S)\nabla\frac{\partial \phi}{\partial t}\Big) &= 0 \text{ on } \Gamma_r \\
\mathbf{u}(t=0) &= \mathbf{u}_0 \\
\dot{\mathbf{u}}(t=0) &= \mathbf{u}_1 \\
\nabla\phi(t=0) &= \nabla\psi_0.
\end{aligned} \tag{1.85}$$

Theorem 1.2. *Assume that the conditions of Theorem 1.1 hold true, further that*

$$-\frac{1}{\omega}\Im(\mathbf{c}^E) \quad \textit{and} \quad -\frac{1}{\omega}\Im(\boldsymbol{\varepsilon}^S) \quad \textit{are positive definite}. \tag{1.86}$$

Then there exists a unique solution $(\mathbf{u}, \phi) \in L^\infty(0,T; H^1_{\mathcal{B}}(\Omega)) \times L^\infty(0,T; H^1_{0,\Gamma}(\Omega))$ *to the problem in (1.39).*

Proof. Since the structure of this proof is similar to the one of Theorem 1.1 we will only discuss the differing parts in detail. The main difference compared to the proof of Theorem 1.1 is that we have now a parabolic equation for the second line in (1.48). Making use of the more compact representation form introduced

e.g. in [43], where with the help of the composed differential operator and material matrices

$$\underline{\mathcal{B}} := \begin{pmatrix} \mathcal{B} & 0 \\ 0 & -\nabla \end{pmatrix}, \ \underline{\mathbf{M}} := \Re \begin{pmatrix} \mathbf{c}^E & -\mathbf{e}^{\mathbf{T}} \\ \mathbf{e} & \boldsymbol{\varepsilon}^S \end{pmatrix}, \text{ and } \underline{\mathbf{C}} := -\frac{1}{\omega}\Im \begin{pmatrix} \mathbf{c}^E & -\mathbf{e}^{\mathbf{T}} \\ \mathbf{e} & \boldsymbol{\varepsilon}^S \end{pmatrix} \tag{1.87}$$

together with

$$\underline{\mathbf{u}} := (\mathbf{u}, \phi_0)^T, \quad \underline{\chi} : (0, \chi)^T \quad \text{and} \quad I_M := \rho \begin{pmatrix} I_{3\times 3} & 0 \\ 0 & 0 \end{pmatrix} \tag{1.88}$$

the weak form given in (1.39) is rewritten as: Find $\underline{\mathbf{u}} \in \mathcal{V} := \{ \begin{pmatrix} \mathbf{u} \\ \phi_0 \end{pmatrix} \in H^1_{\mathcal{B}} \times H^1_{0,\Gamma} \}$ such that

$$\int_\Omega \underline{\ddot{\mathbf{u}}}^T I_M \underline{\mathbf{v}} + (\underline{\mathbf{C}}\,\underline{\mathcal{B}}\,\underline{\dot{\mathbf{u}}})^T \underline{\mathcal{B}}\,\underline{\mathbf{v}} \ + (\underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\mathbf{u}})^T \underline{\mathcal{B}}\,\underline{\mathbf{v}} \, d\Omega = \underbrace{\hat{\phi}^e \int_\Omega (\underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\chi})^T (\underline{\mathcal{B}}\,\underline{\mathbf{v}}) \, d\Omega}_{=:\langle \underline{f}, \underline{\mathbf{v}} \rangle} \tag{1.89}$$

holds $\forall\, \underline{\mathbf{v}} \in \mathcal{V}$. After applying Galerkin's method and testing with discrete $\underline{\dot{\mathbf{u}}}_m$ we obtain

$$\frac{d}{dt}\frac{1}{2}\left(\langle I_M \underline{\dot{\mathbf{u}}}_m, \underline{\dot{\mathbf{u}}}_m \rangle + \langle \underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\mathbf{u}}_m, \underline{\mathcal{B}}\,\underline{\mathbf{u}}_m \rangle\right) + \langle \underline{\mathbf{C}}\,\underline{\mathcal{B}}\,\underline{\dot{\mathbf{u}}}_m, \underline{\mathcal{B}}\,\underline{\dot{\mathbf{u}}}_m \rangle = \langle \underline{f}, \underline{\mathbf{u}}_m \rangle. \tag{1.90}$$

Integration with respect to t gives

$$\begin{aligned} &\langle \underline{\dot{\mathbf{u}}}_m(t), I_M \underline{\dot{\mathbf{u}}}_m(t) \rangle + \langle \underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\mathbf{u}}_m(t), \underline{\mathcal{B}}\,\underline{\mathbf{u}}_m(t) \rangle + \int_0^t \langle \underline{\mathbf{C}}\,\underline{\mathcal{B}}\,\underline{\dot{\mathbf{u}}}_m(s), \underline{\mathcal{B}}\,\underline{\dot{\mathbf{u}}}_m(s) \rangle dt \\ &= \langle \underline{\dot{\mathbf{u}}}_m(0), I_M \underline{\dot{\mathbf{u}}}_m(0) \rangle + \langle \underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\mathbf{u}}_m(0), \underline{\mathcal{B}}\,\underline{\mathbf{u}}_m(0) \rangle + \int_0^t \langle \underline{f}, \underline{\mathbf{u}}_m(s) \rangle ds. \end{aligned} \tag{1.91}$$

By our assumptions on the material tensors both $x^T \underline{\mathbf{M}} x$ and $x^T \underline{\mathbf{C}} x$ are positive for any $x \in \mathbb{R}^9$ since terms involving the piezoelectric coupling $\mathbf{e}$ vanish due to the change of sign in $\underline{\mathbf{M}}$ and also $\underline{\mathbf{C}}$. We can therefore give a lower bound for the left-hand side of (1.91) like in (1.61) and (1.62). An upper bound for the right-hand side of (1.91) is given by the boundedness of $\mathbf{c}^E$ and $\boldsymbol{\varepsilon}^S$ together with Schwarz and Young's inequality. Setting now

$$\eta(t) := ||\underline{\dot{\mathbf{u}}}_m(t)||^2_{L^2(\Omega)} + ||\underline{\mathbf{u}}_m(t)||^2_{H^1_{\underline{\mathcal{B}}}(\Omega)}$$

and

$$\xi(t) := ||\underline{\dot{\mathbf{u}}}_m(0)||^2_{L^2(\Omega)} + ||\underline{\mathbf{u}}_m(0)||^2_{H^1_{\underline{\mathcal{B}}}(\Omega)} + ||\underline{f}||_{L^2(0,T;L^2(\Omega))}$$

and applying Gronwall's Lemma we get

$$\begin{aligned} &||\underline{\dot{\mathbf{u}}}_m(t)||^2_{L^2(\Omega)} + ||\underline{\mathbf{u}}_m(t)||^2_{H^1_{\underline{\mathcal{B}}}(\Omega)} \\ &\leq \underline{C}_1 e^{\underline{C}_1 t} \left(||\underline{\dot{\mathbf{u}}}_m(0)||^2_{L^2(\Omega)} + ||\underline{\mathbf{u}}_m(0)||^2_{H^1_{\underline{\mathcal{B}}}(\Omega)} + ||\underline{f}||_{L^2(0,T;L^2(\Omega))} \right). \end{aligned} \tag{1.92}$$

From this we deduce uniform boundedness of

$$\begin{aligned} \underline{\mathbf{u}}_m \quad &\text{in} \quad L^\infty(0,T;H^1_{\underline{\mathcal{B}}}(\Omega)), \\ \dot{\underline{\mathbf{u}}}_m \quad &\text{in} \quad L^\infty(0,T;L^2(\Omega)). \end{aligned} \tag{1.93}$$

Additionally we get from (1.91) and the positive definiteness of $\underline{\mathbf{C}}$ uniform boundedness of $\underline{\mathbf{u}}_m$ in $L^2(0,T;H^1_{\underline{\mathcal{B}}}(\Omega))$. Fix now any $\underline{\mathbf{v}} \in H^1_{\underline{\mathcal{B}}}(\Omega)$ with $||\underline{\mathbf{v}}||_{H^1_{\underline{\mathcal{B}}}(\Omega)} \leq 1$ and $\underline{\mathbf{v}} := \underline{\mathbf{v}}^1 + \underline{\mathbf{v}}^2$, where $\underline{\mathbf{v}}^1 \in \operatorname{span}\{\underline{\mathbf{v}}_j\}_{j=1}^m$ and $(\underline{\mathbf{v}}^2, \underline{\mathbf{v}}_j) = 0$ for all $1 \leq j \leq m$. With $\underline{\mathbf{u}}_m = \sum_{j=0}^m u_m^j(t)\underline{\mathbf{v}}_j$ we have for the second order derivative $\ddot{\underline{\mathbf{u}}}_m$

$$\begin{aligned} &\int_0^T |\langle \ddot{\underline{\mathbf{u}}}(t)_m, I_m \underline{\mathbf{v}}^1(t)\rangle| dt \\ &\leq \underline{C}_2 \left(||\underline{f}||_{L^2(0,T;L^2(\Omega))} + ||\underline{\mathbf{u}}_m||_{L^2(0,T;H^1_{\underline{\mathcal{B}}}(\Omega))} + ||\dot{\underline{\mathbf{u}}}_m||_{L^2(0,T;H^1_{\underline{\mathcal{B}}}(\Omega))} \right) \text{ and} \\ &\leq \underline{C}_4 \left(||\underline{\mathbf{u}}_{m0}||_{H^1_{\underline{\mathcal{B}}}(\Omega)} + ||\mathbf{u}_{m1}||_{L^2(\Omega)} + ||\phi^e||_{H^1(0,T;H^{\frac{1}{2}}(\Gamma))} \right). \end{aligned} \tag{1.94}$$

By the boundedness of $\underline{\mathbf{u}}_m$ we can select subsequences and pass to their limits $m_l \to \infty$ reminding that $\frac{d}{dt}$ is a closed operator:

$$\underline{\mathbf{u}}_{ml} = \begin{pmatrix} \mathbf{u}_{m_l} \\ \phi_{0m_l} \end{pmatrix} \rightharpoonup \begin{pmatrix} \mathbf{u} \\ \phi_0 \end{pmatrix} \text{ weakly star in } L^\infty(0,T,H^1_B(\Omega)) \times L^\infty(0,T,H^1_{0,\Gamma}(\Omega))$$

and

$$\dot{\underline{\mathbf{u}}}_{m_l} = \begin{pmatrix} \dot{\mathbf{u}}_{m_l} \\ \dot{\phi}_{0m_l} \end{pmatrix} \rightharpoonup \begin{pmatrix} \dot{\mathbf{u}} \\ \dot{\phi}_0 \end{pmatrix} \text{ weakly star in } L^\infty(0,T,L^2(\Omega)) \times L^\infty(0,T,L^2(\Omega)).$$

For the acceleration we have

$$I_M \ddot{\underline{\mathbf{u}}}_{m_l} = \begin{pmatrix} \ddot{\mathbf{u}}_{m_l} \\ 0 \end{pmatrix} \rightharpoonup \ddot{\mathbf{u}} \text{ weakly in } L^2(0,T,H^{-1}_{\mathcal{B}}(\Omega))$$

and deduce existence of the solution.

In order to show that the initial conditions are satisfied we choose $\underline{\mathbf{v}}$ in (1.89) in that way that $\underline{\mathbf{v}}(T) = \dot{\underline{\mathbf{v}}}(T) = 0$ and integrate (1.89) with respect to t to obtain

$$\begin{aligned} &\int_0^T \int_\Omega \left((I_M \underline{\mathbf{u}}(t))^T \ddot{\underline{\mathbf{v}}}(t) - (\underline{\mathbf{C}}\,\underline{\mathcal{B}}\,\underline{\mathbf{u}}(t))^T \underline{\mathcal{B}}\,\dot{\underline{\mathbf{v}}}(t) \; + (\underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\mathbf{u}}(t))^T \underline{\mathcal{B}}\,\underline{\mathbf{v}}(t) \right) d\Omega\, dt \\ &= \langle \underline{f}, \mathbf{v}\rangle - \langle I_M \underline{\mathbf{u}}(0), \dot{\underline{\mathbf{v}}}(0)\rangle - \langle I_M \dot{\underline{\mathbf{u}}}(0), \underline{\mathbf{v}}(0)\rangle + (\underline{\mathbf{C}}\,\underline{\mathcal{B}}\,\underline{\mathbf{u}}(0))^T \underline{\mathcal{B}}\,\underline{\mathbf{v}}(0). \end{aligned} \tag{1.95}$$

Deriving an equation of the same form now for discrete $\underline{\mathbf{u}}_m$, setting $m = m_l$, and considering that $\mathbf{u}_{m_l} \to \mathbf{u}_0$ and $\dot{\mathbf{u}}_{m_l} \to \mathbf{u}_1$ we immediately see that $\mathbf{u}_m(0) = \mathbf{u}_0$ and $\dot{\mathbf{u}}_m(0) = \mathbf{u}_1$. Due to the zero entry in I_M we cannot directly check the initial

condition for the electric part. This however is given by evaluating the last term on the right-hand side in (1.95), namely

$$\nabla\phi_{0m}(0) \to \nabla\phi_0(0) = \nabla\psi_0.$$

Uniqueness can be shown by inspection of (1.91) for the homogeneous case which allows only $\underline{\mathbf{u}} = 0$ as trivial solution.

□

1.2.2 Harmonic and Static Case

In this section we consider harmonically driven piezoelectric transducers whose mathematical description in its weak form is given in (1.47). Two well-posedness theorems will be presented, from which the first one is already published in [77]. In the first theorem we assume that the material tensors are real-valued.

Theorem 1.3. *Let* $\mathbf{c}^E \in \mathbb{R}^{6\times6},\ \mathbf{e} \in \mathbb{R}^{6\times3}$ *and* $\boldsymbol{\varepsilon}^S \in \mathbb{R}^{3\times3}$ *and (1.2), (1.3) hold. Then, there exists a sequence of eigenfrequencies* $(\lambda_n)_{n\in\mathbb{N}}$ *with* $0 < \lambda_1 \leq \lambda_2, ...$ *accumulating only at infinity such that* $\forall\omega \in \mathbb{R}\setminus(\{0\} \cup \{\lambda_1, \lambda_2, ...\})$ *the system of piezoelectric PDEs in (1.38) with boundary conditions given in (1.34) has a unique weak solution which solves (1.47).*

Proof. By assumptions (1.3) on the material coefficients the bilinear form

$$\int_\Omega (\boldsymbol{\varepsilon}^S \nabla\hat{\phi})^T \overline{\nabla w} d\Omega \tag{1.96}$$

is coercive. According to the Lemma of Lax-Milgram there exists one and only one solution $\hat{\phi}_0 = \hat{\phi}_0(\hat{\mathbf{u}}) \in H^1_{0,\Gamma}(\Omega)$ to

$$\int_\Omega \left((\mathbf{e}\mathcal{B}\hat{\mathbf{u}})^T \overline{\nabla w} - (\boldsymbol{\varepsilon}^S \nabla\hat{\phi}_0)^T \overline{\nabla w} \right) d\Omega = \phi^e \int_\Omega (\boldsymbol{\varepsilon}^S \nabla\chi)^T \nabla w \, d\Omega \, \forall w \in H^1_{0,\Gamma}, \tag{1.97}$$

where $\hat{\mathbf{u}} \in H^1_{\mathcal{B}}(\Omega)$ is arbitrarily fixed and $\hat{\phi}_0$ is as in (1.45). The unique solution $\hat{\phi}_0(\hat{\mathbf{u}})$ of (1.97) is given by

$$\hat{\phi}_0(\mathbf{u}) = (-\Delta_\epsilon)^{-1} \mathrm{div}_0(-\mathbf{e}\mathcal{B}\hat{\mathbf{u}} + \phi^e \boldsymbol{\varepsilon}^S \nabla\chi), \tag{1.98}$$

where the elliptic differential operator is now a mapping on complex valued function spaces

$$\begin{array}{rcccl} -\Delta_\epsilon & : & H^1_{0,\Gamma}(\Omega) & \to & H^{-1}_{0,\Gamma}(\Omega) \\ & & v & \mapsto & \left\{ \begin{array}{lccl} -\Delta_\varepsilon v & : H^1_{0,\Gamma}(\Omega) & \to & \mathbb{C} \\ & w & \mapsto & \int_\Omega (\boldsymbol{\varepsilon}^S \nabla v)^T \overline{\nabla w} \, d\Omega \end{array} \right. \end{array}$$

and

$$\begin{array}{rcccl} \mathrm{div}_0 & : & (L^2(\Omega))^3 & \to & H^{-1}_{0,\Gamma}(\Omega) \\ & & \vec{f} & \mapsto & \left\{ \begin{array}{lccl} \mathrm{div}_0 \vec{f} & : H^1_{0,\Gamma}(\Omega) & \to & \mathbb{C} \\ & w & \mapsto & -\int_\Omega \vec{f}^T \overline{\nabla w} \, d\Omega. \end{array} \right. \end{array}$$

Substituting $\hat{\phi}_0$ into the weak form of (1.38) the piezoelectric equations written just for the unknown $\hat{\mathbf{u}}$ now read as

$$\int_\Omega -\omega^2\rho\hat{\mathbf{u}}^T\mathbf{v} + \left((\mathbf{c}^E\mathcal{B}\hat{\mathbf{u}})^T\overline{\mathcal{B}\mathbf{v}} + ((-\Delta_\epsilon)^{-1}\mathrm{div}_0\mathbf{e}\mathcal{B}\hat{\mathbf{u}})^T\mathrm{div}_0\mathbf{e}\mathcal{B}\mathbf{v}\right)d\Omega$$
$$= \phi^e\int_\Omega \left(-(e^T\nabla\chi)^T\mathcal{B}\mathbf{v} + (-\Delta_\epsilon)^{-1}\mathrm{div}_0\boldsymbol{\varepsilon}^S\nabla\chi)^T\mathrm{div}_0\mathbf{e}\mathcal{B}\mathbf{v}\right)d\Omega. \quad (1.99)$$

With this one obtains a problem of the form

$$(L - \rho\omega^2 Id)\mathbf{u}(\omega)) = Uf(\omega) \quad (1.100)$$

where

$$U : H_{\mathcal{B}}^{-1}(\Omega) \to H_{\mathcal{B}}^1(\Omega)$$

is the isometric isomorphism according to the *Riesz Representation Theorem* and $f(\omega)$ is the linear bounded functional on $H_{\mathcal{B}}^1(\Omega)$ given by the right-hand side in (1.99) where the self-adjoint operator L is defined as

$$U^{-1}L : \hat{\mathbf{u}} \mapsto \left(\mathbf{v} \mapsto \int_\Omega (\mathbf{c}^E\mathcal{B}\hat{\mathbf{u}})^T\overline{\mathcal{B}\mathbf{v}} + ((-\Delta_\epsilon)^{-1}\mathrm{div}_0\mathbf{e}\mathcal{B}\hat{\mathbf{u}})^T\mathrm{div}_0\mathbf{e}\overline{\mathcal{B}\mathbf{v}}d\Omega\right). \quad (1.101)$$

Due to (1.2) and (1.3) L is a coercive bilinear form on $H_{\mathcal{B}}^1$. By Theorem 17.11 in [149] L has countably many eigenvalues which accumulate only at infinity. □

In the next theorem complex-valued material tensors are considered. As opposed to (1.87) the composed differential operator $\underline{\mathcal{B}}$ and the material matrix are defined differently, since for the proof we require a symmetric system. The fact that the material matrix is now of saddle-point form has to be compensated by the assumptions, in particular by the sign of the imaginary part of the permittivity tensor. It is an open question whether and how the different sign rules in (1.27) and (1.102) can be justified physically.

Theorem 1.4. *Assume that the material tensors are complex-valued with*

$$\begin{gathered} -\Im(\mathbf{c}^E),\ \Im(\boldsymbol{\varepsilon}^S) \text{ positive definite, and} \\ \forall\nu \in \mathbb{R}^6, \forall\mu \in \mathbb{R}^3 :\ -\mu^T\Im(\mathbf{e})\nu \le \sqrt{-\nu^T\Im(\mathbf{c}^E)\nu}\sqrt{\mu^T\Im(\boldsymbol{\varepsilon}^S)\mu} \end{gathered} \quad (1.102)$$

and let (1.2) hold. Then, for all positive $\omega \in \mathbb{R}$ the system of PDEs (1.38) with boundary conditions (1.34) has a unique weak solution $(\hat{\mathbf{u}}, \phi)$.

Proof. We define the following composed differential operator and material matrix

$$\underline{\mathcal{B}} := \begin{pmatrix} \mathcal{B} & 0 \\ 0 & \nabla \end{pmatrix}, \quad \text{and} \quad \underline{\mathbf{M}} := \begin{pmatrix} \mathbf{c}^E & \mathbf{e}^{\mathbf{T}} \\ \mathbf{e} & -\boldsymbol{\varepsilon}^S \end{pmatrix}.$$

Together with

$$\underline{\hat{\mathbf{u}}} := (\hat{\mathbf{u}}, \hat{\phi})^T, \quad \underline{\chi} : (0, \chi)^T \quad \text{and} \quad I_M := \begin{pmatrix} \rho I & 0 \\ 0 & 0 \end{pmatrix}$$

the weak form given in (1.47) is rewritten as: Find $\underline{\hat{\mathbf{u}}} \in \mathcal{V} := \{ \begin{pmatrix} \hat{\mathbf{u}} \\ \hat{\phi} \end{pmatrix} \in H^1_{\mathcal{B}} \times H^1_{0,\Gamma} \}$ such that

$$\int_\Omega -\omega^2 (I_M \underline{\hat{\mathbf{u}}})^T \overline{\underline{\mathbf{v}}} + (\underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\hat{\mathbf{u}}})^T \overline{\underline{\mathcal{B}}\,\underline{\mathbf{v}}}\; d\Omega = \hat{\phi}^e \int_\Omega (\underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\chi})^T \overline{(\underline{\mathcal{B}}\,\underline{\mathbf{v}})}\; d\Omega \tag{1.103}$$

holds $\forall \underline{\mathbf{v}} \in \mathcal{V}$. Now, the equation (1.103) can be identified as a term of the following form

$$(\underline{L} - \omega^2 I_M)\underline{\hat{\mathbf{u}}} = \underline{U f} \tag{1.104}$$

with the complex symmetric operator

$$\underline{U}^{-1}\underline{L} := \underline{\hat{\mathbf{u}}} \mapsto \left(\underline{\mathbf{v}} \mapsto \int_\Omega (\underline{\mathbf{M}}\,\underline{\mathcal{B}}\,\underline{\hat{\mathbf{u}}})^T \overline{\underline{\mathcal{B}}\,\underline{\mathbf{v}}}\; d\Omega \right),$$

and the isometric isomorphism $\underline{U}$ here is

$$\underline{U} : (H^{-1}_{\mathcal{B}}(\Omega) \times H^{-1}_{0,\Gamma}(\Omega)) \to (H^1_{\mathcal{B}}(\Omega) \times H^1_{0,\Gamma}(\Omega))$$

. The right hand side in (1.104) is defined as

$$\langle \underline{f}, \underline{\mathbf{v}} \rangle := \begin{pmatrix} \int_\Omega (\mathbf{e}^T \nabla \chi)^T \overline{\mathcal{B}\mathbf{v}} d\Omega \\ \int_\Omega (\boldsymbol{\varepsilon}^S \nabla \chi)^T \overline{\nabla w} d\Omega \end{pmatrix}.$$

The following splitting according to the real and complex part of $\underline{\mathbf{M}}$

$$\underline{L} := \underline{A} - j\underline{B}$$

is considered where $\underline{A}$ and $\underline{B}$ are selfadjoint. Let P be the orthogonal projection onto the nullspace $\mathcal{N}(\mathcal{B})$ of the differential operator $\mathcal{B}$ within $H^1_{\mathcal{B}}(\Omega)$ and

$$\underline{P} := \begin{pmatrix} P & 0 \\ 0 & 0 \end{pmatrix}, \quad \underline{I - P} := \begin{pmatrix} I - P & 0 \\ 0 & I \end{pmatrix}.$$

Since ∇ has a trivial nullspace in $H^1_{0,\Gamma}(\Omega)$, $\underline{P}$ is the orthogonal projection onto the nullspace of $\underline{\mathcal{B}}$. Then (1.104) is equivalent to

$$\begin{aligned} (\underline{I - P})(\underline{L} - \omega^2 I_M)(\underline{I - P})\underline{\hat{\mathbf{u}}} &= (\underline{I - P})\underline{U f} \\ -\rho\omega^2 \underline{P}\,\underline{\hat{\mathbf{u}}} &= \underline{P}\,\underline{U f} = 0, \end{aligned} \tag{1.105}$$

since $\underline{P}\,\underline{L} = 0$ and for $\underline{f}$ in (1.104) according to (1.103) one has $\underline{P}\,\underline{U f} = 0$ since for all $\mathbf{v} \in \mathcal{N}(\mathcal{B})$ there holds $U(\mathbf{e}^T \nabla \chi)^T \overline{\mathcal{B}\mathbf{v}} = 0$. The operator $\underline{B}$ is positive definite on the orthogonal complement of $\mathcal{N}(\mathcal{B})$ due to (1.102). Now, assume that $(\underline{I - P})(\underline{L} - \omega^2 I_M)(\underline{I - P})$ is nonregular. Then also

$$(\underline{I - P})(\underline{A} - j\underline{B} - \omega^2 I_M)(\underline{I - P})$$

and

$$((\underline{I-P})\underline{B}(\underline{I-P}))^{-\frac{1}{2}}\,(\underline{I-P})(\underline{A}-\omega^2 I_M)(\underline{I-P})\,((\underline{I-P})\underline{B}(\underline{I-P}))^{-\frac{1}{2}} - jI$$

are nonregular. This however implies that j is in the spectrum of $((\underline{I-P})\underline{B}(\underline{I-P}))^{-\frac{1}{2}}(\underline{I-P})(\underline{A}-\omega^2 I_M)((\underline{I-P})\underline{B}(\underline{I-P}))^{-\frac{1}{2}}$ which is symmetric and possesses a real-valued spectrum. By this contradiction $(\underline{I-P})(\underline{L}-\omega^2 I_M)(\underline{I-P})$ is regular. □

Theorem 1.4 is therefore of more general nature than Proposition 2 in [77]. There an additional postulation of the form $\lambda_{\min}(-\Im(\mathbf{c}^E))\cdot\lambda_{\min}(-\Im(\boldsymbol{\varepsilon}^S)) > \lambda_{max}(\Re(\mathbf{e})^T\Re(\mathbf{e}))$ needs to be fulfilled in order to show existence of the solution for all ω.
In case that $\omega = 0$, i.e. for the static version the equation (1.104)

$$\underline{L}\,\underline{\mathbf{u}} = \underline{U}\underline{f} \tag{1.106}$$

with real-valued material parameters we can immediately formulate a well-posedness result assuming that the compatibility condition

$$\int_\Omega \mathcal{B}^T\vec{\sigma}d\Omega + \int_{\partial\Omega} N^T\vec{\sigma}d\Gamma = 0 \tag{1.107}$$

holds with

$$\vec{\sigma} = \mathbf{c}^E\mathcal{B}\mathbf{u} + \mathbf{e}^T\nabla(-\triangle_\epsilon)^{-1}\mathrm{div}_0\mathbf{e}\mathcal{B}\mathbf{u}.$$

In this case (1.106) can be tested with all $\underline{\mathbf{v}} \in \mathcal{V}$ including in particular those where $\mathbf{v} \in \mathcal{N}(\mathcal{B})$. So the space where the compatibility condition holds can be identified as the orthogonal complement of the nullspace of $\underline{L}$, i.e.

$$\mathcal{N}(\underline{\mathcal{B}}^T\underline{\mathbf{M}\mathcal{B}})^\perp.$$

Corollary 1.1. *Let the material tensors be as in Theorem 1.3. Together with the compatibility condition (1.107) the set of partial differential equations in (1.40) has a unique weak solution defined within an arbitrary rigid displacement.*

For a fundamental solution, see e.g. [4].

1.3 Finite Element Modeling

For the numerical solution of the set of partial differential equations describing the piezoelectric effect in the linear case there exists a natural procedure which proposes to search the solutions in appropriate finite dimensional subspaces of $H^1_{\mathcal{B}}(\Omega)\times H^1(\Omega)_{0,\Gamma}$, which will be denoted by $[S_h]^3\times T_h$. The subscript h will later be referred to as the size of the used finite element mesh. We will demonstrate the finite element analysis for the case of the non-stationary systems without any damping terms in order to keep the representation simple.

Introducing the bilinear forms

$$\begin{aligned} a(\mathbf{u},\mathbf{v}) &:= \int (\mathbf{c}^E \mathcal{B}\mathbf{u})^T \overline{\mathcal{B}\mathbf{v}} d\Omega \\ b(\phi,\mathbf{v}) &:= \int (\mathbf{e}^T \nabla\phi)^T \overline{\mathcal{B}\mathbf{v}} d\Omega \\ c(\mathbf{u},w) &:= \int (e\mathcal{B}\mathbf{u})^T \overline{\nabla w} d\Omega \\ d(\phi,w) &:= \int (\varepsilon\nabla\phi)^T \overline{\nabla w} d\Omega \\ m(\mathbf{u},\mathbf{v}) &:= \int \rho \mathbf{u}^T \overline{\mathbf{v}} d\Omega \end{aligned} \tag{1.108}$$

the pair $(\mathbf{u}_h, \phi_h) \in [S_h]^3 \times T_h$ is a discrete solution of the equations in (1.46) if

$$\begin{aligned} m(\ddot{\mathbf{u}}_h,\mathbf{v}) - (a(\mathbf{u}_h,\mathbf{v}) + b(\phi_h,\mathbf{v})) &= \langle f,\mathbf{v}\rangle \\ c(\mathbf{u}_h,w) - d(\phi_h,w) &= \langle g,\mathbf{v}\rangle \quad \forall\ \mathbf{v} \in [S_h]^3 \text{ and } \forall\ w \in S_h. \end{aligned} \tag{1.109}$$

In particular with n_{eq} indicating the number of nodes with no homogeneous Dirichlet boundary conditions

$$\begin{aligned} &\{\mathbf{v}_1,...,\mathbf{v}_{n_{eq}}\} \quad \text{being a basis of } [S_h]^3,\ \mathbf{v}_i = \begin{pmatrix} v_i & 0 & 0 \\ 0 & v_i & 0 \\ 0 & 0 & v_i \end{pmatrix} \text{ and} \\ &\{w_1,...,w_{n_{eq}}\} \quad \text{being a basis of } T_h, \end{aligned} \tag{1.110}$$

it holds for $i = 1,...,n_{eq}$

$$\begin{aligned} m(\ddot{\mathbf{u}}_h,\mathbf{v}_i) - (a(\mathbf{u}_h,\mathbf{v}_i) + b(\phi_h,\mathbf{v}_i)) &= \langle f,\mathbf{v}_i\rangle \\ c(\mathbf{u}_h,w_i) - d(\phi_h,w_i) &= \langle g,w_i\rangle. \end{aligned} \tag{1.111}$$

With the approximation for the unknowns

$$\begin{aligned} \mathbf{u} &\approx \mathbf{u}_h = \sum_{k=1}^{n_{eq}} \mathbf{u}_m^k \mathbf{v}_k \\ \phi &\approx \phi_h = \sum_{k=1}^{n_{eq}} \phi_m^k w_k \end{aligned} \tag{1.112}$$

where $(\mathbf{u}_h, \phi_h)$ denote the finite element solutions, we arrive at the semidiscrete

linear system of equations

$$\begin{aligned}\sum_{k=1}^{n_{eq}} m(\mathbf{v}_k,\mathbf{v}_i)\ddot{\mathbf{u}}_m^k - \left(\sum_{k=1}^{n_{eq}} a(\mathbf{v}_k,\mathbf{v}_i)\mathbf{u}_m^k + \sum_{k=1}^{n_{eq}} b(w_k,\mathbf{v}_i)\phi_m^k\right) &= \langle f, v_i\rangle \\ \sum_{k=1}^{n_{eq}} c(v_k,w_i)\mathbf{u}_m^k - \sum_{k=1}^{n_{eq}} d(w_k,w_i)\phi_m^k &= \langle g, w_i\rangle, \\ & i = 1,...,n_{eq}\end{aligned} \tag{1.113}$$

which can be rewritten as the following algebraic problem

$$\mathbf{M}\underline{\ddot{\mathbf{u}}} + \mathbf{K}\underline{\mathbf{u}} = \mathbf{F} \tag{1.114}$$

with

$$\mathbf{M} := \begin{pmatrix} M_{uu} & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{K} := \begin{pmatrix} K_{uu} & K_{u\phi} \\ K_{\phi u} & -K_{\phi\phi} \end{pmatrix} \tag{1.115}$$

and

$$\underline{\mathbf{u}} := (\mathbf{u}_m, \phi_m) \text{ and } \mathbf{F} := (f_m, g_m).$$

For any element the values of the mechanical displacement or electric potential according to (1.112) are given by a linear combination of polynomial interpolation functions $\mathbf{v}_k$ and w_k and the quantities $\mathbf{u}_m^k$ and ϕ_m^k. The stiffness and mass matrices are obtained by assembling the entries of the following single matrices

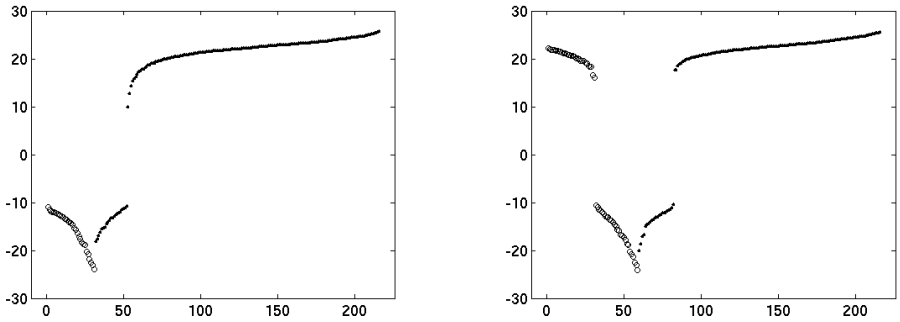
$$\begin{aligned}(K_{uu}^e)_{ik} &= \int_{\Omega_e} (\mathbf{c}^E \mathcal{B}\mathbf{v}_i)^T \mathcal{B}\mathbf{v}_k \, d\Omega_e \\ (K_{u\phi}^e)_{ik} &= \int_{\Omega_e} (\mathbf{e}\mathcal{B}\mathbf{v}_i)^T \nabla w_k \, d\Omega_e \\ (K_{\phi u}^e)_{ik} &= \int_{\Omega_e} (\mathbf{e}^T \nabla w_i)^T \mathcal{B}\mathbf{v}_k \, d\Omega_e \\ (K_{\phi\phi}^e)_{ik} &= \int_{\Omega_e} (\varepsilon \nabla w_i)^T \nabla w_k \, d\Omega_e \\ (M_{uu}^e)_{ik} &= \int_{\Omega_e} (\rho \mathbf{v}_i)^T \mathbf{v}_k \, d\Omega_e \end{aligned} \tag{1.116}$$

using appropriate quadrature formulas for the integration over each finite element Ω_e [17, 62, 80].

For the time harmonic case the second time derivatives transform to a multiplication with $-\omega^2$. This reduces the semidiscrete system in (1.114) to a pure discrete problem for each frequency ω

$$\begin{pmatrix} -\omega^2 M_{uu} + K_{uu} & K_{u\phi} \\ K_{\phi u} & -K_{\phi\phi} \end{pmatrix} \begin{pmatrix} \hat{\mathbf{u}}_m \\ \hat{\phi}_m \end{pmatrix} = \begin{pmatrix} f_m \\ g_m \end{pmatrix}. \tag{1.117}$$

Figure 10 shows the spectra for the system matrix of (1.117). In the pure static case, i.e. $\omega = 0$ the system matrix consists only of the stiffness matrices (left picture). By the properties of the material tensors the system matrix is complex symmetric. Due to the negative sign in front of the electric part, however, one deals with a saddle point problem. In the harmonic case additional contributions from the assembly of the mass term times $-\omega^2$ increases the number of negative eigenvalues (right picture).



static case $\omega = 0\,\mathrm{s}^{-1}$ harmonic case $\omega = 2.0e + 06\,\mathrm{s}^{-1}$

Figure 10: Spectrum of the system matrix for an example 2D problem. The ordinate shows $|\log_{10}(\lambda_k)|$ with $\circ$ marking negative eigenvalues λ_k and $\bullet$ positive ones.

1.3.1 Time Stepping Scheme for Transient Computations

We introduce the damping matrix $\mathbf{C}$ in case of Rayleigh damping

$$\mathbf{C} := \alpha \begin{pmatrix} M_{uu} & 0 \\ 0 & 0 \end{pmatrix} + \beta \begin{pmatrix} K_{uu} & 0 \\ 0 & 0 \end{pmatrix} \tag{1.118}$$

or for the case of the model given in (1.39) by

$$\mathbf{C} := -\frac{1}{\omega} \begin{pmatrix} \Im(K_{uu}) & \Im(K_{u\phi}), \\ \Im(K_{\phi u}) & -\Im(K_{\phi\phi}) \end{pmatrix}, \quad \mathbf{K} := \begin{pmatrix} \Re(K_{uu}) & \Re(K_{u\phi}) \\ \Re(K_{\phi u}) & -\Re(K_{\phi\phi}) \end{pmatrix}. \tag{1.119}$$

Now for the piezoelectric problem of motion the discrete time step $n+1$ reads as

$$\mathbf{M}\underline{\ddot{\mathbf{u}}}_{n+1} + \mathbf{C}\underline{\dot{\mathbf{u}}}_{n+1} + \mathbf{K}\underline{\mathbf{u}}_{n+1} = \mathbf{F}_{n+1}. \tag{1.120}$$

Together with the additional finite difference formulas

$$\underline{\mathbf{u}}_{n+1} = \underline{\mathbf{u}}_n + \Delta t\, \underline{\dot{\mathbf{u}}}_n + \frac{\Delta t^2}{2}((1-2\beta_H)\underline{\ddot{\mathbf{u}}}_n + 2\beta_h \underline{\ddot{\mathbf{u}}}_n) \tag{1.121}$$

$$\underline{\dot{\mathbf{u}}}_{n+1} = \underline{\dot{\mathbf{u}}}_n + \Delta t\, ((1-\gamma_H)\underline{\ddot{\mathbf{u}}}_n + \gamma_H \underline{\ddot{\mathbf{u}}}_{n+1}) \tag{1.122}$$

describing the evolution of the approximate solution, the Newmark scheme can be derived. Here, n denotes the time step counter, Δt the time step size, and β_h and γ_H are integration parameters. Plugging (1.121) and (1.122) into (1.120) one obtains

$$\begin{aligned}&\mathbf{M}\underline{\ddot{\mathbf{u}}}_{n+1} + \mathbf{C}(\underline{\dot{\mathbf{u}}}_n + \Delta t(1-\gamma_H)\underline{\ddot{\mathbf{u}}}_n + \gamma_h\underline{\ddot{\mathbf{u}}}_{n+1})\\ &+ \mathbf{K}(\underline{\mathbf{u}}_n + \delta t\underline{\dot{\mathbf{u}}}_n + \frac{\Delta t^2}{2}(1-2\beta_h)\underline{\ddot{\mathbf{u}}}_n + 2\beta_h\underline{\ddot{\mathbf{u}}}_{n+1}) = \mathbf{F}_{n+1}.\end{aligned}$$

Rearranging terms and introducing a predictor-corrector step one arrives at the following so called effective stiffness formulation where the predictors compute as

$$\begin{aligned}\underline{\tilde{\mathbf{u}}} &= \underline{\mathbf{u}}_n + \Delta t\underline{\dot{\mathbf{u}}}_n + (1-2\beta_H)\frac{\Delta t^2}{2}\underline{\ddot{\mathbf{u}}}_n,\\ \underline{\dot{\tilde{\mathbf{u}}}} &= \underline{\dot{\mathbf{u}}}_n + \Delta t(1-\gamma_H)\underline{\ddot{\mathbf{u}}}_n.\end{aligned} \tag{1.123}$$

With this, in each time step one solves

$$\begin{aligned}\mathbf{K}^*\underline{\mathbf{u}}_{n+1} &= \mathbf{F}_{n+1} - \mathbf{C}\underline{\dot{\mathbf{u}}} + (\frac{1}{\beta_H\Delta t^2}\mathbf{M} + \frac{\gamma_H}{\beta_h\Delta t}\mathbf{C})\underline{\tilde{\mathbf{u}}},\\ \mathbf{K}^* &:= \mathbf{K} + \frac{\gamma_H}{\beta_H\Delta t}\mathbf{C} + \frac{1}{\beta_h\Delta t^2}\mathbf{M}.\end{aligned} \tag{1.124}$$

Performing now one corrector step gives the first and second time derivative at time step $n+1$

$$\begin{aligned}\underline{\ddot{\mathbf{u}}}_{n+1} &= \frac{\underline{\mathbf{u}}_{n+1} - \underline{\tilde{\mathbf{u}}}}{\beta_h\Delta t^2},\\ \underline{\dot{\mathbf{u}}}_{n+1} &= \underline{\dot{\tilde{\mathbf{u}}}} + \gamma_H\Delta t\underline{\ddot{\mathbf{u}}}_{n+1}.\end{aligned} \tag{1.125}$$

1.3.2 Triangulation

Concerning the decomposition of the domain Ω let us assume the following triangulation $\mathcal{T}_h$ where Ω_e and $\Omega_{e'}$ denote two arbitrarily chosen finite elements

i) $\Omega = \bigcup_{e=1}^{n_{el}} \Omega_e$

ii) If $\Omega_e \cap \Omega_{e'}$ consists just of one point, then this is a vertex of both Ω_e and $\Omega_{e'}$

iii) If $\Omega_e \cap \Omega_{e'}$ consists of more than one point, then it is a common edge of Ω_e and $\Omega_{e'}$

iv) $\exists\delta > 0,\ \forall\Omega_e \in \mathcal{T}_h,\ \frac{h_{\Omega_e}}{\rho_{\Omega_e}} < \delta$,
with $h_{\Omega_e} = \operatorname{diam}(\Omega_e)$
and $\rho_{\Omega_e} = \sup\{\operatorname{diam}(S_e),$ where S_e is the inner sphere of $\Omega_e\}$

v) $h = \max_{\Omega_e \in \mathcal{T}_h} h_{\Omega_e}$ tends to zero

vi) Let $\hat{\Omega}_e$ be an arbitrary reference element. Then, there exists an invertible affine function F_{Ω_e} such that

$$F_{\Omega_e} : \hat{y} \in \hat{\Omega}_e \rightarrow F_{\Omega_e}(\hat{x}) = J\hat{x} + b$$

where J denotes the invertible Jacobian and b is a constant vector. To each vertex s_i there exists a vertex $\hat{s}_i$ in the reference element which is obtainable by the mapping

$$F_{\Omega_e}(\hat{s}_i) = s_i, \quad i = 1, ..., n_{el},$$

with this we can write

$$\int_{\Omega_e} \phi(y)dy = \int_{\hat{\Omega}_e} |\det(J)|\hat{\phi}(\hat{y})d\hat{y}.$$

With this one can state more precisely how the finite element spaces T_h look like

$$T_h(\Omega) := \{w \in \mathcal{C}^0(\Omega);\ \forall \Omega_e \in \mathcal{T}_h :\ w_{|\Omega_e} \in \mathbb{P}_k\}.$$

Here $\mathbb{P}_k$ denotes the space of polynomials of degree less than k. For the computations performed here mostly $k = 2$ was used. In particular for harmonic analysis, concerning the mesh size one desires adequate resolutions where as a rule of thumb 8 to 20 finite elements per wavelength are advisory, see e.g. [2].

Concerning the forward problem, the piezoelectric samples operating at radial, thickness, and longitudinal mode are modeled as rotationally symmetric problems. Thin plates vibrating in transversal and shear modes need full three dimensional meshes. Approximately 250 second order quadrilateral elements are used for the rotationally symmetric and 320 second order hexahedron elements for the 3D case. A mesh convergence study has been performed in advance to ensure a sufficient fine discretization. The sparse direct solver PARDISO [128], an efficient solver for large sparse symmetric and unsymmetric linear systems of equations, is employed to solve the arising algebraic system of equations.

FEM Formulations for the Piezoelectric Problem in Literature

Literature on the finite element method applied to piezoelectricity is meanwhile extensive.
The first work giving a detailed description of a 3D piezoelectric finite element procedure was done by Allik and Hughes [6] where the authors extend the variational principle published in Holland and EerNisse [59]. While Allik and Hughes propose a special electroelastic tetrahedral finite element, Lerch [97] developed the finite element theory in a general manner, i.e. independent of a special finite element type and additionally treats transient responses, and discusses non-uniform damping.

An interesting alternative implementation approach is suggested by Landis in [94] where the author proposes not the standard scalar electric potential formulation, but instead uses a vector potential formulation. Higher costs, namely two additional degrees of freedom for the vector potential, are supposed to be compensated by the fact that the resulting stiffness matrix is not only symmetric but also positive definite. For further reading the author refers to the following literature [2,5,61,80,86].

In Figure 12 a numerical solution of a piezoelectric stack actuator is presented.

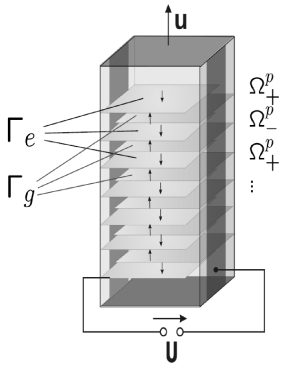


Figure 11: Schematic diagram of piezoelectric stack actuator with alternating polarization within the layers Ω^p_+ and Ω^p_-

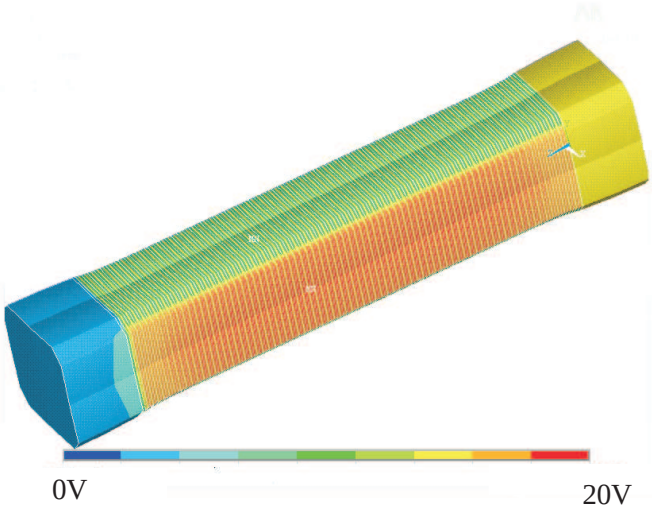


Figure 12: Numerical simulation result of a piezoelectric stack actuator with 200 layers of 100 μm thickness. The electric potential (0-20V) over the deformed shape is presented. The maximal amplitude of the deflection in this model is $50\mu m$.

1.4 Summary Chapter 1

In this chapter the set of PDEs governing linear piezoelectric effects was derived. Appropriate damping terms are introduced in order to model the correct treatment of energy dissipations. Well-posedness results for stationary, non-stationary, and harmonic formulations are given. Here, first of all new results on well-posedness in the transient case including Rayleigh damping are proven. Concerning well-posedness for the harmonic case we can guarantee a solution for all frequencies ω in case of complex-valued material parameters under weaker assumptions on the damping terms than in previous results. As corollaries well-posedness for a transient formulation with stiffness proportional viscous damping for the elastic and electric part of the system, and for the static case could be derived. For the numerical realization the finite element method is shortly introduced.

2 The Inverse Problem – Iterative Regularization of Nonlinear Ill-Posed Problems

The second chapter is mainly devoted to inverse problems in mathematics. It reviews some general concepts from the theory of inverse problems, describes nonlinear iterative regularization algorithms and provides convergence analysis for an iterative multilevel algorithm based on modified Landweber Iterations.

2.1 Preliminaries on Regularization Theory

Inverse or ill-posed problems, including parameter identification, are according to Hadamard [48] those, which violate at least one of the three requirements for being well-posed:

1. For all admissible data, a solution exists.
2. For all admissible data, the solution is unique.
3. The solution depends continuously on the data.

There are a lot of examples in mathematics, physics and real life where problems occur which do not fulfill the conditions given by Hadamard. Among them are for example [36,145]: Numerical differentiation, solving integral equations of the first kind, estimation of probability distributions, Radon inversion (in X-Ray tomography), inverse problems in heat conduction, estimation of coordinates of a seismic event, impedance tomography, inverse scattering problems, and many more.

The mathematical problem of parameter identification considered here is of the form

$$F : D(F) \to Y \text{ with } D(F) \subseteq X \quad \text{and} \quad F(p) = y^\delta, \tag{2.1}$$

where F is a nonlinear operator mapping from the infinite dimensional Hilbert space X into Y. The operator F is assumed to be at least one time Fréchet differentiable. Its argument p denotes the sought-after quantity and y^δ is the noisy input, in general measurements of physical quantities with

$$||y - y^\delta|| < \delta \tag{2.2}$$

where δ is a measure for the noise level and y denotes noisefree data.

The violation of the first Hadamard's criterion can be compensated by relaxing the notion of an 'exact solution', for example by requiring that the solution is a least-squares minimizer. If the second item is not fulfilled one has to think of how to select the 'right solution' from probably several different ones. Choosing, for example, the one with the p^0-minimal norm, i.e. where

$$||p^\dagger - p^0|| = \min_{p^* \in D(F)} \{||p^* - p^0|| \mid F(p^*) = y\}, \tag{2.3}$$

Requirement:	Condition:	Remedy, if not fulfilled:
Solution exists	F is surjective, i.e. $F(D) = Y$	search for least-squares solution
Solution is unique	F is injective, i.e. $\mathcal{N}(F) = \{0\}$	search for p^0 - minimum norm solution
Solution depends continuously on data	F^{-1} is continuous	construct stable neighboring problem, regularize

Table 4: Overview: Well-posedness violations and their remedies

one searches among the solutions the one with minimal distance to the initial guess p^0 assuming however that this minimizer exists. This kind of selection allows to incorporate as much *a priori* information about the physical problem as available.

The violation of the third point in Hadamard's definition causes in particular numerical problems while implementing a scheme to find the solution of the inverse problem. Traditional numerical methods which work fine for well-posed problems become unstable. Regularizing methods tackling these problems may be categorized as

- Nonlinear *Tikhonov method*, i.e $\min_p ||F(p) - y||^2 + \alpha||p - p^0||^2$ with $\alpha > 0$ (regularization by shifting the spectrum) [35, 36, 133].
- Nonlinear iterative methods, i.e $p^{k+1,\delta} = p^{k,\delta} + G(p^k, y^\delta),\ k = 1, 2, ..., k_*$ (regularization by early stopping) [52–54, 79, 123].
- Projection onto finite dimensional subspaces, e.g. in image space, i.e $Y_h \subset Y : \mathcal{Q}_h F(p_h) = \mathcal{Q}_h y_h^\delta$ with a projector $\mathcal{Q} : Y \to Y_h$ (regularization by coarse discretization) [36, 70, 71, 84] .

The values α, k_* and h are so-called regularization parameters. Their correct choice is a demanding task when working with ill-posed problems since they steer the trade-off between regularization and approximation. *A priori* and *a posteriori* parameter choice strategies are called on in order to detect the correct value for α, k_* or h. For example Morozov [104] proposed the discrepancy principle which is an *a posteriori* parameter choice selecting the most stable problem under the restriction that the residual norm is of the level of the data error [104]. For instance for the class of iteratively regularized problems this can be formulated for $\tau > 1$ as

$$||F(p^{k_*(\delta,y^\delta)}) - y^\delta|| \leq \tau\delta \leq ||F(p^{k,\delta}) - y^\delta||, \quad 0 \leq k < k_*(\delta, y^\delta). \tag{2.4}$$

Regarding the nonlinearity, stronger assumptions have to be made than in the well-posed situation. We will here assume that the remainder term of the Taylor approximation of the operator F fulfills the tangential cone condition [54], where

F' denotes a linearization of F

$$||F(p)-F(q)-F'(p)(p-q)|| \leq c_{tc}||F(p)-F(q)|| \quad \text{with} \quad p,q \in \mathcal{B}_\rho(p^0) \subseteq D(F). \tag{2.5}$$

Here, $\mathcal{B}_\rho(p^0)$ denotes an open ball around the initial guess p^0 with radius ρ. Estimates, where the norm of the difference of two arguments appears on the right-hand side of (2.5) do not provide sufficient information in order to establish convergence analyses of iterative methods for ill-posed problems. This is due to the fact that for ill-posed problems the left-hand side of (2.5) may be much smaller than the estimate on its right-hand side [36, 79]. Considering (2.5) one can state a uniqueness result, where $\mathcal{N}$ denotes the nullspace of an operator and $\perp$ the orthogonal complement.

Lemma 2.1. *(Similarly as in [54,79]) Let the tangential cone condition hold. If for any $p, \tilde{p} \in \mathcal{B}_\rho(p^0)$ the equality $F(p) = F(\tilde{p})$ for the images holds and additionally $p - \tilde{p} \in \mathcal{N}(F'(p))^\perp$, then $p = \tilde{p}$.*

Proof. If that the tangential cone condition holds we have

$$\begin{aligned} & c_{tc}||F(p) - F(\tilde{p})|| \\ & \quad \geq \quad ||F(p) - F(\tilde{p}) - F'(p)(p - \tilde{p})|| \\ & \quad \geq \quad ||F'(p)(p - \tilde{p})|| - ||F(p) - F(\tilde{p})|| \end{aligned} \tag{2.6}$$

by the second triangle inequality. Therefore

$$||F'(p)(p - \tilde{p})|| \leq (1 + c_{tc})||F(p) - F(\tilde{p})|| = 0.$$

Due to the nonnegativity of the norm and the fact that $p-\tilde{p} \in \mathcal{N}(F'(p))^\perp$ we have

$$\begin{aligned} ||F'(p)(p - \tilde{p})|| \quad &= \quad 0 \\ &\Leftrightarrow \quad p = \tilde{p}. \end{aligned} \tag{2.7}$$

□

In the sequel it is always assumed that a solution $p^\dagger$ exists and satisfies $p^\dagger - p^0 \in \mathcal{N}(F'(p^\dagger))^\perp$. Further, the initial guess is assumed to be sufficiently close to the solution, i.e

$$p^\dagger \in \mathcal{B}_\rho(p^0), \tag{2.8}$$

so local convergence is considered.

The structure of Chapter 2 is the following: After preluding in most important aspects of inverse or ill-posed problems a review of iterative regularization methods which are applied in this thesis are given. The main focus of this chapter lies in presenting an iterative multilevel algorithm including modified Landweber iterations as regularizing inner methods on each level. Convergence results and regularization properties are proposed and proven. Finally, this chapter contains a harmonic identification example, where the essential nonlinearity condition assumed in the convergence results is shown.

2.2 Regularizing Iterative Methods for Nonlinear Ill-Posed Problems

2.2.1 Inexact Newton Methods

Considering experiences from well-posed problems, *Newton's method* is one of the most powerful algorithms, converging locally with a quadratic rate. Since Newton's method implements a repeated linearization of the nonlinear problem out of a truncated Taylor expansion, in each Newton step a linearized problem needs to be solved. This in case of ill-posed problems is usually as ill-posed as the underlying nonlinear one. However, there are examples [35] where it is shown that the connection between the ill-posedness of a nonlinear operator and its linearization is not as strong as one might think. For linear ill-posed problems the ill-posedness is characterized by the nonclosedness of the range of the operator or by $\dim \mathcal{R}(F) = \infty$ for compact operators [35, 36]. There are examples where e.g. the range of F' of F is finite-dimensional at a dense set of points, i.e. the inversion of the linearized problem is bounded almost everywhere, however the nonlinear problem is everywhere ill-posed [35]. Also in this publication there is an example given where the nonlinear problem is well-posed but the linearization is not.

Nevertheless, for solving the linearized equations in Newton's method, well-known regularizing routines for linear ill-posed problems need to be applied, see among many others [36, 46, 57, 71, 72, 84, 90, 104].

The following algorithm follows ideas of [53, 123] and [31] for well-posed problems:

Algorithm 1. *Regularizing Inexact Newton Method*

CHOOSE p^0;
SET $k = 0$;
WHILE $\|y^\delta - F(p^k)\| \geq \tau\delta$
  SET $i = 0$;
  SET $s_0^{k,\delta} = 0$;
  WHILE $||y^\delta - F(p^{k,\delta}) - F'(p^{k,\delta})[s_i^{k,\delta}]|| \geq \eta_k ||y^\delta - F(p^k)||$ (2.9)
    $s_{i+1}^{k,\delta} = \Phi(F'(p^{k,\delta}), y^\delta - F(p^{k,\delta}), s_i^{k,\delta})$;
    $i = i + 1$;
  $p^{k+1,\delta} = p^{k,\delta} + \theta s_i^{k,\delta}$;
  $k = k + 1$;

The use of iterative methods for the inexact solution of the Newton equation is motivated by the fact, that the regularization parameter, here the number of iteration steps, can easily be determined *a posteriori* by the discrepancy principle (2.9) in Algorithm 1.

The steering parameter $0 < \eta_k \leq \bar{\eta} < 1$ influences the trade-off between convergence and stability of the inner iteration. Optimal choices of η_k are discussed in [114]. Applying additional linesearch strategies steering the parameter $\theta > 0$ enlarges the radius of convergence of Algorithm 1.

The mapping Φ in Algorithm 1 stands representatively for a linear iterative regularizing method. For instance Landweber, $\nu-$ or the conjugate gradient method are suitable and have been successfully tested in this thesis, see Section 3.3.2. Since we will report on results using the $\nu-$methods as inner iterations we briefly describe them at this point. Using $\nu-$methods, which are introduced by Brakhage in [18], the operator Φ is defined via orthogonal polynomials in order to minimize a weighted residual (cf. [36]). The well known three-term recurrence for Jacobi polynomials induces the following two-step iteration for any $\nu > 0$:

Algorithm 2. (ν *- method*)

Set $z_0 = 0$; $s_0^{k,\delta} = 0$; $z_1 = \frac{4\nu+2}{4\nu+1} y^\delta$; $s_1^{k,\delta} = F'(p^{k,\delta})^* z_1$;
set $i = 1$;
while $||y^\delta - F(p^{k,\delta}) - F'(p^{k,\delta})[s_i^{k,\delta}]|| \geq \eta_k ||y^\delta - F(p^{k,\delta})||$ do

$$\xi_i = \frac{(i-1)(2i-3)(2i+2\nu-1)}{(i+2\nu-1)(2i+4\nu-1)(2i+2\nu-3)};$$

$$\omega_i = 4\frac{(2i+2\nu-1)(i+\nu-1)}{(i+2\nu-1)(2i+4\nu-1)};$$

$$z_{i+1} = z_i + \xi_i(z_i - z_{i-1}) - \omega_i(y^\delta - F(p^{k,\delta}) - F'(p^{k,\delta})[s_i^{k,\delta}]);$$

$$s_{i+1}^{k,\delta} = F'(p^{k,\delta})^* z_{i+1};$$

$$i = i + 1;$$

The authors in [79] give an overview of Newton type regularizing methods including convergence results and rates. Among them are

- the *Levenberg-Marquardt method* which minimizes the squared misfit $||F(p_N^{k,\delta}) - y^\delta||^2$ within a trust region which is equivalent to applying Tikhonov regularization to the linearized problem within a Newton iteration.
- the *iteratively regularized Gauss-Newton method*, which is similar to Levenberg-Marquardt, except that an additional regularizing term penalizes large distances of the regularized solution from the initial guess at an early stage of the iteration.
- *Broyden's method* which is a quasi Newton method where $F'(p)$ and its inverse are approximated by means of finite rank corrections.

2.2.2 Nonlinear Landweber Iteration and Modifications

Alternatively, one can implement a nonlinear iterative regularizing method, where the most basic one is the nonlinear *Landweber iteration*. Here during the fixed-point iterations the adjoint of the linearized forward operator is applied to the residual [34, 36, 54, 79, 109]

$$p^{k+1,\delta} = p^{k,\delta} + \omega^{k,\delta} F'(p^{k,\delta})^*(y^\delta - F(p^{k,\delta})), \quad k = 1, \dots . \tag{2.10}$$

For $\omega^{k,\delta} := \omega \ \forall k$ and $||\omega F'|| \leq 1$, this is the classical nonlinear Landweber iteration for which convergence results and rates can be found in [54]. With the special choices of $\omega^{k,\delta}$

$$\omega^{k,\delta} := \frac{||F'(p^{k,\delta})^*(y^\delta - F(p^{k,\delta}))||^2}{||F'(p^{k,\delta})F'(p^{k,\delta})^*(y^\delta - F(p^{k,\delta}))||^2} \tag{2.11}$$

or

$$\omega^{k,\delta} := \frac{||y^\delta - F(p^{k,\delta})||^2}{||F'(p^{k,\delta})^*(y^\delta - F(p^{k,\delta}))||^2} \tag{2.12}$$

(2.10) becomes a steepest descent or minimal error method [109, 129], respectively. The special choice of the relaxation parameters speeds up the convergence remarkably as compared to the classical Landweber iteration, see Figure 13. The choice in (2.11) is motivated by the fact that in the linear case it minimizes the norm of the residual by an optimal choice of the step length parameter $\omega^{k,\delta}$, i.e. $\min_{\omega^{k,\delta}} \frac{1}{2}||F(p^{k,\delta} + \omega^{k,\delta}s^{k,\delta}) - y^\delta||^2$, see Lemma 3.3 in [129]. The second one (2.12) is obtained in the linear case with exact data $(\delta = 0)$ by minimizing the distance between the iterates and the exact solution $\min_{\omega^{k,\delta}} \frac{1}{2}||p^{k,\delta} + \omega^{k,\delta}s^{k,\delta} - p^\dagger||^2$.

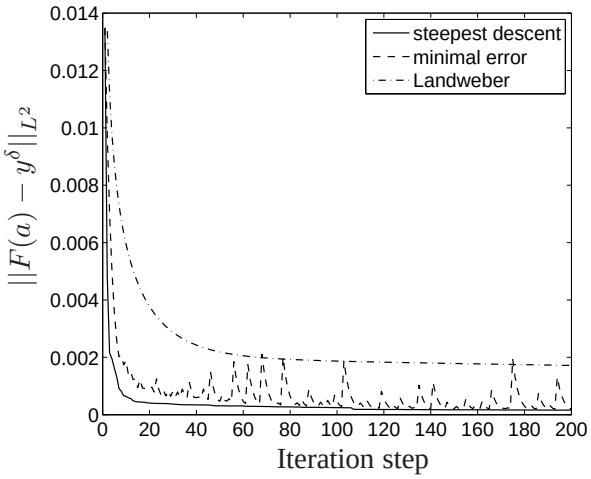


Figure 13: The decrease of the residual is compared for different Landweber methods. They are used to identify a spatially varying diffusion coefficient in a 2D elliptic PDE for $\delta = 0.01$.

2.3 Modified Landweber Methods in an Iterative Multilevel Algorithm

As already mentioned and demonstrated in Figure 13 modified versions of the Landweber method, like the steepest descent and minimal error method, show a remarkable improvement in the speed of convergence due to an appropriate steering

of the relaxation parameter $\omega^{k,\delta}$ at iteration step k. Theorems concerning monotonicity, convergence to the solution in case of exact data and the regularization character of the methods using an *a posteriori* discrepancy principle are presented in [79, 129]. Convergence rates in case of exact data are discussed in [109]. In this thesis the theory will be extended to the case where due to numerical implementations, the unknown is discretized by a finite dimensional approximation. Based on a hierarchical sequence of discretizations we can make use of a multilevel strategy, Scherzer [130]. In particular at the beginning of the identification process, where one might be far away from the true solution a rough discretization of the unknowns, i.e. for example a spline interpolation with few spline breakpoints, brings in additional regularization by coarse discretization [84]. In the course of the iteration while approaching the solution, the discretization is refined in order to reduce the discretization error.

Motivated by the efficiency of a multilevel strategy on one hand and by the speed up of convergence of minimal error and steepest descent method as compared to classical Landweber, on the other hand we aim at carrying over the results by Scherzer [130] on multilevel Landweber to (2.10) with either (2.11) or (2.12). There exists already an extension to [130]in [131], where a multilevel algorithm based on either the conjugate gradient or the Landweber method for linear problems is applied to moment problems.

In the sequel the operator F is assumed to be continuous, differentiable and its Frechét derivative F' to be Lipschitz continuous and normalized such that

$$||F'(p)|| \leq 1, \quad \forall\, p \ \in D(F). \tag{2.13}$$

Further, we assume that $\mathcal{N}(F'(p))$ is trivial for all $p \in \mathcal{B}_\rho(p^0)$.
The numerical realization of the modified Landweber iterations is now considered in a finite dimensional subspace $X_N := P_N X$ of X, where P_N denotes the orthogonal projection onto

$$X_N \subseteq X, \quad \text{with} \quad X_0 \subseteq X_1 \subseteq ... \subseteq X_N \quad \text{and} \quad \overline{\bigcup_{N\in\mathbb{N}} X_N} = X. \tag{2.14}$$

Let the union $\cup_{N\in\mathbb{N}} X_N$ be dense in X. The initial guess at discretization level N is denoted by p_N^0.
Moreover the tangential cone condition (2.5) is assumed to hold only on discrete subspaces X_N

$$\begin{aligned} ||F(p) - F(P_N p^\dagger) - F'(p)(p - P_N p^\dagger)|| &\leq \eta_N ||F(p) - F(P_N p^\dagger)|| \\ \text{for all } p \in X_N \cap \mathcal{B}_{\rho/2}(P_N p^\dagger) \subseteq D(F) &\text{ with } \eta_N \leq \frac{1}{4}. \end{aligned} \tag{2.15}$$

At each level, the iteration now reads as follows

$$p_N^{k+1,\delta} = p_N^{k,\delta} + \omega_N^{k,\delta} s_N^{k,\delta}, \qquad s_N^{k,\delta} := P_N F'(p_N^{k,\delta})^*(y^\delta - F(p_N^{k,\delta})) \tag{2.16}$$

where the coefficients $\omega_N^{k,\delta}$ are chosen either as

$$\omega_N^{k,\delta} := \frac{||s_N^{k,\delta}||^2}{||F'(p_N^{k,\delta})s_N^{k,\delta}||^2} \tag{2.17}$$

or as

$$\omega_N^{k,\delta} := \frac{||y^\delta - F(p_N^{k,\delta})||^2}{||s_N^{k,\delta}||^2}, \tag{2.18}$$

which renders the discrete classical Landweber iteration a discrete version of the steepest descend or minimal error method, respectively. By (2.13) for both choices

$$\omega_N^{k,\delta} \geq 1 \tag{2.19}$$

holds.
The size of the discretization parameter N is rather crucial. A small value of N does not allow for a good approximation. However the iteration might be very sensitive to noise in the data when using a large N. Further, depending on the implementation a large N makes the iterations extremely time consuming, e.g. if one thinks of approximating F' by finite differences.

Concerning the stopping criteria in each level the residual is tested by a combination of an approximation estimate for the current level and the data error level. The approximation error may be estimated either in X or Y which gives the following two criteria

$$||F(p_N^{k+1,\delta}) - y^\delta|| \leq C_1(\delta + ||F(P_N p^\dagger) - F(p^\dagger)||) \tag{2.20}$$

or

$$||F(p_N^{k+1,\delta}) - y^\delta|| \leq \tilde{C}_1(\delta + ||(P_N - I)p^\dagger||). \tag{2.21}$$

Each of them determines a well-defined stopping index $\overline{k}_*(N,\delta)$ at level N , i.e.

$$\overline{k}_*(N,\delta) := \min\{k \in \mathbb{N} \mid \text{where (2.20) holds}\} \tag{2.22}$$

and analogously for (2.21). Note that the case $\overline{k}_*(N,\delta) = \infty$ is possible. In the sequel we assume $C_1 = \tilde{C}_1$.

The stopping rules (2.20) and (2.21) are used for theoretical purposes; in a practical implementation the terms $||F(P_N p^\dagger) - F(p^\dagger)||$ and $||(P_N - I)p^\dagger||$ containing the unknown solution have to be estimated based on *a priori* information on $p^\dagger$, see for example [89, 108, 130, 131]. Obviously, as long as the iterations are not terminated we have for all $k < \overline{k}_*(N,\delta)$

$$||F(p_N^{k,\delta}) - y^\delta|| \quad > \quad C_1(\delta + ||F(P_N p^\dagger) - y||), \tag{2.23}$$

or

$$||F(p_N^{k,\delta}) - y^\delta|| \quad > \quad C_1(\delta + ||(P_N - I)p^\dagger)||). \tag{2.24}$$

Note that the stopping criterion might get active at $k = 0$ so that there is no k satisfying (2.24) (or (2.23)). It is a crucial assumption in [130] and also in some

(but not all) assertions made here, that at least one step is carried out on level N, i.e. (2.24) (or (2.23)) is satisfied for $k = 0$.

Even though the choice of $\overline{k}_*(N, \delta)$ in (2.22) avoids an undesired amplification of the data error, the stopping criterion has a minor drawback since it does not guarantee that the number of iterations is finite in each level. This however is necessary in order to compute the initial guess at any subsequent level. As a consequence we introduce some *a priori* chosen finite maximal number $\tilde{k}_{\max}(N)$ of iterations

$$0 < \tilde{k}_{\max}(N) < \infty. \tag{2.25}$$

A combination of (2.22) and (2.25), i.e.

$$k_*(N, \delta) := \min\{\overline{k}_*(N, \delta), \tilde{k}_{\max}(N)\} < \infty \tag{2.26}$$

gives a well suited, finite stopping index for the inner iterations of the multilevel algorithm.

Refinement of the discretization, i.e. the outer iteration will be performed until an *a posteriori* stopping rule becomes active

$$||y^\delta - F(p_N^{k_*,\delta})|| \leq C_2\delta \leq ||y^\delta - F(p_N^{k,\delta})||, \quad C_2 > 2\frac{1+\eta_N}{1-2\eta_N} > 2. \tag{2.27}$$

Only as long as this global discrepancy principle does not terminate the iterations one continues at the next level $N + 1$ and uses the last approximate from level N as an initial guess at the next finer level, i.e.

$$p_{N+1}^{0,\delta} := p_N^{k_*(N,\delta),\delta}.$$

For this purpose $p_N^{k_*(N,\delta),\delta}$ should be easily expressable by ansatz functions at level $N + 1$.

2.3.1 The Multilevel Algorithm

Algorithm 3. *Iterative multilevel algorithm for modified Landweber itertations:*

SET $p_0^{0,\delta} := P_0 p^0 \in X_0$
SET $N = 0$
SET $k = 0$
SET $k_*(N,\delta) = 0$
CHOOSE $C_1 < C_2$ sufficiently large
WHILE $||F(p_N^{k_*(N,\delta),\delta}) - y^\delta|| > C_2\delta$
  $k = 0$
  If $N > 0$
    $p_{N+1}^{0,\delta} = p_N^{k_*(N,\delta),\delta}$
    $N = N + 1$
  DO WHILE $||F(p_N^{k,\delta}) - y^\delta||$ violates (2.20) or $k > \tilde{k}_{\max}(N)$
    $p_N^{k+1,\delta} = p_N^{k,\delta} + \omega_N^k s_N^{k,\delta}, \quad s_N^{k,\delta} := P_N F'(p_N^{k,\delta})^*(y^\delta - F(p_N^{k,\delta}))$
    where $\omega_N^{k,\delta}$ is either $\omega_N^{k,\delta} := \frac{||s_N^{k,\delta}||^2}{||F'(p_N^{k,\delta})s_N^{k,\delta}||^2}$
    or $\omega_N^{k,\delta} := \frac{||y^\delta - F(p_N^{k,\delta})||^2}{||s_N^{k,\delta}||^2}$
    $k = k + 1$
  $k_*(N,\delta) = k$
$p_{N_*(\delta)} := p_N^{k_*(N,\delta),\delta}$

The algorithm can analogously be formulated with (2.20) replaced by (2.21). The assumption that C_1 and C_2 are sufficiently large can be made more precise with the help of the following definition which is taken from [130]:

Definition 2.2. An operator F is called regular at level X_N in $\overline{U}_N := \overline{X_N \cap \mathcal{B}_\rho(p_0^0)}$ if it is Fréchet-differentiable in $\overline{U}_N := \overline{X_N \cap \mathcal{B}_\rho(p_0^0)} \subseteq D(F)$ and

$$B_{inf}(N, p^\dagger, p_0^0, \rho) := \inf_{p \in U_N, p \neq P_N p^\dagger} \frac{||F(p) - F(P_N p^\dagger)||}{||p - P_N p^\dagger||} > 0.$$

An operator F is called regular at level X_N in U_N with magnitude λ_N if it is regular at level X_N and

$$\lambda_N := B_{inf}(N, p^\dagger, p_0^0, \rho).$$

The locally varying quantity λ_N is a measure for the stability of the solution of (2.1) with respect to perturbations of the right-hand side data on the finite dimensional subspace X_N.

So for the constants the following is assumed if F is regular at level X_N with magnitude λ_N

$$C_1 \;\geq\; 4(1+\eta_N), \quad C_0 < C_1 < C_2 \tag{2.28}$$

with

$$\lambda_N \frac{C_0^2}{8(1+C_0)^2} \leq ||F'(p_N^{0,\delta})||, \quad \text{and } C_0 \geq \sqrt{8} \quad \forall N.$$

2.3.2 Convergence Results

Before we show monotonicity, we investigate the well-definedness of the step-length parameters $\omega_N^{k,\delta}$. One can show, that as long as the stopping rules as defined in (2.20) and (2.21) are not active, the parameter update $s_N^{k,\delta}$ and $F'(p_N^{k,\delta})s_N^{k,\delta}$ are nonzero.

Lemma 2.3. *Let $p^\dagger \in \mathcal{B}_{\rho/2}(p^0)$ be a solution of (2.1) and $\overline{k}_*(N,\delta)$ be as in (2.22) and $p_N^{k,\delta} - P_N p^\dagger \neq 0$ for all $k < \overline{k}_*(N,\delta)$. Then*

$$||s_N^{k,\delta}|| \neq 0 \quad \textit{and} \quad ||F'(p_N^{k,\delta})s_N^{k,\delta}|| \neq 0 \quad \textit{for all } \; k < \overline{k}_*(N,\delta).$$

Proof. We carry out the proof in case of (2.20). The case (2.21) then immediately follows, since by (2.13) the right hand side in (2.21) is greater or equal the one in (2.20) if $\tilde{C}_1 = C_1$.
If $\overline{k}_*(N,\delta) = 0$ we need not to prove anything. Otherwise (2.23) holds. Assume that $k < \overline{k}_*(N,\delta)$. If $F'(p_N^{k,\delta})^*(F(p_N^{k,\delta}) - y^\delta)$ would vanish, then

$$\begin{aligned}
0 \;&=\; \big(F(p_N^{k,\delta}) - y^\delta, F'(p_N^{k,\delta})(p_N^{k,\delta} - P_N p^\dagger)\big) \\
&=\; \big(F(p_N^{k,\delta}) - y^\delta, F(p_N^{k,\delta}) - y^\delta\big) \\
&\quad + \big(F(p_N^{k,\delta}) - y^\delta, y - F(P_N p^\dagger)\big) \\
&\quad + \big(F(p_N^{k,\delta}) - y^\delta, y^\delta - y\big) \\
&\quad - \big(F(p_N^{k,\delta}) - y^\delta, F(p_N^{k,\delta}) - F(P_N p^\dagger) - F'(p_N^{k,\delta})(p_N^{k,\delta} - P_N p^\dagger)\big).
\end{aligned}$$

From this one deduces

$$\begin{aligned}
||F(p_N^{k,\delta}) - y^\delta||^2 \;&\leq\; (\delta + ||y - F(P_N p^\dagger)||)||F(p_N^{k,\delta}) - y^\delta|| \\
&\quad + \eta_N \, ||F(p_N^{k,\delta}) - F(P_N p^\dagger)|| \, ||F(p_N^{k,\delta}) - y^\delta||
\end{aligned}$$

and further

$$\begin{aligned}
||F(p_N^{k,\delta}) - y^\delta|| \;&\leq\; \delta + ||y - F(P_N p^\dagger)|| \\
&\quad + \eta_N \left(||F(p_N^{k,\delta}) - y^\delta|| + ||y - F(P_N p^\dagger)|| + ||y - y^\delta||\right).
\end{aligned} \tag{2.29}$$

So

$$||F(p_N^{k,\delta}) - y^\delta|| \leq \frac{1+\eta_N}{1-\eta_N}(||F(P_N p^\dagger) - y|| + \delta)$$

which however contradicts (2.23).
To show that $||F'(p_N^{k,\delta})s_N^{k,\delta}|| \neq 0$ let us assume again the contrary, i.e. $||F'(p_N^{k,\delta})s_N^{k,\delta}|| = 0$. Then $s_N^{k,\delta} \in \mathcal{N}(F'(p_N^{k,\delta})) = \{0\}$ which is a contradiction to the already shown fact $s_N^{k,\delta} \neq 0$. □

This actually guarantees that the choices of the relaxation parameters in (2.17) and (2.18) are well-defined.

Now, monotonicity of the iteration error at a fixed level N will be proven:

Theorem 2.1. *Let $N \in \mathbb{N}_0$ be fixed such that $||(I - P_N)p^\dagger|| < \rho/2$, where $p^\dagger$ is a solution of (2.1) in $\mathcal{B}_{\frac{\rho}{2}}(p_N^{0,\delta})$ and $p_N^{0,\delta} \in X_N$. Assume that Algorithm 3 carries out at least one inner step at level N, i.e. there exists an iteration index $k \geq 0$ such that (2.23) and (2.24) hold respectively with $C_1 \geq 4(1+\eta_N)$. Then, for $0 \leq k < \overline{k}_*(N,\delta)$*

$$\begin{aligned} p_N^{k+1,\delta} \in \mathcal{B}_{\frac{\rho}{2}}(p^\dagger) \cap X_N &\subseteq \mathcal{B}_\rho(p_N^{0,\delta}) \cap X_N \quad \textit{further} \\ ||p^\dagger - p_N^{k+1,\delta}|| &\leq ||p^\dagger - p_N^{k,\delta}|| \quad \textit{and} \\ ||P_N p^\dagger - p_N^{k+1,\delta}|| &\leq ||P_N p^\dagger - p_N^{k,\delta}||. \end{aligned} \tag{2.30}$$

Proof. Let $p_N^{k,\delta} \in \mathcal{B}_{\frac{\rho}{2}}(p^\dagger)$ for $0 \leq k < \overline{k}_*(N,\delta)$. From the definition of the iteration it follows that

$$\begin{aligned} &||p_N^{k+1,\delta} - p^\dagger||^2 \\ &= ||p_N^{k,\delta} - p^\dagger||^2 + ||\omega_N^{k,\delta} P_N F'(p_N^{k,\delta})^*(F(p_N^{k,\delta}) - y^\delta)||^2 \\ &\quad - 2\omega_N^{k,\delta}\left(F'(p_N^{k,\delta})(p_N^{k,\delta} - P_N p^\dagger), F(p_N^{k,\delta}) - y^\delta\right) \\ &= ||p_N^{k,\delta} - p^\dagger||^2 + ||\omega_N^{k,\delta} P_N F'(p_N^{k,\delta})^*(F(p_N^{k,\delta}) - y^\delta)||^2 \\ &\quad + 2\omega_N^{k,\delta}\left(F(p_N^{k,\delta}) - F(P_N p^\dagger) - F'(p_N^{k,\delta})(p_N^{k,\delta} - P_N p^\dagger), F(p_N^{k,\delta}) - y^\delta\right) \\ &\quad - 2\omega_N^{k,\delta}\left(F(p_N^{k,\delta}) - F(P_N p^\dagger), F(p_N^{k,\delta}) - y^\delta\right). \end{aligned}$$

Incorporating the nonlinearity condition (2.15) one obtains that

$$\begin{aligned} &\left(F(p_N^{k,\delta}) - F(P_N p^\dagger) - F'(p_N^{k,\delta})(p_N^{k,\delta} - P_N p^\dagger), F(p_N^{k,\delta}) - y^\delta\right) \\ &\quad - \left(F(p_N^{k,\delta}) - F(P_N p^\dagger), F(p_N^{k,\delta}) - y^\delta\right) \\ &\leq \eta_N ||F(p_N^{k,\delta}) - F(P_N p^\dagger)||\,||F(p_N^{k,\delta}) - y^\delta|| \\ &\quad - \left(F(p_N^{k,\delta}) - y^\delta, F(p_N^{k,\delta}) - y^\delta\right) \\ &\quad + ||y^\delta - F(P_N p^\dagger)||\,||F(p_N^{k,\delta}) - y^\delta|| \\ &\leq (\eta_N - 1)||F(p_N^{k,\delta}) - y^\delta||^2 + (1+\eta_N)||y^\delta - F(P_N p^\dagger)||\,||F(p_N^{k,\delta}) - y^\delta||. \end{aligned}$$

By (2.23)

$$\begin{aligned}&(1+\eta_N)||y^\delta - F(P_N p^\dagger)||\,||F(p_N^{k,\delta}) - y^\delta|| \\ &\leq (1+\eta_N)\left(||y - F(P_N p^\dagger)|| + \delta\right)||F(p_N^{k,\delta}) - y^\delta|| \leq \frac{1}{4}||F(p_N^{k,\delta}) - y^\delta||^2.\end{aligned} \tag{2.31}$$

Hence we can estimate

$$\begin{aligned}||p_N^{k+1,\delta} - p^\dagger||^2 \;&\leq\; ||p_N^{k,\delta} - p^\dagger||^2 + (\omega_N^{k,\delta})^2||s_N^{k,\delta}||^2 \\ &\quad + 2\omega_N^{k,\delta}(\eta_N - \frac{3}{4})||F(p_N^{k,\delta}) - y^\delta||^2.\end{aligned}$$

Finally, it holds that

$$\begin{aligned}&||p_N^{k+1,\delta} - p^\dagger||^2 + 2\omega_N^{k,\delta}(\frac{1}{4} - \eta_N)||F(p_N^{k,\delta}) - y^\delta||^2 \\ &\quad \leq ||p_N^{k,\delta} - p^\dagger||^2 + (\omega_N^{k,\delta})^2||s_N^{k,\delta}||^2 - \omega_N^{k,\delta}||F(p_N^{k,\delta}) - y^\delta||^2.\end{aligned} \tag{2.32}$$

Inserting now the choice of $\omega_N^{k,\delta}$ as a minimal error method (2.18) one sees, that the difference of the last two terms on the RHS of (2.32) vanishes. For the steepest descent variant (2.17) it holds

$$\omega_N^{k,\delta}||s_N^{k,\delta}||^2 = \frac{\left(F'(p_N^{k,\delta})s_N^{k,\delta}, F(p_N^{k,\delta}) - y^\delta\right)^2}{||F'(p_N^{k,\delta})s_N^{k,\delta}||^2} \leq ||F(p_N^{k,\delta}) - y^\delta||^2,$$

so in both cases

$$||p_N^{k+1,\delta} - p^\dagger||^2 + 2\omega_N^{k,\delta}(\frac{1}{4} - \eta_N)||F(p_N^{k,\delta}) - y^\delta||^2 \leq ||p_N^{k,\delta} - p^\dagger||^2. \tag{2.33}$$

The monotonicity of $||P_N p^\dagger - p_N^{k,\delta}||$ follows by inspection of the estimations of this proof. □

At this point we introduce the following set

$$\mathcal{N}^\delta := \{N \in \mathbb{N} \;:\; ||F(p_N^{0,\delta}) - y|| > C_1(\delta + ||F(P_N p^\dagger) - y^\delta||)\} \tag{2.34}$$

which contains all levels where at least one inner iterations is performed, so that $p_N^{1,\delta}$ is certainly computed. In a next step, weak convergence of the iterative multilevel algorithm with steepest descent and minimal error methods in case of exact data, i.e. $\delta = 0$, is discussed.

Proposition 2.4. *If $\delta = 0$ and $k_*(N,\delta)$ as in (2.26)*

$$F(p_N^{k_*(N,0),0}) \to y \quad \textit{as} \quad N \to \infty. \tag{2.35}$$

Proof. There are two cases to be considered:
1) The iteration stops at a finite level N. In this situation by (2.27) the last iterate is a solution of (2.1).
2) The iteration does not terminate. Because of (2.33) it holds that

$$\sum_{I\in\mathcal{N}^0}\sum_{j=0}^{k_*(I,0)-1} ||y - F(p_I^{j,0})||^2 < \infty. \tag{2.36}$$

For $I \notin \mathcal{N}^0$ the following is true

$$\begin{aligned} ||y - F(p_I^{0,0})|| &\leq ||F(P_I p^\dagger) - y|| \quad \text{or} \\ ||y - F(p_I^{0,0})|| &\leq ||(I - P_I)p^\dagger||. \end{aligned} \tag{2.37}$$

Now, we will make use of a subsequence-subsequence argument applied to

$$a_N := ||y - F(p_N^{0,0})||.$$

Let $(a_{N_i})_{i\in\mathbb{N}}$ be an arbitrary subsequence. We distinguish between the cases:

a) The index set $\mathcal{N}^0 \cap \{N_i\}_{i\in\mathbb{N}}$ is infinite, hence we can denote it by $(N_{i_l})_{l\in\mathbb{N}}$. Due to (2.36) $(a_{N_i})_{i\in\mathbb{N}}$ converges to zero.

b) The set $\mathcal{N}^0 \cap (N_i)_{i\in\mathbb{N}}$ is finite, then $(\mathbb{N}\setminus\mathcal{N}^0)\cap\{N_i\}_{i\in\mathbb{N}}$ must be infinite and with $(N_{i_l})_{l\in\mathbb{N}} := (\mathbb{N}\setminus\mathcal{N}^0)\cap\{N_i\}_{i\in\mathbb{N}}$, we have convergence of $(a_{N_{i_l}})_{l\in\mathbb{N}}$ to zero by pointwise convergence of P_I to the identity in (2.14).

So, in both cases, $(a_{N_i})_{i\in\mathbb{N}}$ has a subsequence converging to zero. This implies that each subsequence of $(a_N)_{N\in\mathbb{N}}$ has a convergent subsequence and the limit of each convergent subsequence of $(a_N)_{N\in\mathbb{N}}$ is zero.

□

With the latter result weak convergence in X_N for exact data can be established:

Proposition 2.5. *Let $\delta = 0$ and the assumptions of Proposition 2.4 hold. Moreover, let F be weakly closed. Then each subsequence of $p_N^{k_*(N,0),0}$ has a weakly convergent subsequence and each weak accumulation point is a solution of $F(p) = y$. If the solution $p^\dagger$ of (2.1) is unique, then $p_N^{k_*(N,0),0}$ converges weakly to $p^\dagger$.*

Proof. In case Algorithm 3 terminates after a finite number $N_*(0)$ of iterations by the stopping criterion (2.27) then $p_{N_*(0)}$ solves (2.1). Otherwise if $k_*(N,0) > 0$ it follows from Theorem 2.1 that

$$||p^\dagger - p_N^{k_*(N,0),0}|| \leq ||p^\dagger - p_0^{0,0}||$$

which trivially holds in the complementary case $k_*(N,0) = 0$. Therefore, we have boundedness of $p_N^{k_*(N,0),0}$ in X and $p_N^{k_*(N,0),0}$ has a weakly convergent subsequence. From (2.35) and the weak closedness of the operator F it follows that for each weak accumulation point $\overline{z}$, the identity $F(\overline{z}) = y$ holds, which proves the assertion. □

The next lemma will show that the first Landweber step at the next finer level under certain assumptions always provides some improvement to the old state of the unknown. Whereas the results so far hold for both stopping rules (2.20) and (2.21), we restrict ourselves to (2.21) with (2.26) now.

Lemma 2.6. *Let F be regular at level X_N with magnitude λ_N in U_N and assume that $\eta_N \leq \frac{1}{8}$, further that (2.24) holds for $k = 0$ at level $N+1$. Then with $\psi := \frac{\hat{C}_1^2}{8(1+\hat{C}_1)^2}$ and $\hat{C}_1 \geq \sqrt{8}$*

$$||p_{N+1}^{1,\delta} - P_{N+1}p^\dagger||^2 \leq (1 - \omega_{N+1}^{0,\delta}\psi\lambda_N^2)||p_{N+1}^{0,\delta} - P_N p^\dagger||^2.$$

Proof. Let us start with some estimates at level N. Suppose $p_N^{k,\delta}, P_N p^\dagger \in \mathcal{B}_\rho(p_0^0)$, then it follows from (2.13) for $k \leq k_*(N,\delta)$

$$\begin{aligned} ||F(p_N^{k,\delta}) - y^\delta|| &\geq ||F(p_N^{k,\delta}) - F(P_N p^\dagger)|| - ||F(P_N p^\dagger) - y|| - \delta \\ &\geq ||F(p_N^{k,\delta}) - F(P_N p^\dagger)|| - (||P_N p^\dagger - p^\dagger|| + \delta). \end{aligned}$$

Now

$$||F(p_N^{k,\delta}) - F(P_N p^\dagger)|| \leq \left(1 + \frac{1}{\hat{C}_1}\right) ||F(p_N^{k,\delta}) - y^\delta|| \tag{2.38}$$

and so together with (2.24) for $k = 0$ and since $\max\{a,b\}^2 \geq \frac{1}{2}a^2 + \frac{1}{2}b^2$

$$||F(p_N^{0,\delta}) - y^\delta||^2 \geq \frac{\hat{C}_1^2}{2(1+\hat{C}_1)^2}||\mathrm{F}(p_N^{0,\delta}) - F(P_N p^\dagger)||^2 + \frac{\hat{C}_1^2}{2}||(P_N - I)p^\dagger||^2. \tag{2.39}$$

Further $\psi \leq \left(\frac{1}{2} - 2\eta_{N+1}\right)\frac{\hat{C}_1^2}{2(1+\hat{C}_1)^2}$ and $\psi_1 := \frac{\hat{C}_1^2}{8} \leq \left(\frac{1}{2} - 2\eta_{N+1}\right)\frac{\hat{C}_1^2}{2}$.
From the proof of monotonicity (2.33) with $p^\dagger$ replaced by $P_N p^\dagger$, due to orthogonality, and since $p_{N+1}^{0,\delta} = p_N^{k_*(N,\delta),\delta} \in X_N \subseteq X_{N+1}$ one obtains

$$\begin{aligned} &||p_{N+1}^{1,\delta} - P_{N+1}p^\dagger||^2 + \omega_{N+1}^{0,\delta}\,(\frac{1}{2} - 2\eta_{N+1})||F(p_{N+1}^{0,\delta}) - y^\delta||^2 \\ \leq\ &||p_{N+1}^{0,\delta} - P_{N+1}p^\dagger||^2 \\ \leq\ &||p_{N+1}^{0,\delta} - P_N p^\dagger||^2 + ||P_{N+1}p^\dagger - P_N p^\dagger||^2 \end{aligned}$$

from which with (2.38) and (2.39) follows

$$\begin{aligned} &||p_{N+1}^{1,\delta} - P_{N+1}p^\dagger||^2 + \omega_{N+1}^{0,\delta}\psi||F(p_{N+1}^{0,\delta}) - F(P_N p^\dagger)||^2 \\ &\qquad + \omega_{N+1}^{0,\delta}\psi_1||P_N p^\dagger - p^\dagger||^2 \\ \leq\ &||p_{N+1}^{0,\delta} - P_N p^\dagger||^2 + ||P_{N+1}p^\dagger - P_N p^\dagger||^2. \end{aligned} \tag{2.40}$$

Now, with the boundedness (2.19) of $\omega_{N+1}^{0,\delta}$ from below as well as $\psi_1 \geq 1$ and

$$\begin{aligned} ||P_N p^\dagger - p^\dagger||^2 &= ||P_N p^\dagger - P_{N+1}p^\dagger + P_{N+1}p^\dagger - p^\dagger||^2 \\ &= ||P_N p^\dagger - P_{N+1}p^\dagger||^2 + \underbrace{||P_{N+1}p^\dagger - p^\dagger||^2}_{\geq 0} \\ &\quad +2\left(\underbrace{P_N p^\dagger - P_{N+1}p^\dagger}_{\in X_{N+1}}, \underbrace{P_{N+1}p^\dagger - p^\dagger}_{\in X_{N+1}^\perp}\right) \\ &\geq ||P_N p^\dagger - P_{N+1}p^\dagger||^2 \end{aligned}$$

we get

$$||(P_{N+1} - P_N)p^\dagger||^2 \leq \omega_{N+1}^{0,\delta}\psi_1||(P_N - I)p^\dagger||^2.$$

Consequently (2.40) reduces to

$$||p_{N+1}^{1,\delta} - P_{N+1}p^\dagger||^2 + \omega_{N+1}^{0,\delta}\psi||F(p_{N+1}^{0,\delta}) - F(P_N p^\dagger)||^2 \leq ||p_{N+1}^{0,\delta} - P_N p^\dagger||^2. \tag{2.41}$$

From the assumption that F is regular at level N with magnitude λ_N and with $p_{N+1}^{0,\delta} = p_N^{k_*(N,\delta),\delta} \in \mathcal{B}_\rho(p_N^{0,\delta})$ one has

$$||F(p_{N+1}^{0,\delta}) - F(P_N p^\dagger)|| \geq \lambda_N||p_{N+1}^{0,\delta} - P_N p^\dagger||.$$

From the latter inequality it follows that

$$||p_{N+1}^{1,\delta} - P_{N+1}p^\dagger||^2 \leq (1 - \omega_{N+1}^{0,\delta}\psi\lambda_N^2)||p_{N+1}^{0,\delta} - P_N p^\dagger||^2. \tag{2.42}$$

□

Corollary 2.7. *Let the assumptions from Lemma 2.6 hold. Then for all* $k < k_*(N,\delta)$*:*

$$||p_{N+1}^{k+1,\delta} - P_{N+1}p^\dagger||^2 \leq (1 - \omega_N^{0,\delta}\psi\lambda_N^2)||p_N^{0,\delta} - P_N p^\dagger||^2.$$

Proof. With the results from Theorem 2.1 and Lemma 2.6 it holds

$$\begin{aligned} ||p_{N+1}^{k+1,\delta} - P_{N+1}p^\dagger||^2 &\leq ||p_{N+1}^{1,\delta} - P_{N+1}p^\dagger||^2 \\ &\leq (1 - \omega_{N+1}^{0,\delta}\psi\lambda_N^2)||p_{N+1}^{0,\delta} - P_N p^\dagger||^2 \\ &= (1 - \omega_{N+1}^{0,\delta}\psi\lambda_N^2)||p_N^{k_*(N,\delta),\delta} - P_N p^\dagger||^2 \\ &\leq (1 - \omega_{N+1}^{0,\delta}\psi\lambda_N^2)||p_N^{0,\delta} - P_N p^\dagger||^2. \end{aligned}$$

□

Now we have all auxiliary means for deriving the main result. It states that the iterates of the proposed multilevel algorithm with least squares or minimal error inner iterations form a monotone sequence.

As opposed to (2.34) we define now the set which contains the indices of those levels where at least one iteration is performed and where (2.27) is not active yet

$$\hat{\mathcal{N}}^\delta := \{N < N_*(\delta) \mid ||F(p_N^{0,\delta}) - y^\delta|| > C_1(\delta + ||(I - P_N)p^\dagger||) =: (N_l)_{l=1}^L \tag{2.43}$$

Theorem 2.2. *Let all conditions of Lemma 2.6 hold. If for all $N \in \hat{\mathcal{N}}^\delta$ the operator F is regular at level N with magnitude λ_N in $U_N = X_N \cap \mathcal{B}_\rho(p_N^0)$, then for all $N < N_*(\delta)$ and all $k \le k_*(N,\delta)$*

$$||p_{N+1}^{k+1,\delta} - P_{N+1}p^\dagger||^2 \le \prod_{\substack{I \in \hat{\mathcal{N}}^\delta \\ I \le N}} (1 - \omega_{I+1}^{0,\delta}\psi\lambda_I^2)||p_0^{0,\delta} - P_0 p^\dagger||^2. \tag{2.44}$$

Especially, if additionally for all $l \in \{1, ..., L_* - 1\}$

$$(1 - \omega_{N_{l+1}}^{0,\delta}\psi\lambda_{N_l}^2)\frac{\phi(\lambda_{N_l})}{\phi(\lambda_{N_{l+1}})} \le 1 \tag{2.45}$$

holds with some function $\phi : \mathbb{R} \to \mathbb{R}$, then

$$||p_{N_*(\delta)} - P_{N_*(\delta)}p^\dagger||^2 \le ||p_0^{0,\delta} - P_0 p^\dagger||^2 \, \frac{\phi(\lambda_{N_{L_*}})}{\phi(\lambda_{N_1})}. \tag{2.46}$$

In case $\hat{\mathcal{N}}^\delta = \{0, ..., N_*(\delta)\}$ this yields

$$||p_{N_*(\delta)} - P_{N_*(\delta)}p^\dagger|| = \mathcal{O}(\sqrt{\phi(\lambda_{N_*(\delta)})}). \tag{2.47}$$

Proof. The result in (2.44) directly follows from Corollary 2.7. Inserting $N := N_*(\delta) - 1,\ k = k_*(N_*(\delta), \delta)$ and (2.45) into (2.44) we get

$$\begin{aligned} &||p_{N_*}(\delta) - P_{N_*(\delta)}p^\dagger||^2 \\ &\le \prod_{l=1}^{L_*}(1 - \omega_{N_{l+1}}^{0,\delta}\psi\lambda_{N_l}^2)||p_0^{0,\delta} - P_0 p^\dagger||^2 \\ &\le \prod_{l=1}^{L_*-1}\frac{\phi(\lambda_{N_{l+1}})}{\phi(\lambda_{N_l})}||p_0^{0,\delta} - P_0 p^\dagger||^2 \\ &= \frac{\phi(\lambda_{N_{L_*}})}{\phi(\lambda_{N_1})}||p_0^{0,\delta} - P_0 p^\dagger||^2. \end{aligned} \tag{2.48}$$

□

As opposed to [130], Lemma 5.4, this can give a convergence rate not only for mildly-ill posed but also for exponentially ill-posed problems for example with $\phi(\cdot) \sim \log(\cdot)$.

2.3.3 Regularization Property

A still open question is whether the iterative multilevel algorithm has regularizing properties or not. In order to answer this question one first needs to show that for fixed iteration index k and fixed level N, the effect of the data error in the iterates vanishes as the data error itself tends to zero.

Theorem 2.3. *Let $\{\delta_n\} \to 0$ for $n \to \infty$. Further, denote by y^{δ_n} a sequence of noisy data and assume that F' is Lipschitz continuous. Then, for fixed $k \in \mathbb{N}$*

$$p_N^{k,\delta_n} \to p_N^k, \quad \text{as } n \to \infty, \tag{2.49}$$

where $p_N^{k,\delta_n} := p_N^{k_*(N,\delta_n),\delta_n}$ for $k \geq k_*(N,\delta_n)$.

Proof. For each pair (δ_n, y^{δ_n}) let us denote by $k_*(N,\delta_n)$ the corresponding stopping index according to (2.27). Similar as in [129] one can define for each level N

$$f_N^k(n) := ||F(p_N^{k,\delta_n}) - y^{\delta_n}||$$

and prove (2.49) by induction for N and k:

1. $N = 0$ and $k = 0$: Since $p_0^{0,\delta_n} = P_0 p^0$ it clearly depends continuously on y^{δ_n}.

2. $N = 0$ and $k \to k+1$. Let $k \in \mathbb{N}$ be fixed and suppose that $p_0^{k,\delta_n} \to p_0^k$ as $\delta_n \to 0$. Assuming that $f_0^k(n)$ is monotone (otherwise consider monotone subsequences) one needs to consider the two cases:

   (a) $f_0^k(n)$ is strictly bounded from below. Then for $n \to \infty$, $f_0^k(n) \to ||F(p_0^k) - y|| > C_1||(I - P_0)p^\dagger|| > 0$ and according to Lemma 2.3 stating that $s_0^k \neq 0$

   $$||F'(p_0^k)F'(p_0^k)^*(F(p_0^k) - y)|| > 0, \quad ||F'(p_0^k)^*(F(p_0^k) - y)|| > 0$$

   which implies continuous dependence of ω_0^k in both cases (2.17) and (2.18). From the definition of the modified Landweber iteration, continuity of F and Lipschitz continuity of F' it follows that $p_0^{k+1,\delta_n} \to p_0^{k+1}$ for $n \to \infty$.

   (b) Let $f_0^k(n) \to 0$: Then $f_0^k(n) \to ||F(p_0^k) - y|| = 0$ for $n \to \infty$, i.e. p_0^k solves (2.1) The following two situations need to be distinguished:
   b.i) $f_0^k(n) > C_2\delta_n$, then $k < \overline{k}_*(0,\delta_n)$ and substituting $p^\dagger$ by p_0^k in (2.30)

   $$||p_0^k - p_0^{k+1,\delta_n}|| \leq ||p_0^k - p_0^{k,\delta_n}||.$$

b.ii) $f_0^k(n) \leq C_2\delta_n$, then the iteration is stopped and $p_0^{k+1,\delta_n} = p_0^{k,\delta_n}$. Together (b.i) and (b.ii) give $p_0^{k+1,\delta_n} \to p_0^k = p_0^{k+1}$ for $n \to 0$.

Now, assume that for some $N > 0$ one has shown that p_N^{k+1,δ_n} depends continuously on the data. Then $p_{N+1}^{0,\delta_n} = p_N^{k_*(N,\delta_n),\delta_n}$ does since $\overline{k}_*(N,\delta_n)$ continuously depends on the data and therefore equals some $k_*(N)$ for all $n \geq n_0$ with n_0 sufficiently large and one can argue as above to carry out the induction step $N \to N+1$. □

The next result shows that the Algorithm 3 with its stopping rules is a regularizing method. For a sequence of data errors which is converging to zero, the regularized solution of (2.1) converges to the exact one $p^\dagger$.

Theorem 2.4. *Assume that δ_n forms a sequence of data errors converging to zero and that the assumptions of Theorem 2.3 are satisfied. Then, the sequence $p_{N_*(\delta_n)}^{\delta_n}$ converges weakly subsequentially (in the sense of Proposition 2.5) to the solution of (2.1).*

Proof. We consider the two cases for $N_*(\delta_n)$:

1.) In the first case we assume that $N_*(\delta_n)$ has a finite accumulation point N for $n \to \infty$. Without loss of generality, $N_*(\delta_n) = N$ for all n sufficiently large. By (2.26) it is assured that $k_*(N,\delta_n)$ has a finite accumulation point. Now, without loss of generality assume that $k_*(N,\delta_n) = k$ for all n sufficiently large. By definition of $k_*(N,\delta_n)$ it follows that

$$||y^{\delta_n} - F(p_N^{k,\delta_n})|| \leq \tau\delta_n. \tag{2.50}$$

Since by Theorem 2.3 $p_N^{k,\delta} \to p_N^k$ for $y^\delta \to y$ as k is fixed now, one has

$$p_N^{k,\delta_n} \to p_N^k \tag{2.51}$$

and

$$F(p_N^{k,\delta_n}) \to F(p_N^k).$$

Taking the limit in (2.50) gives $F(p_N^k) = y$.

2.) In the second case where $N_*(\delta_n) \to \infty$ for $n \to \infty$, then

$$p_M^{k_*(M,\delta_n),\delta_n} = p_{M+1}^{0,\delta_n} \to p_{M+1}^0 \quad \text{for } n \to \infty \tag{2.52}$$

since $k = 0$ is fixed and by an application of Theorem 2.3. Hence Proposition 2.5 yields weak subsequential convergence in this case.

The proof is complete. □

Note, that in the proof of Theorem 2.4 the case that $k_*(N,\delta_n) \to \infty$ as $n \to \infty$ will not occur due to (2.25) and (2.26).

2.3.4 Applicability to a Harmonic Model Identification Problem

In this section we verify the assumptions concerning the nonlinearity (2.5) made during the discussion of the iterative multilevel algorithm for an example. Unfortunately the nonlinearity conditions can most probably not be rigorously shown to hold for the identification of coefficient functions in the fully nonlinear piezoelectric PDEs to be considered in Chapter 4. Hence, as a still infinite dimensional related model, we consider the estimation of a spatially varying diffusion coefficient in a hyperbolic PDE in frequency domain. The example problem is formulated as follows: Find $a \in H^2(\Omega)$ in

$$-\omega^2 \hat{u} - \nabla \cdot (a(x)\nabla \hat{u}) = f \ \in \ \Omega, \quad \hat{u} = 0 \ \text{ on } \ \partial\Omega \tag{2.53}$$

with $f \in L^2(\Omega)$ from given noisy measurements $\hat{u}^\delta \in L^2(\Omega)$ for frequencies $\omega \in \Lambda(a) := \mathbb{R}^+ \setminus \{\lambda_1, \lambda_2, ...\}$ where $(\lambda_n)_{n\in\mathbb{N}}$ are the eigenfrequencies of (2.53). The associated parameter-to-solution mapping is defined as follows

$$\begin{aligned} F : D(F) := \{a \in H^2(\Omega) \mid a(x) \geq \nu > 0\} &\rightarrow L^2(\Omega) \\ a &\mapsto F(a) := \hat{u}(a). \end{aligned} \tag{2.54}$$

The differential operator A is for all $\omega \in \Lambda(a)$ defined as

$$\begin{aligned} A(a) : H^2(\Omega) \cap H_0^1(\Omega) &\rightarrow L^2(\Omega) \\ \hat{u} &\mapsto A(a)\hat{u} := -\omega^2 \hat{u} - \nabla \cdot (a\nabla \hat{u}). \end{aligned} \tag{2.55}$$

With this, the Fréchet derivative of F is calculated from

$$A(a)F(a) \ = \ f.$$

Differentiation with respect to a gives

$$\begin{aligned} A'(a)F(a) + A(a)F'(a) &= 0 \quad \text{and so} \\ F'(a) &= -A(a)^{-1}(A'(a)F(a)), \end{aligned}$$

where the derivative of $A(a)$ is given by the limit

$$\begin{aligned} A'(a)[h]\hat{u} &= \lim_{\epsilon \rightarrow 0} \frac{A(a+\epsilon h)\hat{u} - A(a)\hat{u}}{\epsilon} \\ &= -\nabla \cdot (h\nabla \hat{u}). \end{aligned}$$

Consequently

$$F'(a) = A(a)^{-1}[\nabla \cdot (h\nabla \hat{u}(a)))]. \tag{2.56}$$

Now the question arises whether the inversion of $A(a)$ as a mapping from $\hat{H}^2(\Omega) := H^2(\Omega) \cap H_0^1(\Omega) \rightarrow L^2(\Omega)$ is bounded. The following Lemma gives an answer to this for sufficiently small noneigenfrequencies ω^2.

Lemma 2.8. *The inverse of the operator $A(a)$ defined in (2.55)*

$$A(a)^{-1} : L^2(\Omega) \to \hat{H}^2(\Omega)$$

is bounded for all $a \in D(F)$, $\omega^2 \notin \Lambda(a)$ with

$$\omega^2 < \frac{1}{C},$$

where C is defined by (2.54), (2.58), (2.62), (2.63), (2.66), (2.67), (2.68), and (2.69).

Proof. Since $\omega^2 \notin \Lambda(a)$ the equation $-\omega^2 v - \nabla \cdot (a \nabla v) = f$ is well-posed for all $f \in L^2$, i.e. there exists a unique solution $v \in \hat{H}^2(\Omega) : \ A(a)v = f$.
We begin with an estimation of $\langle A(a)v, v \rangle_{L^2(\Omega)}$

$$\langle A(a)v, v \rangle_{L^2(\Omega)} = \int_\Omega a |\nabla v|^2 - \omega^2 ||v||^2_{L^2(\Omega)}$$

and so

$$||A(a)^{\frac{1}{2}} v||_{L^2(\Omega)} \geq \sqrt{\nu ||\nabla v||^2_{L^2(\Omega)} - \omega^2 ||v||^2_{L^2(\Omega)}}.$$

Together with the interpolation inequality [10]

$$||A(a)^{\frac{1}{2}} v||^2_{L^2(\Omega)} \leq ||A(a)v||_{L^2(\Omega)} ||A(a)^0 v||_{L^2(\Omega)}$$

and with Friedrich's inequality one obtains

$$\begin{aligned} ||A(a)v||_{L^2(\Omega)} &\geq \frac{||A(a)^{\frac{1}{2}} v||^2_{L^2(\Omega)}}{||v||_{L^2(\Omega)}} \\ &\geq \nu \frac{||\nabla v||^2_{L^2(\Omega)}}{||v||_{L^2(\Omega)}} - \omega^2 ||v||_{L^2(\Omega)} \\ &\geq \nu c_2 ||\nabla v||_{L^2(\Omega)} - \omega^2 ||v||_{L^2(\Omega)}, \end{aligned} \tag{2.57}$$

and c_2 is the constant in Friedrich's inequality

$$||\nabla v||_{L^2} \geq c_2 ||v||_{L^2(\Omega)}. \tag{2.58}$$

Now we estimate for the inverse of $A(a)$

$$\begin{aligned} ||A(a)^{-1}||_{L^2(\Omega) \to \hat{H}^2(\Omega)} &= \sup_{w \in L^2, w \neq 0} \frac{||A(a)^{-1} w||_{\hat{H}^2(\Omega)}}{||w||_{L^2}} \\ &= \sup_{A(a)v \neq 0, v \in \hat{H}^2(\Omega)} \frac{||v||_{\hat{H}^2(\Omega)}}{||A(a)v||_{L^2(\Omega)}}. \end{aligned} \tag{2.59}$$

The latter remains bounded if $||A(a)v||^2_{L^2(\Omega)}$ grows as $||v||^2_{\hat{H}^2(\Omega)}$, i.e. we now have to show that $||A(a)v||^2_{L^2(\Omega)} \geq c||v||^2_{\hat{H}^2(\Omega)}$ with some constant $c > 0$. With (2.54) and the second triangle inequality

$$\begin{aligned} |A(a)v| &= |\nabla \cdot (a\nabla v) + \omega^2 v| \\ &= |a\triangle v + \nabla a \nabla v + \omega^2 v| \\ &\geq \nu|\triangle v| - |\nabla a|\,|\nabla v| - \omega^2|v|. \end{aligned} \tag{2.60}$$

Taking norms in L^2

$$||A(a)v||_{L^2(\Omega)} \geq \nu||\triangle v||_{L^2(\Omega)} - ||(\nabla a)(\nabla v)||_{L^2(\Omega)} - \omega^2||v||_{L^2(\Omega)} \tag{2.61}$$

where

$$||\triangle v||_{L^2(\Omega)} \geq c_1||v||_{\hat{H}^2(\Omega)} \tag{2.62}$$

and with Hölder's inequality

$$||(\nabla a)(\nabla v)||_{L^2(\Omega)} \leq ||\nabla a||_{L^6}\,||\nabla v||_{L^3} \leq C_1^{\frac{5}{3}}||a||_{\hat{H}^2(\Omega)}\,||v||^{\frac{2}{3}}_{\hat{H}^2(\Omega)}||\nabla v||^{\frac{1}{3}}_{L^2(\Omega)}.$$

Here C_1 is the norm of the embedding $H^1(\Omega) \to L^6(\Omega)$, i.e.

$$C_1 := ||\mathrm{id}||_{H^1(\Omega)\to L^6(\Omega)}. \tag{2.63}$$

Then by (2.57)

$$\begin{aligned} ||A(a)v||_{L^2(\Omega)} \geq\ & \nu c_1||v||_{\hat{H}^2(\Omega)} - \\ & C_1^{\frac{5}{3}}||a||_{\hat{H}^2(\Omega)}||v||^{\frac{2}{3}}_{\hat{H}^2(\Omega)} \left(\frac{||A(a)v||_{L^2(\Omega)} + \omega^2||v||_{L^2(\Omega)}}{\nu c_2}\right)^{\frac{1}{3}} \\ & -\omega^2||v||_{L^2(\Omega)}. \end{aligned} \tag{2.64}$$

So

$$\begin{aligned} &\left(||A(a)v||_{L^2(\Omega)} + \omega^2||v||_{L^2(\Omega)}\right) \geq \\ &\nu c_1||v||_{\hat{H}^2} - C_1^{\frac{5}{3}}||a||_{\hat{H}^2(\Omega)}||v||^{\frac{2}{3}}_{\hat{H}^2(\Omega)} \left(\frac{||A(a)v||_{L^2(\Omega)} + \omega^2||v||_{L^2(\Omega)}}{\nu c_2}\right)^{\frac{1}{3}}. \end{aligned} \tag{2.65}$$

We define

$$\begin{aligned} z &:= ||A(a)v||_{L^2(\Omega)} + \omega^2||v||_{L^2(\Omega)} \\ x &:= ||v||_{\hat{H}^2(\Omega)} \\ \alpha &:= \nu c_1 \\ \beta &:= \frac{C_1^{\frac{5}{3}}||a||_{\hat{H}^2(\Omega)}}{(\nu c_2)^{\frac{1}{3}}} \end{aligned} \tag{2.66}$$

so (2.65) can be written as

$$z \geq \alpha x - \beta x^{\frac{2}{3}} z^{\frac{1}{3}}$$

i.e. since $A(a)v \neq 0$ and hence $z > 0$

$$1 \geq \alpha \frac{x}{z} - \beta (\frac{x}{z})^{\frac{2}{3}} = \Phi(\frac{x}{z}) \tag{2.67}$$

The function Φ is strictly monotonically increasing on (γ, ∞) where

$$\gamma := \left(\frac{2\beta}{3\alpha}\right)^3 \geq 0. \tag{2.68}$$

So

$$\Phi(\frac{x}{z}) \leq \max\{1, \Phi(\gamma)\}$$

implies

$$x \leq \Phi^{-1}(\max\{1, \Phi(\gamma)\}) z := Cz. \tag{2.69}$$

Then

$$||v||_{\hat{H}^2(\Omega)} \leq C \left(||A(a)v||_{L^2(\Omega)} + \omega^2 ||v||_{L^2(\Omega)} \right)$$

and hence

$$||v||_{\hat{H}^2} \leq \frac{C}{1 - C\omega^2} ||A(a)v||_{L^2(\Omega)}.$$

□

This Lemma generalizes Lemma 2.1 in Kunisch et. al. [27] to the higher dimensional case and to the case of $\omega^2 > 0$.

Theorem 2.5. *Let the assumptions of Lemma 2.8 be satisfied. Then, the forward operator F defined in the identification example (2.53) fullfills (2.5).*

Using Lemma 2.8 we can follow now the lines of the Example 2.14 in [79] to show that the tangential cone condition (2.5) holds for the identification example

(2.53). We set $\tilde{\hat{u}} = F(\tilde{a})$ and formally compute

$$
\begin{aligned}
&||F(\tilde{a}) - F(a) - F'(a)(\tilde{a}-a)||_{L^2(\Omega)} \\
&= \sup_{||w||_{L^2(\Omega)}=1} \langle F(\tilde{a}) - F(a) - F'(a)(\tilde{a}-a), w\rangle_{L^2(\Omega)} \\
&= \sup_{||w||_{L^2(\Omega)}=1} \langle \tilde{\hat{u}} - \hat{u} - A(a)^{-1}\nabla\cdot\left((\tilde{a}-a)\nabla\hat{u}\right), w\rangle_{L^2(\Omega)} \\
&= \sup_{||w||_{L^2(\Omega)}=1} \langle A(a)(\tilde{\hat{u}} - \hat{u}) - \nabla\cdot\left((\tilde{a}-a)\nabla\hat{u}\right), A(a)^{-1}w\rangle_{L^2(\Omega)} \\
&= \sup_{||w||_{L^2(\Omega)}=1} \langle -\omega^2(\tilde{\hat{u}} - \hat{u}) - \nabla\cdot(a\nabla\hat{u}) + \nabla\cdot\left(a\nabla\tilde{\hat{u}}\right) \\
&\qquad\qquad -\nabla\cdot\left((\tilde{a}-a)\nabla\hat{u}\right), A(a)^{-1}w\rangle_{L^2(\Omega)} \\
&= \sup_{||w||_{L^2(\Omega)}=1} \langle \nabla\cdot\left((\tilde{a}-a)\nabla(\tilde{\hat{u}} - \hat{u})\right), A(a)^{-1}w\rangle_{L^2(\Omega)} \\
&= -\sup_{||w||_{L^2(\Omega)}=1} \langle (\tilde{a}-a)\nabla(\tilde{\hat{u}} - \hat{u}), \nabla(A(a)^{-1}w)\rangle_{L^2(\Omega)} \\
&= \sup_{||w||_{L^2(\Omega)}=1} \langle \tilde{\hat{u}} - \hat{u}, \nabla\cdot\left((\tilde{a}-a)\nabla(A(a)^{-1}w)\right)\rangle_{L^2(\Omega)}
\end{aligned}
\tag{2.70}
$$

Now we estimate

$$
\begin{aligned}
&||\nabla\cdot\left((\tilde{a}-a)\nabla A(a)^{-1}w\right)||_{L^2(\Omega)} \\
&\quad\le ||\nabla\cdot(\tilde{a}-a)\nabla\left(A(a)^{-1}w\right)||_{L^2(\Omega)} \\
&\qquad + ||(\tilde{a}-a)\Delta\left(A(a)^{-1}w\right)||_{L^2(\Omega)} \\
&\quad\le ||\nabla\cdot(\tilde{a}-a)||_{L^4(\Omega)}||\nabla\left(A(a)^{-1}w\right)||_{L^4(\Omega)} \\
&\qquad + ||\tilde{a}-a||_{L^\infty(\Omega)}||\Delta\left(A(a)^{-1}w\right)||_{L^2(\Omega)} \\
&\quad\le \left(C_2^2 + C_3\right)||A(a)^{-1}||_{L^2\to\hat{H}^2}|\,|w||_{L^2(\Omega)}||\,||\tilde{a}-a||_{\hat{H}^2(\Omega)},
\end{aligned}
\tag{2.71}
$$

where C_2 and C_3 are the norms of the embeddings $H^1(\Omega) \to L^4(\Omega)$ and $\hat{H}^2(\Omega) \to L^\infty(\Omega)$, respectively. Hence

$$
\begin{aligned}
&||F(\tilde{a}) - F(a) - F'(a)(\tilde{a}-a)||_{L^2(\Omega)} \\
&\quad\le ||F(a) - F(\tilde{a})||_{L^2(\Omega)}(C_2^2 + C_3)||A(a)^{-1}||_{L^2\to\hat{H}^2}||a-\tilde{a}||_{\hat{H}^2(\Omega)}.
\end{aligned}
\tag{2.72}
$$

2.4 Summary Chapter 2

After a short introduction into the theory of nonlinear inverse problems an iterative multilevel algorithm using modified Landweber methods as inner iterations was

proposed. A comprehensive theoretical study concerning convergence and regularizing properties followed. The assumptions on the nonlinearity (2.5) were verified for an identification problem in a linear harmonic PDE.

3 The Inverse Problem – Parameter Identification in Linear Piezoelectricity

Chapter 3 is concerned with the identification of piezoelectric material properties. It reveals the state of the art in parameter identification in piezoelectricity, provides numerical results of the algorithms introduced in Chapter 2 and identification results from real world measurements. Sensitivity analysis and the application of means of optimal experiment design give an insight in the reliability of identified parameters and provide rules how to improve it.

3.1 Motivation

As mentioned before the main discrepancy between mathematical models which are involved in the determination of material tensors using well-known static or resonance methods described in current literature and numerical simulations is the difference in the space dimension assumed. Published data sets are always a collection of parameters determined from a set of mono-modal samples operating in different frequency ranges [1, 24, 32, 33, 60, 91]. However, due to frequency dependency and space reduction, numerical simulations using these data have shown to be never exact enough as required for quantitatively valuable simulation results. We aim at improving this situation within this work.

A further motivation for our efforts comes from the fact that most experimental based determinations of piezoelectric material parameters extract the data in the $\mathbf{d}$-form. For finite element implementations, the $\mathbf{e}$-form is preferred, as we have seen in Section 1.1.6. Each conversion of the data sets from the $\mathbf{d}$- to the $\mathbf{e}$-form distributes error components from single parameters to others, so that the exactness of the parameters decreases.

The proposed inverse scheme allows for an adjustment of given data sets to measured electric impedances and possibly also mechanical deflections and determines directly the parameters in the $\mathbf{e}$-form.

This is why our approach seems to provide more precise results and further seems to be more flexible in its application than most of the techniques which are present in the literature so far.

Even though we cannot prove that our scheme identifies the true physical parameters, we obtain for a selected frequency range a complete and consistent data set, which serves for further precise numerical computations for all possible geometries and applications. The proposed method will even be suited for the case when no manufacturer's data are present, as long as one can provide sufficiently good initial guesses for the sought-for quantities.

The following paragraph reveals the state of the art of parameter identification for the piezoelectric problem for real- and then complex valued material parameters. After a short description on the measurement of the impedance, we mathe-

matically formulate the parameter identification for the piezoelectric problem with a parameter-to-solution mapping which will be shown to be ill-posed. The following sections are then devoted to numerical case studies. First, parameter identification is performed with synthetically generated data in order to study how the proposed method is applicable and in particular observe its behavior in the presence of data noise. Implementation issues like the identification of complex-valued material parameters and scaling are discussed. The subsequent sensitivity analysis provides estimates which allow to destine whether the identified parameter is trustworthy. For the characterization from real-world measurements a rather new and not yet identified material is chosen. Afterwards we treat an identification example where the piezoelectric material is embedded into a compound structure. This chapter closes with an application of means derived from optimal experiment design applied to the optimal selection of measurement frequencies. Two different approaches, one considering a fixed number, the other one a variable number of sampling points, will be described and compared by numerical results.

3.2 Identification Methods - State of the Art

Several scientific groups are involved in the determination of piezoelectric material parameters. The most prominent publications on this topic are certainly the IEEE Standard on Piezoelectricity[1] [1] and the European Standard CENELEC [2] [24]. The procedures suggested there are based upon the analysis of vibrations in piezoelectric materials considering different mode shapes with well-defined geometrical dimensions, i.e. large thickness-length or thickness-radius ratios, that are supposed to allow for reduction to one-dimensional mathematical models. These so called mono-modal resonators are depicted in Figure 14. Identification procedures according to these standards are however restricted to lossless, linear materials.

In parameter identification for piezoelectric materials one may distinguish between quasistatic and resonant methods. Quasistatic measurements are performed far below the first resonant frequency and provide certain information, e.g. on the piezoelectric coupling entry d_{33} which can be obtained from a plate shaped sample excited by an uniaxial stress.

When performing dynamic or resonant techniques the specimens are excited close to their first mechanical resonance. The parameters are then derived from measurements of the electrical impedance Z (Ω) or admittance $Y = 1/Z$ (Ω^{-1}). The concept of the electrical impedance generalizes Ohm's law to AC circuit analysis. The elastic parameters are determined by the resonant frequencies (eigenfrequencies), the density, and the dimensions of the transducers. Piezoelectric coef-

[1] A non-profit organization, IEEE is the world's leading professional association for the advancement of technology. The IEEE name was originally an acronym for the Institute of Electrical and Electronics Engineers, Inc. [64].

[2] CENELEC is the European Committee for Electrotechnical Standardization, a non-profit technical organization set up under Belgian law and composed of the National Electrotechnical Committees of 30 European countries.

ficients are detected evaluating the measured resonance frequencies. Finally, dielectric coefficients (permittivity) are computed from capacitance measurements and geometrical dimensions assuming that the electrodes cover the major surfaces. Analytic formulae exist for different mono-modal resonators directly relating subsets of the parameters to the measured quantities.

The procedure of extracting the material parameters is illustrated exemplarily for the case of a thickness extension mode of a thin disc where one can determine a subset of the parameters as follows [24]

$$c_{33}^D = 4\rho f_p^2 t^2, \quad c_{33}^E = c_{33}^D(1-k_t^2), \quad k_t^2 = \frac{\pi}{2}\frac{f_r}{f_a}\cot\left(\frac{\pi}{2}\frac{f_r}{f_a}\right). \tag{3.1}$$

In case of low losses $f_p \approx f_a$ holds where f_p denotes the parallel resonance (maximum resistance), f_a the antiresonance (zero reactance), and f_r the resonance frequency (zero susceptance). Further t is the thickness and k_t the thickness coupling factor which is related to the three constants

$$\frac{k_t^2}{1-k_t^2} = \frac{e_{33}^2}{\varepsilon_{33}^S c_{33}^E}. \tag{3.2}$$

Alternatively, considering an analytic one dimensional formula of the impedance

$$Z(\omega) = \left(\frac{t}{i\omega\varepsilon_{33}^S A}\right)\left(1 - k_t^2\frac{\tan(\omega/(4\pi f_p))}{\omega/(4f_p)}\right), \tag{3.3}$$

where A denotes the electrode area, one can apply different curve fitting tools in order to obtain the searched for parameters. The formula in (3.3) can be either derived by analyzing appropriate electrical equivalent circuits or solving one dimensional piezoelectric differential equations in frequency domain. Now, evaluating relationships like this either directly or iteratively for all resonators listed in Figure 14 one obtains all entries of the different material tensors for the piezoelectric problem.

New promissing approaches for alternative measurement setups for determining piezoelectric moduli come from Feuillard et. al. [38] where the authors propose apart from impedance measurements either pure acoustic ones which are based on the measurement of the transmission coefficient of a plane wave through a piezoelectric plate or to use the principle of resonance ultrasound spectroscopy of a piezoelectric cube combined with laser detection.

The determination of complex valued material parameters is investigated, among others, by the following groups:
The first work on this topic can be found in Holland and Eer Nisse [60], where the researcher proposes a gain-bandwidth method which bases on the approximation of the complex impedance or admittance by the first term of its partial fraction expansion. According to Du [32] this procedure holds only for materials with moderate loss tangents and low mechanical coupling factors. Smits et. al. [142] propose an iterative method which evaluates admittance values at three different

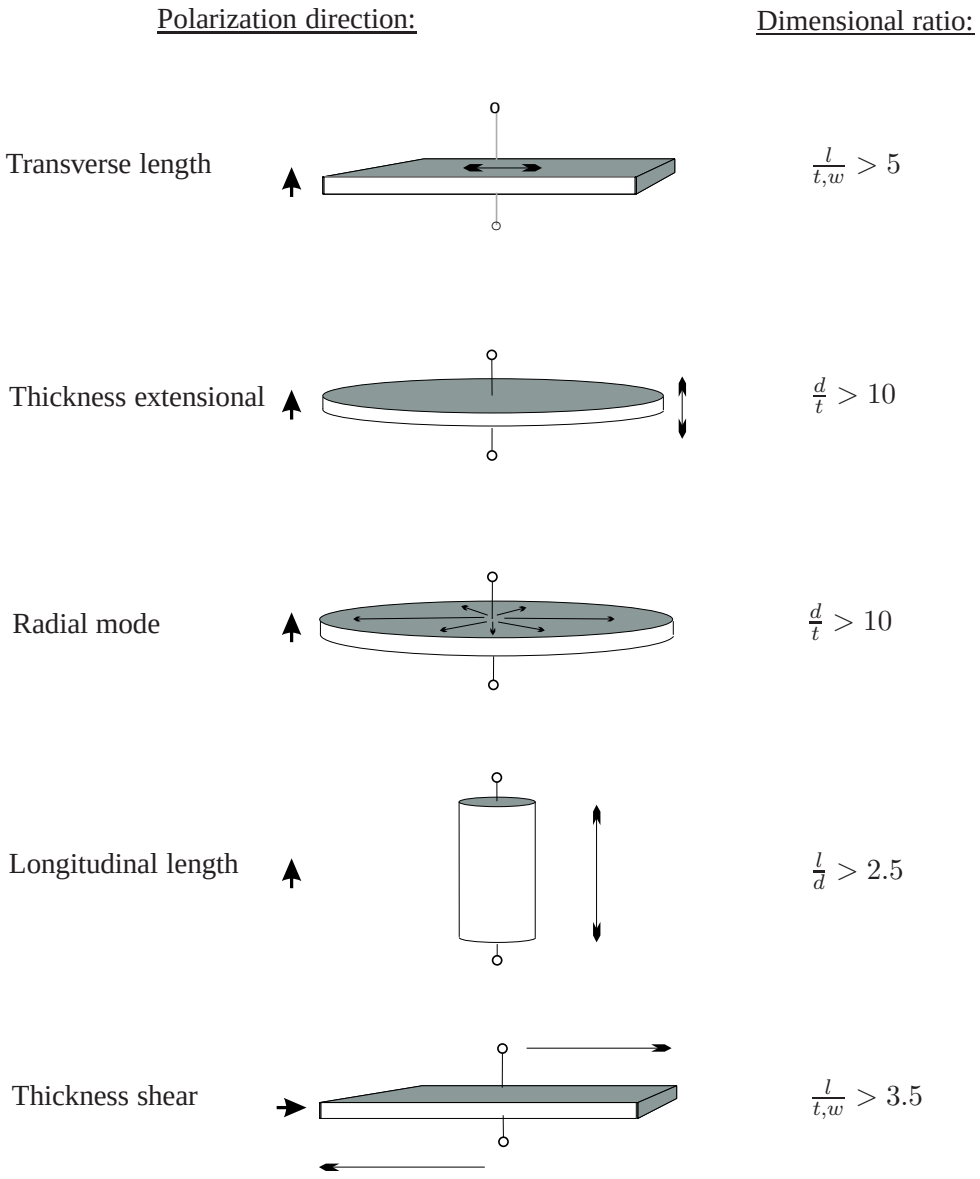


Figure 14: Common resonators. Here $l, t, w,$ and d denote the length, thickness, width, and diameter of the geometry, respectively. The thick arrows on the left indicate the polarization directions, the thin arrows the vibration directions. The figure is according to a source in [24].

frequencies by a fit to the analytical formula of the admittance. Depending on the choice of the frequency points the results are quite sensitive to perturbations in the measurements. A direct method is described in Sherrit et al. [137] by an introduction of the notation of complex parallel and series frequencies. In particular this method is suitable for materials with a low mechanical quality factor, i.e. for materials with high mechanical damping. In Kwok et. al. [91] an iterative curve fitting is suggested. In the most recent works, e.g. by Du et. al. [32] vibration equations are analytically derived for piezoelectric plates and thin rods in arbitrary crystallographic orientations. Based on the analysis of these solutions the complex material parameters are derived from impedance or admittance measurements near the resonance frequencies.

Researchers in Canada [146] have extended and refined these methods and offer a software tool PRAP (Piezoelectric Resonance Analysis Program) which offers full sets of complex valued piezoelectric material parameters.

There are few works which report on identification of piezoelectric material parameters based on numerical simulations formulating the parameter identification process as an optimization problem and using optimization toolboxes [23, 67, 92]. However, these works are also confined to real-valued material parameters and only parameter identifications from one or two waveforms are reported.

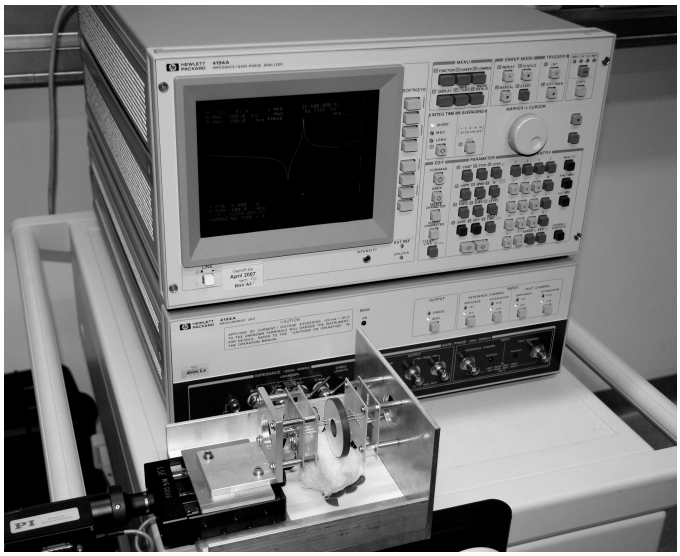


Figure 15: Impedance analyzer with attached test fixture in front. The test fixture is described more detailedly in Figure 16.

As already mentioned the most serious problem one faces when determining consistent parameter sets for piezoelectric media is the frequency dependency. Since the proposed resonators show their fundamental vibration modes at frequency ranges from approximately 60 kHz (longitudinal) up to 2 MHz (thickness)

the parameters hold only for a small frequency range. For $1D$ simplifications this is no hurdle, in three dimensional computations this really is. According to the IEEE Standard [1] "domain wall motion causes energy dissipation, especially at low frequencies, even under low signal conditions when the motion is reversible. It is therefore found that there is considerable variation in permittivity, compliance, and piezoelectric response with frequency. Variations with frequency are most pronounced with ferroelectric materials having relatively lossy permittivity and compliance. Changes may be as high as $5\ \%$ per decade of frequency above frequencies of about 1 Hz. "

Measurement of the Impedance

The auto-balanced-bridge method is the state of the art for the measurement of impedances of piezoelectric resonators. The impedance analyzer used for the measurements in this thesis is the *HP4194A* from Agilent Technologies[3], see Figure 15.

The test fixture is optimized for accurate electric impedance measurements with low electromechanic interaction between the ceramic and the contacts. With

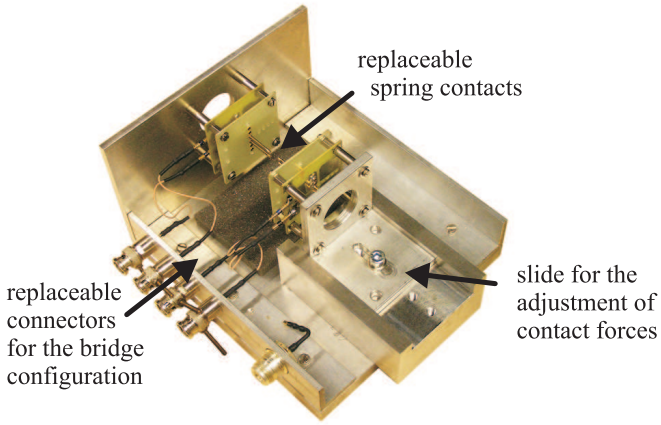


Figure 16: Test fixture for impedance measurements of piezoelectric samples.

tapered electrodes, the contact surface is chosen as small as possible to minimize mechanical influences. The contact positions are aligned to the center of mass, respecting the measured resonance mode. Figure 16 illustrates the test fixture, which is used for the impedance measurements performed at the Department of Sensor Technology in Erlangen.

[3]former Hewlett-Packard

3.3 PDE-Based Parameter Identification

The main task we are concerned with is to adapt all occurring material parameters in the piezoelectric equations in such a way, that simulated results coincide with those received from measurements, namely the electric impedance $Z(\omega)$. Impedance or admittance measurements are very convenient in that the elastic, dielectric, and piezoelectric constants can be determined from just one measurement. This typical parameter identification problem is mathematically formulated with the following parameter-to-solution mapping $\hat{F}$

$$\begin{aligned} \hat{F} : D(F) \subset \mathbb{C}^{n_{\mathrm{par}}} &\rightarrow \mathbb{C}^{n_{\mathrm{freq}}}, \\ p &\mapsto \hat{\mathbf{y}} = (\hat{y}(\omega_l))_{l=1,\ldots,n_{\mathrm{freq}}} \end{aligned} \tag{3.4}$$

mapping from the set of n_{par} complex valued parameters

$$p = (c_{11}^E\,, c_{12}^E\,, c_{13}^E\,, c_{33}^E\,, c_{44}^E\,, e_{15}\,, e_{31}\,, e_{33}\,, \varepsilon_{11}^S\,, \varepsilon_{33}^S)$$

(or subsets of it) to the set of measurements $\hat{\mathbf{y}}$ taken at n_{freq} different frequencies. The measurements $\hat{y}_l = \hat{y}(\omega_l)$ with $\omega_l = 2\pi f_l$ are taken within the interval

$$\begin{aligned} W := \quad & [\omega_1, \min\{\omega_r^{sim}, \omega_r^{meas}\} - \varepsilon_r] \;\cup\; [\max\{\omega_a^{sim}, \omega_a^{meas}\} + \varepsilon_a, \omega_n] \\ & \cup\, [\max\{\omega_r^{sim}, \omega_r^{meas}\} + \varepsilon_r, \min\{\omega_a^{sim}, \omega_a^{meas}\} - \varepsilon_a] \end{aligned} \tag{3.5}$$

where ω_1 and ω_n denote the smallest and largest frequency, respectively, chosen such that a single vibration mode is fully taken into account, compare e.g. with Figure 17. Further, $\varepsilon_r, \varepsilon_a$ are small nonnegative constants and $\omega_a^{\mathrm{meas}}, \omega_a^{\mathrm{meas}}, \omega_r^{\mathrm{sim}}, \omega_a^{\mathrm{sim}}$ denote resonance and antiresonance frequencies of the measured and simulated data, respectively. The gap $W_r := [\omega_1, \omega_n] \setminus W$ is motivated by the fact that the forward problem might be unstable at resonance frequencies in case that sufficiently strong damping cannot be guaranteed. Further, the inverse problem has turned out to be very volatile in this frequency range and easily leads to unusable results. Nevertheless, during the approximation, this gap shrinks, thus at a sophisticated state of the identification almost all available measurements can be taken into account. The measurements may either contain electrical impedances or mechanical displacements, the latter being usually more expensive to obtain in practice, though. Also a combination is possible. In this thesis (except for Section 3.3.5) we only show results from impedance data obtained according to the last subsection of 3.1.

From the finite element solution (see Section 1.3) of the piezoelectric PDEs we obtain directly the mechanical displacement $\hat{\mathbf{u}}$ and electric potential $\hat{\phi}$ at all finite element nodes and by a post-processing step the surface charge on the loaded electrode

$$\hat{q}^e(\omega_i) \;=\; \int_{\Gamma_e} n^T \Big(\mathbf{e}\mathcal{B}\hat{\mathbf{u}}(\omega_i) - \boldsymbol{\varepsilon}^S \nabla\hat{\phi}(\omega_i) \Big)\, d\Gamma_e, \tag{3.6}$$

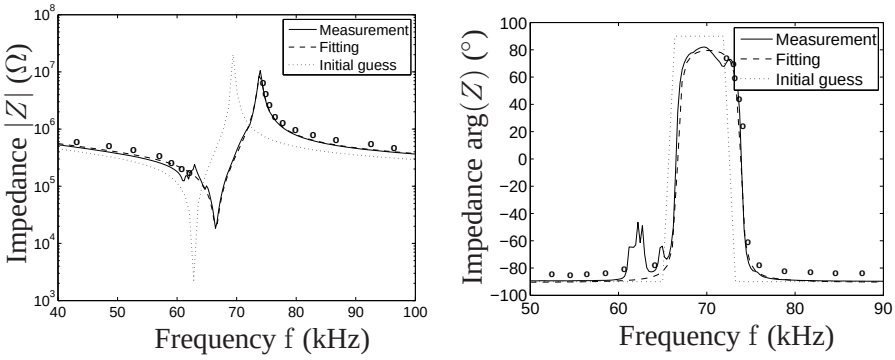


Figure 17: The electrical impedance $|Z|$ and its phase $\arg(Z)$ serve as input for the inverse problem. Small circles denote locations where frequency sampling points are taken.

which on the other hand is related to the measured electric impedance $\hat{Z}(\omega_i)$ via

$$\hat{Z}(\omega_i) \quad = \quad \frac{\phi^e}{j\omega_i \hat{q}^e(\omega_i)}, \quad i = 1, ..., n_{\text{freq}}. \tag{3.7}$$

3.3.1 Ill-Posedness by Rank Deficiency, Linearization of the Forward Operator

The discussion of the Section 3.2 implies that if we used measurements of all test samples depicted in Figure 14 at appropriate non-eigenfrequencies around the resonances (and if there was no frequency dependence) we would be able to uniquely determine all tensor entries. However, since our intention is to provide a method using measurements of only one or a few test samples which can be arbitrarily shaped, we have no guarantee of uniqueness. Hence, the nonlinear system of equations (3.4) is rank deficient in the sense that the system matrix of its linearization does not have numerical full rank at the solution.

Systems of equations with a matrix whose rank is unknown and not full tend to be unstable if the matrix entries are given only approximately [55], as it is the case in our situation. Instability here occurs in the sense that the generalized inverses of a sequence of matrices converging norm-wise to a rank deficient matrix not necessarily converge to the generalized inverse of this matrix.

Additionally, measurements $\hat{\mathbf{y}}^\delta$ in (3.4) are contaminated with data noise, which makes (3.4) an ill-posed problem. As regularizing methods we apply here inexact Newton methods and regularizing modified Landweber iterations as described in Section 2.1 together with the discrepancy principle discussed in (2.4). For both methods, the computation of the Jacobian $\hat{F}'(p)$ and its adjoint $\hat{F}'(p)^*$ is required. Forming for a parameter increment $\underline{dp}$ the difference

$$\hat{F}(p + \underline{dp}) - \hat{F}(p),$$

as a local linearization of $\hat{F}$ at p in direction $\underline{dp}$ one obtains

$$\begin{aligned}\hat{F}'(p)[\underline{dp}] &= \lim_{\Delta\to 0}\frac{\hat{F}(p+\Delta\underline{dp})-\hat{F}(p)}{\Delta}\\ &= \lim_{\Delta\to 0}\frac{\int_{\Gamma_e} n^T\left(\mathbf{e}\mathcal{B}\tilde{\hat{\mathbf{u}}}-\boldsymbol{\varepsilon}^S\nabla\tilde{\hat{\phi}}\right)d\Gamma_e-\int_{\Gamma_e} n^T\left(\mathbf{e}\mathcal{B}\hat{\mathbf{u}}-\boldsymbol{\varepsilon}^S\nabla\hat{\phi}\right)d\Gamma_e}{\Delta}\\ &= \int_{\Gamma_e} e\mathcal{B}\underline{d\hat{\mathbf{u}}}-\boldsymbol{\varepsilon}^S\nabla\underline{d\hat{\phi}}d\Gamma+\int_{\Gamma_e}\underline{d\mathbf{e}}\mathcal{B}\hat{\mathbf{u}}d\Gamma+\int_{\Gamma_e}\underline{d\boldsymbol{\varepsilon}^S}\nabla\hat{\phi}d\Gamma. \quad (3.8)\end{aligned}$$

The terms $(\tilde{\hat{\mathbf{u}}},\tilde{\hat{\phi}})$ are solutions from (1.38) where the parameters are altered by the parameter increment Δp which results in a modification of the material tensors by $\Delta\underline{d\mathbf{c}^E}$, $\Delta\underline{d\mathbf{e}}$, or $\Delta\underline{d\boldsymbol{\varepsilon}^S}$, respectively. Further $(\underline{d\hat{\mathbf{u}}},\underline{d\hat{\phi}})$ solve

$$\begin{aligned}-\rho\omega^2\underline{d\hat{\mathbf{u}}}-\mathcal{B}^T\left(\mathbf{c}^E\mathcal{B}\underline{d\hat{\mathbf{u}}}+\mathbf{e}^T\nabla\underline{d\hat{\phi}}\right) &= \mathcal{B}^T\left(\underline{d\mathbf{c}^E}\mathcal{B}\hat{\mathbf{u}}+\underline{d\mathbf{e}}^T\nabla\hat{\phi}\right)\text{ in }\Omega\\ -\nabla\cdot\left(\mathbf{e}\mathcal{B}\underline{d\hat{\mathbf{u}}}-\boldsymbol{\varepsilon}^S\nabla\underline{d\hat{\phi}}\right) &= \nabla\cdot\left(\underline{d\mathbf{e}}\mathcal{B}\hat{\mathbf{u}}-\underline{d\boldsymbol{\varepsilon}^S}\nabla\hat{\phi}\right)\text{ in }\Omega\end{aligned}$$

with the following boundary conditions

$$\begin{aligned}\mathbf{N}^T\left(\mathbf{c}^E\mathcal{B}\underline{d\hat{\mathbf{u}}}+\mathbf{e}\nabla\underline{d\hat{\phi}}-\underline{d\mathbf{c}^E}\mathcal{B}\hat{\mathbf{u}}-\underline{d\mathbf{e}}\nabla\hat{\phi}\right) &= 0 \quad \text{on } \partial\Omega\\ \underline{d\phi} &= 0 \quad \text{on } \Gamma_g \quad \text{and} \quad \Gamma_e\\ \left(\mathbf{e}\mathcal{B}\underline{d\hat{\mathbf{u}}}-\boldsymbol{\varepsilon}^S\nabla\underline{d\hat{\phi}}-\underline{d\mathbf{e}}\mathcal{B}\hat{\mathbf{u}}+\underline{d\boldsymbol{\varepsilon}^S}\nabla\hat{\phi}\right)\cdot\vec{n} &= 0 \quad \text{on } \partial\Omega\setminus(\Gamma_e\cup\Gamma_g).\end{aligned}$$

After applying the finite element method the discrete $(\underline{d\hat{u}}_m,\underline{d\hat{\phi}}_m)$ are solutions of the following algebraic system

$$\begin{pmatrix}-\omega^2\mathbf{M}_{uu}+\mathbf{K}_{uu} & \mathbf{K}_{u\phi}\\ \mathbf{K}_{u\phi}^T & -\mathbf{K}_{\phi\phi}\end{pmatrix}\begin{pmatrix}\underline{d\hat{u}}_m\\ \underline{d\hat{\phi}}_m\end{pmatrix}=-\begin{pmatrix}d\mathbf{K}_{uu} & d\mathbf{K}_{u\phi}\\ d\mathbf{K}_{u\phi}^T & -d\mathbf{K}_{\phi\phi}\end{pmatrix}\begin{pmatrix}\underline{\hat{u}}\\ \underline{\hat{\phi}}\end{pmatrix}.$$

Here, the matrix on the right-hand side is obtained by substituting the parameter tensors $\mathbf{c}^E,\mathbf{e}$, and $\boldsymbol{\varepsilon}^S$ by $\underline{d\mathbf{c}^E},\underline{d\mathbf{e}}$, and $\underline{d\boldsymbol{\varepsilon}^S}$ in the bilinear forms of (1.116) and applying the finite element assembly. By the identity

$$\langle\hat{F}'(p)[\underline{dp}],q\rangle = \langle[\underline{dp}],\hat{F}'(p)^*[q]\rangle \quad (3.10)$$

the discrete adjoint of $\hat{F}'(p)$ is given by its Hermitian conjugate, i.e. $\hat{F}'(p)^*[q]=\hat{F}'(p)^H[q]$.

3.3.2 Computational Aspects – Reconstruction with Synthetically Generated Data

In order to get an impression how the different iterative methods described in Section 2.2 behave, firstly results concerning convergence in image space, i.e. reduction of the residual and secondly convergence of the parameters in pre-image space

will be presented. The iterative schemes use samples of synthetically generated impedance curves as input. The curves are computed beforehand by the finite element method. Different finite element discretizations for the forward and inverse problem are applied in order to avoid an *inverse crime*. An *inverse crime* occurs when very similar computational methods are employed to synthesize as well as to invert data in an inverse problem. This case must be avoided according to Colton and Kress [28] since it makes the results look overly optimistic. We aim at reconstructing the real parts of the most dominant parameters of a thickness resonator with 0.1 mm thickness and 10 mm radius. Figure 18 compares the decrease of the residuals of different iterative regularization methods introduced in Section 2.2.

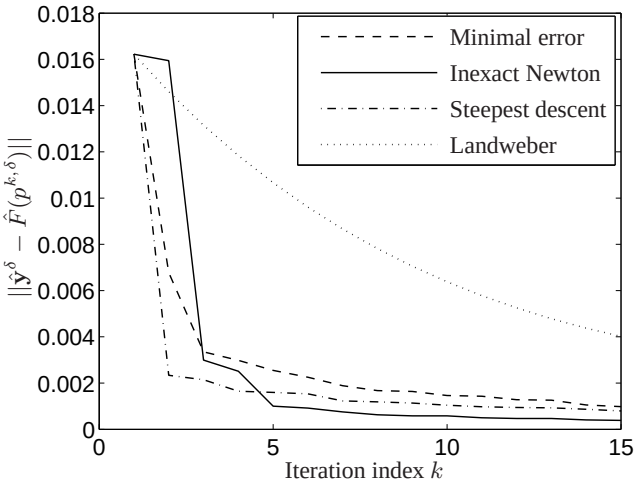


Figure 18: Development of the residual computed for different regularizing methods. The inexact Newton method uses the ν-methods (Algorithm 2) as inner iteration with $\nu = 1.5$. A subset of five parameters is reconstructed from synthetically generated data (impedance) with one per cent data noise.

In the sequel we see the behavior of normed values of the parameters during the computations. As initial guesses a 15% variaton of the assumed exact values is considered. In Figures 19, 20, 21, and 22 parameter reconstructions with inexact Newton methods (see Algorithm 1), the steepest descent method (2.10) with (2.11), the minimal error method (2.10) with (2.12), and the classical Landweber method (2.10) with $\omega^{k,\delta} = \omega$ are computed, respectively for randomly generated noise at different levels. As expected from theory, Newton's method performs well in particular for the noise free case. The modified Landweber iterations need roughly the double amount of iterations in order to reach the same accuracy. Still, compared to the results computed with the classical Landweber method, the modified versions reach an acceptable approximation with a significantly lower number of iterations.

In all results given in Figures 19, 20, 21, and 22 the parameter c_{33}^E shows to be rather easily detectable. The other parameters seem to be less sensitive to the

measurements and are in particular more volatile in the presence of data noise.

A comparison of convergence of selected parameters when applying different inner regularizing methods within the inexact Newton methods is reported in [77].

Our numerical studies show that depending on geometry and type of resonator a large set of parameters might be reconstructed simultaneously, see Figure 23. Of course, when reconstructing all material parameters simultaneously more sophisticated initial guesses are required. Starting values are generated with a ten percent variation for the dominant parameters and a five percent variation for all others. Figure 23 shows convergence of all entries of the mechanical stiffness tensor (except for c_{12}^E, since we perform a rotationally symmetric simulation, see Subsection 1.1.7), and of the the entries e_{33} and ε_{33}^S of the piezoelectric coupling and permittivity tensor, respectively.

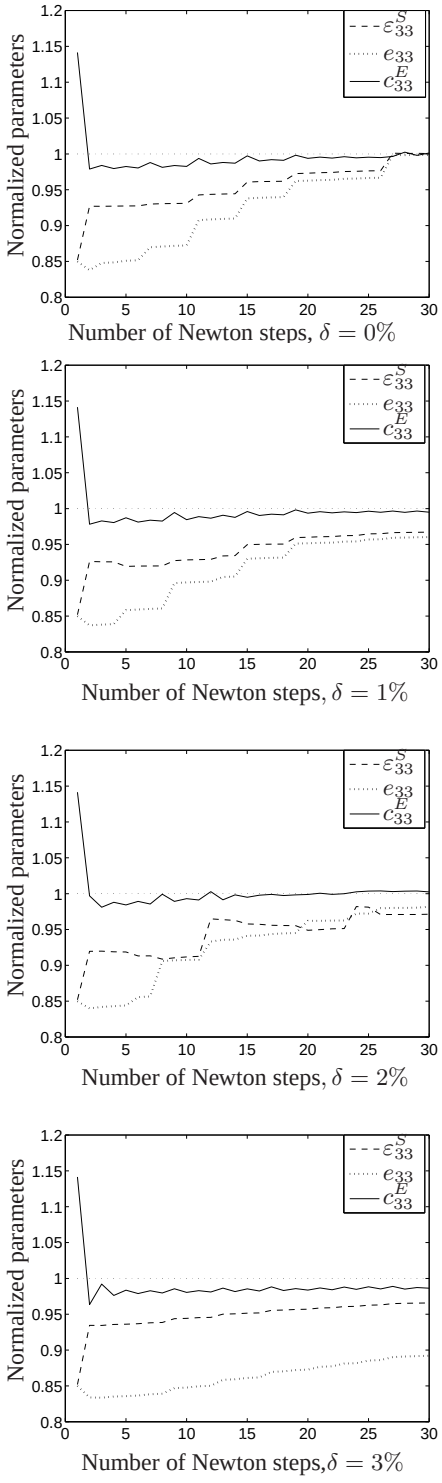


Figure 19: Development of the dominant parameters during their simultaneous reconstruction by Newton's method using ν-methods as inner iterations.

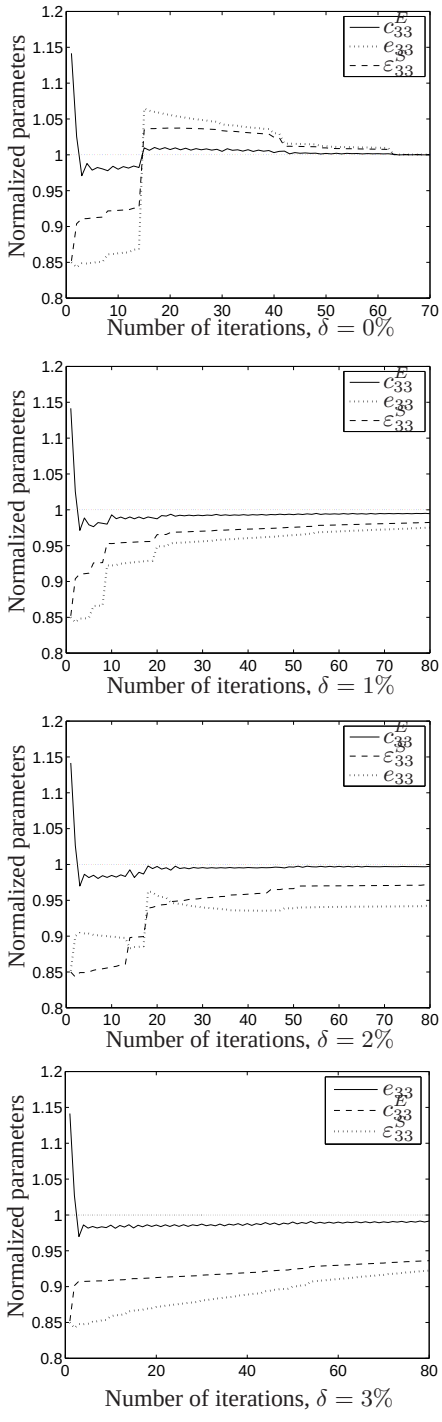


Figure 20: Development of the dominant parameters during their simultaneous reconstruction with the steepest descent method.

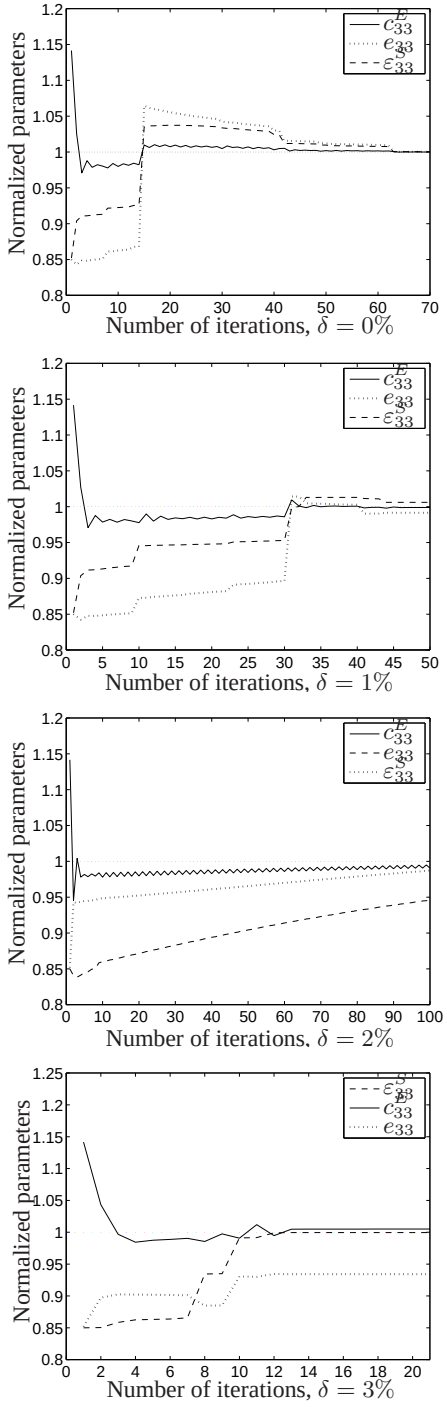


Figure 21: Development of the dominant parameters during their simultaneous reconstruction with the minimal error method.

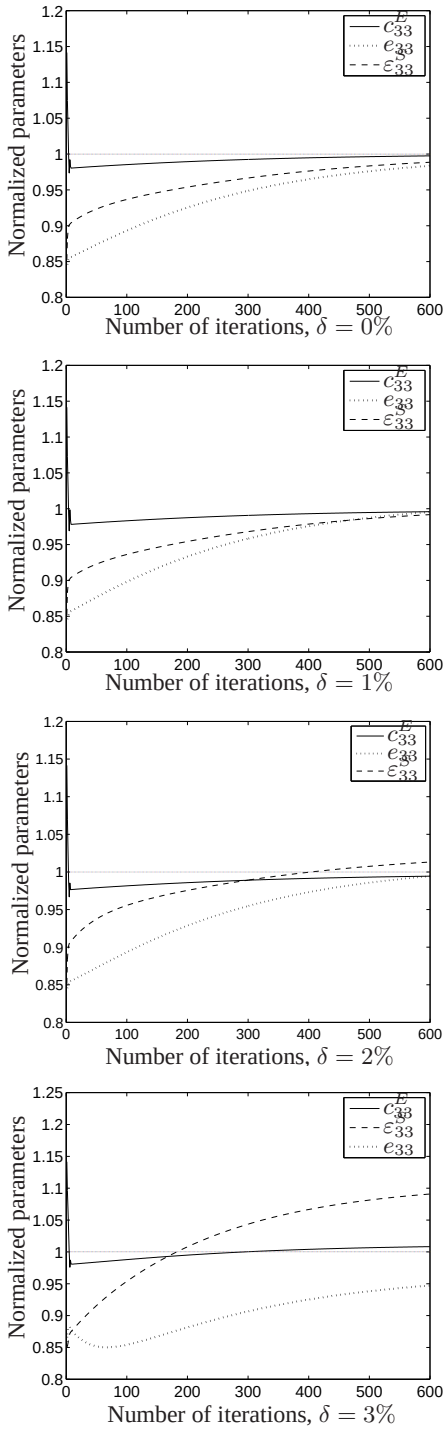


Figure 22: Development of the dominant parameters during their simultaneous reconstruction with the Landweber method.

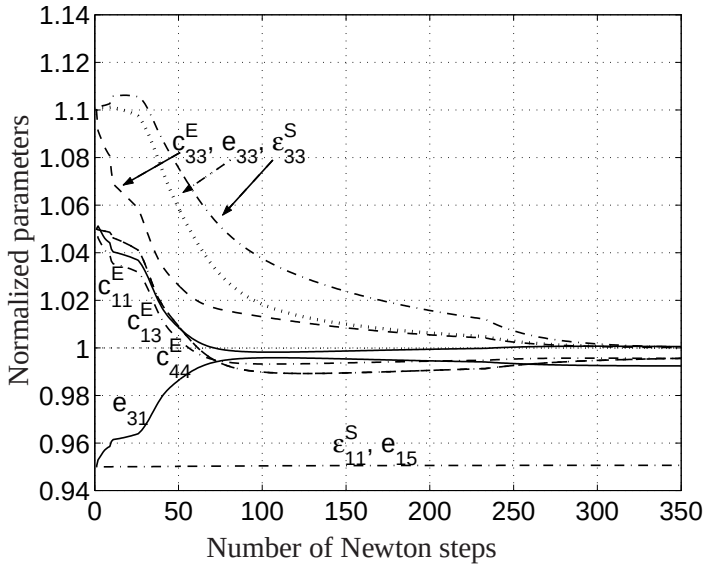


Figure 23: Development of all parameters during their simultaneous reconstruction by Newton's method ($\delta = 0.01\%$). The sample is a thickness resonator of $0.5\mathrm{mm}$ thickness and $10\mathrm{mm}$ radius.

Identification of Complex-Valued Material Parameters

Since the imaginary parts of the parameters mainly steer the height of the magnitude of the impedance in resonance and the "smoothness" of the phase but not the location nor shape of the impedance curve (see e.g. Figure 26 and 27) it is recommendable to proceed in the following manner

1. Adjust $\Re(\mathbf{c}^E), \Re(\mathbf{e})$, and $\Re(\boldsymbol{\varepsilon}^S)$ by fitting $|Z|$, see left graph in Figure 17.
2. Then, additionally adjust $\Im(\mathbf{e})$ and $\Im(\boldsymbol{\varepsilon}^S)$ by fitting $\arg(Z)$, see the right graph in Figure 17 or the left one in Figure 26.
3. Finally, to get $\Im(\mathbf{c}^E)$ (if necessary) fit the amplitude of the mechanical resonance peaks, see Figure 32 in Section 3.3.5.

As initial guesses for the imaginary parts of the complex-valued material parameters one can assume a value being approximately one per mill of its real part. This can be motivated by (1.37) from the Rayleigh coefficients α and β. Typical values for an excitation around $1\mathrm{MHz}$ are for example $\alpha \approx 3.1 \cdot 10^4 \mathrm{s}^{-1}$ and $\beta \approx 7.9 \cdot 10^{10}\mathrm{s}$, see e.g. [80].

In the sequel we report on the simultaneous reconstruction of the imaginary parts of three parameters. The exposition is confined to the steepest-descent method where we assume that the real parts are already sufficiently precisely determined and hold them fixed. Figure 26 plots normalized values of the imaginary parts of the parameters c_{33}^E, e_{33} and ε_{33}^S during their simultaneous reconstruction. Amplitude and phase of the impedance before and after the identification are given in Figure 26.

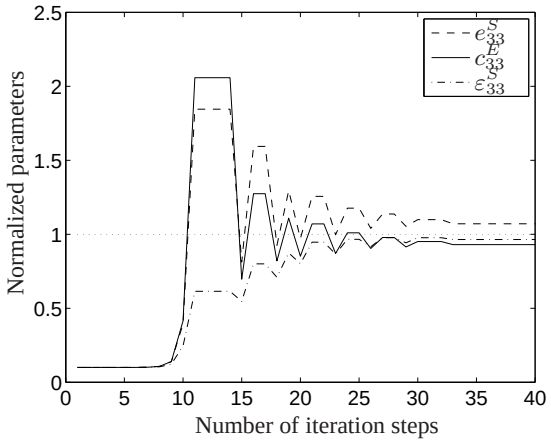


Figure 24: Simultaneous reconstruction of three imaginary parts with the steepest-descent method. Here, the initial guesses have ten per cent of the magnitude of the assumed exact ones. The picture is a reconstruction from noiseless data.

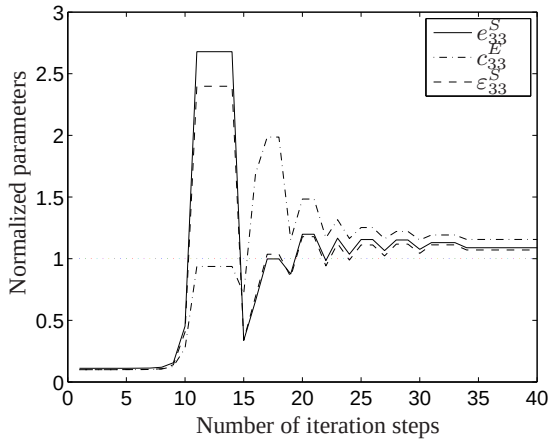


Figure 25: Simultaneous reconstruction of three imaginary parts with the steepest-descent method with three percent data noise.

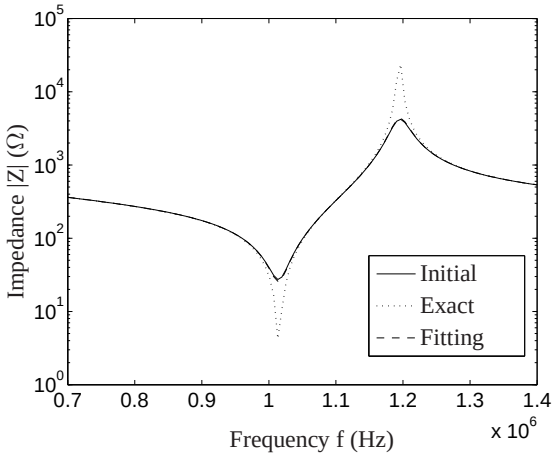


Figure 26: Magnitude of the impedance for exact, initial, and reconstructed data, respectively.

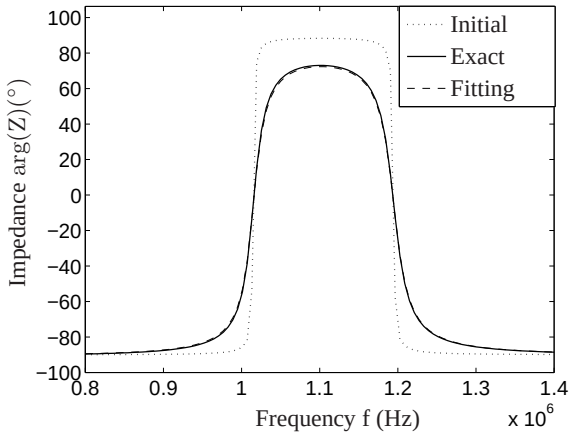


Figure 27: Phase of the impedance for exact, initial, and reconstructed data, respectively.

Scaling

A crucial point for the successful identification is an appropriate scaling of both parameters and measurements. Thus, within the identification methods the parameters are equilibrated, i.e. $\tilde{p} = \xi * p$ where $\xi \approx diag(1/p_1^{typ}, ...1/p_{N_{\text{par}}}^{typ})$ with typical values for the parameters, i.e. for stiffness components $c_{ij}^{E,typ} = 10^{10}$, for the coupling terms $e_{ij}^{typ} = 1.0$ and for the permittivity $\varepsilon_{ij}^{S,typ} = 10^{-08}$. To compare the measured and computed magnitude of the electric impedances, denoted by $\hat{y}_i^\delta$ and $\hat{F}_i(p)$, at different frequencies, we evaluate the following logarithmic norm

$$||\hat{\mathbf{y}} - \hat{F}(p)||_w := \sum_{i=1}^{N_{\text{freq}}} \frac{|\log(\hat{y}_i^\delta) - \log(\hat{F}_i(p))|^2)}{|\log(\hat{y}_i^\delta)|^2}\,.$$

Therewith, we consider the different orders of magnitudes between impedance measurements at various frequencies.

3.3.3 Sensitivity Analysis

Since not all material parameters show a visible impact on the solution of the piezoelectric PDEs, we want to understand, which of them dominate in the different types of probes. These differ for instance in geometry, polarization, and range of excitation frequency. In the following we interpret the linearization $\hat{F}'(p)$ of the nonlinear relation between parameters and observations $\hat{F}(p) = \hat{\mathbf{y}}^\delta$. Radii of confidence intervals for the identified parameters can be estimated by evaluating

	c_{11}^E	c_{33}^E	c_{12}^E	c_{13}^E	c_{44}^E
Radial	0.004	0.012	0.0093	0.02	3.3e+04
Thickness	0.5	0.095	1.9e+02	0.58	2.0e+02
Longitudinal	4.4	0.0015	0.18	0.007	1.9e+05
Transversal	3.4e-03	0.11	0.37	0.12	3.7e+02
Shear	0.014	0.06	0.027	0.033	0.016

	e_{15}	e_{31}	e_{33}	ε_{11}^S	ε_{33}^S
Radial	2.9e+05	0.92	0.29	2.9e+05	0.089
Thickness	2.0e+04	4.1e+02	0.26	4.3	0.05
Longitudinal	9.2e+04	30.1	0.02	4.4e+05	0.62
Transversal	6.7e+04	30.7	0.47	9.0e+05	0.38
Shear	0.68	73.5	11.7	33.3	1.28

Table 5: Computed confidence interval radii of differernt mode shapes for equilibrated parameters.

diagonal entries of the inverse of the information matrix, see e.g. [12, 132]

$$\mathbf{C} = \sum_{i=1}^{n_{\text{freq}}} \left(\hat{F}'(p,\omega_i)^H \hat{F}'(p,\omega_i) \right)^{-1} \in \mathcal{C}^{n_{\text{par}} \times n_{\text{par}}}. \tag{3.11}$$

The sensitivity of each scaled parameter is related to a diagonal entry in the matrix $\mathbf{C}$ and the probability that

$$|p_i^{\text{exact}} - p_i^{\text{computed}}| \leq \sqrt{C_{ii}\chi^2_{n_{\text{par}}}(1-\alpha)},\ i = 1, ..., n_{\text{par}} \tag{3.12}$$

is larger than $(1-\alpha)$, where $\chi^2_{n_{\text{par}}}(1-\alpha)$ denotes the $(1-\alpha)$ quantile of the $\chi^2_{n_{\text{par}}}$ probability distribution. Thus, the smaller the term on the RHS in (3.12), the more reliable the identified parameter and the higher its influence on the transducer's behavior. Table 5 contains upper bounds of the sizes of the confidence intervals using $\alpha = 0.01$ for transducers working in radial, thickness, longitudinal, transversal, and thickness shear mode taking twenty sampling points for each mode into account. Now one may choose some acceptable deviation, e.g. $C_{\text{trust}} \approx 0.1$ and check whether

$$\sqrt{C_{ii}\chi^2_{n_{\text{par}}}(1-\alpha)} \leq C_{\text{trust}}, \quad i = 1, ..., n_{\text{par}}. \tag{3.13}$$

If (3.13) holds, the identified parameter is assumed to be reliable. In the opposite case results from identifications of other modes have to be called on. If the values are close to 0.1 one can still hope to establish sufficiently trusting identification by increasing the number of sampling points or by choosing their location in a more optimal way, see Section 3.4. According to Table 5 a parameter fit with impedances

measured for a radial mode (first line) allows for a confident identification of the parameters $c_{11}^E, c_{33}^E, c_{12}^E, c_{13}^E$ and ε_{33}^S from just one geometry. From this table we also see why in Figure 23 for the simultaneous reconstruction the parameters ϵ_{11}^S and e_{15} were left unchanged and it is obvious that these two parameters are the most difficult to determine.

Studying the last row in Table 5 one sees that for the shear mode almost all confidence intervals are surprisingly small. This is mainly due to the fact that in the same frequency range also higher modes of the thickness vibration are present.

Looking at the single columns in Table 5 identifiability of some parameters is possible by several geometries, e.g. one obtains c_{33}^E by analyzing nearly all mode shapes.

3.3.4 Identification Results for a Newly Developed Ceramic

In this section identification results of our proposed method applied to a new piezoelectric material, *Ferroperm Pz-36*, for which until now no material parameters have been published by the manufacturer are presented, [37].

Geometry	Dimensions
Disc	radius $= 8$ mm, thickness $= 1$ mm
Cylinder	radius $= 2.5$ mm, length $= 18$ mm
Bar	length $= 25$ mm, width $= 4$ mm, height $= 1$ mm

Table 6: Geometry of the used Pz36 samples.

The samples listed in Table 6 have been available for the following investigations. Results of the methods proposed by the IEEE standard are listed in Table 7 which serve as initial guesses for the simulation based parameter identification proposed in this thesis.

c_{11}^E	c_{33}^E	c_{12}^E	c_{13}^E	c_{44}^E	e_{15}	e_{31}	e_{33}	$\varepsilon_{11,r}^S$	$\varepsilon_{33,r}^S$
39.7	35.7	13.9	10.9	13.3	2.51	-0.41	6.2	370.0	457

Table 7: Material parameters of Pz36, as extracted from impedance measurements according to IEEE standard [95]. Units: $c_{ij}^E(\mathrm{GPa})$, $e_{ij}(\mathrm{C/m^2})$, $\varepsilon_{ij,r}^S = \varepsilon_0 \varepsilon_{ij}^S$ with the dielectric constant of vacuum $\varepsilon_0 = 8.854187...10^{-12}(\mathrm{As/Vm})$.

We have performed a set of fittings for different monomodal resonators from which we will discuss two in the sequel more detailedly. The discussion of further modes can be found in [93].

Adjustment of Radial Mode

The first fundamental resonance frequency of the radial mode is located at around 110 kHz, and its antiresonace frequency at 114 kHz, see Figure 28 which displays

the measured and computed electric impedances. The dotted curve in Figure 28

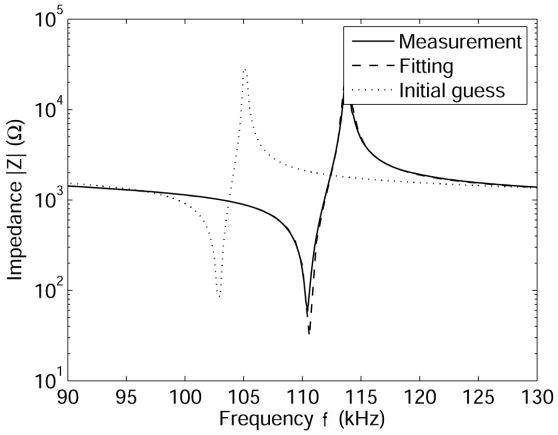


Figure 28: Fitting at radial mode, Pz36.

stems from a simulation with the material parameters given in Table 7 and shows that by the insufficient exactness of the parameters the simulations fail to match the measurements. The most confident (according to Table 5) extracted material parameters obtained by applying the inexact Newton method are listed in Table 8.

	c_{11}^E	c_{33}^E	c_{12}^E	c_{13}^E	c_{44}^E	e_{15}	e_{31}	e_{33}	$\varepsilon_{11,r}^S$	$\varepsilon_{33,r}^S$
Real:	48.8	36.9	12.6	12.3	-	-	-0.41	6.96	-	420.3
Imag.:	0.32	0.3	0.09	0.22	0.11	0.03	-0.004	0.07	4.3	4.5

Table 8: Results radial mode, real and imaginary parts, Pz36.

Adjustment of Longitudinal Mode

To perform the fitting for the longitudinal mode, a cylindrical sample as listed in Table 6 is used. Now, the parameters obtained by identification at the radial mode serve as initial guess. Figure 29 shows the electric impedances and Table 9 displays the obtained trustworth material parameters.

	c_{11}^E	c_{33}^E	c_{12}^E	c_{13}^E	c_{44}^E	e_{15}	e_{31}	e_{33}	$\varepsilon_{11,r}^S$	$\varepsilon_{33,r}^S$
Real:	-	35.5	12.6	10.3	-	-	-	6.01	-	432.6
Imag.:	0.46	0.35	0.1	0.15	0.1	0.03	-0.004	0.06	4.26	4.7

Table 9: Results longitudinal mode, Pz36.

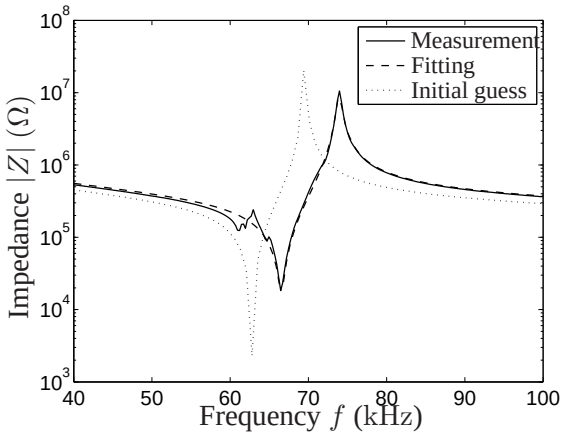


Figure 29: Fitting at longitudinal mode, Pz36.

Now, the procedure is repeated for further geometries and vibration modes, like the thickness, transversal, and the shear mode and allows for the identification of all parameters where at least from one geometry one single parmeter has small confidence intervals, see [93]. By this procedure a consistent set of piezoelectric material parameters can be tabulated, see Table 10. However in Table 10 only the

c_{11}^E	c_{33}^E	c_{12}^E	c_{13}^E	c_{44}^E	e_{15}	e_{31}	e_{33}	$\varepsilon_{11,r}^S$	$\varepsilon_{33,r}^S$
48.7	36.5	12.5	10.3	11.4	2.83	-0.41	6.09	376.1	429.0

Table 10: Consistent data set for Pz36 from FEM based fitting, real parts.

real parts of the entries of the tensors are listed. This is due to the fact that the imaginary parts are strongly frequency dependend and therefore do not allow to build a data set which is consistent for any larger range of frequencies.

Thickness Mode, Verification of Identification Results

In order to show the improvement in the data set and to verify the results, the identified material parameters are applied to a sample, which has not been considered so far, namely a radial disc excited in thickness mode with a thickness of $4\,\mathrm{mm}$ and a radius of $16\,\mathrm{mm}$. Figure 29 shows the simulated impedance curve which gives a reasonable approximation to the measured one and provides an obvious improvement to the one simulated with the initial guesses.

Still, and this is not avoidable, the effects of frequency dependencies are visible in the results. If one really wants to perform an exact three dimensional computer simulation one should carry out a problem specific parameter adaptation to the given transducer in the frequency range actually of interest, for which our developed and implemented scheme can easily be applied.

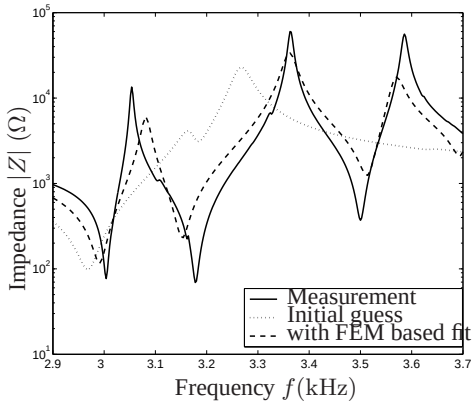


Figure 30: Computed impedance of a specimen not used in the identification of the previously identified parameters compared to the simulation with the initial guess and measurement.

The next section shows the extension of the identification procedure to more complex structures.

3.3.5 Parameter Identification for Piezoelectric Composites

Our experimental results from different applications of piezoelectric transducers have shown that the whole assembly including adaption and protection layers as well as mounting is of decisive influence on the electromechanical coupling properties of the actual sensor or actuator under consideration. Thus, the whole assembly alters the material properties of the piezoelectric elements, mainly due to prestressing, as we will see in the sequel considering as an example a Langevin type transducer. This kind of actuator is for instance used in ultrasonic cleaning applications where acoustic waves in a fluid are excited with frequencies between 20 and $100\mathrm{kHz}$ and one takes advantage of the effects of cavitation to clean contaminated objects in ultrasonic cleaning assemblies. The mounting of the transducer consisting of two oppositionally polarized piezoelectric rings combined with two metallic rods which are hold tight by an inner steel bolt is sketched in Figure 31. The bolt is tightened forcefully which leads to higher oscillation amplitudes of the assembly.

Due to this force and induced deformations of the domains in the ceramics the piezoelectric properties change. Again, in order to perform reliable numerical computations these modified material properties need to be detected which is done by the proposed parameter identification scheme.

In a first step, by combined measurement and simulation based fitting the material properties of the piezoelectric parts without mounting are determined separately, according to the procedures in Section 3.3.4. The material properties of each metal are metrologically specified. The most severe question is now which mate-

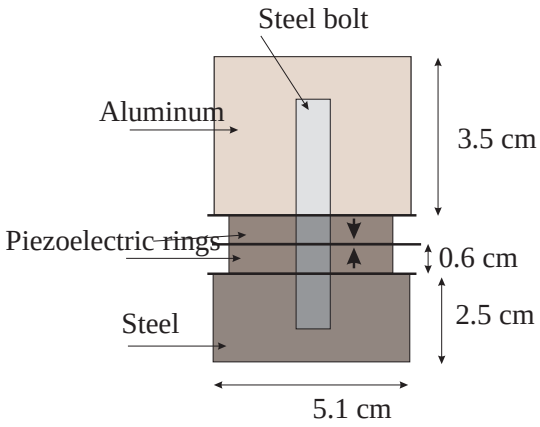


Figure 31: Geometry of the piezoelectric Langevin transducer.

rial properties are the degrees of freedom, i.e. which materials effectively change their values in the mounted state. Several different experiments and parameter fittings revealed that exclusively the properties of the piezoelectric rings change and that the parameters of steel and aluminum remain constant [19]. Additionally, the stiffness parameters, more precisely the Lamé parameters λ and μ of the bolt in the numerical model are treated as unknowns, since the tightness of the bolt connection is not well defined and hence cannot be used in the numerical simulation.

As a result one obtains new sets for the material properties of the piezoelectric rings and the stiffness of the bolt. Impedance measurements combined with mechanical displacements recorded by a laser difference vibrometer are the input for the inverse scheme allowing to identify also appropriate damping. See Figures 32, 33, and 34. For more detailed results consult [19].

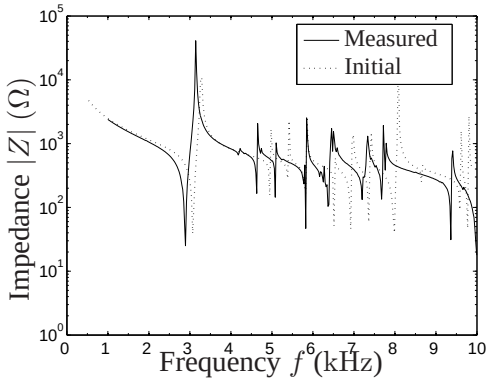


Figure 32: Impedance: Computed with initial guesses (dotted), measured (solid)

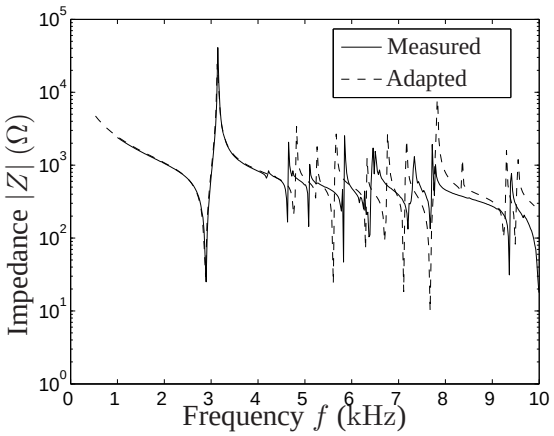


Figure 33: Impedance: Measured (solid) and adapted (dashed).

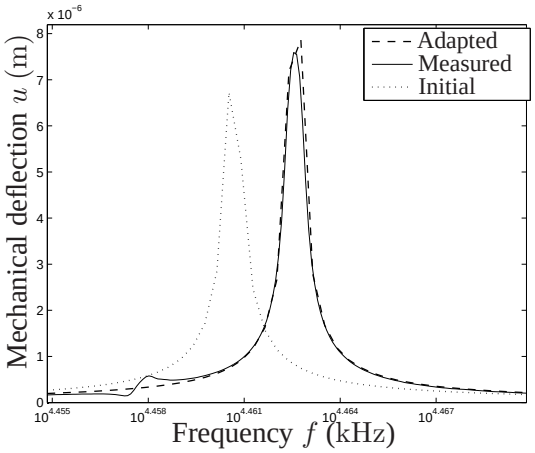


Figure 34: Mechanical displacement of transducer, measured (solid), after material parameter adaption (dashed), before adaption (dotted). In particular the good agreement in absolute deflection comes from the precise adaption of the imaginary parts of the mechanical modulus.

3.4 Optimal Experiment Design

As it has turned out the location of the selected frequency points at which measurements are evaluated have a decisive influence on the successful identification of the different material parameters. The intention of this subsection is to automate the choice of sampling points and improve the reliability of the identified parameters.

3.4.1 Motivation and Approaches

As we have seen in Chapter 3.3.3 there are always some parameters that show hardly any influence on the behavior of the given transducer and are consequently difficult to identify, see Table 5 or Figure 23. However, we aim at improving identifiability of less sensitive parameters as well since it is not always guaranteed that measurements of a large set of differently shaped transducers are available. For those cases in order to provide as satisfying identification results as possible we apply means of *optimal experiment design* [13,69,87] to our identification process. The aim here is to obtain as reliable results as possible with the smallest amount of input (measurements), thus keeping the computational complexity low. The latter point might be in particular important if one thinks of controlling the manufacturing process of ceramics by testing them during the production process whether they fulfill the desired material properties or not. According to producers[4] this is indeed a realistic scenario.

In the sequel it is assumed that the noisy measurements are composed of an exact term plus a normally distributed data noise

$$\hat{y}_i^\delta = \hat{y}_i + \varepsilon_i$$

with

$$\varepsilon_i \sim N(0, W_i^{-1}), \quad W_i \in \mathbb{R}_2^2 \text{ (or } \mathbb{R}\text{) symmetric nonnegative definite.}$$

The noise is assumed to have mean zero and known covariance matrix W_i^{-1} that may also depend on the choice of the frequency points $\omega = (\omega_1, \dots \omega_{n_{freq}})$, i.e., $W_i = W(\omega_i)$. Here $W(\omega_i)$ is set to $(\delta|\hat{y}_i|)^{-2}$. In our application, typically the noise level $\delta \sim 0.01$ which corresponds to one per cent measurement noise. The question arising is how to choose a discrete optimal experiment design. Under an experiment we understand here a set of frequency locations at which measurement data are evaluated, which is optimal in the sense that

- the sensitivity of the measurements is maximal,
- confidence intervals of the parameters are small and
- the result is robust to errors in the measurements.

[4] e.g. CeramTec AG, Lauf (Nuremberg), Germany

Since the mapping $\hat{F}$ from the measurement distribution to the parameters is nonlinear one can only apply local linear sensitivity theory by linearizing $\hat{F}$. A maximization of a certain magnitude measure of the moment or information matrix

$$M := \sum_{i=1}^{n_{freq}} \hat{F}'(p;\omega_i)^H W(\omega_i)\hat{F}'(p;\omega_i)$$

or a minimization of a functional φ of M^{-1}, the so called variance-covariance matrix

$$Cov(p,\omega) = (\sum_{i=1}^{n_{freq}} \hat{F}'(p;\omega_i)^H W(\omega_i)\hat{F}'(p;\omega_i))^{-1} \in \mathbb{C}^{n_{par}}_{n_{par}},$$

which is hermitian nonnegative definite, provides such desired optimal experiment designs. The functional φ is supposed to be antitone, i.e. for $M_1 \succeq M_2 \;\Rightarrow \varphi(M_1) \leq \varphi(M_2)$ in the sense of positive definiteness. Different choices for such a functional $\varphi(M)$ are discussed in the literature [12, 87, 120]

- D - optimality, minimizing the volume of the confidence ellipsoid which is proportional to $\varphi(m) = \frac{1}{det(M)}$,
- A - optimality, minimizing the mean (average) variance of all components, i.e. $\varphi(m) = \frac{1}{n_{par}} trace(M^{-1})$,
- E - optimality, minimizing the largest eigenvalue, i.e. $\varphi(M) = \lambda_{max}(M^{-1})$,
- M - optimality, minimizing the maximal length of the confidence intervals $\varphi(m) = \max_{i=1\ldots n_{par}} \sqrt{(M^{-1})_{ii}}$.

As already mentioned in Section 3.3.3 confidence intervals for the parameters are proportional to the diagonal entries C_{ii} of the covariance matrix $Cov(p,\omega)$. An approximation to the $100 \cdot (1-\alpha)$ per cent confidence region for the parameters is included in the $n_{par}-$ dimensional box with side lengths as given in (3.12), see [87]. The parameter identification problem with noisy data is now defined via a maximum likelihood estimator, i.e. a minimizer of

$$\min_p \frac{1}{2} \sum_{i=1}^{n_{freq}} \|\hat{F}(p,\omega_i) - \hat{y}_i\|^2_{W_i}\,. \tag{3.14}$$

The optimization of the measurement locations is performed in two alternative approaches:

Fixed Number of Measurements

Here a number of measurement locations ($\geq n_{par}$) is defined and one solves the following highly nonlinear optimization problem

$$\min_{\omega} J(\omega) = \min_{\omega} \varphi \left((\sum_{i=1}^{n_{freq}} \hat{F}'(p;\omega_i)^H W(\omega_i) \hat{F}'(p;\omega_i))^{-1} \right) , \tag{3.15}$$

which provides new, optimal (in the sense above), locations for the measurements. In particular fine steering of the parameters is possible with this approach. The computation of the gradient of J which is required for some descent methods is described in [78].

Variable Number of Measurements

Alternatively one can solve the following less nonlinear box-constrained optimization problem where additionally the number of measurements is an optimization parameter. We reformulate the measurement location in form of a measurement density $\varrho(\omega)$ so that the least-squares objective now becomes

$$\min_{p} \frac{1}{2} \int_{\omega_0}^{\omega_f} \varrho(\omega) \, \|\hat{F}(p;\omega) - \hat{y}(\omega)\|^2_{W(\omega)} d\omega \tag{3.16}$$

and the corresponding optimal experiment design requires a solution of

$$\min_{\rho(\omega)\in[0,1]} J(\rho) = \min_{\rho(\omega)\in[0,1]} \varphi \left(\int_{\omega_0}^{\omega_f} \varrho(\omega) \hat{F}'(p;\omega)^H W(\omega) \hat{F}'(p;\omega) d\omega \right)^{-1} , \tag{3.17}$$

where the measurements are weighted by the density function ϱ with $0 \leq \varrho(\omega) \leq 1$ for all $\omega \in [\omega_0, \omega_f]$. After the optimization process one selects those frequencies with large ρ. It is assumed that ρ mainly takes its values at its box constraints, a so called *bang-bang* solution. The selection of frequency locations is subject to an upper bound for the integrated density

$$\int_{\omega_0}^{\omega_f} \varrho(\omega) d\omega \leq M \tag{3.18}$$

which is a second constraint to (3.17) and determines the number of frequencies selected, since whenever for a frequency the weight function exceeds the bound of a defined value $\gamma \in (0,1)$ (typically, $\gamma = 0.8$) it is added to the set of selected locations .Thus, a discrete aspect of optimization is involved in this ansatz. To the best of our knowledge this formulation with a continuous weight function published in [78] is a new approach to experiment design methods. Like in [13] or [87] one generally works with a prescribed number of discrete weights. In a numerical implementation our approach turns out to be more flexible when discretizing adaptively instead of fixing the weighted points a priori.

Combined Measurement Selection and Parameter Identification

A simple sequential one stage optimal experiment design and parameter estimation algorithm looks like this:

Algorithm 4. *One stage algorithm*

Choose $\varrho^0(\omega)$ (or ω^0) and p^0
For $k = 1, 2, 3, \ldots$
    Fix $p = p^k$
    Solve (3.17) (or (3.15)) to obtain ϱ^k (or ω^k)
    Fix $\varrho = \varrho^k$ (or $\omega = \omega^k$)
    Perform an experiment, i.e. evaluation of measurements
    Solve (3.16) (or (3.14)) to obtain p^{k+1}

After the generation of initial guesses (material parameters and frequency locations) nonlinear experiment design and parameter fitting problems are solved in an alternating manner. For the parameter fitting we apply algorithms as discussed in Chapter 3.3 for the modified objective function either in (3.14) or (3.16), respectively.

Both problems, the measurement location optimization, and the parameter identification, are solved iteratively and just until a certain accuracy is reached. As stopping criteria minor changes in the recent computed values $\rho(\omega)$ (or ω) and p are appropriate which avoids time consuming iterations with little improvements in the design variables.

Among the different choices for the functional φ the A - criterion has turned out to be the one providing best results for all parameters. First numerical implementations of the above sketched algorithm provide satisfying results for both choices of the optimization approaches, but also both choices reveal advantages and disadvantages compared with each other. While the choice of frequencies with the algorithm using a variable number of measurements in the first iteration step of Algorithm 4 selects very good starting values (frequencies) without requiring good initial guesses, it does not perform so well in the course of the process and it is rather costly, so it should not be called to often for higher k in Algorithm 4. The choice with a fixed number of frequencies performs very well in fine steering of selected measurement locations. The price to pay is that a very good initial guess is required for this strongly nonlinear problem. Without good initial guesses frequencies might tend to cluster in certain points close to resonance frequencies. Fortunately, its computational cost is much lower.

Out of these experiences we propose a more sophisticated two-stage algorithm which combines the advantages of both strategies.

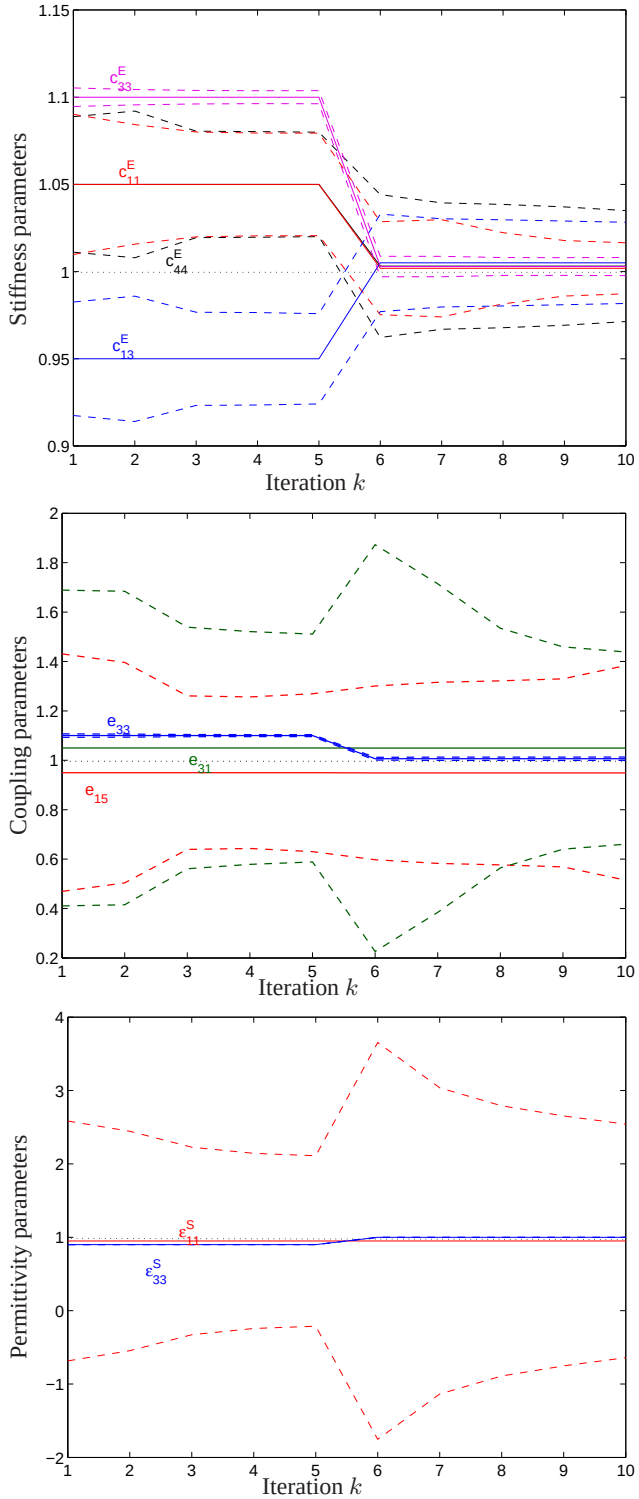


Figure 35: Development of normalized values (solid) and their confidence intervals (dashed) for all parameters during Algorithm 4 for (3.15).

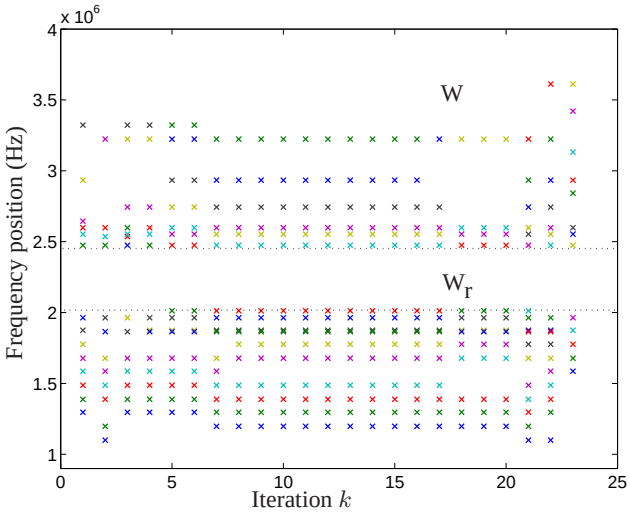


Figure 36: Number and location of selected frequencies by Algorithm 4 with (3.17). Each star denotes here a selected frequency. The number of stars in a column equals the number of selected frequencies for each iteration point k. The sets W and W_r are according to (3.5) and the section below.

Algorithm 5. *Two stage algorithm*

Choose p^0
$k = 0$
Solve (3.17) to get $\rho(\omega^1)$ and from this $\omega^0 = (\omega_1^0, ...\omega_{n(0)}^0)$ for (3.14)
Set $n_{freqs} = n(0)$
Perform an experiment, i.e. solve (3.16) to get p^1
For $k = 1, 2, 3, \ldots$
    Fix $p = p^k$
    Solve (3.15)) to obtain ω^k
    Fix $\omega = \omega^k$
    Perform an experiment, i.e. solve (3.14) to obtain p^{k+1}

We shortly discuss details of the optimal experiment design implementations and refer to [78] for details .

3.4.2 Implementation Issues

Fixed Number of Frequencies - Implementation of Steepest Descent Method

We remind that the identification process only accepts measurements at frequencies which lie in the set W defined in (3.5). Projection onto the feasible set W is done

by the operator P defined by $P\omega = \bar{\omega}$ where

$$\begin{aligned}
\tilde{\omega} &= \min\{\omega_n, \max\{\omega_0, \omega\}\} \\
\omega_r &= \min\{\omega_r^{sim}, \omega_r^{mess}\} \\
\omega_a &= \max\{\omega_a^{sim}, \omega^{mess}\} \\
\omega_c &= \frac{\omega_r + \epsilon_r + \omega_a - \epsilon_a}{2} \\
\bar{\omega} &= \omega_c + \text{sign}(\tilde{\omega} - \omega_c) \max\{|\tilde{\omega} - \omega_c|, \frac{\omega_r + \epsilon_r - \omega_a + \epsilon_a}{2}\}\,. \quad (3.19)
\end{aligned}$$

The minimization problem

$$\min_{\omega \in W} J(\omega) \quad (3.20)$$

is numerically solved by the following projected gradient method combined with a line-search.

Algorithm 6. *Descent method*

Set $\omega^0 \subset W, \lambda^0 > 0, \theta < 1, c_\omega > 0$
While $(||P(\omega^k - \nabla J(\omega^k)) - \omega^k|| > c_\omega)$
$d^k = -\nabla J(\omega^k)$
WHILE $(J(\omega^k + \lambda^k d^k) > J(\omega^k))$
$\lambda^k = \theta \lambda^k$
$\omega^{k+1} = \omega^k + \lambda^k d^k$
$\omega_i^{k+1} = P\omega_i^{k+1} \quad i \in \{1, \ldots, N\}$,
$k = k + 1$

The computation of the gradient can either be done numericaly by finite difference approximation or by analytic formulas which are reported in [78], too, but which have not been implemented here.

Variable Number of Frequencies - Implementation of Optimization

Also the minimization of (3.17) is done by a projected gradient method, where P is the projection on the convex feasible set $[0, 1]$.

Algorithm 7. *Descent method with box constrains*

Set $\rho^0 \in [0,1], \lambda^0 > 0, \theta < 1, \gamma = 0.8, c_\rho > 0$
While $(||P(\rho^k - \frac{\partial}{\partial \rho} J(\rho^k)) - \rho^k|| > c_\rho)$
$d^k = -\frac{\partial}{\partial \rho} J(\rho^k)$
While $(J(\rho^k + \lambda^k d^k) > J(\rho^k))$
$\lambda^k = \theta \lambda^k$
$\rho^{k+1} = \rho^k + \lambda^k d^k$
$\rho = P\rho$
Select $S = \{i \in \{1, \ldots, N\} \mid \rho(\omega^{(i)})^{k+1} > \gamma \max_j \rho(\omega^{(j)})^{k+1}\}$,
Set $\rho(\omega^{(i)})^{k+1} = 0$ for $i \notin S$
$k = k + 1$

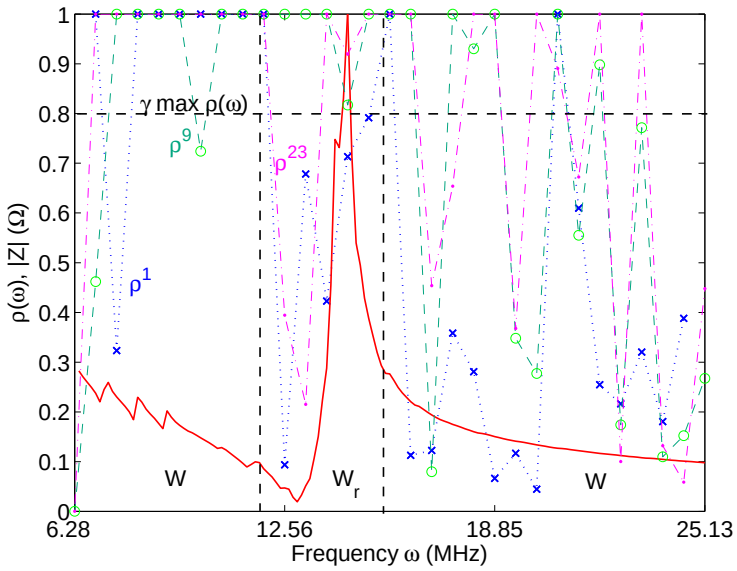


Figure 37: 1st (dotted), 9th (dashed), and 23rd (dash-dotted) iterate of weight function during algorithm 4 with (3.17) and impedance curve (solid) for comparison. Whenever for a frequency the weight function exceeds the bound of 0.8 it is added to the set of selected locations.

As an initial guess we use a constant weight function $\rho^0(\omega) \equiv 0.01$.

3.4.3 Numerical Results

All numerical computations in this section have been performed treating a thickness resonator (Material: Pz27, Ferroperm) with a radius of 10 mm and thickness of 1 mm which according to their resulting confidence intervals is a predestined geometry for the determination of the constants c_{33}^E, e_{33} and ε_{33}^S. The aim here is mainly to improve identifiability for all other parameters involved. For the measurements we assume one percent normally distributed random noise and in the graphics the confidence intervals for 99 per cent reliability are displayed. Fittings are performed to synthetically generated data distorted with 1% data noise.

Optimal Experiment Design - Numerical Results - Variable Number of Frequencies

The results of the constrained optimization of the weight function $0 \leq \rho \leq 1$ over the interval of admissible frequencies W combined with parameter identification as proposed in Algorithm 4 are displayed in Figure 37. Here exemplarily different iterates of the weight function ρ are shown in comparison to the impedance

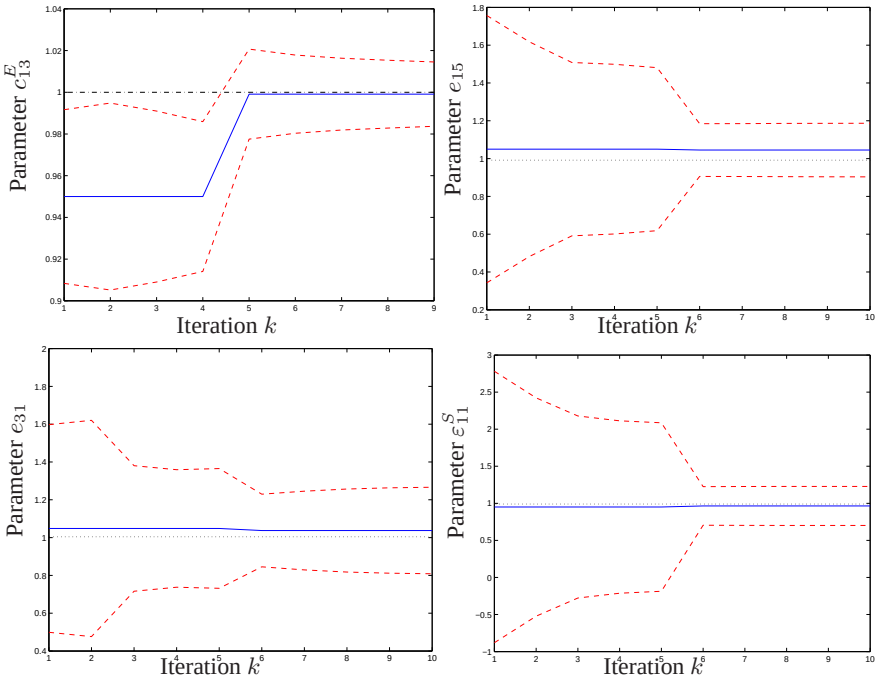


Figure 38: Development of normalized values of selected parameters (solid) and their confidence intervals (dashed) during Algorithm 5.

curve (solid). The discretized function ρ tends to have values either close to one or close to zero, making a distinction of frequencies with high information content easy. In particular lower frequency measurements seem to support the identification process. Figure 36 shows the location where frequencies have been selected. The number of entries in one column equals the number of selected frequencies. The set W_r contains frequencies excluded from consideration due to the instability reasons for the forward problem discussed in Section 3.3.

Optimal Experiment Design - Numerical Results - Fixed Number of Frequencies

Assuming that we have already obtained good initial guesses for the frequencies we now optimize within a set of frequencies with fixed cardinality. Figure 35 shows how the confidence intervals (dashed lines) of the different parameters shrink and how all parameters (solid) converge to their normalized value up to a certain precision corresponding to the data error.

Looking for example at the dielectric constants one sees the big difference in sensitivity between the parameters ε_{11}^S and ε_{33}^S. Since we evaluate the average criterion during the optimization process this approach does not meet the requirements for all parameters. Since our aim is mainly to improve identifiability of less dom-

inant parameters we now restrict the optimization only to those parameters. The plots in Figure 38 now show the improvements for four selected parameters.

3.5 Summary Chapter 3

Chapter 3 is concerned with the identification of the material parameters in linear piezoelectricity, i.e. for small-signal driven ceramics. This chapter comprises the description of the state of the art, limitations of it, and in particular shortcomings with respect to simulations by the finite element method. Mathematical modeling and computation aspects of simulation-based parameter identification is reported including numerical results for both synthetically generated data and real measurements. This flexible methodology has shown to be very effective in characterizing an unknown piezoelectric material as well as a full composite transducer.

Sensitivity analyses and means of optimal experiment design yield information about the identifiability of certain material parameters and give effective rules how to improve it. This is important for the determination of less sensitive parameters and increases the set of trustworthily detectable parameters.

4 Nonlinearities in Piezoelectricity – The Forward and Inverse Problem

This final section is now devoted to both the forward and inverse problem in nonlinear piezoelectricity. Aside from the presentation of nonlinear effects which occur in piezoelectric applications the mathematical components are derived which are necessary to handle the forward and in particular the inverse problem numerically.

4.1 Nonlinear Dependencies in Piezoelectricity

The discussion of nonlinearities in piezoelectricity is an essential task, in particular in actuator applications. Nonlinear models allow for example the design of actuators which are supposed to work in high field ranges. In particular in these actuator applications the piezoelectric materials need to support large mechanical loads and produce high strain output. To accomplish this requirement of higher strains large electric voltages are applied causing high electric fields. Concerning the electric field strengths one can assume linear behavior only in the range of $0.0-0.1\mathrm{kV/mm}$. Above these field strengths nonlinear effects occur. Signals with which piezoelectric actuators are usually driven cause fields of $0.2-0.3\mathrm{kV/mm}$, thus the actuators mainly operate in nonlinear ranges [41]. Due to reductions in the thickness of piezoceramic layers, e.g. in piezoelectric stack actuators higher strains are generated by excitation with moderate voltage levels, making the correct treatment of nonlinearities even more necessary [49].

In broad terms three different types of nonlinear phenomena exist which describe nonlinearities in piezoelectric materials:

1) The first one is thermodynamically motivated by considering higher order terms in the constitutives. The model allows to describe nonlinear effects for moderate electric fields staying below the coercive field strength. In this model the nonlinear effects are assumed to be reversible, i.e. no depolarization of the crystals is expected.

   Effects like a nonlinear relation between applied voltage and displacement, jump phenomena in the response spectra, and softening of the material are already visible at weak fields [125, 127]. These effects can be taken into account by allowing higher order expressions in the thermodynamical energy potentials which leads to constitutive equations with parameters being functions of the field quantities. This kind of nonlinearity is often quoted as *material nonlinearities*.

2) A second type includes irreversible effects caused by high fields, exceeding the coercive fields. *Hysteresis* effects occur which are caused by reorientation of the domains in the materials and which are a typical phenomenon for ferroelectric materials [56, 81, 83, 141].

3) A third kind of nonlinearity is derived from pure mechanics and handles effects coming from large mechanical stresses and deflections.
   Thus, *geometric nonlinearities* are also considered for the piezoelectric problem, important for the theoretical description of piezoelectric workpieces subject to large strains [139, 140, 151].

Since, as it is also pointed out in [125, 126], only few publications exist where nonlinear effects "under weak electric fields" are considered, we aim at investigating the first model in more detail. It is suitable for actuator applications where the transducers are unipolarly excited with moderate signals, i.e. by excitations with purely positive signals where hysteresis effects are as far as possible avoided.

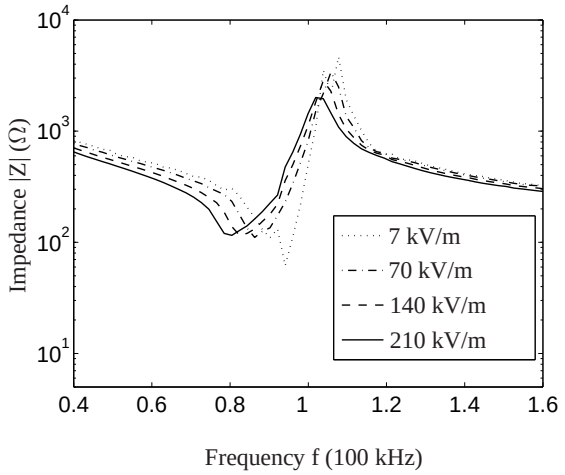


Figure 39: Changes in fundamental resonance due to different excitations. Obviously, the piezoelectric coupling increases with higher fields (larger bandwidth), damping effects are more visible and most remarkably, the resonance frequency moves left, i.e. material softens under increasing fields. Measurements are performed at the Department of Sensor Technology in Erlangen.

Further, this first approach seems to be of rather general nature and the ideas developed here might be easily transferred to other nonlinear problems in physics, like e.g. the determination of the permeability as a function of the magnetic field intensity in the nonlinear relationship between the magnetic field intensity and the magnetic flux density [76].

All three kinds of nonlinearity will be described briefly in the following subsections in order to point out the major differences between them. After this, we turn to the mathematical treatment of the first model introduced, namely material nonlinearities, where the focus lies in the inverse problem. In order to apply the iterative multilevel algorithm presented in Chapter 2 the adjoint operator of the linearized parameter-to-solution map for the nonlinear piezoelectric problem is derived. Numerical identification results close this chapter.

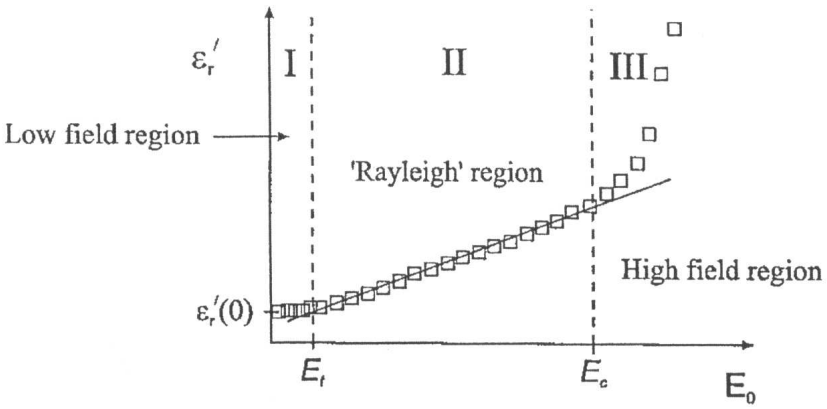


Figure 40: Field dependence of the real part of the relative permittivity ϵ_r. The classification is by Hall and Stevenson [50]. The fields of interest lie in the "Rayleigh region" below the coercive field strength E_c. Below E_t linear theory holds.

Material Nonlinearities

Regarding experimental results, piezoelectric ceramics exhibit different types of nonlinearities already under weak or moderate fields as for example jump phenomena of the displacement amplitude in frequency sweeps, presence of higher harmonics and nonlinear relations between applied electric voltage and mechanical displacement [125, 127]. As it is shown there or in [14], the consideration of higher order terms in the constitutive equations allows for an exact modeling of these effects. In particular this holds for high-power actuators which are driven close to or in their resonance frequencies and are used for instance in ultrasonic cleaning, piezo-vibrators or machine cutters.

The treatment of piezoelectric nonlinearities by higher order terms is valid as long as no depolarizations of the ceramic is present, i.e. the applied or induced electric field strength remains below the coercive field strength. Figure 40 shows the field dependence of the dielectric permittivity in ferroelectric ceramics over a wide range of field strength and visualizes the different ranges regarding the electric field strength where either linear, nonlinear reversible, and nonlinear irreversible effects occur. Evidently, the boundaries between these regions are not sharp. While there is already considerable progress in measurement techniques for determining field dependent material parameters, there are only few approaches on the mathematical side, see e.g. [11, 73, 125]. Considering nonlinearities in frequency domain a multiharmonic approach for the forward problem along with an identification scheme is proposed in [75], where higher harmonics are taken into account and parameter curves are effectively reconstructed by an inversion scheme.

Beginning with [107] and [68] and continuing with [11, 14, 47] and [8, 66]

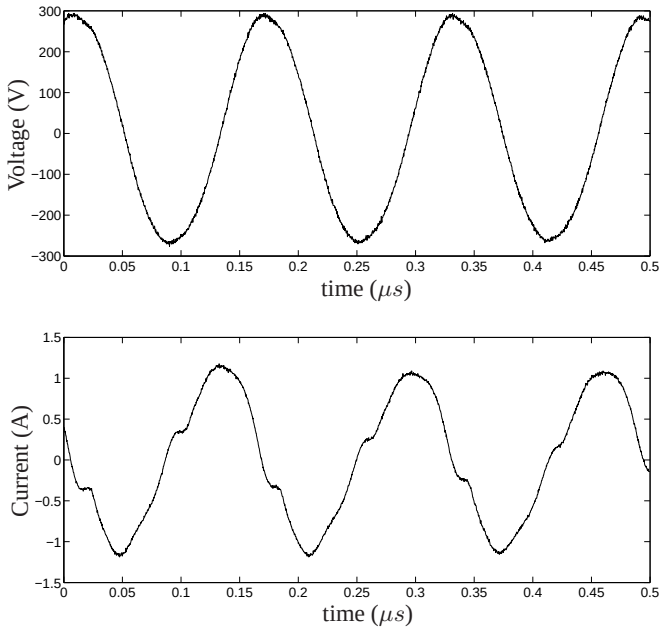


Figure 41: Effects of higher harmonic generation by linear voltage excitation (top) on current (bottom) of a radial resonator at $388\mathrm{kHz}$ measured at the Department of Sensor Technology Erlangen.

among others, nonlinear effects are taken into account considering nonlinear thermodynamical potentials, for example by evaluating the nonlinear electric enthalpy density function H_{nonlin} [125, 126]. Therewith one considers constitutive relations with higher order terms, i.e. the model is realized by truncating the Taylor series expansion in (1.7) not after the first but after the second or third term

$$\begin{aligned}
\sigma_{ij} &= \left.\frac{\partial^2 H}{\partial S_{ij}\partial S_{kl}}\right|_E S_{kl} - \frac{\partial^2 H}{\partial E_k \partial S_{ij}} E_k + \frac{1}{2}\left.\frac{\partial^3 H}{\partial S_{ij}\partial S_{kl}\partial S_{mn}}\right|_E S_{kl}S_{mn} \\
&+ \frac{\partial^3 H}{\partial E_m \partial S_{ij}\partial S_{kl}} S_{kl}E_m - \frac{1}{2}\frac{\partial^3 H}{\partial E_m \partial E_m \partial S_{ij}} E_k E_m + \dots , \qquad (4.1) \\
D_i &= \frac{\partial^2 H}{\partial E_i \partial S_{kl}} S_{kl} - \left.\frac{\partial^2 H}{\partial E_i \partial E_k}\right|_S E_k + \frac{1}{2}\frac{\partial^3 H}{\partial E_i \partial S_{kl}\partial S_{mn}} S_{kl}S_{mn} \\
&- \frac{\partial^3 H}{\partial E_i \partial E_k \partial S_{mn}} E_k S_{mn} + \frac{1}{2}\left.\frac{\partial^3 H}{\partial E_i \partial E_k \partial E_m}\right|_S E_k E_m + \dots . \qquad (4.2)
\end{aligned}$$

The material law is consequently extended to

$$\begin{aligned}
\sigma_{ij} &= c^E_{ijkl}S_{kl} - e_{ijk}E_k + \frac{1}{2}c^E_{ijklmn}S_{kl}S_{mn} + e_{ijklm}S_{kl}E_m - \frac{1}{2}\zeta_{ijkm}E_k E_m + \dots \\
D_i &= e_{ikl}S_{kl} + \varepsilon^S_{ik}E_k + \frac{1}{2}e_{iklmn}S_{kl}S_{mn} - \zeta_{ikmn}E_k S_{mn} + \frac{1}{2}\varepsilon^S_{ikm}E_k E_m + \dots
\end{aligned} \qquad (4.3)$$

where c^E_{ijklmn} is the higher order elastic modulus, e_{mijkl} the higher order piezoelectric coupling, ζ_{kmij} the electrostrictive modulus and ε^S_{ikm} is the permittivity of higher order.

In particular the tensor e_{ijklm} describes the dependence of the linear elastic modulus on the electric field strength as well as the dependence of the piezoelectric coupling on the mechanical strain. The electrostrictive parameters ζ_{ijkm} are derived from the dependency of the permittivity on the mechanical strain [14]. However, for these newly introduced material tensors of higher order the determi-

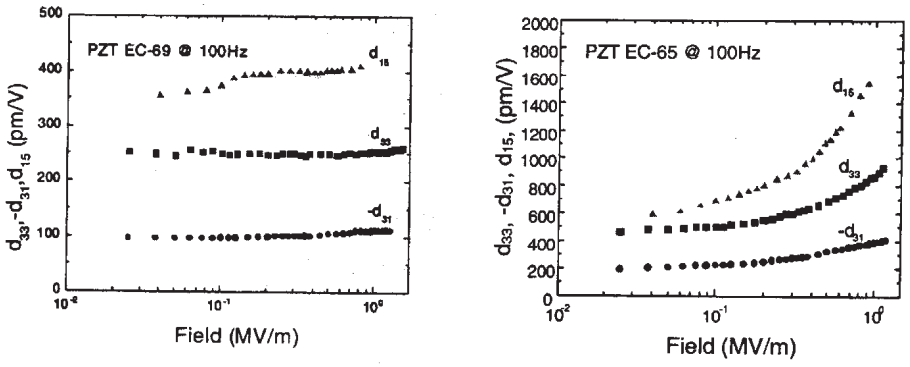


Figure 42: Variation of the piezoelectric constants as a function of the applied field. Left for a hard PZT and right for a soft PZT. The results which are reported in [106] are obtained by an experiment driving the ceramic with a high AC voltage and measuring the displacements by laser Doppler interferometry

nation of their components is a nontrivial task both from the measurement side and from the mathematical point of view. In particular for the latter a stable dependency of the parameters from measurements might be hard to establish. Literature provides only few results, see e.g. [14, 68, 124, 125]. However, the terms in (4.3) suggest to combine all pure mechanical, all pure electric and the mixed ones and express them by functional dependencies as shown in Table 11. The material law

	mechanical strain S	electric field strength E
elasticity modulus	$\mathbf{c}^E(\vec{S})$	
piezoelectric coupling	$\mathbf{e}(\vec{S},\vec{E})$	$\mathbf{e}(\vec{S},\vec{E})$
permittivity		$\boldsymbol{\varepsilon}^S(\vec{E})$

Table 11: Functional dependencies of piezoelectric material parameters [140].

which is now expressed with functional dependencies of the tensors by the field quantities reads as [85]

$$\sigma_{ij} = c^E_{ijkl}(\vec{S})S_{kl} - e^T_{ijk}(\vec{S},\vec{E})E_k \tag{4.4}$$

$$D_i = e_{ikl}(\vec{E},\vec{S})S_{kl} + \varepsilon^S_{ik}(\vec{E})E_k. \tag{4.5}$$

The following setting shows that expressions with higher order tensors are just a special case of the functional dependencies [140]

$$c^E_{ijkl}(\vec{S}) = c^E_{ijkl} + \frac{1}{2}c^E_{ijklmn}S_{mn} + ... \tag{4.6}$$

$$e_{ijk}(\vec{S},\vec{E}) = e_{ikl} - e_{ijklm}S_{kl} + \frac{1}{2}\zeta_{ijmk}E_m + ... \tag{4.7}$$

$$\varepsilon^S_{ik}(\vec{E}) = \varepsilon^S_{ik} + \frac{1}{2}\varepsilon^S_{ikm}E_m + ... \tag{4.8}$$

The task is now not to determine single constant parameters but parameter curves depending on electric field and mechanical strain which will be the main topic of Section 4.3.

There are several possible techniques to determine the functional dependencies by appropriate measurement setups which are discussed in Simkovic [140].

The following subsections describe further nonlinearities and dependencies which are present in piezoelectric applications. However, neither new mathematical models nor new cognitions will be presented, so the reader can easily proceed with Section 4.2 without losing the central idea.

Hysteresis

Piezoelectric media, driven at high voltages or charges, may tend to change their state of polarization if the applied electric field is strong enough which is a typical phenomenon for ferroelectric materials. Therefore, if a large alternating voltage drives the ceramic a switching of the polarization direction has to be taken into account. This is often done by considering reversible and irreversible effects in an additive manner. The reversible ones are modelled by the already discussed linear model. There are well-developed thermodynamically consistent models (see Kamlah [81–83] and references therein) which can take the irreversible phenomena into account while respecting the current state of polarization. Besides these there are phenomenological ones such as Preisach or Jiles-Atherton operators [56, 88, 122, 140]. So called hysteresis or butterfly curves make these effects visible, see Figure 43 from [141]. Here in point A for a sufficiently large positive electrical field all domains are aligned in the direction of the applied field. The material acts as a single domain. Point B is referred to as the point of positive remanence. The applied field is zero, remanent polarization and strain are present. Small changes of the fields respond with linear, reversible behavior. Decreasing the electric field below the negative coercive field $-E_c$ the polarization quickly begins to change with 180° domain switches. Negative strains in Figure 43.b are due to 90° switches. Reducing the field further to point D, all dipoles are again oriented in a parallel but opposite direction as in point A. Strains are equal to the ones in A since distortions are the same for positive and negative polarizations. Negative remanence is present in point F, again a state where small field changes cause linear, reversible responses. Increasing again the applied field involves 180° switching of the domains, point

G [141]. Negative strains are again induced by $90°$ switchings. After the field exceeds E_c, the coercive field, we return to point A with a uniform alignment of all dipoles.

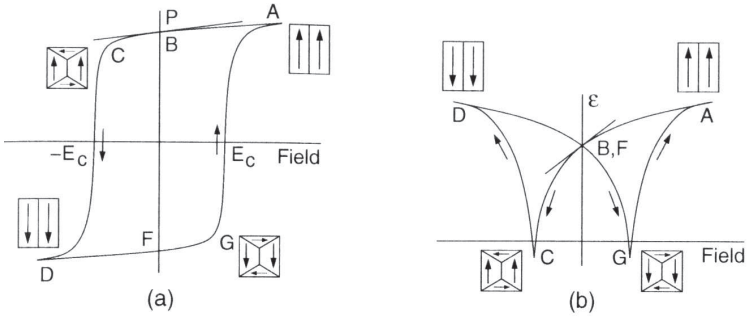


Figure 43: Typical $\vec{E} - \vec{P}$ and $\vec{E} - \vec{S}$ $(= \varepsilon)$ curve due to hysteresis. [141]

A second effect based on the load history is the ferroelastic behavior. Here, due to mechanical stresses a change in polarization happens by $90°$ switching of the domains. Thus, the macroscopic polarization decreases. Figure 44 illustrates the

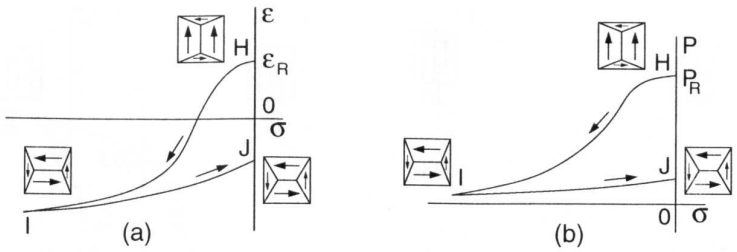


Figure 44: Typical $\vec{\sigma} - \vec{S}$ $(= \varepsilon)$ and $\vec{\sigma} - \vec{P}$ curve due to ferroelasticity. [141]

ferroelastic effect. Point H is the point of positive remanence equivalent to the point B in Figure 43. Confronting the piezoelectric workpieces with a compressive stress $90°$ switchings are invoked, causing negative strains and reducing the polarization. In point I the mechanical modulus is approximately linear. Reducing the applied stress to zero, point J, leaves reversible and almost linear changes in polarization and strains.

The correct treatment of the transducer's load history and numerical simulation combined with identification routines for occurring model parameters involved is a present topic of research, but not subject of this thesis. For further reading consult, e.g. [56, 74, 83].

Geometric Nonlinearities

Well-known models from mechanics which treat effects of geometric nonlinearities are included into the mathematical description of piezoelectric ceramics. This,

for example is in particular expedient if one quantitatively analyzes thin shells, cantilevers, bimorphs, piezo-laminated plates or partially multilayered stack actuators. In particular for low mass and highly flexible structures geometric nonlinear effects due to large deformations are apparent [105]. The approach in realizing nonlinear geometric effects consits of a substitution of the linear mechanical stress and strain tensors by the second Piola-Kirchhoff stress tensor and the Green-Lagrangian strain tensor, respectively. The resulting nonlinear relation between stress and strain requires an iterative linearization where one distinguishes between the total Lagrangian formulation referring to the initial state and the updated Lagrangian formulation referring to the deformed state, respectively. For further reading, see e.g. [80, 115, 121, 140, 150].
Here we consider applications with comparatively small deformations and therefore do not take geometric nonlinearities into account.

In literature sometimes deviant behavior of piezoelectric materials due to further effects, like e.g. heating or frequency dependency are also considered as nonlinear effects. We briefly specify the different effects in the sequel since their impact on specimen driven by large signals is even more severe than in the small-signal, i.e. the linear case.

Frequency Dependency

As we have already seen in the results of Section 3.3.4 the material parameters for the different monomodal resonators show remarkable differences. This is mainly due to the fact that these resonators vibrate at different frequency ranges ($60\mathrm{kHz} - 2\mathrm{MHz}$) and the effect of frequency dependency becomes obvious [49] . Usually one determines material parameters at the first (fundamental) resonant mode. But as soon as the transducer is driven at some different frequency range the parameters lose their exactness. In particular the dielectric constant ε shows a remarkable decrease in higher frequency ranges, approximately 10% per decade. In order to determine the frequency dependency, measurements of higher modes are required. Characterization of the material parameters by the method proposed in Section 3.3 considering each range of frequencies where a higher resonance of a single vibration mode occurs gives a set of parameters as a function of frequency [106]. Also one should have in mind that nonlinearities induced by hysteresis phenomena differ remarkably at different excitation frequencies and decrease at higher rates of oscillations. Parameter curves for the nonlinear case are therefore functions of the amplitude and frequency of the applied field.

Temperature Dependency

Due to the physical effects pyroelectricity and thermal expansion as well as piezo- and electrocaloric effects the transducers are markably influenced by inner and surrounding temperature changes. Usually, with higher temperatures the piezoelectric

specimen shows a damped behavior. Difficulties in detecting nonlinear parameters lie in particular in a self-heating of the probes which occurs in high power applications [106].

4.2 The Nonlinear Forward Problem

We remind that for both the forward and inverse problem the parameter curves do not depend on the unknown values electric potential and mechanical displacement but on their spatial derivatives. However, for modeling the material dependency it is common not to distinguish between the several entries in the field quantities (six for strain and three for electric field), but to confine the representation to either a dependency on the amplitude of the fields $|\vec{S}|, |\vec{E}|$ or on the most dominant components. The latter is a good approximation for the case of uniaxial loading which holds in many actuator applications. For example for a thickness resonator we mainly observe changes in $\phi_{|3} := E_3 = \partial\phi/\partial x_3$ and $u_{|33} := S_3 = \partial u_3/\partial x_3$. If one considers to evaluate the norm of the field components one has to be aware that non-differentiability is introduced causing non-smooth results. In this section we confine the represention to lossless materials, i.e. no damping is considered in order to shorten the notation as far as possible. So the set of piezoelectric PDEs in case of material nonlinearities for time dependent problems now reads

$$\rho\frac{\partial^2\mathbf{u}}{\partial t^2} - \mathcal{B}^T\Big(\mathbf{c}^E(u_{|xx})\mathcal{B}\mathbf{u} + \mathbf{e}^T(u_{|xx},\phi_{|x})\nabla\phi\Big) = 0 \quad \text{in } \Omega \tag{4.9a}$$

$$-\nabla\cdot\Big(\mathbf{e}(u_{|xx},\phi_{|x})\mathcal{B}\mathbf{u} - \boldsymbol{\varepsilon}^S(\phi_{|x})\nabla\phi\Big) = 0 \quad \text{in } \Omega\,, \tag{4.9b}$$

where x stands for an arbitrary space dimension. For the numerical treatment of the nonlinearities in the piezoelectric stiffness matrices a damped fixed-point iteration is carried out during each time step, i.e.

$$\underline{\mathbf{u}}_{n+1}^{k+1} = \underline{\mathbf{u}}_{n+1}^{k} + \eta\Delta\underline{\mathbf{u}}_{n+1}^{k} \tag{4.10}$$

where $\Delta\underline{\mathbf{u}}_{n+1}^{k}$ solves

$$\mathbf{K}^*(\underline{\mathbf{u}}_{n+1}^{k})\Delta\underline{\mathbf{u}}_{n+1}^{k} = \left(f_{n+1} + \frac{1}{\beta_H\Delta t^2}\mathbf{M}\tilde{\underline{\mathbf{u}}} - \mathbf{K}^*(\underline{\mathbf{u}}_{n+1}^{k})\underline{\mathbf{u}}_{n+1}^{k}\right), \tag{4.11}$$

where the predictor $\tilde{\underline{\mathbf{u}}}$ is computed as in (1.123) and

$$\mathbf{K}^*(\underline{\mathbf{u}}_{n+1}^{k}) := \mathbf{K}(\underline{\mathbf{u}}_{n+1}^{k}) + \frac{1}{\beta_h\Delta t^2}\mathbf{M}.$$

The corrector step now reads as

$$\begin{aligned}\ddot{\underline{\mathbf{u}}}_{n+1}^{k+1} &= \frac{\underline{\mathbf{u}}_{n+1}^{k+1} - \tilde{\underline{\mathbf{u}}}}{\beta_h\Delta t^2}\\ \dot{\underline{\mathbf{u}}}_{n+1}^{k+1} &= \dot{\tilde{\underline{\mathbf{u}}}} + \gamma_H\Delta t\ddot{\underline{\mathbf{u}}}_{n+1}^{k+1}.\end{aligned} \tag{4.12}$$

The relaxation parameter $0 < \eta \leq 1$ is steered by a simple line-search.

4.3 The Inverse Problem

We formulate the inverse problem of identifying the material parameter curves from electrical or mechanical measurements with the following parameter-to-solution map F which in time domain in its most general form is defined as

$$\begin{aligned} F : \mathcal{D}(F) \subseteq X_{par} &\rightarrow Y_{meas} \\ (\mathbf{c}^E, \mathbf{e}, \boldsymbol{\varepsilon}^S) &\mapsto y^\delta(t), \quad t \in [0,T], \end{aligned} \tag{4.13}$$

where $(\mathbf{c}^E, \mathbf{e}, \boldsymbol{\varepsilon}^S)$ are parameter curves depending on the physical field quantities electric field and mechanical strain. In order to reduce the complexity of computations and the representation of the Gâteaux derivative of F and its adjoint we consider only the dependency of the 33 components of $\mathbf{e}$ and $\boldsymbol{\varepsilon}^S$ on $\vec{E}_3 = (0,0,\phi_{|3})^T$ and assume that the constant material parameters are known (e.g. from identification at small signals according to Chapter 3). The parameter-to-solution map reduces and specifies in case of electric charge measurements to

$$\begin{aligned} F : \mathcal{D}(F) \subseteq \left(H^2(\mathbb{R})\right)^2 &\rightarrow L^2([0,T]) \\ (e_{33}, \varepsilon_{33}) &\mapsto \int_\Gamma \vec{D}(t) \cdot \vec{n} d\Gamma \\ &= \int_\Gamma \left(\mathbf{e}_3(\phi_{|3})\mathcal{B}\mathbf{u}(t) - \boldsymbol{\varepsilon}_3^S(\phi_{|3})\nabla\phi(t)\right) \cdot \vec{n} d\Gamma, \end{aligned} \tag{4.14}$$

where the pair $(\mathbf{u}, \phi)$ solves

$$\begin{aligned} \rho\frac{\partial^2 \mathbf{u}}{\partial t^2} - \mathcal{B}^T\left(\mathbf{c}^E\mathcal{B}\mathbf{u} + \mathbf{e}_3(\phi_{|3})^T\nabla\phi\right) &= 0 \quad \text{in } \Omega \times [0,T] \\ -\nabla \cdot \left(\mathbf{e}_3(\phi_{|3})\mathcal{B}\mathbf{u} - \boldsymbol{\varepsilon}_3^S(\phi_{|3})\nabla\phi\right) &= 0 \quad \text{in } \Omega \times [0,T]\,, \end{aligned} \tag{4.15}$$

combined with the boundary and initial conditions

$$\begin{aligned} \mathbf{N}^T(\mathbf{c}^E\mathcal{B}\mathbf{u} + \mathbf{e}_3(\phi_{|3})^T\nabla\phi\Big) &= 0 \quad \text{on } \partial\Omega \times [0,T] \\ \phi &= 0 \quad \text{on } \Gamma_g \times [0,T] \\ \phi &= \phi^e \quad \text{on } \Gamma_e \times [0,T] \\ \Big(\mathbf{e}_3(\phi_{|3})\mathcal{B}\mathbf{u} - \boldsymbol{\varepsilon}_3^S(\phi_{|3})\nabla\phi\Big) \cdot \vec{n} &= 0 \quad \text{on } \Gamma_r \times [0,T] \\ \mathbf{u}(\cdot,0) &= \mathbf{u}_0 \quad \text{on } \Omega \\ \mathbf{u}_t(\cdot,0) &= \dot{\mathbf{u}}_0 \quad \text{on } \Omega \end{aligned} \tag{4.16}$$

and the tensors $\mathbf{e}_3(v)$ and $\boldsymbol{\varepsilon}_3^S(v)$ are defined as

$$\begin{aligned} \mathbf{e}_3(v) &= \begin{pmatrix} 0 & 0 & 0 & 0 & e_{15} & 0 \\ 0 & 0 & 0 & e_{15} & 0 & 0 \\ e_{31} & e_{31} & e_{33}(v) & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \\ \boldsymbol{\varepsilon}_3^S(v) &= \begin{pmatrix} \varepsilon_{11} & 0 & 0 \\ 0 & \varepsilon_{11} & 0 \\ 0 & 0 & \varepsilon_{33}(v) \end{pmatrix}. \end{aligned} \tag{4.17}$$

The derivatives of the terms in (4.17) are defined as

$$\mathbf{e}_3'(v) := \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & e_{33}'(v) & 0 & 0 & 0 \end{pmatrix} \text{ and } \boldsymbol{\varepsilon}_3^{S\prime}(v) := \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \varepsilon_{33}^{S\,\prime}(v) \end{pmatrix}.$$

The choice of a second order Sobolev space as a pre-image space is motivated by the continuous differentiability of the parameter curves that is required for carrying out Newton's method in forward computations. Since we only have measurements of zero order derivatives available, i.e. electric charge or mechanical displacements and not any values of higher order derivatives with respect to time (velocity, acceleration or electric current) the data space is

$$Y = L^2[0,T]. \tag{4.18}$$

For the sought-after quantities, the parameter curves, as already mentioned, we assume spaces

$$\mathcal{D}(F) := X \subseteq (H^2(\mathbb{R}))^2 \tag{4.19}$$

in order to obtain C^1 curves by *Sobolev's embedding theorem*. The operator F actually maps into $C^2[0,T]$. So we have a difference in the regularity of these spaces which corresponds to an ill-posedness of twice numerical differentiation for the parameter curves reconstruction, see e.g. Example 1.1 and 1.6 in [36].

4.3.1 The Adjoint Operator

The following is devoted to the computation of the adjoint operator of the linearized problem.

Proposition 4.1. *The adjoint operator $F'(e_{33}, \varepsilon_{33}^S)^*$ of the linearization of F defined in (4.14) is given by*

$$\begin{aligned} &(F'(e_{33}, \varepsilon_{33}^S)^*[z])(\lambda) \\ &= \int_0^T \int_\Omega \Phi(\lambda - \phi_{|3}) \begin{pmatrix} (\mathcal{B}\mathbf{u})_3 \tilde{z}_{|3} - \phi_{|3}(\mathcal{B}\tilde{v})_3 \\ \phi_{|3}\tilde{z}_{|3} \end{pmatrix} d\Omega\, dt, \end{aligned} \tag{4.20}$$

where Φ is defined by

$$\Phi(a) = \frac{1}{2}\sqrt{\frac{\pi}{2}}(1+|a|)e^{-|a|}. \tag{4.21}$$

The values $(\tilde{v}, \tilde{z})$ are here obtained by solving the adjoint set of differential equations

$$\begin{aligned}
\rho\tilde{v}_{tt} - \mathcal{B}^T(\mathbf{c}^E\mathcal{B}\tilde{v} + \mathbf{e}_3(\phi_{|3})^T\nabla\tilde{z}) &= 0 \text{ on } \Omega\times[0,T]\\
-\nabla\cdot\Big((\mathbf{e}_3(\phi_{|3}) + \mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v} + \mathbf{e}_3'(\phi_{|3})\mathcal{B}\mathbf{u}\tilde{z}_{|3}&\\
- (\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\tilde{z}\Big) &= 0 \text{ on } \Omega\times[0,T]\\
\mathbf{N}^T\Big(\mathbf{c}^E\mathcal{B}\tilde{v} + \mathbf{e}_3(\phi_{|3})^T\nabla\tilde{z}\Big) &= 0 \quad \text{on } \partial\Omega\times[0,T]\\
\tilde{z} &= z(t) \quad \text{on } \Gamma_e\times[0,T]\\
\tilde{z} &= 0 \quad \text{on } \Gamma_g\times[0,T]\\
\Big((\mathbf{e}_3(\phi_{|3}) + \mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v} + \mathbf{e}_3'(\phi_{|3})\mathcal{B}\mathbf{u}\tilde{z}_{|3}&\\
- (\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\phi_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\tilde{z}\Big)\cdot\vec{n} &= 0 \quad \text{on } \Gamma_r\times[0,T]\\
\tilde{v}(\cdot,t=T) = \tilde{v}_t(\cdot,t=T) &= 0 \quad \text{on } \Omega.
\end{aligned} \tag{4.22}$$

Proof. We define the parameter increment tensors $\underline{\mathbf{de}}_3(v)$ and $\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(v)$ by

$$\underline{\mathbf{de}}_3(v) := \begin{pmatrix} 0&0&0&0&0&0\\ 0&0&0&0&0&0\\ 0&0&\underline{de}_{33}(v)&0&0&0\end{pmatrix} \text{ and } \underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(v) := \begin{pmatrix} 0&0&0\\ 0&0&0\\ 0&0&\underline{d\varepsilon}_{33}^S(v)\end{pmatrix}.$$

Let us consider the following neighboring problem with the incremented material parameters $e_{33} + \delta\underline{de}_{33}$ and $\varepsilon_{33}^S + \delta\underline{d\varepsilon}_{33}^S$

$$\begin{aligned}
\rho\frac{\partial^2\tilde{\mathbf{u}}}{\partial t^2} - \mathcal{B}^T\Big(\mathbf{c}^E\mathcal{B}\tilde{\mathbf{u}} + (\mathbf{e}_3 + \delta\underline{\mathbf{de}}_3)^T(\tilde{\phi}_{|3})\nabla\tilde{\phi}\Big) &= 0 \text{ in } \Omega\times[0,T]\\
-\nabla\cdot\Big((\mathbf{e}_3 + \delta\underline{\mathbf{de}}_3)(\tilde{\phi}_{|3})\mathcal{B}\tilde{\mathbf{u}} - (\boldsymbol{\varepsilon}_3^S + \delta\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S)(\tilde{\phi}_{|3})\nabla\tilde{\phi}\Big) &= 0 \text{ in } \Omega\times[0,T]\\
\mathbf{N}^T\Big(\mathbf{c}^E\mathcal{B}\tilde{\mathbf{u}} + (\mathbf{e}_3 + \delta\underline{\mathbf{de}}_3)^T(\tilde{\phi}_{|3})\nabla\tilde{\phi}\Big) &= 0 \text{ on } \partial\Omega\times[0,T]\\
\tilde{\phi} &= 0 \text{ on } \Gamma_0\times[0,T]\\
\tilde{\phi} &= \phi^e \text{ on } \Gamma_e\times[0,T]\\
\Big((\mathbf{e}_3 + \delta\underline{\mathbf{de}}_3)(\tilde{\phi}_{|3})\mathcal{B}\tilde{\mathbf{u}} - (\boldsymbol{\varepsilon}_3^S + \delta\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S)(\tilde{\phi}_{|3})\nabla\tilde{\phi}\Big) &= 0 \text{ on } \Gamma_r\times[0,T]\\
\tilde{\mathbf{u}}(\cdot,0) &= \mathbf{u}_0 \text{ on } \Omega\\
\tilde{\mathbf{u}}_t(\cdot,0) &= \dot{\mathbf{u}}_0 \text{ on } \Omega.
\end{aligned} \tag{4.23}$$

Formally, the derivative of the forward F operator in direction $[\underline{de}_{33}, \underline{d\varepsilon}_{33}]$ is given

by

$$
\begin{aligned}
&F'(e_{33},\varepsilon_{33}^S)[\underline{de}_{33},\underline{d\varepsilon}_{33}^S]\\
&= \lim_{\delta\to 0}\frac{F(e_{33}+\delta\underline{de}_{33},\varepsilon_{33}^S+\delta\underline{d\varepsilon}_{33}^S)-F(e_{33},\varepsilon_{33}^S)}{\delta}\\
&= \int_{\Gamma_e}\lim_{\delta\to 0}\left(\frac{\mathbf{e}_3(\tilde{\phi}_{|3})\mathcal{B}\tilde{\mathbf{u}}-\boldsymbol{\varepsilon}_3^S(\tilde{\phi}_{|3})\nabla\tilde{\phi}-\mathbf{e}_3(\phi_{|3})\mathcal{B}\mathbf{u}+\boldsymbol{\varepsilon}_3^S(\phi_{|3})\nabla\phi}{\delta}\right)\cdot\vec{n}\,d\Gamma\\
&= \int_{\Gamma_e}\Big(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}}+\mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u}-(\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3}+\boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\underline{d\phi}\Big)\cdot\vec{n}\,d\Gamma\\
&+\int_{\Gamma_e}\underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u}\cdot\vec{n}\,d\Gamma+\int_{\Gamma}\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\nabla\phi\cdot\vec{n}\,d\Gamma,
\end{aligned}
$$

where $(\underline{\mathbf{du}},\underline{d\phi})$ solve the following linear system in $\Omega\times[0,T]$

$$
\begin{aligned}
\rho\frac{\partial^2}{\partial t^2}\underline{\mathbf{du}}-\mathcal{B}^T\Big(\mathbf{c}^E\mathcal{B}\underline{\mathbf{du}}+(\mathbf{e}_3(\phi_{|3})+\mathbf{e}_3'(\phi_{|3})\phi_{|3})^T\nabla\underline{d\phi}\Big)&\\
=\ \mathcal{B}^T(\underline{\mathbf{de}}_3(\phi_{|3})\nabla\phi)&\\
-\nabla\cdot\Big(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}}+\mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u}-(\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3}+\boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\underline{d\phi}\Big)&\\
=\ \nabla\cdot(\underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u}+\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\nabla\phi),&
\end{aligned}
\tag{4.24}
$$

with boundary and initial conditions

$$
\begin{aligned}
\mathbf{N}^T\Big(\mathbf{c}^E\mathcal{B}\underline{\mathbf{du}}+(\mathbf{e}_3(\phi_{|3})+\mathbf{e}_3'(\phi_{|3})\phi_{|3})^T\nabla\underline{d\phi}+\underline{\mathbf{de}}_3(\phi_3)\Big)&=0\ \text{ on }\partial\Omega\times[0,T]\\
\underline{d\phi}&=0\ \text{ on }\Gamma_0\times[0,T]\\
\underline{d\phi}&=0\ \text{ on }\Gamma_e\times[0,T]\\
\big(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}}+\mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u}&\\
-(\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3}+\boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\underline{d\phi}&\\
+\underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u}+\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\big)\cdot\vec{n}&=0\ \text{ on }\Gamma_r\times[0,T]\\
\mathbf{u}(\cdot,0)&=0\ \text{ on }\Omega\\
\mathbf{u}_t(\cdot,0)&=0\ \text{ on }\Omega.
\end{aligned}
\tag{4.25}
$$

With the Gâteaux derivative of F we can compute the adjoint $F'(\cdot)^*$, since it holds

$$
\begin{aligned}
&\forall\,\underline{de}_{33},\underline{d\varepsilon}_{33}^S\in\left(H^2(\mathbb{R})\right)^2,\ z\in L^2[0,T]\\
&\langle F'(e_{33},\varepsilon_{33}^S)[\underline{de}_{33},\underline{d\varepsilon}_{33}^S],z\rangle_{L^2[0,T]}\ =\ \langle[\underline{de}_{33},\underline{d\varepsilon}_{33}^S],F'(e_{33},\varepsilon_{33}^S)^*z\rangle_{(H^2(\mathbb{R}))^2}.
\end{aligned}
\tag{4.26}
$$

The inner product in the left-hand side of (4.26) is given by

$$\int_0^T \int_\Gamma \big(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}} + \mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u} - (\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\underline{d\phi} + \underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u} + \underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\big)\cdot\vec{n}\, z\, d\Gamma\, dt. \tag{4.27}$$

Due to the boundary conditions in (4.25) we can extend the surface integral in (4.27) to the complete boundary $\partial\Omega$ of Ω and applying the *divergence theorem* we get for (4.27)

$$\begin{aligned}
&\int_0^T \int_\Omega \nabla\cdot\big(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}} + \mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u} \\
&\quad -(\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\underline{d\phi} + \underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u} + \underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\big)\tilde{z}\, d\Omega\, dt \\
&+\int_0^T \int_\Omega \big(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}} + \mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u} \\
&\quad -(\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\underline{d\phi} + \underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u} + \underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\big)^T\nabla\tilde{z}\, d\Omega\, dt.
\end{aligned} \tag{4.28}$$

Now, the following identity holds since $\underline{d\phi} = 0$ on Γ_e and Γ_g and due to the homogeneous Neumann boundary condition on Γ_r in (4.22)

$$\begin{aligned}
0 &= \int_0^T \int_{\partial\Omega} \big((\mathbf{e}_3(\phi_{|3}) + \mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v} + \mathbf{e}_3'(\phi_{|3})\tilde{z}_{|3}\mathcal{B}\mathbf{u}\big)\cdot\vec{n}\,\underline{d\phi}\, d\Gamma\, dt \\
&\quad - \int_0^T \int_{\partial\Omega} \Big((\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\tilde{z}\Big)\cdot\vec{n}\,\underline{d\phi}\, d\Gamma\, dt \\
&= \int_0^T \int_\Omega \nabla\cdot\big((\mathbf{e}_3(\phi_{|3}) + \mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v} + \mathbf{e}_3'(\phi_{|3})\tilde{z}_{|3}\mathcal{B}\mathbf{u}\big)\,\underline{d\phi}\, d\Omega\, dt \\
&\quad - \int_0^T \int_\Omega \nabla\cdot\Big((\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\tilde{z}\Big)\,\underline{d\phi}\, d\Omega\, dt \\
&\quad + \int_0^T \int_\Omega \big((\mathbf{e}_3(\phi_{|3}) + \mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v} + \mathbf{e}_3'(\phi_{|3})\tilde{z}_{|3}\mathcal{B}\mathbf{u}\big)\,\nabla\underline{d\phi}\, d\Omega\, dt \\
&\quad - \int_0^T \int_\Omega \Big((\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3} + \boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\tilde{z}\Big)\,\nabla\underline{d\phi}\, d\Omega\, dt.
\end{aligned} \tag{4.29}$$

Subtracting (4.29) from (4.28) we obtain

$$
\begin{aligned}
&\langle F'(e_{33},\varepsilon_{33}^S)[\underline{de}_{33},\underline{d\varepsilon}_{33}^S],z\rangle_{L^2[0,T]}\\
&\quad=\int_0^T\int_\Omega \nabla\cdot\left(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}}+\mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u}\right)\tilde{z}\,d\Omega\,dt\\
&\qquad-\int_0^T\int_\Omega \nabla\cdot\left(\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3}+\boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\underline{d\phi}\right)\tilde{z}\,d\Omega\,dt\\
&\qquad-\int_0^T\int_\Omega \nabla\cdot\left(\underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u}+\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\right)\tilde{z}\,d\Omega\,dt\\
&\qquad+\int_0^T\int_\Omega \left(\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}}+\mathbf{e}_3'(\phi_{|3})\underline{d\phi}_{|3}\mathcal{B}\mathbf{u}\right)^T\nabla\tilde{z}\,d\Omega\,dt\\
&\qquad-\int_0^T\int_\Omega \left((\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3}+\boldsymbol{\varepsilon}_3^S(\phi_{|3}))^T\nabla\underline{d\phi}\right)^T\nabla\tilde{z}\,d\Omega\,dt\\
&\qquad-\int_0^T\int_\Omega \left(\underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u}+\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\right)^T\nabla\tilde{z}\,d\Omega\,dt\\
&\qquad-\int_0^T\int_\Omega \nabla\cdot\left((\mathbf{e}_3(\phi_{|3})+\mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v}+\mathbf{e}_3'(\phi_{|3})\tilde{z}_{|3}d\mathcal{B}\mathbf{u}\right)\underline{d\phi}\,d\Omega\,dt\\
&\qquad+\int_0^T\int_\Omega \nabla\cdot\left((\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3}+\boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\tilde{z}\right)\underline{d\phi}\,d\Omega\,dt\\
&\qquad-\int_0^T\int_\Omega \left((\mathbf{e}_3(\phi_{|3})+\mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v}+\mathbf{e}_3'(\phi_{|3})\tilde{z}_{|3}\mathcal{B}\mathbf{u}\right)^T\nabla\underline{d\phi}\,d\Omega\,dt\\
&\qquad+\int_0^T\int_\Omega \left((\boldsymbol{\varepsilon}_3^{S\prime}(\phi_{|3})\underline{d\phi}_{|3}+\boldsymbol{\varepsilon}_3^S(\phi_{|3}))\nabla\tilde{z}\right)^T\nabla\underline{d\phi}\,d\Omega\,dt,
\end{aligned}
\tag{4.30}
$$

which with (4.22) and (4.24) simplifies to

$$
\begin{aligned}
&\int_0^T\int_\Omega \left(\underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u}+\underline{\mathbf{d}\boldsymbol{\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\right)^T\nabla\tilde{z}\,d\Omega\,dt\\
&-\int_0^T\int_\Omega (\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}})^T\nabla\tilde{z}\,d\Omega\,dt\\
&+\int_0^T\int_\Omega \left((\mathbf{e}_3(\phi_{|3})+\mathbf{e}_3'(\phi_{|3})\phi_{|3})\mathcal{B}\tilde{v}\right)^T\nabla\underline{d\phi}\,d\Omega\,dt.
\end{aligned}
\tag{4.31}
$$

Partial integration of each term containing $\mathbf{e}_3(\phi_{|3})$ in (4.31) gives with (4.22) and (4.24)

$$
\begin{aligned}
&-\int_0^T \int_\Omega (\mathbf{e}_3(\phi_{|3})\mathcal{B}\underline{\mathbf{du}})^T \nabla\tilde{z}\, d\Omega\, dt \\
&= \int_0^T \int_\Omega (\mathcal{B}^T \mathbf{e}_3(\phi_{|3})^T \nabla\tilde{z})^T \underline{\mathbf{du}}\, d\Omega\, dt - \int_0^T \int_{\partial\Omega} (\mathbf{e}_3(\phi_{|3})^T \nabla\tilde{z})^T \mathbf{N}\underline{\mathbf{du}}\, d\Gamma\, dt \\
&= \int_0^T \int_\Omega (\rho\tilde{v}_{tt} - \mathcal{B}^T \mathbf{c}^E \mathcal{B}\tilde{v})^T \underline{\mathbf{du}}\, d\Omega\, dt + \int_0^T \int_{\partial\Omega} (\mathbf{c}^E \mathcal{B}\tilde{v})^T \mathbf{N}\underline{\mathbf{du}}\; d\Gamma, dt \\
&= \int_0^T \int_\Omega \rho\tilde{v}_{tt}\underline{\mathbf{du}}\, d\Omega\, dt + \int_0^T \int_\Omega (\mathbf{c}^E \mathcal{B}\tilde{v})^T \mathcal{B}\underline{\mathbf{du}}\; d\Omega\, dt,
\end{aligned}
\tag{4.32}
$$

and

$$
\begin{aligned}
&\int_0^T \int_\Omega (\mathbf{e}_3(\phi_{|3})\mathcal{B}\tilde{v})^T \nabla \underline{d\phi}\, d\Omega\, dt \\
&= -\int_0^T \int_\Omega (\mathcal{B}^T \mathbf{e}_3(\phi_{|3})^T \nabla \underline{d\phi})^T \tilde{v}\, d\Omega\, dt + \int_0^T \int_{\partial\Omega} (\mathbf{e}_3(\phi_{|3})^T \nabla \underline{d\phi})\mathbf{N}\tilde{v}\, d\Gamma dt \\
&= -\int_0^T \int_\Omega \left(\rho\underline{\mathbf{du}}_{tt} - \mathcal{B}^T \left(\mathbf{c}^E \mathcal{B}\underline{\mathbf{du}} + \mathbf{e}_3'(\phi_{|3})^T \phi_{|3} \nabla \underline{d\phi}\right)\right)^T \tilde{v}\, d\Omega\, dt \\
&\quad - \int_0^T \int_{\partial\Omega} (\mathbf{c}^E \mathcal{B}\underline{\mathbf{du}} + \mathbf{e}_3'(\phi_{|3})^T \phi_{|3} \nabla \underline{d\phi})\mathbf{N}\tilde{v}\, d\Gamma dt \\
&\quad + \int_0^T \int_\Omega (\mathcal{B}^T \underline{\mathbf{de}}_3(\phi_{|3})^T \nabla\phi)^T \tilde{v}\, d\Omega\, dt - \int_0^T \int_{\partial\Omega} (\underline{\mathbf{de}}_3(\phi_{|3})^T \nabla\phi)\mathbf{N}\tilde{v}\; d\Gamma\, dt \\
&= -\int_0^T \int_\Omega \rho\underline{\mathbf{du}}_{tt}\tilde{v}\, d\Omega\, dt - \int_0^T \int_\Omega (\mathbf{c}^E \mathcal{B}\underline{\mathbf{du}})^T \mathcal{B}\tilde{v}\, d\Gamma dt \\
&\quad - \int_0^T \int_\Omega \left(\mathbf{e}_3'(\phi_{|3})^T \phi_{|3} \nabla \underline{d\phi}\right)^T \mathcal{B}\tilde{v}\, d\Omega\, dt - \int_0^T \int_\Omega (\underline{\mathbf{de}}_3^T(\phi_{|3})\nabla\phi)^T \mathcal{B}\tilde{v}\, d\Omega\, dt.
\end{aligned}
\tag{4.33}
$$

Substituting (4.32) and (4.33) in (4.31) we obtain after partial integration with respect to time

$$
\begin{aligned}
&\langle F'(e_{33}, \varepsilon_{33}^S)[\underline{de}_{33}, \underline{d\varepsilon}_{33}^S], z\rangle_{L^2[0,T]} = \\
&\int_0^T \int_\Omega (\underline{\mathbf{de}}_3(\phi_{|3})\mathcal{B}\mathbf{u})^T \nabla\tilde{z} - \underline{\mathbf{de}}_3(\phi_{|3})\phi_{|3}\mathcal{B}\tilde{v} + \underline{\mathbf{d\varepsilon}}_3^S(\phi_{|3})\phi_{|3}\nabla\tilde{z}\, d\Omega\, dt.
\end{aligned}
\tag{4.34}
$$

Taking advantage of the *sifting property* of the Dirac delta distribution δ_0 one ob-

tains

$$\begin{aligned}
&\langle[\underline{de}_{33}, \underline{d\varepsilon}^S_{33}], (F'(e_{33}, \varepsilon^S_{33})^*[z])\rangle \\
&\quad= \int_0^T \int_\Omega \int_{\mathbb{R}} \delta_0(\lambda - \phi_{|3})(\underline{\mathbf{de}}_3(\lambda)\mathcal{B}\mathbf{u})^T \nabla\tilde{z}\, d\lambda\; d\Omega\, dt \\
&\qquad- \int_0^T \int_\Omega \int_{\mathbb{R}} \delta_0(\lambda - \phi_{|3})\underline{\mathbf{de}}_3(\lambda)\phi_{|3}\mathcal{B}\tilde{v}\, d\lambda\; d\Omega\, dt \\
&\qquad+ \int_0^T \int_\Omega \int_{\mathbb{R}} \delta_0(\lambda - \phi_{|3})\underline{\mathbf{d}\boldsymbol{\varepsilon}}^S_3(\lambda)\phi_{|3}\nabla\tilde{z}\, d\lambda\; d\Omega\, dt \\
&\quad= \int_{\mathbb{R}} \underline{de}_{33}(\lambda) \int_0^T \int_\Omega \delta_0(\lambda - \phi_{|3})(\mathcal{B}\mathbf{u})_3\tilde{z}_{|3}\, d\Omega\, dt\, d\lambda \\
&\qquad- \int_{\mathbb{R}} \underline{de}_{33}(\lambda) \int_0^T \int_\Omega \delta_0(\lambda - \phi_{|3})\phi_{|3}(\mathcal{B}\tilde{v})_3\; d\Omega\, dt\, d\lambda \\
&\qquad+ \int_{\mathbb{R}} \underline{d\varepsilon}^S_{33}(\lambda) \int_0^T \int_\Omega \delta_0(\lambda - \phi_{|3})\phi_{|3}\tilde{z}_{|3}\, d\Omega\, dt\, d\lambda\,. \qquad (4.35)
\end{aligned}$$

By applying the *theorem of Parseval* the inner product from (4.35) in Fourier space is

$$\begin{aligned}
&\langle[\underline{de}_{33}, \underline{d\varepsilon}_{33}], (F'(e_{33}, \varepsilon^S_{33})^*[z])\rangle = \\
&\qquad\int_{\mathbb{R}} (\mathcal{F}_\lambda[\underline{de}_{33}, \underline{d\varepsilon}^S_{33}])(\omega)\overline{\mathcal{F}_\lambda\left(\int_0^T \int_\Omega \delta_0(\cdot - \phi_{|3})\Psi(x,t)\, d\Omega\, dt\right)}(\omega)d\omega, \qquad (4.36)
\end{aligned}$$

where

$$\Psi(x,t) = \begin{pmatrix} (\mathcal{B}\mathbf{u})_3\tilde{z}_{|3} - \phi_{|3}(\mathcal{B}\tilde{v})_3 \\ \phi_{|3}\tilde{z}_{|3} \end{pmatrix}. \qquad (4.37)$$

Now computing the Fourier transformation of the δ_0 function

$$\begin{aligned}
\mathcal{F}(\delta_0(\cdot - \alpha))(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\omega\lambda}\delta_0(\underbrace{\lambda - \alpha}_{:=\tilde{\lambda}})d\lambda \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\omega(\tilde{\lambda}+\alpha)}\delta_0(\tilde{\lambda})d\tilde{\lambda} \\
&= e^{-i\omega\alpha}(\mathcal{F}\delta_0)(\omega) \\
&= e^{-i\omega\alpha} \qquad (4.38)
\end{aligned}$$

and with $||\cdot||^2_{H^2} = \int_{\mathbb{R}}(1 + 2\omega^2 + \omega^4)|\mathcal{F}(\cdot)|^2(\omega)d\omega$ we obtain

$$\begin{aligned}
&\langle[\underline{de}_{33}, \underline{d\varepsilon}^S_{33}], F'(e_{33}, \varepsilon^S_{33})^* z\rangle_{(H^2(\Lambda))^2} = \\
&\qquad\int_{\mathbb{R}}(1+\omega^2)^2(\mathcal{F}_\lambda[\underline{de}_{33}, \underline{d\varepsilon}^S_{33}])(\omega)\overline{\int_0^T \int_\Omega \tfrac{1}{(1+\omega^2)^2} e^{-i\omega\phi_{|3}}\Psi(x,t)\, d\Omega\, dt}d\omega. \qquad (4.39)
\end{aligned}$$

From this

$$F'(e_{33},\varepsilon_{33}^S)^* = \mathcal{F}_\lambda^{-1}\left(\int_0^T\int_\Omega \frac{e^{-i\omega\phi_{|3}}}{(1+\omega^2)^2}\Psi(x,t)\,d\Omega\,dt\right) \tag{4.40}$$

which requires a solution of the following integral

$$\int_{-\infty}^{\infty}\frac{e^{i\omega(\cdot-\phi_{|3})}}{(1+\omega^2)^2}d\omega. \tag{4.41}$$

We will evaluate the integral in (4.41) by expressing it as a limit of contour integrals along the contour C which goes along the real line from some $-a$ to a and then counterclockwise along a semicircle centered at 0 from a to $-a$. Considering a to be greater than 1 encloses the imaginary units $\pm i$ within the curve. Regarding the complex plane the only singularities lie at $\omega = \pm i$ with order two. The residue of (4.41) is given by

$$\begin{aligned}\mathrm{Res}(f(\omega),\pm i) &= \lim_{\omega\to\pm i}\frac{\partial}{\partial\omega}(\omega\pm i)^2\frac{e^{i\omega(\cdot-\phi_{|3})}}{(i+\omega)^2(i-\omega)^2}\\ &= \frac{i}{4}(1+(\cdot-\phi_{|3}))e^{\pm(\cdot-\phi_{|3})}\end{aligned} \tag{4.42}$$

By the *residue theorem* the contour integral along the contour C which forms a *Jordan curve* computes e.g. for $\omega = i$ as

$$\oint_C f(z)dz = 2i\pi\sum_{k=1}^{2}Res_{(f(z),i)} = \frac{\pi}{2}(1+(\cdot-\phi_{|3}))e^{-(\cdot-\phi_{|3})}. \tag{4.43}$$

Dividing the contour into a part which is just the straight line in the complex plane for $i = 0$ and going to the limit for $a \to \infty$ gives the required solution since the integral over the rest of the contour tends to zero for this limit. Thus, distinguishing between $(\cdot-\phi_{|3}) > 0$ and $(\cdot-\phi_{|3}) < 0$ we have

$$\int_{-\infty}^{\infty}\frac{e^{i\omega(\cdot-\phi_{|3})}}{(1+\omega^2)^2}d\omega = \frac{1}{2}\sqrt{\frac{\pi}{2}}(1+|(\cdot-\phi_{|3})|)e^{-|(\cdot-\phi_{|3})|}. \tag{4.44}$$

With this, the adjoint operator computes as

$$(F'(e_{33},\varepsilon_{33}^S)^*[z])(\lambda) = \int_0^T\int_\Omega \Phi(\lambda-\phi_{|3})\Psi(x,t)\,d\Omega\,dt. \tag{4.45}$$

□

4.3.2 The Degree of Ill-Posedness

By the analytic formula of the adjoint the smoothing character of the adjoint and so the ill-posedness of the linearized problem can be quantified by the following theorem:

Proposition 4.2. *Assume that the solution operator $S : z \mapsto (\tilde{z}, \tilde{v})$ of the adjoint equation in (4.22) is bounded in its norm, i.e.*

$$||S||_{(H^\sigma(0,T))^* \to (H^\sigma(0,T;H^1_{0,\Gamma}(\Omega)^*))^* \times (H^\sigma(0,T;H^1_{\mathcal{B}}(\Omega)^*))^*} < \infty. \tag{4.46}$$

If further $\mathbf{u}, \phi \in C^\infty([0,T] \times \Omega)$, then the operator $F'(e_{33}, \varepsilon^S_{33})^*$ can be extended to a continuous linear operator mapping from $(H^\sigma[0,T])^*$ to $(H^2(\mathbb{R}))^2$ for any $\sigma \in (0, 3/2)$.

Remark: The assumptions in (4.46) are somehow "plausible", the space C^∞ for $\mathbf{u}$ and ϕ is chosen for simplicity.

Proof. For any $z \in L^2[0,T]$ and $\sigma \in (1, \frac{3}{2})$ we have

$$\begin{aligned}
&||F'(e_{33}, \varepsilon^S_{33})^*[z]||_{(H^2(\mathbb{R}))^2} \\
&= \int_{\mathbb{R}} (1+\omega^2)^2 \left| \int_0^T \int_\Omega e^{-i\phi_{|3}\omega} (\mathcal{F}\Phi)(\omega) \Psi(x,t)\, d\Omega\ dt \right|^2 d\omega \\
&= \int_{\mathbb{R}} \left| \int_0^T \int_\Omega \frac{e^{-i\phi_{|3}\omega}}{1+\omega^2} \Psi(x,t)\, d\Omega\ dt \right|^2 d\omega \\
&= \int_{\mathbb{R}} \left(\frac{1}{1+\omega^2} \right)^2 \left(\left| \int_0^T \int_\Omega e^{-i\phi_{|3}\omega} \left((\mathcal{B}\mathbf{u})_3 \tilde{z}_{|3} - \phi_{|3} (\mathcal{B}\tilde{v})_3 \right) d\Omega dt \right|^2 \right) d\omega \\
&\quad + \int_{\mathbb{R}} \left(\frac{1}{1+\omega^2} \right)^2 \left(\left| \int_0^T \int_\Omega e^{-i\phi_{|3}\omega} \phi_{|3} \tilde{z}_{|3}\, d\Omega dt \right|^2 \right) d\omega \\
&\leq \int_{\mathbb{R}} \left(\frac{1}{1+\omega^2} \right)^2 \big(2||\tilde{z}_{|3}||^2_{(H^\sigma([0,T];L^2(\Omega)))^*} ||e^{-i\phi_{|3}\omega} (\mathcal{B}\mathbf{u})_3||^2_{H^\sigma([0,T];L^2(\Omega))} \\
&\qquad + 2||(\mathcal{B}\tilde{v})_3||^2_{(H^\sigma([0,T];L^2(\Omega)))^*} ||e^{-i\phi_{|3}\omega} \phi_{|3}||^2_{H^\sigma([0,T];L^2(\Omega))} \\
&\qquad + 2||\tilde{z}_{|3}||^2_{(H^\sigma([0,T];L^2(\Omega)))^*} ||e^{-i\phi_{|3}\omega} \phi_{|3}||^2_{H^\sigma([0,T];L^2(\Omega))} \big)\ d\omega.
\end{aligned} \tag{4.47}$$

We now analyze single components of (4.47), where

$$\begin{aligned}
||\tilde{z}_{|3}||_{(H^\sigma([0,T];L^2(\Omega)))^*} &\leq ||\tilde{z}||_{(H^\sigma([0,T];H^1_{0,\Gamma}(\Omega)^*))^*} \quad \text{and} \\
||(\mathcal{B}\tilde{v})_3||_{(H^\sigma([0,T];L^2(\Omega)))^*} &\leq ||\tilde{v}||_{(H^\sigma([0,T];H^1_{\mathcal{B}}(\Omega)^*))^*}.
\end{aligned} \tag{4.48}$$

The values in (4.48) are, by our assumption on the solution operator of the adjoint equation, bounded by some constant times $||z||_{(H^\sigma([0,T]))^*}$. The terms in (4.47)

with norms in $H^\sigma([0,T],L^2(\Omega))$ including the function $e^{-i\phi_{|3}\omega}$ can be estimated with the help of the following interpolation inequality

$$\forall v \in H^2[0,T] \ : \quad ||v||_{H^\sigma[0,T]} \leq ||v||^{2-\sigma}_{H^1[0,T]} ||v||^{\sigma-1}_{H^2[0,T]}, \tag{4.49}$$

which is a Nirenberg-Gagliardo type interpolation inequalitiy [40,110] of the form

$$||\nabla^{(k)} f||_{L^2} \leq ||f||^{1-k/m}_{L^2} \, ||\nabla^{(m)} f||^{k/m}_{L^2}$$

with $\nabla^{(l)} f = (D^\alpha f)_{|\alpha|=l}$, together with the choices $m=1$ and $k=\sigma-1$ and setting $f=\nabla v$. Now, the first and second time derivatives of $e^{-i\phi_{|3}\omega}\phi_{|3}$ are

$$\begin{aligned}
\frac{d}{dt}e^{-i\phi_{|3}\omega}\phi_{|3} &= e^{-i\phi_{|3}\omega}\left(-i\phi^2_{|3}\omega\frac{d}{dt}\phi_{|3}+\frac{d}{dt}\phi_{|3}\right)\\
\frac{d^2}{dt^2}e^{-i\omega\phi_{|3}}\phi_{|3} &= e^{-i\phi_{|3}\omega}\left(-\omega^2\phi^3_{|3}\left(\frac{d}{dt}\phi_{|3}\right)^2-3i\phi_{|3}\omega(\frac{d}{dt}\phi_{|3})^2\right)\\
&\quad + e^{-i\phi_{|3}\omega}\left(-i\phi_{|3}\omega\frac{d^2}{dt^2}\phi_{|3}+\frac{d^2}{dt^2}\phi_{|3}\right).
\end{aligned} \tag{4.50}$$

Thus with $\mathbf{u}$ and $\phi \in C^\infty$

$$\begin{aligned}
&\left(\frac{1}{1+\omega^2}\right)^2 ||e^{-i\omega\phi_{|3}}\phi_{|3}||_{(H^\sigma[0,T];L^2(\Omega))}\\
&= \mathcal{O}\left(\left(\frac{1}{1+\omega^2}\right)^2\right)\mathcal{O}\left((\omega^{(2-\sigma)})^2\right)\mathcal{O}\left(((\omega^2)^{\sigma-1})^2\right)\\
&= \mathcal{O}\left(\left(\frac{1}{1+\omega^2}\right)^2\right)\mathcal{O}\left(\left(\frac{1}{1+\omega^2}\right)^{-2+\sigma}\right)\mathcal{O}\left(\left(\frac{1}{1+\omega^2}\right)^{2-2\sigma}\right)\\
&= \mathcal{O}\left(\frac{1}{1+\omega^2}\right)^{2-\sigma}.
\end{aligned} \tag{4.51}$$

We can argue analogously for $||e^{-i\omega\phi_{|3}}(\mathcal{B}\mathbf{u})_3||_{(H^\sigma[0,T];L^2(\Omega))}$. Finally this gives together with (4.47)

$$||F'(e_{33},\varepsilon^S_{33})^*[z]||_{(H^2(\mathbb{R}))^2} \leq C||z||_{(H^\sigma[0,T])^*}\int_{\mathbb{R}}\left(\frac{1}{1+\omega^2}\right)^{2-\sigma} d\omega.$$

The integral here over $\mathbb{R}$ remains finite as long as $\sigma < 3/2$. Since $L^2[0,T]$ is dense in $(H^\sigma[0,T])$ it follows that $F'(e_{33},\varepsilon^S_{33})^*$ can be extended to a continuous linear operator mapping from $(H^\sigma[0,T])^*$ to $(H^2(\mathbb{R}))^2$ for any $\sigma \in (0,3/2)$. □

This allows for the interpretation of the adjoint operator being smoothing of order $\frac{3}{2}$.

As it is similarly shown in [76]

$$\begin{aligned} &||F'(e_{33},\varepsilon_{33}^S)[s]||_{H^\sigma[0,T]} \\ &\qquad = \sup_{z\in Y}\frac{(F'(e_{33},\varepsilon_{33}^S)[s],z)_{L^2[0,T]}}{||z||_{(H^\sigma[0,T])^*}} \\ &\qquad = \sup_{z\in Y}\frac{(s,F'(e_{33},\varepsilon_{33}^S)^*[z])_{(H^2(\mathbb{R}))^2}}{||z||_{(H^\sigma[0,T])^*}} \\ &\qquad \leq ||F'(e_{33},\varepsilon_{33}^S)^*||_{(H^\sigma[0,T])^*\to(H^2(\mathbb{R}))^2}||s||_{(H^2(\mathbb{R}))^2} \end{aligned}$$

for any $s \in X$ and $\sigma \in (0,3/2)$. Hence, the range $\mathcal{R}(F'(e_{33},\varepsilon_{33}^S))$ of the linearization of the forward operator F is nonclosed in Y which shows even the ill-posedness of the linearized problem.

4.3.3 Numerical Results

In this section numerical identification results will be presented, however the exposition is limited to synthetically generated data only. Identification from real measurements is the subject of future work.

The numerical results comprise both the identification of one single parameter as a function of the electric field and the simultaneous reconstruction of two parameters from given charge signals over a finite time interval.

The main intention of this section is to test the iterative multilevel algorithm, described in detail in Algorithm 3. The discretization of the parameter curves is implemented with either cubic splines or piecewise linear functions. The refinement of discretization when going from level N to level $N+1$ is done by bisection of each subinterval. The coarsest level always consists of three grid points only. For the implementation of the inner stopping rule a term including information about the size of the discretization intervals, the data error and appropriate scaling factors is evaluated according to

$$||F(e_{33}^k, \varepsilon_{33}^{S,k}) - y^\delta|| \leq C_1(c_1\delta + c_2\frac{1}{2^N}). \tag{4.52}$$

The discretization error using splines may be estimated by $\inf_{S\in\mathbb{S}_h^k} ||f-S||_{H^j(\Omega)} \leq ch^{k-j}||f||_{H^k(\Omega)}$ for any spline S of order k with uniform knots and width h [144].

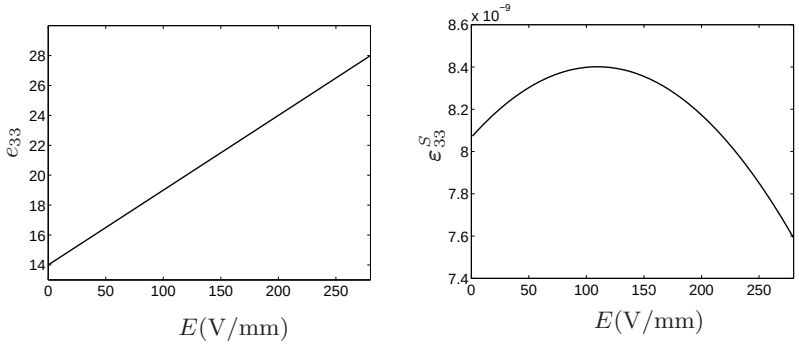


Figure 45: Nonlinear case: Parameter e_{33} and ε_{33}^S as functions of the electrical field. Values are obtained from [7] for a Pz27 ceramic.

In order to derive realistic parameter curves for our numerical experiments, considering the relationship of the material tensors

$$\boldsymbol{\varepsilon}^S = \boldsymbol{\varepsilon}^\sigma - \mathbf{d}^T(\mathbf{s}^E)^{-1}\mathbf{d} \tag{4.53}$$

we can compute the functional dependency of ε_{33}^S on the electrical field, for example, via the one of d_{33} and ϵ_{33}^σ which are measurable from transversal and longitudinal resonators. With the data published in Andersen [7] one gets the functional dependency of ε_{33}^E and e_{33} as depicted in Figure 45.

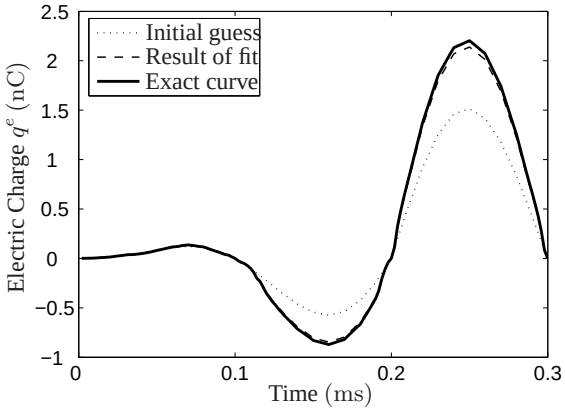


Figure 46: Charge response with constant (initial guess, dotted) and nonlinear permittivity (exact and fitted data, solid and dashed).

In a first step we report on reconstructing just the permittivity as a function of the electric field. As initial guess we use a constant function with the value obtained in the small signal case, see Figures 48-50. As excitation a special signal with zero charge and zero derivative for $t = 0$ is chosen. The electric charge serves as measurements, see Figure 46

Figure 47 shows the development of the residual during the multilevel algorithm for different choices of the damping parameter ω_N^k. Along the abscissa the accumulated inner iteration index is plotted, vertical lines show a transition between two levels. The horizontal line shows the quantity $C_2\delta$. As we see, both the steepest descent and minimal error variant behave rather similarly and proceed (by coincidence) to the next finer level after the same amount of inner iterations.

Three different types of functions have been reconstructed, see Figures 48-50, where the linear and parabolic curve are assumed to be physically reasonable, the third one rather serves to demonstrate the efficiency of the algorithm even to detect curves with more oscillating behavior.

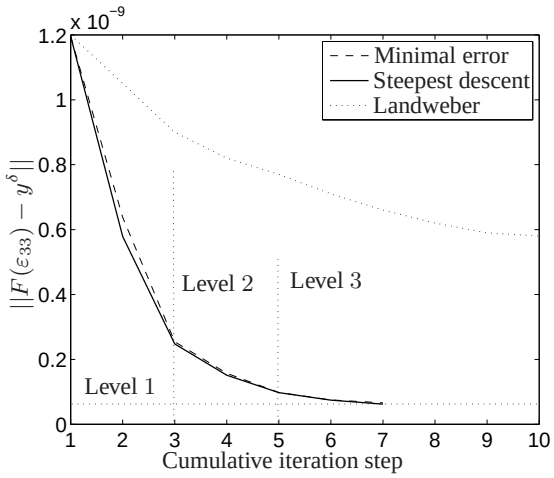


Figure 47: Development of residual norm for different choices of ω during reconstruction of the permittivity (for the curve in Figure 49) with one percent data noise.

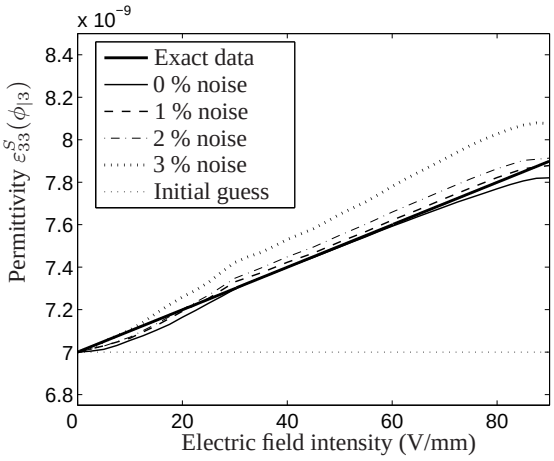


Figure 48: Reconstruction of a linear parameter curve with different amounts of data noise.

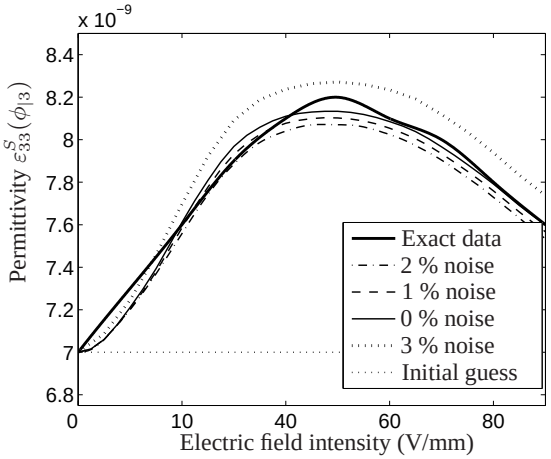


Figure 49: Reconstruction of a parabolic parameter curve with different amounts of data noise.

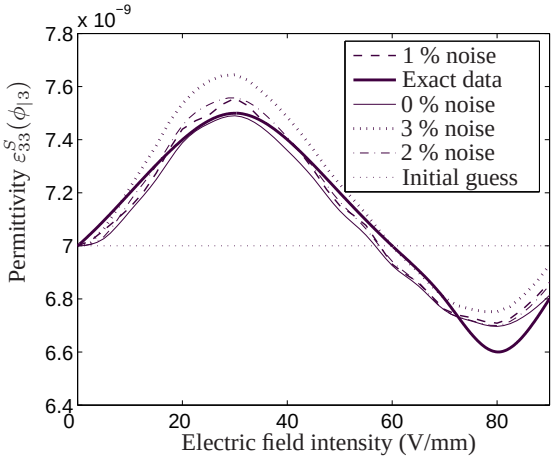


Figure 50: Reconstruction of a curve having two extrema with different amounts of data noise.

Simultaneous Reconstruction of two Parameter Curves

Results of the simultaneous reconstruction of the two parameter curves e_{33} and ε_{33}^S are given in Figure 51. The reconstruction seems to be robust with respect to errors in the data. However, in particular for lower field intensities one sees less satisfying reconstruction results which might be due to the fact that the material parameters mutually influence each other.

Finally, we observe that not only the shape of the curve or the amount of data error is responsible for a satisfying data fit, but also the discretization method and with this the given space, see Figure 52. Here, a comparison between the identification results with either cubic splines in H^2 or picewise linear functions in L^2 is presented. According to these plots piecewise linear functions seem to be closer to the exact sought-for quantity, however lacking the desired regularity.

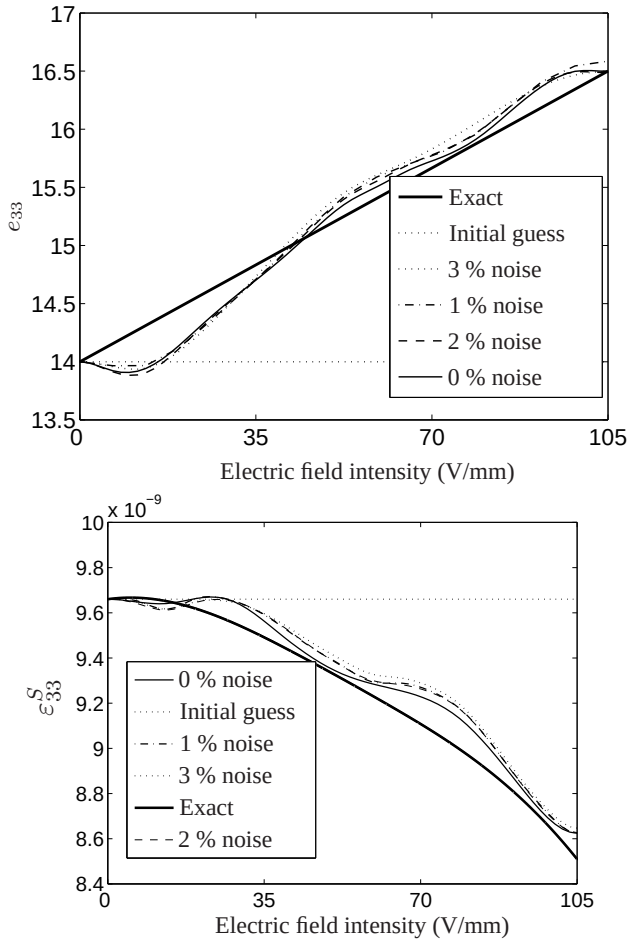


Figure 51: Simultaneously reconstructed parameter curves, with different amounts of data noise.

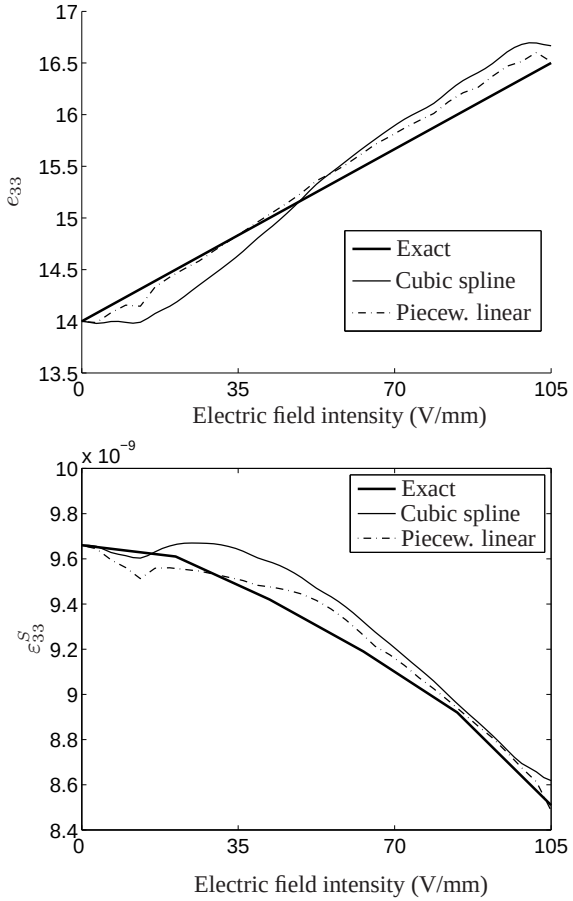


Figure 52: Simultaneously reconstructed parameter curves, noise free. Comparison: Interpolation either with cubic splines or piecewise linear functions.

The graphic in Figure 53 again shows the charge signal, now after the simultaneous reconstruction of two parameter curves for a thickness resonator. In Figure 54 the signals are computed for a longitudinal specimen.

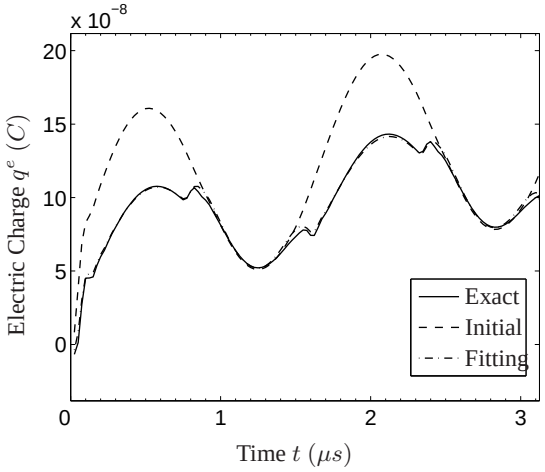


Figure 53: Charge response. Simulation with constant coefficients (dashed), with parameter curves as in Figure 45 and after reconstruction (dash-dotted), thickness resonator.

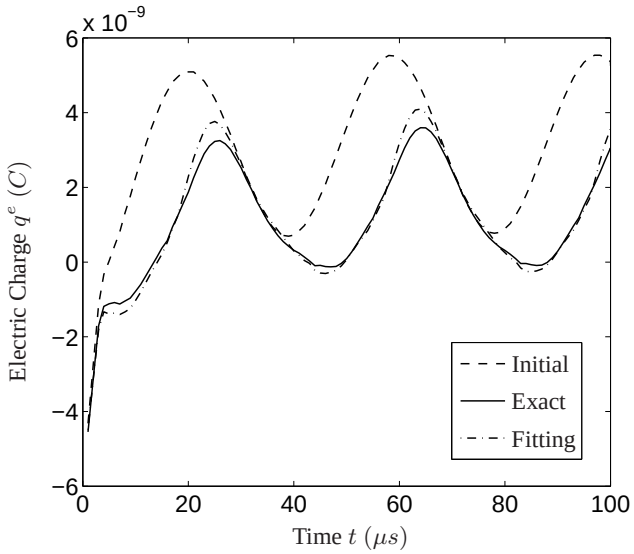


Figure 54: Charge response. Simulation with constant coefficients (dashed), with parameter curves as in Figure 45 and after reconstruction (dash-dotted) for longitudinal resonator.

4.4 Summary Chapter 4

This final chapter extends the piezoelectric problem to large-signal applications where different types of nonlinearities occur. After a detailed discussion of different nonlinear effects, we confine our investigations to material nonlinearities by considering field dependencies of the material parameters. For the parameter identification the iterative multilevel algorithm introduced in Chapter 2 is applied and a detailed description of the computation of the adjoint operator $F'(e_{33}, \varepsilon_{33}^S)^*$ is presented for the dependency of two parameters on the electric field strength. The smoothing property of the adjoint operator and the ill-posedness of the linearized problem are shown. Numerical results applied to synthetically generated data showing robustness and effectiveness of the multilevel algorithm close this chapter.

5 Summary and Outlook

5.1 Summary

The thesis at hand provides detailed results for both the mathematical forward and inverse problem in piezoelectricity. The theoretical results for the forward problem comprise, apart from the derivation of appropriate sets of partial differential equations for the piezoelectric problem, which permit the modeling of possibel losses, well-posedness results in the linear case for transient, harmonic, and static formulations. As we see, in particular for harmonic computations the range where the forward problem is well-posed is enlarged to the complete frequency space by modeling losses with complex-valued material parameters.

Being familiar with conditions under which the piezoelectric partial differential equations are well-posed, the attention of the reader is now directed to inverse problems in mathematics, mainly with the ulterior motive of parameter identification. After recalling briefly some well analyzed regularizing iterative methods for nonlinear problems, an iterative multilevel frame, once proposed by Scherzer [130], is extended to include modified Landweber iterations which by their special choice of the relaxation parameter feature much faster algorithms than the classical Landweber iteration. Convergence results of the inner modified Landweber iterations are established for finite dimensional levels considered during the iterative multilevel Algorithm 3. The assumptions on the nonlinearity comprise the tangential cone condition. Further certain regularity of the nonlinear problem at each level is additionally required in order to establish the transition from one to the next finer level. The convergence result is established for mildly and even severely ill-posed problems.

The essential nonlinearity condition which we assumed could be verified for a harmonic identification example.

Algorithm 3 together with *a posteriori* regularization parameter choices, which contain estimates on the data noise level and additionally an evaluation of the approximation error, is shown to be a regularization method, i.e the solutions obtained by Algorithm 3 converge to the exact solution of the identification problem for a series of measurements contaminated with data noise level converging to zero.

Well-known iterative regularizing methods turn out to be a robust choice for the identification of material tensors in linear piezoelectricity. However, a sensitivity analysis questions the reliability of the identification results showing large differences in credibility for the single parameters in different mode shapes. Thus, even though the fitted impedance curves give impressions of a “perfect” identification result, some parameters are not reliable even up to the sign. This demands fittings with either a higher number of sampling points or analyzing additional mode shapes. In particular the lack of reliability encourages the application of optimal experiment design methods, which by an improved choice of sampling points reduces the confidence intervals of all or a selected subset of parameters. To the best of our knowledge a new approach with a weight function is proposed, as well as

the combination of two approaches, where one considers on the one hand a fixed and on the other hand a variable number of measurement locations.

An additional adjustment of the imaginary parts of complex-value material parameters in harmonic analysis allows for a mathematical design of piezoelectric transducers including mechanical and electrical losses as well as imperfect piezoelectric coupling.

However, in practical applications most of the piezoelectric actuator applications are driven under large signals requiring appropriate nonlinear models to allow for a computer-based design or implementing efficient controlling devices. Among the different kinds of nonlinearities this thesis is confined to material nonlinearities, i.e. we model the entries of the material tensors as functions of the mechanical strain and the electric field. Mainly, the derivation of the iterative multilevel algorithm discussed in Chapter 2 is motivated by the present demand of recovering these functional dependencies. Thus we aim at detecting infinite dimensional sought-for quantities which are however approximated with increasing exactness during the identification scheme.

Numerical results of chapter 4 show efficiency of such a multilevel approach applied to the ill-posed problem of parameter curve identification. The implemented strategy profits from the inherent regularizing effects of coarse discretization but also allows for adequate fine resolution of the sought-after parameter functions at an advanced state of the algorithm as shown by the reconstruction reconstruction results for synthetically generated data.

5.2 Outlook

According to our current perception this thesis answers a wide spectrum of topics of direct and inverse problems in piezoelectricity. However, this treatise also raises new scientific questions which are for both, the mathematical and engineering side, of particular interest.

Concerning the well-posedness of the different partial differential equations for the piezoelectric problem it remains open to which extent the results can be generalized to the nonlinear case considering functional dependencies on spatial derivatives of the quantities electric potential and mechanical displacement.

During the identification of linear material parameters no attention has yet been paid to the state of polarization of the single ceramics. Polarized piezoelectric ceramics however alter in time, i.e. the grade of polarization decreases gradually. By recording the time passed after the polarization one may detect the parameters as functions of time for both the unstressed and stressed case.

The multilevel scheme discussed in Chapter 2 may serve as a framework in which the inner method may be exchanged by arbitrary regularizing iterative schemes, e.g. (inexact) Newton type methods. Convergence results, after establishing monotonicity of the iterates at fixed level, are to be derived. In particular, propositions including error estimates of the first iterate on a new level compared to estimates on a prior one are encountered to be the key steps in the outstanding analyses.

Further, the iterative multilevel algorithm allows only for a refinement when switching from one level to the next. However, optional coarsening may reduce potentially occurring oscillations in the sought-for quantities. V- or W- cycles known from multigrid methods might be applied.

Certainly, discretization of the measured data, which corresponds to finite numbers of time steps or frequency locations, provides an amount of regularization which may not be underestimated. Thus, one may also consider a variable discretization in image-space correlated to the process of identification.

Until know, the determination of functional dependencies with simulation-based identification routines has not been applied to real world measurements. It is open yet, if the functional dependencies can be reduced to one or two unknowns as it is done in this thesis and how appropriate measurements need to be conducted in order to mask interactions with other tensor entries.

It is also conceivable, that one may try to detect parameter curves, in particular for the piezoelectric coupling and permittivity, for increasing and decreasing signals, respectively, so that “weak hysteresis effects” can partially be taken into account.

5.3 Zusammenfassung

Die vorliegende Arbeit offeriert ausführliche Ergebnisse zu sowohl dem mathematischen Vorwärts- wie auch dem inversen Problem in der Piezoelektrik. Die theoretischen Ergebnisse für das Vorwärtsproblem umfassen neben der Herleitung von geeigneten partiellen Differentialgleichungen für das piezoelektrische Problem, welches die Modellierung von Verlusten erlaubt, Ergebnisse zur Wohlgestelltheit im linearen Fall für zeitabhängige, harmonische und statische Formulierungen. Wie man sieht, kann speziell für harmonische Rechnungen der Bereich in dem das Problem wohlgestellt ist auf den ganzen Frequenzraum durch die Modellierung von Verlusten mit komplexwertigen Materialparametern erweitert werden.
Nachdem man mit den Bedingungen, unter denen die piezoelektrischen partiellen Differentialgleichungen wohlgestellt sind, vertraut ist, kann die Aufmerksamkeit des Lesers auf das inverse Probleme in der Mathematik gelenkt werden; dies hauptsächlich mit dem Hintergedanken der Parameteridentifizierung. Nachdem ein knapper Überlick über gut analysierte, regularisierende iterative Verfahren für nichtlineare Probleme gegeben wurde, wird das Gerüst eines iterativen Multilevel Verfahrens, welches einst von Scherzer [130] entwickelt wurde, dahin erweitert, dass es modifizierte Landweber Iterationen beinhaltet. Diese stellen durch ihre spezielle Wahl des Relaxationsparameters deutlich schnellere Algorithmen im Vergleich zur klassischen Landweber Iteration dar.
Konvergenzresultate der modifizierten Landweber Iterationen werden auf jedem betrachteten endlich-dimensionalen Level des Multilevelalgorithmus 3 entwickelt. Die Annahmen an die Nichtlinearitätsbedingung enthalten die Tangentialkegelbedingung. Zusätzlich muss eine gewisse Regularität des nichtlinearen Problems auf jedem Level gefordert werden, um einen Übergang von einem zum nächsten Level zu ermöglichen. Die Konvergenzergebnisse sind für schwach und sogar stark schlecht gestellte Probleme entwickelt. Es wird gezeigt, dass der Algorithmus 3 zusammen mit *a posteriori* Regularisierungsparameterstrategien, welche Abschätzungen des Fehlerniveaus der Daten sowie zusätzlich eine Bewertung des Approximationsfehlers beinhalten, eine regularisierende Methode darstellt. Dies bedeutet, dass für eine Reihe von Messungen mit einem gegen Null strebenden Fehlerniveau die Lösung des Algorithmus 3 gegen die exakte Lösung des Identifizierungsproblems konvergiert.
Die Annahme an die Nichtlinearität kann anhand eines harmonischen Identifikationsproblems verifiziert werden.
Wohlbekannte iterative Regularisierungsverfahren zeigen sich als eine stabile Wahl bei der Identifizierung von Materialtensoren in linearer Piezoelektrik. Jedoch klärt eine Sensitivitätsanalyse über die Verlässlichkeit der identifizierten Parameter auf, welche große Unterschiede in der Vertrauenswürdigkeit der einzelnen Parameter für verschiedene Schwingungsformen aufzeigt. So, obwohl die angepasste Impedanzkurve den Eindruck eines “perfekten“ Fittings vermittelt, können einige Parameter sogar bis auf das Vorzeichen nicht vertrauenswürdig sein. Dies erfordert Identifizierungen mit einer grösseren Anzahl von Messwerten oder die Analyse von

weiteren Schwingungsformen. Speziell der Mangel an Vertrauenswürdigkeit motiviert die Anwendung von Methoden der optimalen Versuchsplanung, welche bei einer verbesserten Auswahl von Messpunkten die Konfidenzintervalle aller oder die einer Teilmenge der Parameter reduziert. Ein unserer Erkenntnis nach unveröffentlicher Ansatz mit einer Gewichtsfunktion wird vorgestellt, sowie die Kombination von zwei verschiedenen Ansätzen, in denen einerseits von einer festen und andererseits von einer variablen Anzahl von Messpunkten ausgegangen wird.
Eine zusätzliche Anpassung der imaginären Anteile der komplexwertigen Materialparameter erlaubt harmonische Analysen piezoelektrischer Wandler, wobei mechanische, elektrische Verluste sowie nichtperfekte piezoelektrische Koppelungen berücksichtigt werden.
In praktischen Anwendungen werden jedoch die meisten piezoelektrischen Aktoren unter großen Signalen betrieben, was entsprechende nichtlineare Modelle erfordert, die ein rechnergestütztes Design oder die Programmierung von effizienten Kontrolleinheiten erlaubt.
Unter den verschiedenen Arten von Nichtlinearitäten beschränken wir uns in dieser Arbeit auf Materialnichtlinearitäten, was bedeutet, dass wir die Einträge der Materialtensoren als Funktionen der mechanischen Dehnung und der elektrischen Feldstärke betrachten.
Die Entwicklung des iterativen Multilevelalgorithmus von Kapitel 2 ist hauptsächlich durch die gegenwärtige Notwendigkeit der Identifizierung dieser funktionalen Abhängigkeiten motiviert. Somit sind wir hier bestrebt, unendlichdimensionale Unbekannte zu detektieren, welche mit wachsender Genauigkeit während des Identifikationsprozesses diskretisiert werden.
Die numerischen Ergebnisse des Kapitels 4 zeigen die Effizienz eines solchen Multilevelansatzes, welcher auf das schlechtgestellte Problem der Identifizierung von Parameterkurven angewandt wird. Die implementierte Strategie profitiert von den inhärenten regularisierenden Effekten durch grobe Diskretisierungen einerseits, erlaubt aber auch eine adäquate Auflösung der gesuchten Parameterfunktionen zu einem fortgeschrittenen Zeitpunkt des Algorithmus, was Rekonstruktionsergebnisse von synthetisch generierten Daten zeigen.

5.4 Ausblick

Unserer derzeitigen Erkenntnis nach beantwortet diese Dissertation ein weites Spektrum von Themen, die direkte und inverse Probleme der Piezoelektrizität betreffen. Jedoch wirft diese Abhandlung neue wissenschaftliche Fragen auf, die sowohl für die mathematischen wie auch von ingenieurstechnischer Seite von speziellem Interesse sind.
Es bleibt offen, bis zu welchem Grade die Wohlgestelltheit der verschiedenen partiellen Differentialgleichungen für das piezoelektrische Problem auf den nichtlinearen Fall verallgemeinert werden können, wobei funktionale Abhängigkeiten der Materialtensoren von den örtlichen Ableitungen des elektrischen Potentials und der mechanischen Verschiebung betrachtet werden sollen.

Während der Identifizierung der linearen Parameter wurde bislang der Zustand der Polarisation der einzelnen Keramiken nicht berüchsichtig. Polarisierte Keramiken hingegen ”altern“ im Laufe der Zeit, sprich der Grad der Polarisierung nimmt stetig ab. Die Zeit messend, welche seit der Polarisierung verstrichen ist, könnte es erlauben, die Parameter als Funktionen der Zeit zu identifizieren, sowohl für den unbelasteten wie auch den belasteten Fall.
Der Multilevelalgorithmus, welcher in Kapitel 2 diskutiert wird, kann als Rahmen dienen, in welchem die innere Methode durch beliebige regularisierende iterative Methoden ersetzt werden kann, zum Beispiel (inexakte) Newton-artige Verfahren. Konvergenzaussagen, nachdem Monotonie der Iterierten auf einem fixen Level garantiert wird, müssen gezeigt werden. Hier werden insbesondere Aussagen, welche Fehlerabschätzungen der ersten Iterierten auf einem neuen Level verglichen mit den Iterierten auf einem vorhergegangenen Level betreffen, als wichtigste Schritte der ausstehenden Analyse angesehen.
Weiterhin erlaubt der iterative Multilevelalgorithmus nur Verfeinerungen beim Übergang von einem zum anderen Level. Jedoch können optionale Vergröberungen mögliche auftretende Oszillationen in den gesuchten Größen reduzieren. V- oder W-Zyklen, wie man sie von Multigrid Methoden her kennt, könnten angewendet werden.
Sicherlich stellt auch die Diskretisierung der gemessenen Daten, was einer endlichen Anzahl von Zeitschritten oder Frequenzpunkten entspricht, eine nicht zu unterschätzende Regularisierung dar. Somit könnte man ebenso eine variable Diskretisierung im Bildbereich betrachten, die mit dem Fortschritt der Identifizierung korreliert.
Bislang wurde die Bestimmung der funktionalen Abhängigkeiten mit simulationsbasierten Identifikationsroutinen noch nicht auf real gemessene Daten angewandt. Es sind bislang offene Fragen, ob die funktionalen Abhängigkeiten auf ein oder zwei Unbekannte reduziert werden können, wie es in dieser Arbeit getan wurde und wie geeignete Messungen durchgeführt werden müssten, um Abhängigkeiten von weiteren Tensoreinträgen möglichst auszublenden.
Ferner wäre denkbar, dass man versucht, die Parameterkurven, speziell die der piezoelektrischen Spannungskonstante und der Permittivität für ansteigende und fallende Signale getrennt zu bestimmen, so dass ein ”schwacher Hystereseeffekt“ teilweise mit in das Modell aufgenommen werden kann.

References

[1] IEEE Standard on Piezoelectricity, 1985. IEEE/ANSI.

[2] N. Abboud, G. L. Wojcik, D. K. Vaughan, J. Mould, D. Powell, and L. Nikodym. Finite element modeling for ultrasonic transducers. *Proc SPIE Int. Symp. Medical Imaging*, 1998.

[3] M. Akamatsu and G. Nakamura. Well-posedness of initial-boundary value problems for piezoelectric equations. *Taylor and Francis Ltd, Applicable Analysis*, 81(1):129–141, 2002. DOI: 10.1080/000368102320334348.

[4] M. Akamatsu and K. Tanuma. Greens's function of anisotropic piezoelectricity. In *Proc. R. Soc. Lond. A*, volume 453, pages 473–487, 1997.

[5] H. Allik and T. J. R. Hughes. Finite element formulation for piezoelectric continua by a variational theorem. *GD/Electric Boat division report P411-68-056*, 1968.

[6] H. Allik and T. J. R. Hughes. Finite element method for piezoelectric vibration. *International Journal for Numerical Methods in Engineering*, 2:151–157, 1970.

[7] B. Andersen, E. Ringgaard, T. Bove, A. Albareda, and R. Perez. Performance of piezoelectric ceramic multilayer components based on hard and soft PZT. In Brogmann, editor, *Proc. of Actuator*, pages 419–422, 2000.

[8] M. Arafa and A. Baz. On the nonlinear behavior of piezoelectric actuators. *Journal of Vibration and Control, Sage Publications*, 10:387–398, 2004. DOI: 10.1177/1077546304033365.

[9] E. Arratia-Avendano. Numerical simulation of thermopiezoelectricity using the Finite Element Method. Master's thesis, Friedrich-Alexander-Universität Erlangen-Nürnberg, 2008.

[10] J.P. Aubin. *Applied Functional Analysis*. John-Wiley & Sons, New York, Chichester, Brisbane, Toronto, 1979.

[11] N. Aurelle, D. Guyomar, C. Richard, P. Gonnard, and L. Eyraud. Non linear behavior of an ultrasonic transducer. *Ultrasonics*, 34:187–191, 1996.

[12] H. Bandemer and W. Näther. *Theorie und Anwendung der optimalen Versuchsplanung II, Anwendung.* Berlin: Akademie-Verlag, 1979.

[13] I. Bauer, H.G. Bock, S. Körkel, and J.P Schlöder. Numerical methods for optimum experimental design in DAE systems. *Journal of Computational and Applied Mathematics*, 1-2:1–25, 2000.

[14] H. Beige and G. Schmidt. Electromechanical resonances for investigating linear and nonlinear properties of dielectrics. *Ferroelectrics*, 41:39–49, 1982.

[15] A. M. Bolkisev. Dissipation of energy during harmonic loading of a viscous-elastic piezoceramic material. *International Applied Mechanics*, 23(3):248–252, 1987.

[16] Bosch-Umwelt, Robert Bosch GmbH, 70049 Stuttgart, Germany. Presse-Information, Zukunftspreis 2005. online: http://www.bosch-umwelt.de/up/de/downloads/ PI5145_Zukunftspreis.pdf, 2007.

[17] D. Braess. *Finite Elemente*. Springer Verlag Berlin Heidelberg New-York, 2 edition, 2003.

[18] H. Brakhage. On ill-posed problems and the method of conjugate gradients. In *H.W. Engl, H.W. and C.W. Groetsch, eds., Inverse and Ill-Posed Problems*, 1987.

[19] N. Bretz, T. Lahmer, M. Kaltenbacher, E. Leder, and R. Lerch. Material parameters identification of piezoelectric transducers including the whole assembly. In *Proceedings of the IEEE Ultrasonics Symposium*, pages 804–807. Vancouver, Kanada, 2006.

[20] T. Buchukuri, O. Chkadua, and D. Natroshvili. Mathematical modelling and analysis of interaction problems for metallic-piezoelectric composite structures with regard to thermal stresses. Technical report, Georgian Technical University / Georgian Academy of Science, 2006.

[21] W. G. Cady. *Piezoelectricity - An Introduction to the Theory and Applications of Electromechanical Phenomena in Crystals*, volume 1. Dover Publications, INC, 1964.

[22] W. G. Cady. *Piezoelectricity - An Introduction to the Theory and Applications of Electromechanical Phenomena in Crystals*, volume 2. Dover Publications, INC, 1964.

[23] L. Carcione, J. Mould, V. Pereyra, D. Powell, and G. Wojcik. Nonlinear inversion of piezoelectrical transducer impedance data. *Journal of Computational Acoustics*, 9(3):899 – 910, 2001.

[24] CENELEC. European standard prEN, European Committee for Electrotechnical Standardization - Brussels. 50324-1:1998, 50324-2:1998, 50324-3:2001, 1998-2001.

[25] M. Ciarletta and A. Scalia. Thermodynamic theory for porous piezoelectric materials. *Meccanica*, 28:303–308, 1993.

[26] G. Cimatti. The piezoelectric continuum. *Annali di Matematica*, 183:495–514, 2004. DOI: 10.1007/s10231-004-0102-z.

[27] F. Colonius and K. Kunisch. Stability for parameter estimation in two point boundary value problems. *Journal für Mathematik*, 370:1–29, 1986.

[28] D. Colton and R. Kress. *Inverse acoustic and electromagnetic scattering theory*. Number 93 in Applied Mathematical Sciences. Springer-Verlag, Berlin, 1992.

[29] Computers and Structures, Berkeley, CA 94704 USA. Damping And Energy Dissipation, 2008.

[30] J. Curie and Curie P. Développement, par pression, de l'électricité polaire dans les cristaux hémièdres à faces inclinées. *Comptes rendus hebdomadaires des séances de l'Académie des sciences*, 91:294, 1880.

[31] R. S. Dembo, S. C. Eisenstat, and T. Steihaug. Inexact Newton methods. *SIAM J. Numer. Anal.*, 19(2):400–408, 1982.

[32] X.H. Du, Q.M. Wang, and K. Uchino. An accurate method for the determination of complex coefficients of single crystal piezoelectric resonators I: Theory. *IEEE UFFC*, 51:227–235, 2004.

[33] X.H. Du, Q.M. Wang, and K. Uchino. An accurate method for the determination of complex coefficients of single crystal piezoelectric resonators II: Design of measurements and experiments. *IEEE UFFC*, 51:238–248, 2004.

[34] B. Eicke, A. Louis, and R. Plato. The instability of some gradient methods for ill-posed problems. *Numer. Math*, pages 129–134, 1990.

[35] H. Engl, K. Kunisch, and A. Neubauer. Convergence rates for Tikhonov regularisation of non-linear ill-posed problems. *Inverse Problems*, 5:523–540, 1989.

[36] H. W. Engl, M. Hanke, and A. Neubauer. *Regularization of Inverse Problems*. Kluwer, Dordrecht, 1996.

[37] Ferroperm, Piezoceramics A/S, Kvistgård, Denmark. online: www.ferroperm-piezo.de, 2008.

[38] G. Feuillard, E. Le Clezio, F. Levassort, L.P. Tran Huu Hue, T. Delaunay, and M. Lethiecq. New tools for electromechanical characterisation of piezoceramics. *Electroceram*, 19:419–426, 2007.

[39] Fraunhofer-Institut für Betriebsfestigkeit und Systemzuverlässigkeit. Noise-absorbing windows. online: http://www.fraunhofer.de/EN/press/pi/2007/04/Mediendienst42007Thema2.jsp, 2007.

[40] E. Gagliardo. Ulteriori proprieta di alcune classi di funzioni in piu variable. *Ricerche di Mat.*, 8:24–51, 1959.

[41] G. H. Gautschi. *Piezoelectric Sensorics*. Springer-Verlag, Berlin, Heidelberg, New York, 2006.

[42] W. Geis. *Piezoelectricity in composites -Modelling, analysis and simulation-*. PhD thesis, Faculty of Mathematics and Physics, University of Stuttgart, Germany, 2007.

[43] W. Geis, G. Mishuris, and A. M. Sändig. Piezoelectricity in multilayer actuators. Modelling and analysis in two and three dimensions. Technical report, University of Stuttgart, 2003.

[44] C. Gerthsen. *Gerthsen Physik*, volume 21. Springer - Verlag Berlin, Heidelberg, New York, 2002.

[45] A. Görnandt and U. Gabbert. Finite element analysis of thermopiezoelectric smart structures. *ACTA MECHANICA*, pages 129–140, 2002.

[46] C. W. Groetsch. *Inverse Problems in the Mathematical Sciences*. Braunschweig: Vieweg, 1993.

[47] D. Guyomar, N. Aurelle, C. Richard, P. Gonnard, and L. Eyraud. Nonlinearities in Langevin transducers. In *Proceedings of the 1994 IEEE Ultrasonics Symposium*, volume 2, pages 925–928, 1994.

[48] J. Hadamard. Lectures on Cauchy's Problem in linear partial differential equations, 1923.

[49] D. A. Hall. Review nonlinearity in piezoelectric ceramics. *Journal of Material Science*, 36:4575 – 4601, 2001.

[50] D. A. Hall and P. J. Stevenson. High field dielectric behavior of ferroelectric ceramics. *Ferroelectrics*, 223:319–328, 1999.

[51] J. F. Hall. Problems encountered from the use (or misuse) of Rayleigh damping. *Earthquake Engng. Struct. Dyn.*, 35:525–545, 2006.

[52] M. Hanke. A regularizing Levenberg - Marquardt scheme, with applications to inverse groundwater filtration problems. *Inverse Problems*, 13:79–95, 1997.

[53] M. Hanke. Regularizing properties of a truncated Newton–CG algorithm for nonlinear inverse problems. *Numer. Funct. Anal. Optim.*, 18:971–993, 1997.

[54] M. Hanke, A. Neubauer, and O. Scherzer. A convergence analysis of the Landweber iteration for nonlinear ill-posed problems. *Numer. Math*, 72:21–37, 1995.

[55] P. C. Hansen. *Rank-Deficient and Discrete Ill-Posed Problems: Numerical Aspects of Linear Inversion*. SIAM Monographs on Mathematical Modeling and Computation. SIAM, 1998.

[56] T. Hegewald, B. Kaltenbacher, M. Kaltenbacher, and R. Lerch. Efficient modeling of ferroelectric behavior for the analysis of piezoceramic actuators. In *15th International Symposium on Applications of Ferroelectrics*. ISAF, 2006.

[57] B. Hofmann. *Mathematik inverser Probleme*. B. G. Teubner Stuttgart-Leipzig, 1999.

[58] R. Holland. Representation of dielectric, elastic, and piezoelectric losses by complex coefficients. *IEEE Transactions on UFFC*, 14:18–20, 1967.

[59] R. Holland and E. EerNisse. Variational evaluation of admittances of multielectroded three-dimensional piezoelectric structures. *IEEE Transactions on Sonics and Ultrasonics*, SU-15:119–132, 1968.

[60] R. Holland and E. EerNisse. Accurate measurement of coefficients in a ferroelectric ceramic. *IEEE Transactions on Sonics and Ultrasonics*, SU-16(4):173–181, 1969.

[61] J. A. Hossack and G. Hayward. Finite-element analysis of 1-3 composite transducers. *Ultrasonics, Ferroelectrics and Frequency Control, IEEE Transactions on*, 38(6):618–629, 1991.

[62] T. J. R. Hughes. *The Finite Element Method - Linear Static and Dynamic Finite Element Analysis*. Dover Publications, Inc, 1987.

[63] D. Ieşan. Reciprocity, uniqueness and minimum principles in the linear theory of piezoelectricity. *Int. J. Engng. Sci.*, 28:1139–1149, 1990.

[64] IEEE - IEEE Corporate Office, New York, USA. online: - http://www.ieee.org, 2008.

[65] IEEE-UFFC - IEEE Corporate Office, New York, USA . A history of the quartz crystal industry in the USA. online: http://www.ieee-uffc.org/fc_history/bottom.html, 2007.

[66] T. Ikeda. *Fundamentals of Piezoelectricity*. Oxford University Press, 1990.

[67] H.-W. Joo, C.-H. Lee, and H.-K. Jung. Identification of the piezoelectric material coefficients using the finite element method with an asymptotic waveform evaluation. *Ultrasonics*, pages 13–19, 2004.

[68] S. P. Joshi. Non-linear constitutive relations for piezoceramic material. *Smart Materials and Structures*, 1:80–83, 1992. doi:10.1088/0964-1726/1/1/012.

[69] J. P. Kaipio, A. Seppänen, E. Somersalo, and H. Haario. Posterior covariance related optimal current patterns in electrical impedance tomography. *Inverse Problems*, 20:919–936, 2004.

[70] B. Kaltenbacher. A projection-regularized Newton method for nonlinear ill-posed problems with application to parameter identification problems with Finite Element discretization. *SIAM J. Numer. Anal.*, pages 1885–1908, 2000.

[71] B. Kaltenbacher. Regularization by projection with a posteriori discretization level choice for linear and nonlinear ill-posed problems. *Inverse Problems*, 16:1523–1539, 2000. DOI:10.1088/0266-5611/16/5/322.

[72] B. Kaltenbacher. On the regularizing properties of a full multigrid method for ill-posed problems. *Inverse Problems*, 17:767–788, 2001. DOI:10.1088/0266-5611/17/4/313.

[73] B. Kaltenbacher. Identification of nonlinear coefficients in hyperbolic PDEs, with application to piezoelectricity. In *Optimal Control of Coupled Systems of PDEs, Hrsg.: Kunisch, Leugering, Sprekels, Tröltzsch*. Springer, 2006.

[74] B. Kaltenbacher and M. Kaltenbacher. *Modelling and iterative identification of hysteresis via Preisach operators in PDEs*. In RICAM book series, 2007.

[75] B. Kaltenbacher, M. Kaltenbacher, M. Hofer, and R. Lerch. Piezoelectric material nonlinearity identification via multiharmonic finite elements. *Ultrasonics Symposium, 2004 IEEE*, 2:1149– 1152, 2004.

[76] B. Kaltenbacher, M. Kaltenbacher, and Reitzinger S. Identification of nonlinear B-H curves based on magnetic field computations and multigrid methods for ill-posed problems. *European Journal of Applied Mathematics*, 14:15–38, 2003. DOI: 10.1017/S0956792502005089.

[77] B. Kaltenbacher, T. Lahmer, and M. Mohr. PDE based determination of piezoelectric material tensors. *European Journal of Applied Mathematics*, 17(4):383–416, 2006. DOI: 10.1017/S0956792506006474.

[78] B. Kaltenbacher, T. Lahmer, and V. Schulz. Optimal experiment design for piezoelectric material tensor identification. *Inverse Problems in Science and Engineering*, to appear, 2007.

[79] B. Kaltenbacher, A. Neubauer, and O. Scherzer. *Iterative Regularization Methods for Nonlinear Ill-Posed Problems*. To appear.

[80] M. Kaltenbacher. *Numerical Simulation of Mechatronic Sensors and Actuators*. Springer Verlag Berlin-Heidelberg, 2004.

[81] M. Kamlah. Ferroelectric and ferroelastic piezoceramics - modeling of electromechanical hysteresis phenomena. *Continuum Mechanics and Thermodynamics, Springer Berlin / Heidelberg*, 13(4):219–268, 2001.

[82] M. Kamlah and U. Böhle. Finite Element analysis of piezoceramic components taking into account ferroelectric hysteresis behavior. *International Journal of Solids and Structures*, 38:605–633, 2001.

[83] M. Kamlah and Z. Wang. A thermodynamically and microscopically motivated constitutive model for piezoceramics. Technical report, Forschungszentrum Karlsruhe in der Helmholtz Gesellschaft, 2003.

[84] A. Kirsch. *An Introduction to the Mathematical Theory of Inverse Problems*. Springer - New York, 1996.

[85] E. Kittinger, E. Tichy, and W. Friedel. Nonlinear piezoelectricity and electrostriction of alpha quartz. *Journal of Applied Physics*, 60(4):1465–1471, 1986.

[86] Kocbach. *Finite Element Modeling of Ultrasonic Piezoelectric Transducers - Influence of geometry and material parameters on vibration, response functions and radiated field*. PhD thesis, University of Bergen, Department of Physics, 2000.

[87] S. Körkel and E. Kostina. Numerical methods for nonlinear experimental design. *In H. G. Bock, E. Kostina, H. X. Phu and R. Rannacher (eds.), Modeling, Simulation and Optimization of Complex Processes, Proceedings of the International Conference on High Performance Scientific Computing,*, pages 255–272, March 10-14, 2003.

[88] K. Kuhnen. Modelling, identification, and compensation of complex hysteretic and log(t)-type creep nonlinearities. *Control and Intelligent Systems*, 33:134–147, 2005.

[89] K. Kunisch and G. Geymayer. Convergence rates for regularized nonlinear illposed problems. In *Book Series Lecture Notes in Control and Information Sciences*, volume 154, pages 81–92. Springer Berlin / Heidelberg, 1991. DOI: 10.1007/BFb0044485.

[90] K. Kunisch and J. Zou. Iterative choices of regularization parameters in linear inverse problems. *Inverse Problems*, 14:1247–1264, 1998.

[91] K. W. Kwok, H. L. W. Chan, and C. L. Choy. Evaluation of the material parameters of piezoelectric materials by various methods. *IEEE Transactions on ultrasonics, ferroelectric, and frequency control*, 44(4):733–742, 1997.

[92] D. Kybartas and A. A. Lukoševiučius. Determination of piezoceramic parameters by the use of mode interaction and fitting of impedance characteristics. *ULTRAGARSAS*, 45(4), 2002.

[93] T. Lahmer, M. Kaltenbacher, R. Lerch, B. Kaltenbacher, and E. Leder. FEM based determination of real and complex elastic, dielectric and piezoelectric moduli in piezoceramic materials. *IEEE Transactions on ultrasonics, ferroelectric, and frequency control*, 55(2):465–475, 2008.

[94] C. M. Landis. A new finite element formulation for electromechanics. In Christopher S. Lynch, editor, *Smart Structures and Materials 2002: Active Materials: Behavior and Mechanics*, volume 4699 of *Proc. SPIE*, pages 31–39, 07 2002.

[95] E. Leder, M. Kaltenbacher, B. Kaltenbacher, T. Lahmer, and M. Mohr. Determination of material parameters by inverse methods. In *In IWPMA, 2nd International Workshop on Piezoelectric Materials and Applications in Actuators, Proceedings*, pages 63–70, Heinz Nixdorf Institute, Paderborn, Germany, 2005.

[96] R. Lerch. Vorlesungsskript Sensorik, Department of Sensor-Technology, University or Erlangen-Nuremberg.

[97] R. Lerch. Simulation of piezoelectric devices by two- and three-dimensional finite elements. *IEEE Transactions on ultrasonics, ferroelectric, and frequency control*, 37(3):233–247, 1990. ISSN 0885-3010.

[98] W. P. Mason. *Piezoelectric Crystals and Their Application to Ultrasonics*. New York: Van Nostrand, 1950.

[99] W. P. Mason. Piezoelectricity, its history and applications. *The Journal of the Acoustical Society of America*, 70:1561–1566, 1981.

[100] V. N. Melnik. Existence and uniqueness theorems of the generalized solution for a class of non-stationary problem of coupled electroelasticity. *Soviet Mathematics, Izvestiya VUZ. Mathematika*, 35(4):24–32, 1991.

[101] D. Mercier and S. Nicaise. Existence, uniqueness and regularity results for piezoelectric systems. *SIAM J. Math. Anal.*, 37:651–672, 2005.

[102] B. Miara. Contrôlabilité d'un corps piézoélectrique. In *C. R. Acad. Sci. Paris, Problèmes mathématiques de la mécanique / Mathematical Problems in Mechanics*, 1, pages 267–270. 2001.

[103] D. R. A. Mohammad, N. U. Khan, and V. Ramamurti. On the role of Rayleigh damping. *Journal of Sound Vibration*, 185(2):207–218, 1995.

[104] V. A. Morozov. On the solution of functional equations by the method of regularization. *Soviet Math. Dokl*, 7:414–417, 1966.

[105] A. Mukherjee and S. Chaudhuri. Piezolaminated beams with large deformations. *International Journal of Solids and Structures*, 39:4567–4582, 2002.

[106] B.K. Mukherjee, W. Ren, S-F. Liu, A.J. Masys, and G. Yang. Non-linear properties of piezoelectric ceramics. In C. Lynch, editor, *Active Materials*, volume 4333 of *Smart Structures and Materials*, 2001.

[107] D. F. Nelson. Theory of nonlinear electroacoustics of dielectric, piezoelectric, and pyroelectric crystals. *The Journal of the Acoustical Society of America*, 63(6):1738–1748, 1978.

[108] A. Neubauer and O. Scherzer. Finite-dimensional approximation of tikhonov regularized solutions of non-linear ill-posed problems. *Numerical Functional Analysis and Optimization*, 11(1&2), 1990.

[109] A. Neubauer and O. Scherzer. A convergence rate result for a steepest descent method and a minimal error method for the solution of nonlinear ill-posed problems. *Z. Anal. Anwend.*, 14:369–377, 1995.

[110] L. Nirenberg. An extended interpolation inequality. *Annali della Scuola Normale Superiore di Pisa - Classe di Scienze Sér 3*, 20(4), 1966.

[111] W. Nowacki. Some general theorems of thermopiezoelectricity. *Journal of Thermal Stresses*, 1:171–182, 1978.

[112] American Institute of Physics. online: http://aip.org/, 2008.

[113] H. Parkus. *Mechanik der festen Körper*, volume 2. Springer - Verlag, Wien - New York, 1995.

[114] M. Pernice and H. F. Walker. NITSOL: A Newton iterative solver for nonlinear systems. *SIAM J. Sci. Comput.*, 19:302–318, 1998.

[115] A. Pica, R. D. Wood, and E. Hinton. Finite element analysis of geometrically nonlinear plate behaviour using a Mindlin formulation. *Computers and Structures*, 11:203–215, 1980.

[116] V. Piefort. *Finite Element Modelling of Piezoelectric Active Structures*. PhD thesis, Université Libre de Bruxelles, Department of Mechanical Engineering and Robotics, 2000-01.

[117] PIEZO SYSTEMS, INC. Cambridge, MA 02139, USA. History of piezoelectricity. online: http://www.piezo.com/tech4history.html, 2007.

[118] Piezoelectrics, Micromechatronics Inc, State College, Pennsylvania, USA. Piezoelectric history. online: http://www.piezoelectrics.net, 2007.

[119] D. J. Powell, J. Mould, and G. L. Wojcik. Dielectric and mechanical absorption mechanism for time and frequency domain transducer modeling. Technical report, Weidlinger Associates Inc, Los Altos, CA 94022, 1998.

[120] F. Pukelsheim. *Optimal design of experiments*. Wiley & Sons, New York, 1993.

[121] O. Rabinovitch. Geometrically nonlinear behavior of piezoelectric laminated plates. *Smart. Mater. Struct.*, 14:785–798, 2005.

[122] U. Reiländer. *Das Großsignalverhalten piezoelektrischer Aktoren*. PhD thesis, Technische Universität München, 2003.

[123] A. Rieder. On the regularization of nonlinear ill-posed problems via inexact Newton iterations. *Inverse Problems*, pages 307–327, 1999.

[124] M. K. Samal, P. Seshu, and B. K. Dutta. An analytical formulation in 3D domain for the nonlinear response of piezoelectric slabs under weak electric fields. *International Journal of Solids and Structures*, 44(14-15):4656–4672, 2007.

[125] M. K. Samal, P. Seshu, S. Parashar, U. von Wagner, P. Hagedorn, and B. K. Dutta. A finite element model for nonlinear behavior of piezoceramics under weak electric fields. *Finite Elements in Analysis and Design*, 41:1464–1480, 2005.

[126] M. K. Samal, P. Seshu, S. Parashar, U. von Wagner, P. Hagedorn, B. K. Dutta, and H.S. Kushwaha. Nonlinear behaviour of piezoceramics under weak electric fields. Part I: 3D Finite Element Formulation. *International Journal of Solids and Structures*, 43:1422–1436, 2006.

[127] M. K. Samal, P. Seshu, S. Parashar, U. von Wagner, P. Hagedorn, B. K. Dutta, and H.S. Kushwaha. Nonlinear behaviour of piezoceramics under weak electric fields. Part II: Numerical results and validation with experiments. *International Journal of Solids and Structures*, 43:1437–1458, 2006.

[128] O. Schenk and K. Gärtner. Solving unsymmetric sparse systems of linear equations with PARDISO. *Future Generation Computer Systems, available online www.sciencedirect.com*, pages 475–487, 2004.

[129] O. Scherzer. A convergence analysis of a method of steepest descent and a two-step algorithm for nonlinear ill-posed problems. *Numer. Funct. Anal. and Optimiz.*, pages 197–214, 1996.

[130] O. Scherzer. An iterative multi level algorithm for solving nonlinear ill-posed problems. *Numer. Math.*, pages :579–600, 1998.

[131] O. Scherzer and T. Strohmer. A multi-level algorithm for the solution of moment problems. *Numerical Functional Analysis and Optimization*, 19(3&4):353 – 375, 1998.

[132] V. Schulz. Compact course on nonlinear optimization and inverse problems, University of Trier. Lecture Notes, 2005.

[133] T. I. Seidman and C. R. Vogel. Well posedness and convergence of some regularisation methods for non-linear ill posed problems. *Inverse Problems*, 5:227–238, 1989.

[134] N. Setter. *ABC of Piezoelectricity and Piezoelectric Materials*, chapter 1. Ceramics Laboratory, EPFL, 2002.

[135] N. Setter, editor. *Wersing: Applications of Piezoelectric Materials: An Introductory Review*, chapter 2. Siemens AG, Corporate Technology, Munich, Germany, 2002.

[136] F. Shang, M. Kuna, and M. Scherzer. Development of finite element techniques for three-dimensional analyses of thermo-piezoelectric materials. *Transactions of the ASME, Journal of Engineering Materials and Technology*, 125:18–21, 2003.

[137] S. Sherrit, H. D. Wiederick, and B. K. Mukherjee. Non-iterative evaluation of the real and imaginary material constants of piezoelectric resonators. *Ferroelectrics*, 134:111–119, 1992.

[138] S. Sherrit, H. D. Wiederick, and B. K. Mukherjee. A complete characterization of the piezoelectric, dielectric, and elastic properties of Motorola PZT 3203 HD, including losses and dispersion. *SPIE Proceedings*, 3037-22:158–169, 1997.

[139] R. Simkovic, H. Landes, M. Kaltenbacher, and R. Lerch. Nonlinear finite element analysis of piezoelectric transducers. In *IEEE Ultrasonics Symposium*, pages 1057–1060, 1999.

[140] R. Simkovics. *Nichtlineares piezoelektrisches Finite-Elemente-Verfahren zur Modellierung piezokeramischer Aktoren*. PhD thesis, Friedrich-Alexander Universität Erlangen-Nürnberg, Germany, 2002.

[141] R. C. Smith. *Smart material systems, model development*. Frontiers in applied mathematics. SIAM, 2005.

[142] Jan G. Smits. Iterative method for accurate determination of the real and imaginary parts of the materials coefficients of piezoelectric ceramics. volume 6, SU-23 of *IEEE Transactions on Sonics and Ultrasonics*, November 1976.

[143] O. Sosnicki, F. Barillot, M. Leger, and F. Claeyssen. Active damping of vibrations applied on ski structures. Technical report, Cedrat Technologies S.A., Meylan and Skis Rossignol, Voiron, France, 2006.

[144] Yu.N. Subbotin. Extremal functional interpolation and approximation by splines. *Mat. Zametki*, 15(5):843–854, 1974.

[145] A. Tarantola. *Invere Problem Theorey and Methods for Model Parameter Estimation*. SIAM - Philadelphia, 2005.

[146] TASI - Technical Software, Kingston, Ontario, Canada. Prap technical information. online: http://www.tasitechnical.com/praptechspectra.html, 2008.

[147] N. Turbé and G. A. Maugin. On the linear piezoelectricity of composite materials. *Mathematical Methods in the Applied Sciences*, 14:403–412, 1991.

[148] Woldemar Voigt. *Lehrbuch der Kristallphysik*. Leipzig, 1910.

[149] J. Wloka. *Partielle Differentialgleichungen*. Teubner, Stuttgart, 1982.

[150] L. Q. Yao, J. G. Zhang, L. Lu, and M. O. Lai. Nonlinear extension and bending of piezoelectric laminated plate under large applied field actuation. *Smart Mater. Struct.*, 13:404–414, 2004.

[151] Y. H. Zhou and J. Wang. Vibration control of piezoelectric beam-type plates with geometrically nonlinear deformation. *International Journal of Non-Linear Mechanics*, 39:909–920, 2004.

Lebenslauf



Zur Person

Tom Lahmer

geboren am **03. Januar 1978** in Bonn

ledig, ohne Kinder

Wohnanschrift: Luitpoldstr. 50
91052 Erlangen

E-Mail: **tom@lahmer.com**

Ausbildung

2004 - 2008	Promotion an der Friedrich-Alexander -Universität Erlangen-Nürnberg am Lehrstuhl für Sensorik, Thema: *Direct and Inverse Problems in Piezoelectricity.*
2005	Teilnahme am *Special Radon Semester on Computational Mechanics* am Johann Radon Institute for Computational and Applied Mathematics (RICAM), Linz.
1998 - 2004	Studium der angewandten Mathematik an der Technischen Universität Bergakademie Freiberg. Vertiefungsrichtung: Modellierung, wissenschaftliches Rechnen. Nichtmathematisches Nebenfach: Kommunikationstechnologie und Wirtschaftswissenschaften. Diplomarbeit: *Gebietszerlegungsmethoden für schwach nichtlineare Gleichungssysteme*.
2000 - 2001	Auslandsaufenthalt an der Università degli Studenti di Roma, La Sapienza, in Rom.
1997-1998	Zivildienst an der katholischen Hochschulgemeinde in Köln.
1984-1997	Besuch der Grundschule und des Nicolaus-Cusanus Gymnasiums in Bonn-Bad Godesberg.

Praktische Erfahrungen

Praktikant und Werkstudent bei der inform. Consult GmbH (Lotus Notes Dienstleister) in Köln in den Zeiträumen Februar-März 1999, August-September 1999, August-Oktober 2001.

Studentische Hilfskraft an der Fakultät für Mathematik und dem akademischen Auslandsamt in Freiberg.

Teilnahme an Wissenschaftlichen Tagungen

2005	Jahrestagung der Gesellschaft für Angewandte Mathematik und Mechanik, Luxemburg
2005	Miniworkshop "Direct and Inverse Problems in Piezoelectricity", Linz
2006	Jahrestagung der Gesellschaft für Angewandte Mathematik und Mechanik, Berlin
2006	17th International Conference on Domain Decomposition Methods, St. Wolfgang/Strobl
2006	Second International Workshop "Direct and Inverse Problems in Piezoelectricity", Hirschegg (Kleinwalsertal)
2007	Seminar Vortrag am Centre for Advanced Computing and Emerging Technologies, University of Reading, U.K
2007	6th International Congress on Industrial and Applied Mathematics, Zurich
2007	Chemnitz Symposium on Inverse Problems
2007	Third International Workshop "Direct and Inverse Problems in Piezoelectricity", Unteroewisheim

Mitgliedschaft

seit 2004	GAMM (Gesellschaft für Angewandte Mathematik und Mechanik)

Erlangen, 24. Juni 2008